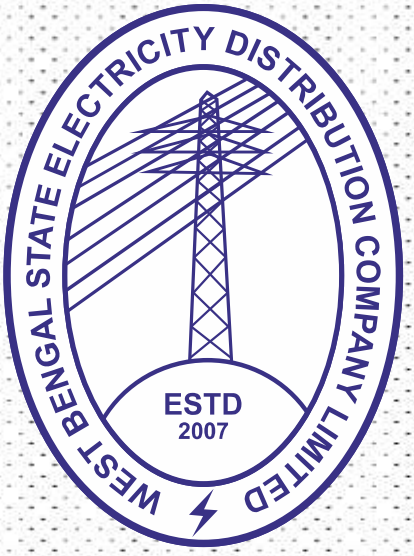
WBSEDCL

SAFETY MANUAL 2025

WEST BENGAL STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED

Vidyut Bhavan, Bidhannagar, Kolkata - 700091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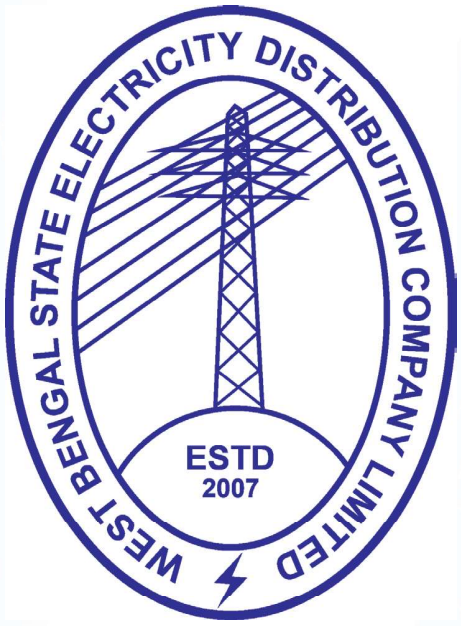

Corporate Safety Cell
HR Directorate

September, 2025

WBSEDCL

SAFETY MANUAL 2025

FOREWARD

Electricity - one of the most vital enablers of modern society, is inherently dangerous to handle without utmost care and precision. It demands the highest standards of caution and competence in its handling. Its generation and distribution are high-risk operations that not only demand technical excellence but also a strong organization-wide commitment to safety.

Over the years, WBSEDCL has shown consistency, professionalism and dedication in serving the community, often working under challenging and hazardous situations. It is our duty as an organization to ensure that the working environment is governed by the highest standards of safety and operational discipline.

We must also acknowledge that unsafe infrastructure, poor workmanship, and negligence at any level significantly increase risks—not only to individuals but to the stability and reliability of our power network. Quality and safety must go hand-in-hand and can never be compromised in pursuit of speed or cost-efficiency.

I am pleased to present this revised edition of the **Safety Manual**—a testament to our unwavering commitment to the health, safety, and wellbeing of every individual who contributes to the functioning of our organization as well as the community at large. This revised manual has embedded updated regulatory guidelines, technological advancements, and field-level learnings in it to provide clear guidelines and procedures to ensure a safe and healthy working environment, reduce accidents and minimize occupational hazards. It is designed not only as a reference document but as a daily guide—supporting each of us in making safe, informed and responsible decisions by fostering a culture of safety and ensuring compliance with all relevant laws and regulations. We aim to protect our workplace and promote well-being in the workplace.

I urge every employee, contractor, agency worker and partner of our organization to familiarize themselves with this revised safety manual and incorporate its principles into their daily work.

Let us commit ourselves to a work culture where no task is so urgent or important that we cannot take the time to do it safely and do it right.

24/9/25.

(Santanu Basu)
CHAIRMAN & MANAGING DIRECTOR
WBSEDCL

SAFETY MANUAL 2025

FOREWARD

THE SAFETY OF THE PEOPLE SHALL BE THE HIGHEST LAW'
Marcus Tullius Cicero, Roman Philosopher

WBSEDCL as a premier public power distribution utility has a long-standing commitment to both workplace and public safety, with the singular goal of 'zero harm'. Whether in the field or in the office, we reinforce our 'safety-first' message through comprehensive training and procedures, while also help to create a culture of caring. Our passion for safety reaches beyond our workforce. We proactively engage with and educate consumers we serve about electrical safety.

WBSEDCL's updated Safety Manual published under the Human Resource Directorate incorporating the latest Environment Health & Safety (EHS) regulations, along with relevant procedures, for occupational and community safety will be a ready source for safety compliance information and safe work practices.

A company's safety environment is a reflection of the organisation's overarching culture and the people who work in it. Creating a safe ambience is an integral part of loss control efforts, empowering the organisation to protect its employees, safeguard assets and control costs. The Safety Manual will help instill a set of values, beliefs, and behaviors influencing how employees. prioritise and perform tasks in a safe and reliable manner. A strong safety custom will eliminate accidents, injuries and losses as well as improve customer satisfaction, employee morale, and regulatory compliance.

I extend rich compliments to everyone who has put in sincere effort to compile and publish the Safety Manual, which will certainly go a long way in improving safety culture through leadership development, workplace training, safety governance, workshops and improved awareness programmes.

(Avijit Kumar Latua)
DIRECTOR (HUMAN RESOURCE)
WBSEDCL

SAFETY MANUAL 2025

FOREWARD

It gives me immense satisfaction to see that the new edition of Safety Manual, prepared with the objective of strengthening a culture of safety across our organization is going to be published. Electricity, while being the lifeline of modern society, also carries inherent risks. For those of us entrusted with its generation, transmission, and distribution, safety is not just a regulation to be followed—it is a responsibility, a commitment, and a way of life.

This manual is intended to serve as a practical guide for all employees, contractors, and stakeholders who are engaged in our operations. It lays down standard operating procedures so that every task, however routine or complex, is carried out with utmost care and precaution.

Let us remember that safety is not a one-time effort, but a continuous discipline. Each of us has a role to play in preventing accidents, protecting lives, and preserving assets. By adhering to the guidelines in this manual, we not only safeguard ourselves but also contribute to the efficiency, reliability, and reputation of our company.

I urge everyone to study this manual carefully, follow it diligently, and cultivate the habit of "safety first" in every activity. Together, let us ensure that our workplaces remain safe, our people remain protected, and our company continues to grow with responsibility and pride.

22/09/25

(Sumit Mukherjee)
DIRECTOR (Distributation)
WBSEDCL

SAFETY MANUAL 2025

FOREWARD

Safety of the employees, assets and the general people is of paramount importance of the Company. WBSEDCL has strengthened the Distribution Network to great extent maintaining the declared Safety Norms of the Company.

In view of the growing complexity of the electrical network, it is the need of the time to upgrade the Safety Manual which is in vogue since 2021 by incorporating the current Safety Norms guided by CEA Regulations,2023, OSH Code 2020 and PCB Regulation requirement 2016.

I would urge all concerned to follow the Safety Manual 2025 to maintain highest standard of Safety Norms, adherence to the safety practices and develop the behavioral safety approach using practical techniques to teach and improve observable skills and behavior, focusing workplace conduct, communication, emotional intelligence and leadership.

I am sure the Safety Manual 2025 will help a great deal to enhance the standard of Safety in construction activities being undertaken through different projects schemes.

(Parth Pratim Mukherjee)
DIRECTOR (PROJECTS)
WBSEDCL

SAFETY MANUAL 2025

FOREWARD

WBSEDCL is committed to ensure "Zero Accident" in all its offices and workplaces. It has its own safety manual published long ago. With the passage of time, the business field of WBSEDCL has expanded to a large extent. Simultaneously the risk factors in the offices and workplaces have also increased manifold. To fulfil the commitment of ensuring Zero accident as well as supplying 24x7 electricity to its consumers in line with growing demand of power, WBSEDCL has taken up updating the safety manual and incorporating measures to cope up with the new risk factors.

Safety should be the topmost priority for everyone involved in the work, be it our employee, our contractual workers, our casual employees and last but not the least our consumers. Safety is not just avoiding accidents or injuries. It is also about creating a safe and healthy work environment for everyone involved. It creates a culture of safety so that everyone feels comfortable in the workplace. It should be kept in mind that that by prioritizing safety we are not only protecting ourselves and our colleagues but we are also contributing to the success of our work.

I congratulate the entire team involved in the most timely and precious work of revising the safety manual. I appeal to all employees, contractual workers and all concerned person to follow the safety manual and ensure a safe working environment.

(Subhasankar Debsarma Biswas)
DIRECTOR (GENERATION)
WBSEDCL

SAFETY PLEDGE

By

WBSEDCL

WBSEDCL is committed to provide a safe & healthy work place to all employees. We further pledge to consider worker and customer safety in every decision we make. Our mission as a company is to protect customers, other people of West Bengal from any danger arising out of distribution and use of electricity as well as for prevention of accidents & protection of environment in the interest of ourselves, our family, our community & the Nation at large.

SAFETY PLEDGE BY EMPLOYEE OF WBSEDCL

I, Sri/Smt …… Designation…… an employee of WBSEDCL, in order to promote an uncompromising safety environment and to ensure the well-being of all employees including contractor employees working towards continuous improvement in safety excellence, do hereby make the following commitment. I will

a) Work safely as a condition of employment & accept responsibility for my own safety & others.

b) Always put safety first in completing my duties.

c) Inspect the equipment prior to the start of work, report any unsafe condition & never operate unsafe equipment

d) Report all safety incidents immediately.

e) Know & practise all site safety rules.

f) Take all safety precautions in performing my duties.

g) Never sacrifice or compromise safety to perform a job quicker or easier.

h) Encourage safe work habits & behaviour that support the company's vision & values.

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LIST OF ABBREVIATIONS

AB Cable	Aerial Bunched Cable
AC	Alternating Current
ACDB	Alternating Current Distribution Box
ACES INFOTECH	Advanced Computer Education & Software INFOTECH
ACM	Assistant Charge Man
AE	Assistant Engineer
AIMA	All India Management Association
ALDC	Area Load Despatch Centre
AMF	Automatic Mains Failure
ASCI	Administrative Staff College of India
ATT & CK	Adversarial Tactics, Techniques, and Common Knowledge
CBIP	Central Board of Irrigation and Power
CCC	Customer Care Centre
CCTV	Closed Circuit Television
CERT	Computer Emergency Response Team
CM	Charge Man
CPRI	Central Power Research Institute
CT	Current Transformer
DCDB	Direct Current Distribution Box
DCP	Dry Chemical Powder
DG	Diesel Generator
DP	Double Pole
DTR	Distribution Transformer
EETI	Electricity Employees' Training Institute
EHV	Extra High Voltage
ELCB	Earth Leakage Circuit Breaker
ERLDC	Eastern Regional Load Despatch Centre
ESCI	Engineering Staff College of India
ESI	Employees' State Insurance
FTA	Fault Tree Analysis
GFCI	Ground Fault Circuit Interrupter
GI	Galvanized Iron
GVT	Guide Vane Transducer
HAZOP	Hazard and Operability Study
HFL	Highest Flood Level
HIRA	Hazard Identification and Risk Assessment
HPS	Hydel Power Station

HQ	Head Quarter
HRDD	Human Resource and Development Department
HT	High Tension
HV	High Voltage
HVAC	Heating, Ventilation, and Air Conditioning
ID	Identification Document
IE Rules	Indian Electricity Rules
IEEE	Institute of Electrical and Electronics Engineers
IIM	Indian Institute of Management
IISWBM	Indian Institute of Social Welfare and Business Management
IR	Insulation Resistance
IS	Indian Standard
IT-OT	Information Technology - Operational Technology
KJTC	Kalyani Junior Training Centre
LA	Lightning Arrester
LED	Light Emitting Diode
LOI	Letter of Intent
LOTO	Lockout /Tagout
LT	Low Tension
LTOH	Low Tension Overhead
LV	Low Voltage
MFA	Multi Factor Authentication
MITRE	Massachusetts Institute of Technology Research Establishment
MV	Medium Voltage
NIELIT	National Institute of Electronics and Information Technology
NPTI	National Power Training Institute
NSC	National Safety Council
NTPC	National Thermal Power Corporation
OH	Overhead
OHE	Overhead Electrification
PCB	Polychlorinated Biphenyls
PCC	Prestressed Cement Concrete
PF	Provident Fund
PMI	Power Management Institute
PPE	Personal Protective Equipment
PPSP	Purulia Pumped Storage Project
PRV	Pressure Relief Valve
PT	Potential Transformer

PTR	Power Transformer
PVC	Polyvinyl Chloride
RCC	Reinforced Cement Concrete
RCCB	Residual Current Circuit Breaker
RCD	Residual Current Device
RECIPMT	REC Institute of Power Management & Training
RMU	Ring Main Unit
RVT	Runner Vane Transducer
SCADA	Supervisory Control and Data Acquisition
SF6	Sulphur Hexafluoride
SLDC	State Load Despatch Centre
SPGD	Solar Power Generation Department
T&AA	Technical and Administrative Approval
UG	Under Ground
UIDAI	Unique Identification Authority of India
UPS	Uninterrupted Power Supply
USB	Universal Serial Bus
VCB	Vacuum Circuit Breaker
VCS	Visible Cut-Off Switch
WPB	Wide Parallel Beam
XLPE	Cross-linked Polyethylene
XLRI	Xavier Labour Relations Institute

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CHAPTER-I

SAFETY POLICY, OBJECTIVE, PURPOSE & SCOPE

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CHAPTER-1

SAFETY POLICY, OBJECTIVE, PURPOSE & SCOPE

1. PROLOGUE

Safety in electrical distribution systems is essential to prevent severe injuries and fatalities from shocks and arc flashes, reduce fire hazards by controlling faults, ensure reliable operation by preventing equipment failure, and comply with regulations.

This Safety Manual reflects the unwavering commitment of West Bengal State Electricity Distribution Company Limited (WBSEDCL) to the health and safety of its employees and the community. This manual is established to provide clear guidelines and procedures to ensure a safe and healthy working environment, reduce accidents, and minimize occupational hazards. We believe that safety is a shared responsibility, and this document outlines the roles and expectations for both management and employees in achieving these critical objectives. By fostering a culture of safety and ensuring compliance with all relevant laws and regulations, we aim to protect our workforce and promote well-being in the workplace.

Accordingly, this corporate safety manual has been revised in line with the latest environment, health and safety laws and regulations of Government of India including The Central Electricity Authority (Measures Relating to Safety and Electric Supply) Regulations, 2023; Health and Safety (H&S) Code, 2020 and Polychlorinated Biphenyls (PCB) Regulations, 2016, etc.

In the preparation of this manual every effort has been made to make it complete yet practical while compiling the safety norms, procedures and rules. Attempt has been made to make it comprehensive and compliable with all applicable current laws and standards. The responsibilities of the employees at all level are also emphasized.

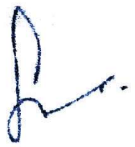
1.1 SAFETY POLICY

WBSEDCL is fully committed to meet its responsibilities under all applicable legal & other requirements. To comply with the statutory and common law duties, the company has arranged insurance against liability for death, and medical assistance for injuries & diseases suffered by any of its employees arising out of and in course of employment. Company employees agree to comply with their individual duties under both the Health & Safety policy.

The Company will comply with its duties towards employees for health & safety and working conditions so far as is reasonably practicable by:

- Providing and maintaining plant and systems of work that are safe and without risks to health, a safe place of work, a safe system of work.
- Ensuring the safety & removing risks to company employee, members of public in general and consumer, contractor employee & animals during distribution of electricity.
- Providing sufficient information (regarding generation & distribution), instructions (operation &maintenance), training and supervision as may be necessary to ensure the health & safety at work of its employees.
- Doing Hazard Identification & Risk assessment before starting any routine or non-routine activities & accordingly taking appropriate mitigating/ protective measures.
- Providing employees suitable personal protective measures as necessary.

This policy is applicable to company employees, contractual employees, general public and lawful visitors to the company's premises. All are requested to adhere to this policy.



Chairman & Managing Director

1.2 OBJECTIVES OF SAFETY POLICY

Main objectives of safety policy include:

a) The intent of the policy is to provide the safety norms to be adopted for the activities being carried out by WBSEDCL and thus prevent injury, occupational diseases, loss of life and damage to assets.

b) Promoting of a healthy culture to ensure and be responsible for safety and health of employees, workers and other persons on the work premises.

c) Complying with all applicable safety rules and regulations.

d) Identifying hazards that can cause accidents.

e) Providing Safety Training to employees and contractors' workers.

f) Creation of Safety Committees, taking necessary actions to implement safety plans and to communicate the findings to the top management.

g) Integrating safety procedures with every field of work of the Company.

h) Monitoring safety performance by tracking and following up accident enquiries & taking preventive & corrective actions and addressing hazardous situation or practices.

i) Engagement of contractors who are committed to adopt the safety rules & procedures in their works.

1.3 PURPOSE & SCOPE

The importance of safety in today's context of large-scale Construction, Operation, and Maintenance of electrical installations cannot be avoided. There is a need for fundamental change in our approach towards work practice and culture. All employees are required to be brought under safety. It is to be remembered that "SAFETY IS FOR EVERYONE, OF EVERYONE AND BY EVERYONE".

Safety has to be started and practiced by all. But at the same time, it has to be driven from the top at all levels of the Organization. The implementation of safe practices has to be recognized as an inseparable part of our activities. Accidents can kill or permanently injure employees if they work carelessly. At the same time, all employees of the Company have legal, personal, and economic reasons for reducing the accidents at their workplace.

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CHAPTER-II

ROLES AND RESPONSIBILITIES

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CHAPTER-2

ROLES & RESPONSIBILITIES

2.1 TOP MANAGEMENT

(i) Provide leadership to achieve policy objectives to ensure health & safety of the employees, customers, contractor's worker & general public.

(ii) Ensure compliance with all relevant norms, rules, regulations & legislations enforcing safety.

(iii) Ensure that workplace is free from hazards which cause or are likely to cause injury or occupational disease to the employees;

2.2 HEAD OF THE DEPARTMENTS

(i) Issue procedures & instructions to maintain safety in workplace to keep it accident free.

(ii) Establish objectives/targets and implementing safety action plan.

(iii) Ensure resources for meeting safety commitments.

(iv) Recognize and reward good safety performance.

(v) Ensure disposal of hazardous and toxic waste including disposal of e-waste;

(vi) Provision of such information, instruction, training and supervision as are necessary to ensure the health and safety of all employees at work;

(vii) Provision and maintenance of plant and systems of work to keep the workplace safe and without risk to health;

(viii) Provision and maintenance of means of access to, and egress from, such places as are safe and without such risk;

2.3 ENGINEERS, OFFICERS, SITE IN-CHARGES & SHIFT IN-CHARGES

(i) Ensure strict compliance of Safety rules & regulations at work place.

(ii) Ensure Trainings & information for all categories of employees & contract workers regarding safety rules & norms.

(iii) Ensure sound condition of safety equipment through periodic checking.

(iv) Conduct accident enquiries, investigations, preparation of accident reports & submit suggestions for remedial measures.

(v) Maintain Accident records.

(vi) Co-ordinate with other departments/Company regarding Safety.

(vii) SITE IN-CHARGES: Arrangement in the workplace for ensuring safety and absence of risk to health in connection with the use, handling, storage and transport of articles and substances;

(viii) SITE IN-CHARGES: Ensure that workplace is free from hazards which cause or are likely to cause injury or occupational disease to the employees;

(ix) SITE IN-CHARGES: Provide and maintain, as far as is reasonably practicable, a working environment that is safe and without risk to the health of the employees;

(x) SITE IN CHARGES: Maintenance or monitoring of work environment in the workplace to keep it safe and without risk to health.

2.4 SUPERVISORS

(i) Ensure accident-free workplace.

(ii) Ensure use of PPEs for all workers.

(iii) Regular checking of safety equipment.

(iv) Report unsafe acts/conditions to the departmental Heads.

(v) Take part in accident enquiries, investigations & suggest for remedial measures.

2.5 GENERAL EMPLOYEES (DEPT.)

(i) Maintain safety rules, procedures &code of practices during work.

(ii) Show active interest in safety issues.

iii) Always use PPEs during hazardous work.

2.6 DESIGNATED IN-CHARGES FOR SUPERVISING ELECTRICAL INSTALLATIONS EXCEEDING 250 kW CONNECTED LOAD

(i) For every electrical installation with more than 250 kW connected load, the concerned officer/s shall ensure that the owner of the electrical installation shall carryout recommended periodic tests as per the relevant standards. He/She shall also ensure inspection of such installations at least once in a year.

(ii) Keep record of these inspection reports in Form I or Form II or Form III or Form IV, as the case may be, of Schedule II of the **CENTRAL ELECTRICITY AUTHORITY (MEASURES RELATING TO SAFETY AND ELECTRIC SUPPLY) REGULATIONS, 2023** test reports and

(iii) Keep record of recommendations in regard with safety duly acknowledged by owner and compliances made thereafter; and

(iv) Make these records available to the Electrical Inspector, as and when required.

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CHAPTER-III

GENERAL SAFETY PRECAUTIONS AND SAFETY RULES

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CHAPTER-3

GENERAL SAFETY PRECAUTIONS & SAFETY RULES

3.1 INSPECTION OF ELECTRICAL EQUIPMENT & SAFETY DEVICES

All equipment & apparatus used for working on overhead lines shall be surveyed every month by a responsible official and he shall take random check on the equipment and satisfy himself that the equipment are in good condition, paying special attention to the safety devices such as safety belts, hand gloves, ropes used for hoisting etc. as per schedule. Any replacement due to wear and tear shall be made immediately. Every authorized person/in-charge of a working party before commencing work shall ensure that all equipment being used, are in safe condition and not weakened by deterioration, abrasion etc. He shall not permit the work to be carried out if for any reason he is in doubt that the equipment is unsuitable or deteriorated to the extent that it is likely to cause a hazard.

3.2 SAFETY MEASURES IN WORKING ON LV & MV LINE, DEAD MAINS AND APPARATUS

Only authorized persons are allowed to work on low and medium voltage mains and apparatus. All mains and apparatus to be worked upon shall be isolated from all sources of supply before starting the work. For earthing and short-circuit only appropriate methods (earthing chains, earthing rods etc.) should be followed. Measures shall be taken against the inadvertent energizing of the mains and the apparatus.

CONNECTING DEAD MAINS TO LIVE MAINS

When dead mains are connected to live mains, all connections to the live parts shall be made last, and in all cases the phase sequence should be checked to ensure that only like phases are connected together. Before inserting fuses or links in a

feeder or distribution pillar controlling the cable on which a fault has been cleared, each phase shall first be connected through a test fuse.

3.3 SAFETY MEASURES IN WORKING ON HIGH VOLTAGE MAINS

a) All high voltage mains and apparatus shall be regarded as live and a source of danger and treated accordingly, unless it is confirmed to be dead and earthed.

b) No person shall work on, test or earth high voltage mains or apparatus unless covered by a Permit to Work (PTW) and after proving the mains dead except for the purpose of connecting the testing apparatus etc. which is specially designed for connecting to the live parts.

c) The operations of proving dead, earthing and short-circuiting of any mains shall be carried out only by an authorized person under the instructions of the person in-charge of the work.

d) While working on mains, the following precautions shall be taken:

   (i) No person, after receiving a permit to work, shall work on, or in any way interfere with, any mains or conduits containing a live main except under the personal instructions and supervision, on the site of work, of competent person.

   (ii) The earths and short circuits, specified on the permit to work shall not be removed or interfered with except by authority from the person in charge of the work.

   (iii) Unauthorized entry of any person /cattle should be prohibited in the substation & switchyard.

   (iv) During operation of any equipment; gossiping/using personal gazettes etc., i.e, remaining inattentive is strictly prohibited.

   (v) Alcohol & drug are allowed neither within company office premises nor at worksites.

3.4 SAFETY CLEARANCES & CENTRAL ELECTRICITY AUTHORITY (MEASURES RELATING TO SAFETY AND ELECTRIC SUPPLY) REGULATIONS, 2023

No person shall work within minimum working distance from the exposed live mains and apparatus. The minimum working distance depends upon the

actual voltages. Exposed live equipment in the vicinity shall be guarded off. The guarding shall be done in such a way that it does not hinder the movement of the maintenance personnel. If necessary, a person for observing safety could be posted.

All barriers, shutters etc. of high voltage equipment must always be kept locked except when required for carrying out work under a permit to work. Keys of controlling locks, except those in the possession of specified officials, shall be kept in safe place in control room. The controlling I movement of these keys shall only be maintained by authorized persons/ site in-charges.

3.4.0. STATUTORY CLEARANCES AS PER CEA REGULATIONS

3.4.1 MINIMUM SAFETY WORKING CLEARANCES IN METRE

The following minimum safety working clearances shall be maintained for the bare conductors or live parts of any apparatus in outdoor sub-stations, excluding overhead lines of installations of voltage exceeding 650 V. **[sub-regulation (2)(iii) of regulation 46]**

Highest system voltage (kV)	Safety working clearance (M)
12	2.6
36	2.8
72.5	3.1
145	3.7
245	4.3
420	6.4
800	10.3

3.4.2 MINIMUM CLEARANCES IN METRE BETWEEN LINES CROSSING OR APPROACHING EACH OTHER– AS PER CEA REGULATIONS

Where an overhead line crosses another overhead line, minimum clearances (in metre) as provided in the table below shall be ensured. **[regulation 71 (iv)]**

Nominal System Voltage (kV)	11-66 kV	110-132 kV	220 kV	400 kV	765 kV	1200 kV
Below 11	2.44	3.05	4.58	5.49	7.94	10.44
11-66	2.44	3.05	4.58	5.49	7.94	10.44

3.4.3 MINIMUM CLEARANCES IN METRE OF LINES AND SERVICE LINES FROM BUILDINGS – AS PER CEA REGULATIONS

The minimum vertical and horizontal clearance of LV and HV lines and service lines from a building in metre shall be as per the following table. **[regulation 62, 63]**

System Voltage	Vertical Clearance from the highest point (M)	Horizontal clearance from the nearest point [on the basis of maximum deflection due to wind pressure] (M)
Up to 650 V	2.5	1.2
Above 650 V & up to 11 kV	3.7	1.2
Above 11 kV & up to 33 kV	3.7	2.0
Above 33 kV	3.7 (+0.3 for every additional 33 kV or part thereof)	2.0 (+0.3 for every additional 33 kV or part thereof)

3.4.4 MINIMUM CLEARANCE IN AIR ABOVE GROUND FOR THE LOWEST CONDUCTOR OF AN AC OVERHEAD LINE INCLUDING SERVICE LINE OF NOMINAL VOLTAGE SYSTEM

The minimum clearance above ground of the lowest conductor of an ac overhead line, including service line of nominal voltage shall have the following values in metre. **[sub-regulation (1) of regulation 60]**

Nominal voltage of system	Clearance above ground (M)		
	Across Street	Along Street	Elsewhere
Up to 650 V	5.8	5.5	4.6
11 kV	6.5	5.8	4.6
33 kV	6.5	5.8	5.2

3.5 PERSONAL SAFETY

a) Head Protection: Every individual entering the construction/maintenance work premises shall wear Safety Helmet.

b) Hand Protection: Appropriate Hand Gloves shall be used for materials handling, welding, grinding, gas cutting, chemical handling & electrical work.

c) Eye Protection: Suitable Goggles, Face Shield/Visor shall be used by all personnel deployed in welding, grinding, gas cutting, etc.

d) Ear Protection: Employees engaged in the noisy areas like operating areas of compressor, pneumatic vibrators, breakers, DG sets etc. shall use suitable Ear protection, like, Ear Plug.

e) Body Protection (Fall Protection): All employees working at a height above 1.8 m from ground or floor level shall use Safety Harness. All employees using the Safety Harness shall anchor its lanyard to a fixed structure in such a manner that the harness works for fall restraint service. Fall Arrest Devices shall be used in conjunction with safety harness, wherever required.

f) Foot Protection: Suitable foot protection, like Gum Boot, Safety Shoe shall be worn by the personnel entering into the site.

All employees shall follow the instructions from their immediate supervisors. If instructions are not clear or are confusing, the employee has a responsibility to get it clarified by the supervisor and obtain clear instructions before commencement of work.

3.6 SAFEGUARDING PUBLIC

a) Every effort should be made to protect the public at all times where company's work is in progress by the use of signs, barricades or personal warning.

b) When working on customer's premises or public property, every effort should be made to avoid hazards to persons or unnecessary property damage.

c) Barriers shall be placed around all exposed open ditches and excavations.

d) Authorized visitors shall not be left to find their own way.

e) Public shall be encouraged to report dangerous situation which may come to their notice.

f) Visitors shall be provided proper Personnel Protective equipment during visit

PROTECTIVE BARRIER (TEMPORARY)

When the work is conducted along streets or highways, pedestrian and traffics shall be warned by signs and flags at day time and red lights or flares at night. Wherever necessary, signalman should be provided. The public shall be protected against hazards of tree trimming along streets & highways by placing traffic cones, danger signs, and signals.

3.7 TREE TRIMMING

a) Before climbing a tree, it is to make sure that all the power lines adjacent to the tree is dead / without any power.

b) Before climbing a tree, the trimmer should look it over carefully to decide how best to climb on it. The tree branches should be carefully inspected to make sure that it could hold the trimmer's weight.

c) Axes should not be used. Always Saw or Billhooks to be used.

d) All tools should be raised & lowered by hand lines to avoid accidents.

e) The tree branches should be cut off up to sufficient height to avoid touching electric lines. Where there is danger that the tree may fall & damage property, block & tackle should be used to control the direction of fall.

f) Felling operation once started, should be finished before the crew leaves.

3.8 SAFETY AND PROTECTIVE DEVICES

Every overhead line erected over any part of street or other public place or in any factory or on any consumers' premises shall be protected through protecting relays, cradle guards, safety devices and/or shields for rendering the line electrically harmless in case it breaks. Adequate arrangements should be made to prevent unauthorized persons from ascending any of the supports of any overhead lines which can be easily climbed upon without the help of a ladder or special appliances. Rails, reinforced cement concrete poles, tubular poles, wooden supports without steps, shall be deemed as supports which cannot be easily climbed upon for the purpose of this rule. The barbed wire should be wrapped on the supports to avoid any unauthorized climbing on the OH line support.

3.9 DOS & DON'TS

Dos	Don'ts
General safety precaution	
To preach and practice safety at all times	Not to remove any "Danger" notice or any other safety sign or interfere with safety barriers or go beyond them until & unless specifically needed / instructed / safety is not compromised
Always to report to the person in-charge immediately after observing any dangerous working practice	Not to touch a circuit with bare fingers or hand or other makeshift devices to determine whether it is live or not

Dos	Don'ts
Always to remain careful and not to take any chance against any possible accident. To at once attend all injuries however slight those may be.	
Always to obey safety instructions given by the person-in-charge of the worksite	
Electric shock	
At first switch off / disconnect the power supply or make it disconnected calling the local office. Then remove the casualty from the danger using insulating item like rescue stick and then render First Aid and send to a doctor or Hospital for checkup & treatment	Not to leave the casualty in contact with live apparatus and not to forget to switch off power supply or inform the local office to switch off immediately
To report each & every accident, whether minor or major, non-fatal or fatal immediately to the person-in-charge	Not to attempt to disengage a person in contact with live apparatus which cannot be switched off. To insulate oneself from earth before attempting to remove the casualty. Not to touch the body of the person but to push him with a piece of dry wood.
Whenever possible, to use one hand only while working on an electrical circuit even though it is supposed to be dead.	

Fire	
To disconnect the power supply immediately in case of fire near electric apparatus	Not to throw water on live electrical equipment
To make sure that when using water hose the jet of water does not come into contact with live electrical apparatus. Extinguish electrical fire using CO_2 or Mechanical Foam type fire extinguisher after switching off all power sources.	

3.10 SAFETY PROVISIONS FOR ELECTRICAL INSTALLATIONS AND APPARATUS OF VOLTAGE NOT EXCEEDING 650V

Test of insulation resistance.– Where any electric supply line for use at voltages not exceeding 650 V has been disconnected from a system for the purpose of addition, alteration or repair, such electric supply line shall not be reconnected to

the system until the supplier or the owner has carried out the test. **[Ref: regulation 42 of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]**

Residual Current Device (RCD) - The use of electricity to electrical installation, shall be controlled by a residual current device to disconnect the supply having rated residual current and duration as per the relevant standards. **[Ref: regulation 44 of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]**

3.11 SAFETY PROVISIONS FOR ELECTRICAL INSTALLATIONS AND APPARATUS OF VOLTAGE EXCEEDING 650V

Approval by the Electrical Inspector and self-certification – Every electrical installation of notified voltage and below shall be inspected, tested and self-certified by the owner or supplier or consumer, as the case may be, of the installation before commencement of supply or recommencement after shutdown for six months or more for ensuring observance of safety measures. **[Ref: regulation 45 of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]**

Use of electricity at voltage exceeding 650V - The supplier, where the supply voltage is equal to or below the notified voltage but exceeds 650 V, shall not commence supply or recommence the supply where supply has been discontinued for a period of six months or more, to any consumer. **[Ref: regulation 46 of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]**

Inter-locks and protection for use of electricity at voltage exceeding 650 V – To ensure safety, the owner shall ensure that different systems / subsystems shall be inter-locked. Isolators & the controlling circuit breakers shall be interlocked. Isolators & corresponding earthing switches shall be interlocked. **[Ref: regulation 47 of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]**

Testing, Operation and Maintenance. – No new apparatus, cable or supply line of voltage exceeding 650 V shall be commissioned unless such apparatus, cable or supply line are subjected to site tests as per relevant standards. Also, no apparatus, cable or supply line of voltage exceeding 650 V which has been kept disconnected for a period of six months or more from the system for alterations or repair, shall be connected to the system until such apparatus, cable or supply line are subjected to the site tests as per relevant standards. **[Ref: regulation 48 of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]**

Precautions to be taken against excess leakage in case of metal sheathed electric supply lines. – Some precautions are to be taken in case of electric supply lines other than overhead lines, for use at voltage exceeding 650 V. **[Ref: regulation 49 of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]**

3.12 SPECIFIC SAFETY MEASURES RELATED TO HYDEL GENERATION

Necessary safety protocols must be followed for Operation & Maintenance of Turbine Generator Sets, Governor System, Static Excitation Equipment, Battery, Air Cooler, Oil Cooler, Lathe Machine,132 kV Switchyard, Oil supply plant to Bearing, SF6 Circuit Breaker, Vacuum Circuit Breaker, Compressor, Isolators, Transformers, and different Equipment. Proper communication to be maintained with other substations after prior approval is received from

SLDC and ERLDC. Necessary communication related to the generation to be established with ALDC.

A. Following Inspections may be done on daily, weekly, monthly, quarterly and annually in case of Pelton turbine-generator Units:

Daily Inspection

a) *Stator winding Temperature*: Checking of the temperature of cooling water and cold air from air cooler.

b) *Journal / Thrust bearing Temperature*: If abnormality is found, following need to be checked after stopping the units:

(i) Load unbalance/ increase

(ii) Abnormal vibration.

(iii) Foreign material (rust, spatter water etc.

(iv) Abnormal operation of cooling water system

(v) Abnormal operation of oil cooler.

(vi) Wear of metal surface

c) *Cold air temperature from air cooler*: Checking the air flow and cleaning of cooling tube required if abnormality persist.

d) *Stator core noise*: checking of finger plate or core in case of abnormality.

e) *Stator frame vibration:* Checking of finger plate for core looseness. Tightening the coupling bolts for core looseness.

f) *Brake track:* Checking of brake shoe, (Crack, abnormal wear and tear setting condition)

g) *Collector ring Surface & brushes:* Checking for perfect bedding between collector surface and brush surface. Adjusting brush pressure if not perfect.

h) *Oil tank and piping for Bearing oil circulation:* Checking for oil level and oil leakage before starting the machine.

i) *Brake system:* Checking for air leakage.

Weekly Inspection

(i) *Collector ring Surface & brushes:* Checking for perfect bedding between collector for surface and brush surface.

(ii) *Insulation Resistance:* Checking for the ovality of the collector ring. Cleaning of dust and dirt. Dry after cleaning if IR is less than 0.5 Megaohms.

Monthly Inspection

(i) *Stator frame:* Checking finger plate and for core looseness tighten the coupling bolts at frame split.

(ii) *Holder & pigtail of brush-gear:* Replace brush and/or brush holder, if abnormality is found. Cleaning the brush holder for smooth movement of brush

Quarterly Inspection

a) *Stator*

(i) Stator Core Pressing flanges: Retighten the core studs if found loose.

(ii) Stator joint: Retighten the coupling bolts at joints.

(iii) Insulation Resistance of Winding: Clean and dryout.

b) *Collector & Brushgear*

(i) Brush pressure: Set the brush between 0.13 to 0.18 kg/cm^2.

(ii) Run-out at ring surface: Re-machining if run-out of ring is beyond specified limits. Change polarity if wear of positive ring is more

c) *Brakes*

(i) Brake shoe wear: Reduce brake air pressure.

(ii) Brake piston movement: Dismantling of brake assembly and cleaning of piston, cylinder and brake pipings. Adjusting spring tension if felt necessary.

(iii) Limit switch operation: Checking& adjustment, if required.

(iv) Fastening bolts for tightness: Tightening the bolts, if required.

d) *Stator frame split:* Retightening.

e) *Rotor current carrying Leads clamp*: Tightening the clamp fixing bolts etc.

f) *Journal bearing Check for clearance against shaft at both ends of each bearing:* Reset, if necessary.

(i) Lubrication oil: Checking as per recommendation of oil supplier.

(ii) Bearing insulation: Taking of corrective measure if result is less than 0.5 Mega Ohm.

g) *Air Cooler*

(i) Fastening bolts for Tightness: If found loose, tightening required.

(ii) Prevention against sweating: Reduce water supply

h) *Space heaters:* for damage on wiring, rectify or replace wiring as required.

Annual Inspection

a) *Stator winding*

(i) Coil overhangs for crack or damage of Insulation: Blow with compressed air& repairing if required

(ii) Series & pole to pole connector insulation: Wipe/clean with non-lining cloth soaked in white spirit or turpentine.

(iii) End winding packing for tightness: Repair & paint with epoxy red gel coat.

b) *Rotor Coil*

(i) Inter pole connections & their support: Checking for their perfectness.

(ii) Insulation resistance: Measuring IR values & ensuring it to be within limit.

c) *Journal / Thrust Bearing Housing*

(i) Mounting bolts: Tightening required.

(ii) Welded potion cracks: Repairing required

d) *Brakes*

(i) Dust accumulation: Cleaning required

(ii) Clearance against Brake Track: Re-adjustment required

e) *Shaft vibration:* Check for run-out at slip ring surface and re-adjusted if found more than the specified value.

B. Following Inspections may be done on daily, weekly, monthly and annually in case of Francis & Kaplan-Bulb turbine-generator Units:

Daily Inspection

a) Turbine

(i) Checking of leakage of Oil from Casing & Stay Vane, Outer & Inner Guide Vane Ring, PRV & Shaft Seal is to be done & corrective measures is to be taken.

(ii) Checking of leakage of Oil from Guide Vane Servo Motor is to be done.

(iii) Checking of Shaft seal Water system is to be done

(iv) Checking of temperature of oil & its flow for Guide Bearing Oil reservoir & Lubricating Oil and Vibration & noise of Guide bearing is to be done.

b) Governor

(i) Checking of leakage of Oil from Converter, Distributing Valve, Oil filter, Solenoid Valve, Pressure Switch, Control Valve of Generator Brake is to be done.

(ii) Checking of operating condition of Limit switch, Quick Stop Valve, Opening Director(GVT & RVT) is to be done.

c) Pressure Oil Supply System

(i) Checking of Oil level in the Pressure Tank& leakage of oil & Air from the tank is to be checked. Checking of oil leakage, voltage & current, Vibration & Noise, Temperature of Oil Pump & Motor is to be done.

(ii) Checking of leakage of oil from Safety Valve is to be done.

(iii) Checking of oil level, oil temp, oil leakage is to be done for Oil Sump Tank. Checking of Oil Leakage & Operation for Unloader & Control Valve, Oil Level Control Device, Oil Level Indicator, Pressure Switch are to be done. Checking of Oil & Water Leakage & flow of Cooling Water for Oil Coolers are to be checked. Cleaning of Oil Coolers is to be done daily.

d) Air Compressor

(i) Checking of Air Leakage & temperature of Cylinder Head, Piston & Valve Plate is to be done. Checking of oil leakage from Crank Case Lubricating Oil system and operation of Pressure Controller are to be done. Checking

of leakage of Air from Safety Valve and "After Cooler & Central Cooler" are to be done. Checking of operation of Drain Valve, Air Valve and belt is to be done. Checking of Noise, Vibration & looseness of belt is to be done.

(ii) Checking of Current & Voltage, Temperature and Noise & Vibration of Compressor Motor is to be done.

e) Lubricating Oil Supply System

(i) Checking of leakage of Oil, Voltage & Current, Noise & Vibration for Pump & Motors of Oil Sump Tank is to be done. Checking of Oil Level, Oil temperature & Oil Leakage for Oil Sump Tank is to be done.

(ii) Checking of Oil Leakage from Safety Valve for Pump, Pressure gauge, Oil Flow relay, oil Cooler, Oil Gravity Tank, Oil Level Indicator, solenoid Valve, Oil Sump Tank, Oil Return Tank and Oil Pipe line is to be checked.

Weekly Inspection

(i) Checking of Condition of Water basin of Shaft seal is to be done. Checking of oil Leakage & oil Level of Guide Bearing Oil Reservoir & Lubricating Oil is to be done. Checking of operating Condition of Distributing Valve of Governor is to be done.

Monthly Inspection

(i) Checking of color of oil in the pressure tank, oil Sump Tank, IR Value of Motor is to be done. Flashing through Drain Valve is to be done. Cleaning of Suction Filter is to be done. Calibration of level Indicator is to be checked.

Annual Inspection

(i) Adhesive Condition of Concrete, Wear & Tear, Looseness & deformation related to Casing & Stay Vane of Turbine are to be checked.

(ii) Tightness of Fitting Nuts, Wear & tear, Looseness & deformation of Outer & Inner Guide Vine Ring are to be checked.

(iii) Tightness of Fitting Nuts, Wear & tear, Looseness & deformation of Pressure Release Device are to be checked.

(iv) Wear & tear, Looseness & deformation, Painting for Draft Tube of Turbine are to be checked.

(v) Wear & damage, cavitations, pitting, clearance of mating surface, side clearance of Guide Vane are to be checked.

(vi) Guide Vane bearing, Guide vane spherical bearing, Guide vane Servo Motor are to be thoroughly checked.

(vii) Thorough checking of Shaft seal, runner, servo motor, all parts of Governor are to be checked.

(viii) All the parts of Pressure Oil Supply system, Air Compressor, Cooling water System, Drainage & Dewatering System are to be checked thoroughly.

(ix) The Generator, excitation system, breakers, C&R panels, all switchgears are to be checked thoroughly.

(x) Checking of all ACDB, DCDB, Battery & Chargers are to be done thoroughly.

(xi) The illumination & ventilation System is to be checked thoroughly.

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CHAPTER-IV

PERMIT TO WORK SYSTEM

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CHAPTER-4

PERMIT TO WORK (PTW) SYSTEM

4.1 INTRODUCTION

Safety should be ensured by following certain well-defined policies, procedures and methods during construction, operation & maintenance. The objective of safety documentation system is to ensure that written instructions & guidelines are available and followed. Safety document system is an essential part of safety management by following safety rules/procedures & precautions so that the safe act & safe condition can be ensured. These rules/procedures & precautions are documented and the documents are issued by the shift in-charge with strict instructions to be followed up at appropriate stage. These safety documents are also useful in investigation of accident.

Following written documents are to be preserved in the substations of all voltage grades including hydel:

a) Permit to Work

b) Authorization

c) Office Records

(i) Log Book/Log Sheet

(ii) Site Test Report

(iii) Tripping Register

(iv) Equipment maintenance register/Maintenance schedules/Safety Audit

(v) Inspection Register/Record

(vi) Shutdown Register

(vii) Temporary Alteration Register

(viii) Incident Register

4.2 ELECTRICAL INSPECTOR'S APPROVAL

As per sub-regulation (6) of CEA Safety regulation 45, the owner of any installation who makes any addition or alteration to his installation shall not connect to the supply his apparatus or electric supply lines, comprising the said alterations or additions, unless and until such alteration or addition has been approved in writing by the Electrical Inspector or self-certified by the owner of the installation, as the case may be.

4.3 DESIGNATED PERSON TO OPERATE AND CARRY OUT THE WORK ON ELECTRICAL LINES AND APPARATUS

As per sub-regulation (3) of CEA Safety regulation 3, the supplier or consumer, or owner of electrical installation or a contractor who has entered into a contract with a supplier or a consumer, or owner of electrical installation, to carry out duties incidental to the generation, transformation, transmission,

conversion, distribution or use of electricity shall designate persons for the purpose to operate and carry out the work on electrical lines and apparatus. He shall maintain a record, in paper or electronic form, wherein the names of the designated persons and the purpose for which they are designated, shall be entered.

No person shall be designated to operate and carry out work on electrical lines and apparatus unless he possesses a certificate of competency or electrical work permit, issued by the Appropriate Government;

4.4 PERMIT SYSTEM – GENERAL INSTRUCTIONS

a) The written document 'Permit To Work' shall be issued for safe working & precautions to be taken at work spot.

b) Permits to work shall be applied by an authorized person to take line clear and shall be issued by the Engineer (or competent person in-charge of operation) in writing in the form prescribed.

c) The work spot may be far away from the switching sub-station. Hence the PTW shall be issued with the necessary instructions for precautions to be taken at the work spot.

d) In extraordinary circumstances such as for the purpose of saving life or time in the event of an emergency, it may become necessary to start the work without being able to obtain the necessary permit to work. In such cases the action taken shall be reported to the person-in-charge as soon as possible. But all safety precautionary measures are to be taken prior to start of work.

e) No employee shall carryout or attempt any work on live apparatus or work in proximity of open conductor unless the supervisor has necessary permit to work, issued by competent authority.

f) The special instructions to provide safe working conditions shall be rigidly complied with by all persons and at all times when working on a job with a permit-to-work. Any one disregarding this instructions or behaving in a manner likely to cause danger, either to himself or to anyone else, should be properly punished.

g) The person-in-charge of work shall explain the nature and duration of work to be carried out to the permit issuing authority, and obtain from him a permit-to work before commencing any work.

h) Application for Pre-arranged shutdown: Except for emergencies, all works for repairs, maintenance and construction on or in close proximity of live apparatus and mains, shall be prearranged and programmed. Accordingly, applications for prearranged shutdowns shall be submitted by the Supervisor to the officer in charge, in the prescribed form, which when duly approved, will be presented to the concerned Permit Issuing Officer for switching off the apparatus and issuing of permit-to-work. These applications shall be made sufficiently in advance to enable the permit Issuing Officer to carry out necessary load management, if any and other operations connected with the work. The duration and nature of the work must be clearly explained to the Permit Issuing Officer before getting a permit.

i) Issue and return of permit shall only be made in person.

4.5 SPECIAL INSTRUCTIONS TO SUPERVISORS

The person-in-charge, before allowing any workman to commence work on the mains and apparatus/line, shall take the following precautions:

a) Explain the nature of work and the precautions taken to ensure the safety of the workmen and also the precautions to be taken by them during the progress of the work;

b) The circuit breaker or line apparatus must be opened correctly and line equipment must be isolated, discharged & earthed properly wherever possible and tested with a discharge rod and then danger notice plates should be placed at appropriate position.

c) Warn the workmen and public from the danger that exists in the vicinity of the area covered by the permit.

d) Where work is to be carried out on live or medium voltage mains or apparatus, the following additional instructions shall be complied with.

e) No employee shall be ordered or permitted to carry out any work, other than that for which he is specially authorized.

f) The Supervisor in charge of a work shall keep with him a complete list of all persons who are working on the particular job. All other persons shall be warned to keep away from the area.

g) The supervisor, before allowing any workman to commence work on the lines, mains or apparatus, create a safety zone by short circuiting together all the conductors and adequately earthing on both sides of the place of work.

h) The instructions shall be read out and explained to the workmen in the language they understand, and copies shall be pasted on the various notice boards. Ignorance of the instructions shall not be accepted as an excuse for non-compliance with them.

i) On completion of the work, remove all earthing devices so that the mains and apparatus are fit in all respects for charging; withdraw all workmen and warn them that it is no longer safe to touch or approach the mains and apparatus, and thereafter return the keys and the permit to the shift in charge for cancellation.

j) Prior to issuance of any PTW, information in connection with the permit jurisdiction, if any, recorded in Temporary Alteration register, must be shared with the receiver of the PTW. Both the issuer of the PTW and its receiver will have to sign with date & time against that relevant information as documented evidence that information was passed on to the workforce duly prior to commencement of work.

4.6 PRECAUTIONS TO BE TAKEN WHEN WORKING WITH A PERMIT

a) On Equipment in the Substations and inside powerhouse of the hydel wings

It is also the duty of the permit holder to check whether the equipment/line has been properly earthed. The staff member shall be informed:

(i) On which equipment/part of equipment it is safe to work,

(ii) The area where it is dangerous to work shall be barricaded by ropes & flags.

When men have to climb up structures or apparatus which bring them under reduced clearance to live parts, the authorized person shall be present at site and direct the work.

b) On Overhead line

(i) Before touching the conductors, each conductor shall be discharged and connected well to earth by means of discharge rods & earthing chain.

(ii) This shall be done at two points on each side of the place of work. In addition, the conductors should be earthed on poles (or towers) on which work is actually carried out, as per standard practice.

(iii) The work on one of the circuits of a double circuit line shall be carried out in accordance with the following regulations:

The minimum distance from the conductors of the live circuit and where the men are working on the dead circuit shall be 1.5 m & above.

4.7 TRANSFER OF PERMIT

a) Transfer of permit to work from one Supervisor to another is strictly prohibited. If there are more than one working party, separate permits should be issued to the Supervisor in charge of each working party and a written record should be kept of the number of such permits issued for each work.

b) If work is of such a nature and duration that it has to be carried out continuously but under the supervision of more than one Supervisor in shift duty, the "Permit to work" form shall be endorsed by the Permit Issuing Officer cancelling the name of the supervisor to whom

it was originally issued and substituting the name of the second or subsequent supervisor to whom the permit will now become valid. The time of each endorsement should be noted on the "Permit to work" form and it's duplicate.

4.8 LOGGING OF PERMIT ISSUE & RETURN

a) The permit issuing person shall carefully log the various operations performed by him in connection with the issue and return of permit in the daily main station logbook and the sequence in which the operations are done.

b) The operation reported to have been done by other side stations or in connection with the issue or return of a permit shall also be logged.

The issue and return of the permit shall be logged in the book in red ink.

c) When change of shift occurs during the pendency of a permit, the outgoing in charge shall specifically orally inform the incoming in-charge operator as well as by logging in handing over note, regarding the permit which is pending. The incoming in-charge operator shall also note the pendency of the permit.

d) No line or equipment should be made again live until the permit issued on it, is returned.

e) The same person who takes the permit must return the permit.

f) If the duration of permit is considerably extended or shortened due to any circumstances, then proper reason must be stated in the said permit by the in-charge.

4.9 GENERAL SHUTDOWN PROCEDURE FOR MAINTENANCE OF EQUIPMENT IN PUMPED STORAGE PROJECT POWERHOUSE:

a) Authorized person of WBSEDCL can only take work permit for maintenance activity.

b) The Control Engineer or his Authorized Representative will check the switching sequence in the Work Permit. After that representatives of Control room and maintenance team will jointly complete the switching activity and ensure proper de-energisation of concerned Electrical and Mechanical Circuit. Then the Control Engineer or his authorized representative will allow the shutdown and hand over the Work Permit to Maintenance Team.

c) At the time of withdrawing the shutdown after completing the Maintenance Activity, the switching activity is to be completed jointly by the Control Room and Maintenance Team. In case of multiple simultaneous shutdowns, the common switching activities, i.e., the cross permits, if any, are also to be checked at the time of withdrawing the shutdown. The work permit is to be returned and the shutdown is to be withdrawn only after ensuring that all persons, maintenance tools, and tackles have been removed from the Maintenance Site.

WEST BENGAL STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED
SAFETY INDUCTION FORM

(Safety Induction for contractor)

1	Name & Address of the Contractor	
2	LOI / T&AA No.	
3	Name of Division/CCC/Substation	
4	Date of starting the work	
5	Date of completing the work	

I shall follow all Safety precautions/instructions given to me and shall be responsible for the safety of my staff/ employees / workmen.

I certify that safety appliances including Safety belts, gloves, helmets, discharge rod etc. have been issued to each of my workmen(..... in total no.) and the same are being used. All other necessary safety aids/ appliances of prescribed standard will be used as and when required.

Signature of Contractor

(Stamp)

1. Necessary safety precautions have been explained.
2. Contractor has been advised that safety appliances including full body harness, gloves, helmets, discharge rod, etc. must be used by all of his workers, as also other safety aids/appliances when required.

Signature of AE/CM/ACM/LM

WEST BENGAL STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED

DESIGNATED PERSON TO OPERATE AND CARRY OUT THE WORK ON ELECTRICAL LINES AND APPARATUS

(As per sub-regulation (3)of regulation 3)

Name ……

Age ….....…… Designation ……..

Sri ….....…… is hereby ……..

Authorised to work in the Substation .....……..

Overhead lines ……

kV From…….. To……

His duties are:

1. To operate the H.V. Switches,
2. To clean the Switches Circuit Breakers when they are made absolute dead and properly earthed.
3. He is only to upkeep the control room floor and the switchyard.
4. He is to work on the flowerbeds inside the switchyard.
5. He is to change H.V. fuses on where:
   a) Dropout switches are used, by properly earthing of the isolator handled & using suitable hand gloves,
   b) Where links and fuses are used, cutting off the links by operating handle.
6. He is to operate isolators, rectify armouring, binding and other maintenance work on dead Feeders and Distributors which have been properly earthed and short circuited using full body harness while working on O.H. supports.

7. He is to patrol the O.H. lines from the ground. Under no circumstances he should climb up the pole.
8. He is to replace L.T. aerial fuses on dead line only.
9. He is to fill up or replace transformer oil, or clean bushing etc. after short circuiting the Primary side & Secondary side.

DIVISIONAL MANAGER

N.B. The Divisional Manager is authorized to issue the letter. Items for which the worker is authorized would be ticked in RED ink.

Any type of unusual changes in network & associated switchgear (breaker) must be kept in record. The information must be exchanged between substation operator & person, requisitioning PTW. May be made part of PTW.

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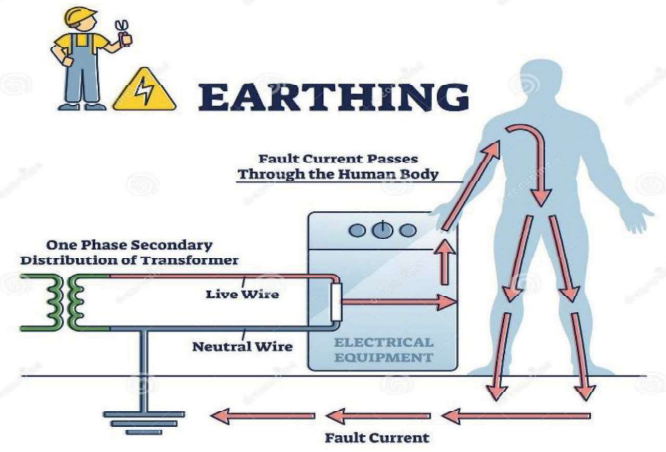
CHAPTER-V

EARTHING



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CHAPTER-5

EARTHING

5.1 REASONS FOR EARTHING

a) To create a reference voltage,

b) To create a safe zero potential surface,

c) To create a return path for neutral currents,

d) To reduce transients,

e) To reduce harmonics

5.2 EARTHING PIT

a) Earth surface has high resistance and therefore we have to create a pit which maintains low resistance bonding with earth and brings out a low resistance connection which can be distributed.

b) Neutral has to be earthed either directly or through grounding resistor to create reference voltage otherwise lines can assume very high voltages which can be a serious safety threat.

5.3 EARTHING IN HARD ROCK AREA

Following options may be adopted as per site conditions:

a) Locate earthing near a surface or deep water body and connect earthing mat to it.

b) Bore through granite up to underground water body and sink metal plates in to it and connect earthing mat to it.

c) Chemically treat soil to increase conductivity.

5.4 REQUIREMENTS FOR EARTHING

a) For substations and industrial uses, recommended earthing resistance should be between 0.1-1.0 ohm.

b) If resistance is higher, saline water should be added to the pit.

c) Many pits should be connected together to create a low resistance path for large short circuit current without creating large voltage drop.

d) Every motor, panel and switchgear should be earthed at two places.

e) Neutral of generators and transformers should be connected to two earthing pits which are independent of body earthing pits. (** UG Earthing)

5.5 SOME IMPORTANT NOTES

a) Water pipes should not be used for earthing,

b) Lead sheathing and armor of cables should be earthed,

c) Wherever RCC poles are used as in substations, four poles can be tied with earthing strip and then connected to earthing pit,

d) Earthing electrode should not get corroded. GI or tinned copper is preferred. Aluminum can be used above ground connections.

5.6 SUBSTATION SHOCK HAZARD

a) **Metal to metal touch voltage –**A situation when two metal structure within substation are simultaneously in contact with body.

b) **Step Voltage –** Difference in ground surface potential experienced by a person bridging 1m gap with his feet without contacting any other earthed object.

c) **Touch Voltage –** Voltage difference between earthed object which a person is touching and ground on which a person is standing during earth fault condition.

d) **Mesh Voltage –** The maximum touch voltage within a mesh of the ground grid.

e) **Transferred Voltage -** A special case of touch voltage where a voltage is transferred in to or out of a substation from or to a remote point outside.

5.7 SUBSTATION EARTHING

Criteria:

a) Low resistance,

b) Low step potential,

c) Low touch potential

5.8 RECOMMENDED EARTHING RESISTANCE VALUE

EHV AC Installations<0.01ohm
HV Installation>33 kV <0.5ohm
MV installation between1-33kV<0.5ohm
LV Installation upto1.0kV<2ohm
Large residential buildings<2ohm
Individual residential buildings <4ohm

5.9 EARTHING IN ELECTRICAL SYSTEM

Earthing is a vital part of all electrical system. A well designed earthing system is necessary to ensure safety of equipment and personnel and for correct operation of protective devices during–

a) Earth fault in electrical system,

b) Lightning strokes on equipment/structures,

c) Occurrence of induced voltage and current on equipment, conductors, cables & structures etc.

5.10 GENERAL GUIDELINE FOR EARTHING

An effective grounding system must satisfy the following conditions:

a) Provide a low impedance path to ground during fault.

b) Withstand and dissipate repeated faults & surge current.

c) Limit the touch & step potential to a tolerable value.

d) Limit the ground potential rise to a tolerable value so as to prevent danger due to transfer of potential through ground, earth wire, cable sheath, fence, pipelines etc.

5.11 EARTHING OF OVERHEAD LINES, SUBSTATIONS, DISTRIBUTION TRANSFORMER

a) All metal supports /PCC /RCC poles and metallic fittings attached there to shall be permanently & effectively earthed.

b) Metallic wire used for supporting insulated wires/AB cables should be efficiently earthed.

c) Stay wires shall be insulated through GUY insulators not below 3 and above 3.5 meters from the ground.

d) Neutral of Distribution transformers & LAs shall be earthed separately &effectively.

e) All metallic noncurrent carrying parts of the 33/11 kV sub stations shall be grounded through earth mat of the substation. Earthing shall effect as per code of practice for earthing. For substation provided with earthing mat, IEEE guidelines for substation earthing shall be followed.

5.12 TEMPORARY EARTHING & SHORT CIRCUITING HIGH VOLTAGE MAINS

a) High voltage mains shall not be worked upon unless they are discharged to earth after making them dead and are earthed and short circuited with earthing and short circuiting equipment adequate to carry possible short-circuit currents and which are specially meant for the purpose. All earthing switches wherever installed should be locked up.

b) If High voltage cable is required to be cut, a steel wedge shall be carefully driven through it at the point where it is to be cut or preferably by means of a spiking gun of approved design.

c) After testing of the high voltage cable with DC voltage, the cable shall be discharged through a 2 megaohm resistance and not directly, owing to dielectric absorption which is particularly prominent in the DC voltage testing of high voltage cables. The cables shall be discharged for a sufficiently long period to prevent rebuilding up of voltage.

d) The earthing device when used shall be first connected to an effective earth & the other end of the device shall then be connected to the conductors to be earthed.

e) The earthing and short-circuiting device shall remain connected for the duration of the work.

f) When it is necessary to cut a line bus bar or loop or to repair a broken conductor or damaged loop, earth shall be placed on both sides of the working zone.

g) Before working on underground cables, they shall be disconnected from the source, discharged and then earthed. To discharge them use an earthed wire and make contact with each terminal repeatedly.

5.13 REMOVAL OF EARTH CONNECTION

a) On completion of work, removal of the earthing & short-circuiting devices shall be carried out in the reverse order to that adopted for placing them, that is, the end of the earthing devices attached to the conductor of the earthed mains or apparatus shall be removed first and the other end connected to earth shall be removed last. The conductor shall not be touched after the earthing device has been removed from it.

b) After this, if we notice that, some work is still left or some Tools &Tackles etc. is still left on the line, then **Do Not climb** on the line unless the line is discharged again.

5.14 SAFETY PRECAUTIONS IN EARTHING

a) Examine earthing device periodically and always prior to their use.

b) Earths shall never be attached or removed with bare hands. Rubber hand gloves, gauntlets or approved protective equipment shall always be used.

c) Verify that the circuit is dead by means of discharging rod or potential indicator of approved type.

d) Earthing should be done in such a manner that the persons doing the job are protected by earth connections on both sides of their working zone.

e) All the conductors including neutral should be properly earthed and short circuited even though work is to be done on one phase only.

5.15 SAFE EARTHING PROVISIONS FOR ELECTRICAL INSTALLATIONS & APPARATUS OF VOLTAGE NOT EXCEEDING 650V –

The conditions which apply to the connection with earth of systems at voltage exceeding 50 V but not exceeding 650 V have been elaborated in regulation 43. **[Ref: regulation 43 of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]**

5.16 SAFE EARTHING PROVISIONS FOR ELECTRICAL INSTALLATIONS & APPARATUS OF VOLTAGE EXCEEDING 650V -

The entire switchyard or substation equipment and buildings including all non-current carrying metal parts associated with an installation shall be effectively

earthed to an earthing system or mat which shall limit the touch and step potential to tolerable values and limit the earth potential rise to tolerable values as per relevant standards. This shall also maintain the resistance of the earth connection to such a value as to make operation of the protective device effective.

[Ref: regulation 50 of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

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CHAPTER-VI

SAFETY IN CONSTRUCTION WORK

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CHAPTER-6

SAFETY IN CONSTRUCTION WORK

RURAL ELECTRIFICATION, PROJECT WORKS

6.1 SAFETY RULES & NORMS FOR CONTRACTOR

a) The Contractor shall ensure proper safety of all the workmen, materials, plant and equipment belonging to him or to Employer/ Owner or to others, working at the Site. The Contractor shall also be responsible for provision of all safety notices and safety equipment required both by the relevant legislations and the Engineer as he may deem necessary.

b) All equipment used in construction and erection by Contractor shall meet Indian / International Standards and where such standards do not exist, the Contractor shall ensure these to be absolutely safe.

c) Periodical examinations and all tests for all lifting or hoisting machines and lifting tackles shall be carried out in accordance with the relevant legal & other requirements. A register of such examinations and tests shall be properly maintained by the Contractor and will be promptly produced as and when desired by the Engineer or by the person authorized by him.

d) The Contractor shall provide suitable safety equipment of prescribed standard to all employees and workmen according to the need, as may be directed by Engineer who will also have right to examine the safety equipment to determine their suitability, reliability, acceptability and adaptability.

e) The Contractor shall provide safe working conditions to all workmen and employees at the Site including safe means of access, railings, stairs, ladders scaffoldings etc. The scaffoldings shall be created under the control and supervision of an experienced and competent person. For

creation, material of good and standard quality only shall be used by the Contractor.

f) The Contractor shall not interfere or disturb electric fuses, wiring and other electrical equipment belonging to the Owner or other Contractor under any circumstances, whatsoever, unless expressly permitted in writing by Engineer to handle such fuses, wiring or electrical equipment.

g) No Electric cable in use by the Contractor or Owner will be disturbed without prior permission. No weight of any description will be imposed on any cable and no ladder or similar equipment will rest against or attached to it.

h) No repair work shall be carried out on any live equipment and line. The equipment must be declared safe by the Engineer and a permit to work shall be obtained before any repair work is carried out by the Contractor.

i) The Contractors shall employ necessary number of qualified, fulltime electricians/ Electrical Supervisors to maintain his temporary electrical installations.

j) The Contractor employing more than 250 workmen whether temporary, casual, probationer, regular or permanent or on contract, shall employ at least one full time officer exclusively as safety officer to supervise safety aspects of the equipment and workmen, who will coordinate with the Project Safety Officer. In case of work being carried out through subcontractors, the subcontractor's workmen or employees will also be considered as the Contractor's employees / workmen for the above purpose.

The name and address of such Safety Officer of Contractor will be promptly informed in writing to Engineer with a copy to Safety In-charge before he starts work or immediately after any change of the incumbent is made during currency of the Contract.

k) In case any accident occurs during the construction / erection or other associated activities undertaken by the Contractor thereby causing any minor or major or fatal injury to his employees due to any reason, whatsoever, it shall be the responsibility of the Contractor to promptly inform to the Engineer in prescribed form and also to all the authorities envisaged under the applicable laws.

l) It is mandatory for the Contractor to observe during the execution of the works, requirements of safety rules which would generally include but not be limited to the following:

(i) Each employee shall be provided with initial introduction regarding safety by the Contractor, so as to enable him to conduct his work in a safe manner.

(ii) No employee shall be given a new assignment of work unfamiliar to him without proper introduction as to the hazardous incident thereto, both to himself and his fellow employees.

(iii) Under no circumstances shall an employee hurry or take unnecessary chance when working under hazardous conditions.

(iv) Employees must not leave naked fires unattended. Smoking shall not be permitted around fire prone areas and adequate firefighting equipment shall be provided at crucial locations.

(v) Employees under the influence of any intoxicating substance, even to the slightest degree shall not be permitted to remain at work.

(vi) There shall be a suitable arrangement at every work site for rendering prompt and sufficient first aid to the injured.

(vii) The staircases and passage ways shall be adequately lighted.

(viii) The employees when working around moving machinery must not be permitted to wear loose garments. Safety shoes are recommended when working in places where materials or tools are likely to fall. Only experienced workers shall be permitted to go behind guard rails or to clean around energized or moving equipment.

(ix) The employees must use the standard protection equipment intended for each job. Each piece of equipment shall be inspected before and after it is used.

(x) Requirements of ventilation in underwater working to licensed and experienced divers, use of gum boots for working in slushy or in inundated conditions are essential requirements to be fulfilled.

(xi) In case of rock excavation, blasting shall invariably be done through licensed blasters and other precautions during blasting and storage/ transport of charge material shall be observed strictly.

m) The Contractor shall follow and comply with all State Safety Rules, relevant provisions of applicable laws pertaining to the safety or workmen, employees, plant and equipment as may be prescribed form time to time without any protest or contest or reservation. In case of any discrepancy between statutory requirement and State Safety Rules referred above, the later shall be binding on the Contractor unless the statutory provisions are more stringent.

n) If the Contractor fails in providing safe working environment as per State Safety Rules / Regulations or continues the work even after being instructed to stop work by the Engineer as provided above, the Contractor shall promptly pay to Engineer on demand by the Owner compensation at the rate of Rs. 5000/- (five thousand) per day or part thereof till the instructions are so complied with and so certified by the Engineer. However, in case of accident taking place causing injury to any individual, the provisions contained shall also apply in addition to compensation mentioned.

o) Permanent disablement shall have same meaning as indicated in Workmen's Compensation Act. The compensation mentioned above shall be in addition to the compensation payable to the workmen/employees under the relevant provisions of the Workmen's Compensation Act and rules framed hereunder or any other applicable laws as applicable from time to time. In case the Owner is made to pay such compensation then the Contractor is liable to reimburse the Owner such amount in addition to the compensation indicated above.

p) If the Contractor observes all the Safety Rules and Codes, Statutory Laws and Regulations during the currency of Contract awarded by the Owner and no accident occurs then Owner may consider the performance of the Contractor and award suitable "ACCIDENT FREE SAFETY MERITORIOUS AWARD" as per scheme as may be announced separately from time to time.

6.2 WORK PERMIT SYSTEMS

General Rules:

(i) Before doing any work in a restricted area a work permit should be issued.

(ii) The receiver of the work permit must keep the permit posted at the job location.

(iii) The appropriate permit shall be obtained before any job is started.

(iv) Work permits should be valid for the duration as mentioned therein.

(v) If the work is not started or is stopped due to safety considerations the permit is required to be revalidated by the issuing authority before the work is started or resumed.

(vi) If the work is not completed during the time for which the permit was issued the person responsible for performing the work shall get it revalidated.

(vii) Safety officer or any other responsible officer is authorized to suspend a work and withdraw the work permit if safety regulations being violated and for any of the work permit conditions are not being abided by and / or conditions around work site have changed which may lead to an imminently hazardous situation.

(viii) The receiver of a work permit must keep a copy in his possession or within view of the job site for duration of the job so that it may be presented upon request.

(ix) When the job is completed or at the end of the shift both issuer and receiver must close out each work permit.

Types of Work Permit:

The following Work Permits may be used in order to control the construction site activities and to prevent accidents:

a) Working at height permit

For carrying out activities at workplaces that are 2 meters or more, necessary work permit is required to be issue. The Site In charge should be appointed as responsible person for issue and enforcing this system.

b) Hot (High Temperature) work permit

Hot work would mainly comprise - Welding, gas cutting, use of open flame or open flames or other sources of fire in a fire prone place containing inflammable substances, explosives &/or other such highly combustible material susceptible to spontaneous ignition &/or explosion.

c) Confined space entry permit

To identify safe work practices related to working inside a confined space or a vessel or a plant / equipment, this work permit is necessary.

d) Safety work permit for excavation, trenching and earth removal

To identify safe work practices related to excavation work, trenching work and earth removal activities, this work permit is necessary.

e) Electrical isolation permit

This work permit is necessary to do job in an area where the electrical hazards exist.

f) Permit to work on plant, machinery & other power driven equipment

This work permit is necessary for performing job with the help of any plant, machinery & power driven equipment.

g) Blasting work permit

This work permit is necessary for performing any job with the help of blasting work.

6.3 SAFETY NORMS TO BE OBSERVED FOR HYDRO-ELECTRIC POWER PROJECTS I POWER STATIONS

Hazardous Features at Hydro-electric Power Projects

a) Spillways

Spillways are provided at various locations such as Forebay Tank, Barrage, Dam, etc. In case of Dams, un-gated overflow spillways are particularly dangerous because they may be difficult to recognize from reservoir and anything/any person can easily be drawn over them. Also, in case of Barrages, gated spillways may be hazardous depending on their operation time. In some gate opening positions, dangerous whirlpools develop upstream of gates. In case of Forebay Spillways, the danger lies in sudden surges due to instantaneous machine stoppage or due to excess water in Forebay coming from intakes during monsoon.

b) Penstock intakes

In general, water is fed into penstock pipes from Forebay tanks and trash racks are installed at the inlet of penstocks to prevent any trash from entering. Water level at Forebay should be regularly checked else bubbles can enter the Penstock Pipes, which in turn may cause accidents in Penstock Pipes.

c) Tailrace channels

Tail Race channels convey water from the power house to river. Sudden increases in tailrace flows caused when the generators go online are often very hazardous to anything/any person that are near the shorelines or wading in the

tailwater areas. The degree of the hazard varies depending on such factors as the volume of flow, the rate of change in flow, the turbulence created, and the size of the Tailrace area. At some projects with large tailrace areas, eddy currents can be quite dangerous because they can draw an object or unwary victim into turbulent water below spillway or powerhouse discharge areas. To avoid any untoward incidents, proer fencing should be maintained along the Tail Race Channel edge.

d) Forebay

Forebay is basically a water reservoir from where penstock pipes draw water to feed Power House Turbines. The depth of water can be really high and any carelessness can cause people drowning in the deep water. Also, sometimes Forebay is used as surge tank; so, in case of instantaneous machine stoppage, there can be sudden surge of water causing damages in surrounding areas if proper safety is maintained. Proper fencing is required along the periphery of Forebay Structure.

e) Canals

Canals are a part of many hydro projects such as Canal Fall projects. Concrete or other hard surface linings (whether vertical or sloped) are common and create hazardous conditions. It is very difficult for a person caught in the water of a lined canal to get out without help because the water, algae, mud, etc. make the linings very slick. Other structures associated with canals may also be dangerous.

f) Intake areas

The inlets to conduits, tunnels, ducts are extremely dangerous. From the surface, such areas may offer little visible evidence of the dangerous undercurrents. Head gate structures near entrances to canals often create hazards as water flows under or through the gate openings or trash racks. Also in some run-of-the-river projects, de-silting chambers if not protected properly can cause hazards. Also, in hilly regions, often flash floods during monsoon cause enormous damages to different project components. The hilly rivers being narrow in width in intake areas, the height of water during flash floods may reach deceptively higher (7-8m) and may bring tree logs, debris, boulders, etc. If proper safety is not followed people may get swept away.

g) Natural channels

Hydro projects are constructed to take advantage of unique natural topography. This inherently results in projects being located in steep terrain, canyon, or near channels that are used primarily for power production. These areas are naturally hazardous even without the hydro facility. However, since a project could make

access easier, this factor should be considered in any overall plan for preventing public access or providing safety devices. Often the spillway, powerhouse, and other project facilities can be obscured from view by the natural topography. Therefore, individuals could unknowingly enter dangerous areas if access is allowed or adequate safety devices are not provided.

i) Substations and powerlines

The electrical shock hazards associated with the substations are obvious and such facilities should be well protected with fences and signs. High voltage power lines associated with the powerhouses and substations should be so located that safe vertical and side clearances are available.

j) Other project structures

Project structures accessible to the public can present many safety problems. There is danger of falling from dams, wing walls, or head gate structures into reservoirs or tailrace areas. Large boulders and other rip-rap material on dams and along dikes can be hazardous to the public. Powerhouses and other project buildings have many inherent hazards. Therefore, public access to these areas should be restricted. Catwalk structures and slippery spillway surfaces are also hazardous areas that should be avoided. Public access to intake area should be restricted.

k) Natural and other hazards

Whether man-made or natural, consideration should be given to marking hazards with signs.

l) Recreation areas within project

There are hazardous conditions at many public recreation areas and scenic overlooks within the project area that may require special consideration. Recreation areas should be away from sudden drop-offs, swift currents, or other dangers. Playgrounds should be located away from heavily traveled roads. Adequate fencing should be used in high use areas. Public safety at recreation areas is under the jurisdiction of State and local agencies but can be enhanced by the support of the project authority. Potential safety problems are to be identified by the project authority and brought to the attention of the appropriate authorities. Adequate safety measures in the Recreation Plan of the project should be considered with.

6.4 OPERATIONS AND OTHER FACTORS AFFECTING PUBLIC SAFETY

a) Many hazardous aspects of projects are not present at all times and they may not be readily observed during a brief inspection of the project. Therefore, it is important to consider the full range of spillway and plant operations that could cause hazardous conditions.

b) Peaking operations are more hazardous than run-of-river operations, since tail waters are normally calm and low flows occur between periods of generation. When generation begins, the tail waters could rise rapidly and become swift and hazardous in a very short time.

c) Tailrace areas are particularly dangerous when spillway gates are opened quickly and without warning, discharging flows into dry or calm areas below spillways. Remote operation or automation of hydro projects may increase the chances of accidents at the projects by eliminating the observations, judgments, and warnings of an operator. Therefore, un-manned, remotely-controlled facilities may require more safety devices to adequately warn and protect the public.

d) Weather factors, such as rain, ice and fog, can make project surfaces slick and obscure visibility, resulting in signs being less effective. Relatively safe areas in the daytime may become dangerous at night when neither signs nor the hazardous areas can be observed. During windy or high flow conditions, public may not be able to adequately react to warning signs or lights if they are forced into hazardous areas. The potential for these situations should be taken into account when considering site-specific public access restrictions and safety requirements.

e) At some projects recreational use is very heavy during special seasons. Such concentrations of recreationists can create special safety problems as they compete for the use of areas that provide the better recreational opportunities and likely tend to cause recreationists to take more risks.

f) The proximity of recreation areas to dams and powerhouses should be scrutinized. If they are too close to these facilities, special safety considerations may be necessary. For instance, during periods of high flows, the area should be physically closed to the public, but provisions should be considered for rescue purposes.

g) Public respect for high dams and spillways is generally greater than it is for low structures. Low head and small diversion dams can be especially hazardous due to the indifference with which the

public views them. Very dangerous reverse or "keeper" flows are often created on the downstream side of small overflow dams, and even those with as little as two feet of head can be very dangerous. Therefore, all the dams should have some sort of safety protection regardless of the size of the dam and reservoir,

h) It is possible that, at times, several of these factors could simultaneously occur at a project, making otherwise safe areas dangerous. Therefore, it is necessary that consideration be given to these and other factors when analyzing the hazardous aspects and determining the safety needs of a project.

6.5 SAFETY DEVICES AND MEASURES

As a general rule, all projects will require some types of safety devices, warning systems or other measures. The amount of protection necessary increases as public exposure to the hazards increases. For example, one canal may require only a warning sign, while another may require safety nets, escape ladders, several warning signs, and the need to be enclosed with a chain link fence.

Safety devices and measures can be divided into five basic categories: Educating and informing the public, Visual and audible warnings of hazardous areas, Physical restraining devices, Escape devices, Procedures for safer project operations

6.6 EDUCATION AND INFORMATION

Project Authority to take the initiative to educate and inform the public of the specific hazards near its hydro projects and of the general rules that should be followed to be safety conscious. Where appropriate, information could be disseminated in recreational brochures, company literature, video tapes, television or radio announcements and in newspaper articles and advertisements. If feasible, as part of its public relations program, project authority should make every effort to meet with the public at schools, CIVIC organizations, etc., communicate with the public through the media, and distribute literature on safety practices.

6.7 WARNING DEVICES

Warning devices include such items as danger and warning signs, audible warning devices, lights and illumination, beacons and strobe lights, buoys, and verbal warnings. These devices are required where necessary to warn of hazardous spillways, powerhouse intake areas, tailraces, and other hazardous areas and conditions.

Danger and Warning Signs – Each dam should have adequate danger and warning signs. Properly located and spaced signs can be an effective method of preventing persons from entering hazardous areas. It is important to locate signs so that persons entering an area from any direction can see one or more of the signs. The size of lettering and the signs themselves should be of sufficient size that persons (even those with less than perfect eyesight) would not have to enter the danger zone to read the signs. Signs should convey a message that clearly advises the reader of the real danger. In addition to "No Trespassing" or "Keep Out" signs, it would be informative to have signs that state: "Danger- Dam Ahead': "Danger of Drowning': or "Stay Alive by Staying Out': Signs should be kept in good repair and fading signs should be repainted. Plants and grasses that obstruct warning signs should be removed. Contrasting colors should be used for sign lettering and background. A regular sign inspection program should be developed and documented to ensure that all signs are maintained in good condition. It is particularly important to inspect signs after severe weather conditions.

Power and Communication Lines- Power and communication lines require special signing to warn public of clearance heights.

Audible Devices- Audible devices, such as sirens, horns, or buzzers, are generally used to warn of sudden changes in the rate of flow, usually in tail water areas of spillways or powerhouses. It is also important to provide warnings of sudden changes in operation, such as a gate opening to pass flows that have been cut-off due to a plant shutdown. It may also be prudent to have a warning sounded shortly before the releases are made. Signs advising of the meaning of the audible devices should be posted along all access points near hazardous tail water and other areas.

Lights- Lights can be used to illuminate signs, the dam itself, and other hazardous areas. It is desirable that at least some of the warning signs around dams be illuminated at night. Lighting should be considered at dams, tailrace areas, substations etc. for night visibility. Specially designed signs, safety devices, including lighting may often be necessary so that safety devices, including lighting may often be necessary so that safety devices are effective under adverse weather conditions. A regular lighting inspection program should be developed to ensure that lights are functional.

Beacon- Beacons and strobe lights can be utilized near spillway gates and overflow spillways and, if appropriate, they should be activated to provide visual warning when water is being discharged. Flashing strobe lights can also effectively compliment audible warning devices.

Verbal Announcements- Recorded or direct verbal warnings can be used at projects to warn public in tailrace areas that gates are going to be opened.

6.8 RESTRAINING DEVICES

Restraining devices include boat restraining barriers, fences, guardrails, natural barriers, trash racks and other similar devices.

Boat Barriers- Boat restraining barriers as well as warning devices should be provided at those projects where boaters are exposed to hazardous spillways, tailrace areas or intake areas. Protection must be provided in such areas if there is recreational activity near the dam/barrage/reservoir so that accident can be avoided.

In addition to the boat barriers, warning signs are required on the dams/ barrages to identify and warn of the hazards. These signs help the public to understand why they are being restricted by the boat barriers. It is important to determine the proper location of the barriers at each dam/barrage/reservoir. They should be located an adequate distance upstream to be free from the heavy currents upstream of open gates and intake areas.

Fences- Fences and guardrails are required at most projects to prevent public access to hazardous areas such as dams, powerhouses, substations, intake areas, wing walls, etc. Fences, together with signs and locked gates, are probably the most effective means of prohibiting land based access to hazardous project features. At some projects, buildings, high cliffs and other natural barriers can serve, along with fences, to limit public access.

Trash racks- Trash racks are located upstream of most powerhouse intake structures and occasionally in other areas. They are primarily installed for the purpose of catching debris. It is not desirable for the public to have access to trash rack areas because they are dangerous, particularly where there is high velocity flow. Trash racks do prevent those who have fallen in the water from being drawn into turbines, penstocks, siphons, or other enclosed water passages. As such, they serve as last chance safety devices. However, adequate safety devices and warning signs should be the first line of protection because the probability of survival if someone is caught by a trash rack is doubtful.

Guards- Uniformed guards and watchmen are employed in some heavily used public areas to enforce regulations and warning signs and minimize trespassing and vandalism.

6.9 ESCAPE DEVICES

Escape devices include such items as life preservers, safety ropes, escape nets, escape ladders etc. These items provide a means of escape for persons who are otherwise unable to get out of hazardous conditions.

Life Preservers- It is desirable that life rings and safety ropes be provided near dams, powerhouses, and canals, etc., where it is likely that someone would be available to use them to aid a victim. These devices should be in readily accessible locations and well identified.

Escape Ladders- When project waters flow through open but confined channels, such as lined canals and concrete lined intake and tailrace areas, self escape devices such as ladders may be considered. Their location or use should be carefully considered to avoid unsafe use of such devices. If these devices would create a more unsafe condition, then they should not be employed.

Safety Nets- When canals terminate at hazardous structures such as powerhouse and penstock inlets, or spillways, etc., safety cables, booms, or safety nets are often necessary for a victim to grab when caught in midstream. If they can be safely installed and used, escape ladders should be located on each side of the safety booms or safety nets to adequately provide for an effective means of escape. Use of safety ladders should always be governed by whether or not they will, in fact, increase public safety.

6.10 PROJECT OPERATING PROCEDURES

In certain situations, changing project operating procedures can improve safety conditions at a project. For instance, altering or imposing ramping rates for generating units can prevent sudden increases in tail water levels. Modifying gate opening procedures may reduce or eliminate sudden surges in flows or may be used to direct flows to less hazardous areas. It should be encouraged to review operating procedures as they affect public safety and to propose any necessary improvements, consistent with good public safety practices and the primary purposes of the project.

6.11 VANDALISM, ENFORCEMENT AND OTHER PROBLEMS

At remote projects, especially located in isolated areas, acts of vandalism are common. Fences and gates may be tom down or cut; safety signs may be painted over, tom down or even stolen; life rings may be removed and safety ropes and cables may be cut or destroyed. While such acts are difficult to prevent, efforts can be made to reduce them. Wire cables can be substituted for ropes and heavier

and stronger gauge materials can be used. If possible, signs can be stenciled on concrete surfaces. Frequent surveillance of vandal-prone areas may be effective. Safety devices should be freshly painted and well maintained to be effective. Back-up systems should be installed where feasible. Facilities damaged by acts of vandalism should be promptly restored.

Consultation with officials of local authorities to assist in enacting laws or ordinances prohibiting public from entering dangerous areas can improve the effectiveness of the safety rule. This approach is desirable and provides local enforcement so that the public does not misuse or defeat the purpose of the safety barriers. There should be adequate enforcement of safety laws and regulations.

6.12 BASIC PUBLIC SAFETY MEASURES

Each project should be reviewed for public safety needs on a case-specific basis. It is extremely important to assess the number and type of public safety measures at any project based on the public use patterns at a project. The following is a partial listing of public safety measures that should be considered when undertaking a case by-case review of public safety at a project:

a) Boat restraining barriers upstream of overflow, gated spillways, powerhouse and canal intake areas, and upstream of natural channels that extend to project structures.

b) Warning signs in tailrace areas.

c) Clearly visible and legible warning signs at an appropriate distance upstream of the dam and facing the reservoir of each dam.

d) Fences at substations and restricted access to hazardous areas around dams and other project structures.

e) Audible warning devices, together with signs to explain their meaning, at those projects with sudden changes in operation that result in large flows and rapidly changing tail water levels.

f) Restricted public access to powerhouses, intakes, and other operating structures.

g) If determined necessary, signs (at least one) facing the reservoir on each dam with a hazardous spillway, with lighting for night time visibility.

h) Signs posted at surge chambers to warn of sudden discharges.

i) Warning signs posted to warn boaters that may have to pass under low bridges.

j) Trash racks in the intake areas of powerhouses.

k) Where safe installation can take place, escape devices installed at about 250 foot intervals in steep-sided or concrete lined canals.

l) Power line clearances in accord with appropriate codes.

m) Spillways, intake areas, and tailrace areas sufficiently lighted at night (i.e. perimeter lighting) to be recognizable from the shoreline and the reservoir, if appropriate.

The public safety aspects of each project should be periodically evaluated by the project authority. The Project Authority should have a well-conceived public safety plan. A review of all public safety devices and measures should also be made to identify the need for additional devices or improvements to existing devices. Working cooperatively, the goal of maximizing public safety at all the projects can be achieved.

6.13 SAFETY IN OTHER CIVIL CONSTRUCTION WORKS (POWER PROJECTS)

Scaffolding

a) Posts shall be accurately spaced, erected on suitable bases, and maintained properly.

b) Runners shall be erected along the length of the scaffolding located on both the inside and the outside posts at even height. Runners shall be interlocked with the posts. Runners shall be interlocked to form continuous lengths and coupled to each post. The bottom runners shall be located as close to the base as possible. Runners shall be placed not more than 6.5 feet on centers.

c) Bearers shall be installed transversely between posts and shall be securely coupled to each post bearing on the runner coupler. When directly coupled to the runners, the coupler shall be as close to the post as possible.

d) Bearers shall be at least 4 inches but not more than 12 inches longer than the post spacing or runner spacing.

e) Double Lanyard Safety Harness shall be used by all workmen engaged in scaffold erection works.

f) Ladders & scaffolds specification is to be followed IS 3696 (part 1 & 2).

g) Safety net is to be provided below the scaffolding to trap the falling materials.

h) Handrail & Toe board is to be provided in scaffolding platform.

i) Ladders are to be provided to every scaffolding.

6.14 FORMWORK SYSTEM FOR CIVIL CONSTRUCTION

a) Wall / Column formwork may be used when it is required.

b) Proper Slinging and Rigging Practices should be followed while lifting the formwork assembly.

Suspended Personnel Platform

The use of a crane suspended Personnel Platform / Man-cage shall be limited as far as possible unless is no safer, practical, conventional means of access to an elevated work area.

Handling and use of Explosives during Blasting Operation in Civil Construction

The covers of the explosive cases or packages shall be replaced every time after taking out part of the contents as long as any explosives are left in them. IS 4081: 1986-Safety Code for blasting drilling operations.

Excavation

(i) No excavation or earthwork below the level of any foundation of building or structure shall be commenced or continued unless adequate steps are taken to prevent danger to any person employed, from collapse of the structure or excessive settlement, which is detrimental to structure.

(ii) Foundations, adjacent to and below which excavation is to be made, shall be supported by shoring, as long as the trench remains open.

(iii) Under Cutting / Cave-in should be avoided during excavation.

(iv) Borrowing or mining shall not be allowed. In any trench where such methods have been followed, the cavities left shall be eliminated by cutting back the bank slope before removing any further material from the section of the trench or movement of any heavy vehicle or crane operation.

Means of Access and Escape from Excavation

(i) Stairs, ladders or ramps must be provided where workers are required to enter trench excavations four feet or more in depth.

(ii) Crossover gangway should be fixed in crossover points.

Edge of Excavation

(i) Heavy equipment, such as excavating machinery and road traffic shall be kept back from the excavated sides at a distance not less than the depth of trench or at least 6m for trench deeper than 6m.

Fences, Guards, Lighting & Warning Signals

(i) Protection to Public.

(ii) At every of a trench likely to be frequented by the public, fences, guards or barricades shall be provided to prevent to persons or livestock from falling into the trench.

Harmful Gases and Fume

(i) It shall be ensured that no harmful or fumes are present in the trench to such a degree as may endanger the health or safety of persons working in them.

6.15 INDIAN STANDARDS IN CONSTRUCTION SAFETY

Indian Standards have been developed to provide necessary guidance for carrying out tasks at work places providing safe and healthy work environment. Following is a gist of Indian standards that are widely used in construction site;

I. IS 818:1968 - Safety and health requirements in electric and gas welding and cutting operations

Scope: This code lays down requirements for the protection of Persons from injury and illness, and properties (including equipment) from damage by fire and other causes, arising from electric and gas welding and cutting equipment, its installation, operation and maintenance. The general provisions of this code of practice are applicable to welding and cutting of metals and other allied processes.

II. IS 1161:2014 Steel tubes for structural purpose

Scope: This standard covers the requirements for hot finished welded (HFW), hot finished seamless (HFS), and electric resistance welded (ERW) or High frequency induction welded (HRIW) plain carbon steel tubes for structural purposes.

Ill. IS 2750:1964 -Specification for steel scaffoldings

Scope : This standard lays down the requirements for materials, fabrication and performance of steel scaffoldings constructed with tubes, fittings and/or prefabricated frames, suitable for use in normal building construction work.

IV IS 3016:1982 Fire Protections in welding and cutting operations (First revision)

Scope : This standard covers the recommendations for the practices for prevention of loss of life and/or property by fire or explosions arising out of the use of gas and electrical-equipments for the purposes of welding, cutting and similar operations. General precautions recommended cover the normal work area designated for the fabrication and maintenance activities using welding, cutting and allied processes and also special situations having potential fire hazards.Special precautions recommended cover the situations or locations that are susceptible to fire or explosion hazards arising out of the presence of naked flame or sparks, capable of igniting the combustible materials present in the area.

V. IS 3696:1987 Scaffolds & ladders (Part I - Scaffolds)

Scope : This standard (Part 1) lays down the safety requirements for the erection, use and dismantling of scaffolds for providing access and for supporting workmen, equipment and materials for any construction work including maintenance, repairs and demolition. The requirements of the various components

of the scaffold, the care in their use and storage are also covered.

VI. IS 3696:1991 Scaffolds & ladders (Part II - Ladders)

Scope : This standard (Part 2) lays down the safety requirements for ladders used for the various jobs in general construction work including maintenance and demolition.

VII. IS 3764: 1992 Excavation work

Scope : This standard lays down the requirements for carrymg out safety the excavation work, such as trenches, test pits, cellars, borrow pits, cuttings for rail, canal and road formations and all excavations on which the sides of excavations are not trimmed simultaneously to a stable slope. The requirements laid down in this standard do not apply to the following:

Any part of a trench where the depth is less than 1.5 m; b)A trench into which no person is required to enter for any purpose; and c) Any part of a trench made for a pipeline or conduit if the trench is mechanically excavated, the sections

of the pipeline or conduit are permanently assembled before being mechanically placed in the trench and the trench is mechanically back-filled.

VIII. IS 4014:1967 Steel tubular scaffolding (Part I and Part 11)

Scope: This code (Part I) covers common definitions and general guidance in regards to selection of materials for tubular scaffoldings.

IX. IS 4081:2013 - Blasting and related drilling operations

Scope : This standard lays down the safety requirements for blasting and related drilling operations in locations other than mines.

X. IS 4082: 1996 --Recommendations on Stacking and storage of Construction materials and components at site

Scope : This standard provides general guidance regarding stacking and storage of construction materials and components at site.

XI. IS 4130: 1991 - Demolition of buildings (Second revision)

Scope : This standard lays down the safety requirements for carrying out safely the demolition/dismantling of all types of buildings, for example, residential building (load-bearing structure, multistoried framed structures), public buildings and factories.

XII. IS 4138:1977 Working compressed air (First revision)

Scope : This standard lays down the safety requirements for work in compressed air undertaken in construction works, such as foundations of bridges, docks and tunnels.

XIII. IS 4435 : 1967 Trestles and Ladders (Part II)

Scope : This standard covers the requirements of wooden ladders and trestles of the following types:

a) Single-section ladders (standing ladders and step ladders),

b) Extending ladders,

c) Steps (swing back), and

d) Folding trestles.

XIV IS 4756: 1978 - Tunneling works (first revision)

Scope : This standard lays down the safety requirements for tunneling and underground excavations in rocks and soft strata. This code does not cover tunnels made in connection with mining operations, gassy tunnels, and also tunnels made in running ground, where special methods like shield tunneling are adopted.

Note - 'Gassy tunnel' wherever used in this code shall mean a tunnel in any part of which flammable gas is present or is likely to occur including a tunnel which is adjacent to or is approaching any such place.

XV IS 4912:2024 Safety requirements for floor and wall openings, railings and toe boards

Scope : This standard deals with the safety requirements to be provided in buildings where there is danger of persons or materials falling through floor or wall openings, or from stairways, or catwalks. The recommendations laid down in this standard apply to buildings under construction as well as to buildings after their completion.

XVI. IS 5121: 2013 Piling and other Deep foundations

Scope : This standard lays down the safety requirements for piling and other deep foundations as stated below:

a) Safety measures while working with a piling rig, and

b) Safety measures while preparing other deep foundations.

XVII. IS 5916:2013 - Construction involving use of hot bituminous materials

Scope : This standard lays down the safety requirements for constructions involving the use of hot bituminous materials.

XVIII. IS 7293:1974 -Working with Construction machinery

Scope : This standard lays down the essential requirements for safety in the operation and maintenance of earth moving, lifting and hoisting, transporting and other construction machinery. It also includes some features of design and construction, which are essential for safe working of construction machinery.

XIX. IS 7969:1975 - Handling & Storage of Building materials

Scope : This standard lays down the safety requirements to be observed in handling and storage of building materials at building sites and receiving depots.

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CHAPTER-VII

SAFE WORK PRACTICE

Safe work practice is not a choice, it's the rule

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CHAPTER-7

SAFE WORK PRACTICE

7.1 GENERAL SAFE WORKING PRACTICES:

Some of the practices to be adopted to avoid risks associated with working with electricity are briefed hereunder:

a) Tool Box Talk (TBT):

Prior to start of daily work, TBT is to be conducted by the workers to discuss on hazards associated with their assigned job(s), safe working practices, and the use of personal protective equipment (PPE) relevant to the specific tasks of the day. Relevant information from Incident Register is to be shared in the meeting. These briefings help raise awareness, bring alertness, refresh workers' knowledge and help in maintaining safe work culture. Regular TBT with documented agenda and attendance would ensure consistence and effective communication of safety practices.

b) Creating Safe Working Zone:

Work is to be done creating Safe Working Zones so as to avoid accidents due to back fed power or accidental lifting of shutdown or line charging etc. Effectively Earthed Discharge Rods, High Voltage Detectors are to be used for ensuring creation of Safe Working Zone.

c) Worksite Barricading:

Where there would be any necessity felt to keep the worksite inaccessible to everybody, not part of the working team, to avoid any distraction caused to the workforce or accident due to close proximity to hazardous infrastructure, some area surrounding the worksite should be kept barricaded. Traffic cone & nylon rope along with required caution board would be used to cordon off the area.

d) Arranging Local Earthing:

Earthing is to be arranged suitably at worksite as per prevailing guideline using earth rod / pipe / plate of appropriate quality, size & material and inserting it into ground as prescribed.

e) Working with PPE & Safety Devices:

Work is to be done using appropriate PPE & Safety Devices.

f) Strict adherence to Safety Norms:

Working in presence of & as per guidance of supervisor, working with properly trained persons having requisite authorization, working with Permit to Work or Field Memo as per applicability, taking proper shutdown to make the working infrastructure dead, to discharge the electrically disconnected installation prior to start of work, to visually check isolator contacts to ascertain proper isolation of working infrastructure from live part, to draw out, lockout & tag out switchgears, to make effective earthing arrangements, to religiously keeprecord of all alterations in infrastructure and / or device settings made temporarily accessibleto all concerned – are some of the safety norms to be followed.

7.2 SAFE HANDLING PROCEDURES FOR SOME ELECTRICAL INSTALLATIONS & EQUIPMENT:

Safe handling procedures for some electrical installations / equipments targeted to avoid any untoward incident, damage to system or its collapse are given hereunder:

a) Power Transformers (PTR)

(i) Before commencing any work on a PTR Permit To Work (PTW) need be obtained.

(ii) Switchgears of both primary (33 kV) and secondary (11 kV) side are to be turned off firstly. Then isolator on 33 kV side is to be opened and truck of 11 kV switchgear is to be downed to ensure visual disconnection from live line.

(iii) In case Potential Transformer (PT) is connected, that also need be isolated and its low voltage fuse is to be taken out to avoid possibility of transformer getting power through synchronizing or voltmeter plug.

(iv) In case the transformer neutral is connected to common earth, prior to commencement of work that connection has to be removed. If the transformer is solidly earthed separately, that need not be disconnected.

(v) The Transformer is to be discharged to ground before commencing any work. The incoming side of the PTR is to be shorted at isolator or fuse terminal and to be connected with shunting earth. HV & LV terminals are also to be shorted and earthed.

(vi) It is always to be presumed that explosive gases are present on top of oil inside the oil tank and as such prior to working with the tank sufficient air circulation is to be ensured.

(vii) Open fire, lamps with open flame, matchsticks, lighted cigarette etc. must not be brought in close proximity of the transformer to avoid any fire hazard. Smoking should also be strictly prohibited near the worksite.

(viii) Persons working on transformers shall not carry any loose article like key bunches/mobile etc. in their pocket.

(ix) A list of all the requisite & available tools is to be prepared. All the tools are to be wrapped in cotton tape and get it tied on waist. After completion of work all the items are to be checked with the list to ensure that no loose tool is left on top of the transformer.

(x) While working on any one of two transformers connected in parallel, both sides i.e, HV & LV side of it are to be disconnected from live line and work is to be commenced after ensuring proper earthing.

(xi) Transformer oil level is to be checked regularly.

(xii) Transformer oil should be tested six-monthly with its proper documentation.

(xiii) For PTRs installed before the year 2016, PCB testing for its oil is highly recommended.

b) Circuit Breaker

While doing maintenance work on circuit breaker, make sure:

(i) Line and bus isolators are open.

(ii) Main fuses at the switchboard/panel should be taken out and D.C. voltage should be disconnected from breaker mechanism.

(iii) Tools and equipment should be kept in safe working condition.

(iv) In outdoor HV or EHV breakers, work must be done from ladders or platforms alongside the breaker. Energized overhead conductors are dangerous for workmen to stand on the breakers. Essential work may be done from the top of the breaker only if protective barriers have been installed.

(v) Be sure that auto-re-closure feature has been by-passed during maintenance work.

(vi) Ensure Trip Circuit and Spring Charge are in healthy condition and all relays associated with the breakers are in reset condition.

(vii) Body of all instrument transformers shall be earthed.

(viiii) Secondary of the current transformer shall not be opened when it is on.

c) Isolator & Earth Switch Operation

(i) The isolators should not be operated (open or close) on load in any case.

(ii) The isolators should not be operated (open or close) on charged line/ transformer in any case.

(iii) After opening isolator be sure that there is no pole stuck and all three poles of the isolator have been opened fully.

(iv) After closing isolator be sure that all three poles of the isolator have been closed.

(v) The isolator should be locked with pad lock and key after obtaining PTW and should be closed when PTW is returned properly.

(vi) When PTW is given for work on substation, the earthing switch is closed after opening the isolator and both should be locked in position till return of PTW.

d) 11 kV Metal Clad Switchgear

(i) While working on panel mounted circuit breakers a "DANGER" notice shall be placed on the handle.

(ii) When the work is to be carried out on busbar spouts, the bus bar on which work is to be carried out shall be made dead and isolated from all points of supply.

(iii) The bus bar shall be earthed with approved earthing equipment at a panel other than at which work is to be done on the isolated section of busbar.

(iv) Rack out the entire breaker connected with the bus and isolate the bus from the breaker.

e) 30V Battery & its charger

(i) Battery charger room should be well ventilated.

(ii) While preparing electrolyte, always add acid to water and not water to acid. Smoking, open flames or the use of tools or any other devices that are liable to cause sparks, shall be avoided in battery room.

(iii) When handling acid or batteries; use proper tools, rubber aprons &hand gloves.

(iv) Portable fire extinguisher should be kept inside the room.

(v) Hard rubber or other insulating materials shall be mounted as insulating support if storage battery cells are not composed of glass.

(vi) If the acid is splashed in the eyes, immediately flush eyes with water.

f) Capacitor Bank

(i) The capacitor is an electrical device which can store energy even if the power supply is withdrawn. One interlock should be provided so that it cannot be energized within 10 minutes after switching off the circuit.

(ii) Before attending the capacitor bank discharge rod should be used in all 3phases.

(iii) In unloaded condition the capacitor bank should not be put into service.

(iv) The capacitor bank should be put in to circuit on monitoring the voltage level and power factor level of the related circuit. It should be switched off when the voltage exceeds 12 kV.

(v) If two banks are controlled by single 11 kV panel & if there is a problem in one bank, the other bank should not be kept in operation.

(vi) If capacitor bank trips due to earth fault or overcurrent or unbalance, it should not be kept in operation.

g) Underground Cable

(i) For isolation of cables open at least one of disconnecting switches or fuses in every source through which the cables can be made alive including leads to the cables of potential transformer, then discharge the cable to earth.

(ii) Cable route indicators should be provided and records of cable route should be maintained to assess correctly the particulars of all underground cables in the vicinity of the faulty cable.

(iii) Use of sharp-edged crow bars or picks axes to be avoided while excavating the earth to locate faulty cables.

(iv) All cables in the vicinity of the fault area shall be exposed and identified to establish the identity of the faulty cable.

(v) Before high voltage cable is cut, the senior authorized person shall make definite checks to identify the cable to ensure that the cable has been made dead and earthed. He shall then spike the cable in an approved manner at a point where the cut is to be made.

(vi) Before any high voltage joint or chamber is to be opened it is not desirable to spike the cable

(vii) entering the joint or chamber, the senior authorized person shall satisfy from cable route records and if necessary by approved tests that the joint or chamber is associated with the particular cable which has been made dead and on which it is safe to work.

(viii) Employees shall not step on live cables even though they are insulated and enclosed in a lead sheath. Tools and materials shall not be rested against the sheath of the cable.

(viii) After measuring IR value of the cable using Megger, the cable must be discharged by connecting to any local earth point through 2 megaohm resistance and not directly.

- **Laying of cables –** As per CEA regulation, 2023, sub regulation 79,

1) No underground power cable of voltage level up to 33 kV shall be laid without a minimum underground depth from ground surface to top of the cable as per relevant standards:

Provided that the underground power cable meant for use exceeding 33 kV shall be laid with a minimum underground depth of 1.2 metre.

2) No underground telecommunication cable shall be laid without a minimum separation distance of 0.3 metre and 0.6 metre to the underground power cable of voltage up to 33 kV and exceeding 33 kV, respectively.

h) Distribution Transformer

(i) Any work shall be carried out under a 'permit to work' only.

(ii) Periodical health check-up of DTR is required.

(iii) Separate earthing other than transformer neutral earthing should be fixed.

(iv) Transformer bushing condition should be free from dirt and crack.

(v) Before changing or replenishing oil or painting, all exposed live parts of the transformers shall be disconnected.

(vi) While working on poles that have lightning arresters installed on them, the workman shall avoid touching lightning arrester and lightning arrester jumpers

(vii) If there is common neutral earthing in multi DTR structure, the neutral of the DTR under maintenance must be disconnected prior to commencement of work.

7.3 HANDLING OF AERIAL BUNCHED (AB) CABLE, ITS ACCESSORIES & ASSOCIATED INFRASTRUCTURE FOR SAFE DRAWING & MAINTENANCE

a) Cable drums shall not be thrown or dropped from the carrier truck. Loading as well as unloading is to be done using a crane, or hoist, or by raising a platform at the level of the carrier truck body.

b) The drums shall be stored on metal floor or floor made of bricks. Flanges shall be in vertical position with space in between the drums as unsystematic keeping of the cable drums will result in damage to the cable.

c) Drum shall be rolled in the direction of arrow marked on the drum and unwinding of the cable from the drum shall be done by loading the drum on the jacks and rotating it or by putting it on turn table, flat on the flange.

d) Before laying AB cable drum number and its length need be checked, screen continuity to be checked wherever available of each phase and it is to make sure that required size & number of pulleys are available for pulling cable from the drum for laying.

e) Stringing of AB cables shall be done in a proper manner ensuring insulated conductors do not get damaged during installation. Dragging the ABC on the ground is not permitted. Pulleys installed on poles shall be used to pull AB cables.

f) The unreeling of AB cables shall be continuous so as to prevent formation of loops between the supports.

g) The poles are to be erected first and then the pulleys are to be tied on it.

h) The pulley system should be developed to draw the cables directly from the drums aerially and from other side the stretch should be given through a developed mechanism.

i) Sufficient number of trained manpower need be deployed to avoid damage of the cable during laying.

j) Thorough visual inspection of pole need be done prior to taking up the job of AB cable drawing or replacing bare conductor with AB cable. In case any damage is identified visibly, or, by any other effective means, that pole must be replaced with a new one prior to climbing on it or working on conductor replacement.

k) Temporary support stays (guy ropes) should be provided to prevent the pole from falling.

l) In case existing service cables are jointed after bare to AB cable conversion work or for any other inevitable reason, that information has to be brought to the notice of concerned custodian field unit to let them replace those service cables with new ones.

m) It is extremely important that Phase & neutral wires are identified properly from its markings in order to avoid any possibility of phase & neutral wire interchange.

o) Extra cables after bare conductor replacement, if remain, are to be looped / tied carefully to avoid abrasion and consequent insulation damage even in long term.

p) The cables drawn up to the distribution boxes piercing it from the main cables, need be handled with utmost care to avoid any damage to its insulation due to continuous abrasion with sharp metallic edges or with any rough surface.

q) Minimum clearance above ground to line shall be maintained. Sag tension charts for installing AB cables shall be developed by the Contractor taking into consideration of cable characteristics, maximum / minimum temperatures and maximum wind pressure as per service conditions. Based on this, clear guidelines shall be provided to ensure bare/insulated messenger is pulled at appropriate tension so that;

   (i) Required ground clearances are maintained, and

   (ii) Messenger conductor tension is maintained well below its breaking load at all temperatures.

r) Dynamometer method or sag method may be used to ensure appropriate tension of neutral messenger during installation. Over tensioning of neutral messenger should be avoided to ensure its tension does not exceed permissible loading limits at low temperatures. Loose spans of AB cable should be avoided to maintain permissible maximum sag at high temperatures. Loose spans may only be allowed for short spans in special cases. This applies when there are practical difficulties to install necessary stays or struts as required at a t-off point. A short loose span of AB cable may be used in this case to transfer the stay / strut support point upstream or downstream of the line.

s) Stringing of AB cable shall be done using proper equipment such as stringing blocks with plastic coated pulleys, pulling (come along)

clamp, cable hoist and pulling tool, dynamometer etc. Proper equipment recommended by the ABC manufacturer shall be used to avoid any damage to the cable during installation. Temporary stays or strut poles shall be employed as necessary during stringing operation to ensure safety of personnel and equipment. Phasing of insulated conductors shall be identified by one, two and three ridges on the XLPE insulation. Same phasing shall be maintained accordingly through the line. Interchanging of phasing at any connection point is not permitted.

t) Mid-span joints shall be generally avoided by properly planning stringing work. In exceptional cases where mid span joint is required, pre-insulated compression connectors shall be used. The joints for each phase shall be staggered along the cable. No mid-span joints are allowed for AB cable sections running across a street.

u) Neutral cum messenger wire earthing is to be done effectively at poles to keep the Ground Potential Rise within tolerable limit during standard fault clearing time. and quality of earthing wire to be used at poles & other locations need be maintained diligently.

v) Earthing shall generally be carried out in accordance with the requirements of latest CEA regulations (as amended from time to time) and the relevant regulations of the Electricity Supply Authority. The Contractor shall ensure every 6th pole of ABC line including neutral messenger and any metallic hardware is earthed with spike earth (20x2500 mm) as per existing practice of Employer. (for normal soil).

w) During neutral cum messenger wire earthing effort must be taken to keep earth resistance as low as practicable and for that all the joints must be tightened.

x) Necessary Permit is to be taken during maintenance. Site-in charge is responsible for ensuring isolation of the faulty cable at both ends & in providing the PTW. It is to be ensured that the cable is de-energized, isolated & grounded.

y) The in-charge should ensure the availability of proper tools and accessories, safety kits, spares and consumables required for carrying out the work.

z) AB cable feeder is to be patrolled to identify & remove unauthorized tappings, if any, before commencement of the work.

aa) In case the AB cable is drawn through areas covered with trees/ vegetation, tree branches may be trimmed effectively to avoid abrasion on AB cables.

ab) AB cable joints are to be made properly. AB cable should not be jointed temporarily. In case of emergency also, proper joints must be made immediately. For jointing of messenger wire, mid span jointing sleeve should be used instead of binding wires.

ac) OH distribution box (3-sigma/1-sigma) should be mounted on pole for giving service connection.

ad) The incoming and outgoing cables must pass through the rubber grommets supplied with the service junction box.

ae) Incoming cable of suitable size for Distribution box should be jointed with AB Cable by Insulation Piercing Connector (IPC).

af) The service wire is to be hanged on supportive GI wire between pole support and the house.

ag) Earthing procedure as recommended for safe erection & maintenance need be followed religiously.

7.4 SAFE PROCEDURE FOR PREVENTIVE/ CORRECTIVE MAINTENANCE OF AB CABLE

Type of fault/abnormality	Action to be taken
Flashover due to phase to phase & phase to earth insufficient clearance at cable termination	Ensure sufficient clearance
ABC /Termination failure due to improper connection between Earth Braid at End termination & Messenger wire	Ensure use of 1Φ Al conductor PVC insulated unarmored cable with thimble for connection between Earth Braid & Messenger wire
Termination failure due to contaminated surface of terminations	Ensure cleanliness of terminations
Cable failure due to Cable testing at high voltage (DC)	Ensure testing at recommended Voltage
Stress developed due to Improper Bending radius	Maintain Proper Bending radius
Stress/Abrasion developed due to improper clearance between poles & AB cable suspended on suspension clamps	Ensure proper clearance between poles & AB cable
Stress developed due to improper span length	Ensure proper span length
Insulation degradation due to Hot spot at joints and terminations	Ensure proper tightening/ crimping of Lugs & Ferrules
Insulation degradation due to overloading	Ensure recommended loading

Type of fault/abnormality	Action to be taken
Insulation degradation due to improper earthing connections	Ensure ABC & Messenger earthing connected to earth/ground
Insulation Damaged during installation due to improper laying practices	Ensure use of Rollers/Pulleys and rope to discontinue improper Laying practices.
Water ingress leakage current through the cable insulation/cable failure, corrosion in cable conductor & armour due to ingress of water	Ensure lug seal to be intact at outdoor termination

7.5 SAFE PRACTICES FOR POLE ERECTION:

a) A safe pole erection procedure involves a risk assessment, ensuring all workers use appropriate Personal Protective Equipment (PPE) and that the work area is clear of hazards and public access.

b) It requires using approved and certified lifting equipment with properly trained operators, as well as securing the pole with temporary supports like guy ropes before applying conductor tension.

c) Finally, all tools and equipment must be inspected, and a Permit-to-Work system should be implemented for high-risk activities, especially near energized power lines.

d) The work of erection of poles can be classified in two segments as "Before the Work'' and ''During the Work'' which are elaborated hereunder.

A. Before the work:

(i) Risk Assessment: Conduct a thorough risk assessment to identify potential hazards and develop safe working procedures.

(ii) Permit to Work: Secure a Permit to Work (PTW) system for any high-risk activities, ensuring competent personnel are assigned.

(iii) Area Preparation: Inspect the work area for hazards, and ensure it's free from public access by using barriers and warning signs.

(iv) Equipment Inspection: Inspect all tools, lifting equipment, and machinery to ensure they are approved, fit for purpose, and in good working condition.

B. During the work:

(i) Personal Protective Equipment (PPE): Ensure all ground workers wear safety helmets and shoes, and those working at elevated positions use safety harnesses.

(ii) Proper Shutdown Procedures: For any installation or maintenance, follow proper shut down procedures, including notifications and lockouts, to prevent hazards.

(iii) Secure the Area: Properly manage and secure the construction site to prevent unauthorized access.

(iv) Communication: Establish full-proof communication using walkie-talkies or mobile phones for effective coordination between workers and the operator.

(v) Controlled Lifting: Use approved lifting equipment and have a qualified, authorized operator to lift the pole.

(vi) Base support of poles: The steel bottom plate shall be used in steel tubular poles, and cement concrete reinforced plate shall be used as base plate for PCC/WPB Poles.

(vii) Backfilling: Pits shall be backfilled and compacted completely with sufficient added backfill piled above grade to prevent depressions being created by natural compaction.

(viii) Temporary Support: Secure the erected pole with temporary support stays (guy ropes) before applying tension to the conductors to prevent it from falling.

(ix) Final Inspection: Inspect the erected pole for stability and proper position, ensuring it is aligned and perpendicular.

7.6 SAFETY IN CONSTRUCTION & MAINTENANCE OF HT & LT OVERHEAD LINE

a) The erection or maintenance work shall not be carried out during high wind, thunderstorms and heavy rainfall, which would make the work hazardous. However, in emergency situations if restoration is inevitable under extreme weather condition special and effective measures are to be taken to work with safety.

b) The designated officer shall have to keep all the equipment, tools and plant in safe operating conditions.

c) Wherever work is carried out at roadsides or highway crossings etc. road traffic need be controlled effectively. Needful permissions from the concerned

authorities, such as department of public works, highway authorities, police etc., shall have to be obtained before commencement of work.

d) Similarly, for erection or maintenance work at railway crossing, the permission from railway authority for Railway block shall have to be obtained before commencing the work.

e) The designated officer shall ensure the required clearance to be maintained while equipment are moving under or near the energized lines. While moving under or near an energized line, moving equipment need be earthed by earthing chain(s) etc. and it is to be ensured that no part of earthed equipment is at a height of 2.44 m (i.e. 8 feet) above the ground. While operating equipment from a fixed position, the section safety working clearance shall have to be maintained from operating floor in/on earthed equipment.

f) Before commencing the stringing operations or removal of conductors a briefing shall be held by the supervisor with the workmen setting forth the following:

(i) Plan of operation,

(ii) The type of equipment, tools, and plants to be used,

(iii) Grounding devices and procedures to be followed,

(iv) Crossover methods to be employed,

(v) The Clearance authorization required.

g) Wherever there is a possibility of the conductor being handled coming in contact with an energized conductor, or there is a possibility of a dangerous voltage build up due to induction, the conductor being handled shall be grounded, unless a provision is made to insulate or isolate the employee. If the existing line is deenergized, a PTW shall be obtained and the line shall be grounded on both sides of the crossover. In case the PTW cannot be obtained, the line shall be considered as energized for all practical purposes. While executing the work of crossing over an existing line, suitable guard structures with rope nets shall be installed to isolate the conductors and workmen coming within the required minimum clearances specified for the voltage. If there is any auto reclosure installed on the energized line, the same shall be made inoperative. In addition, the line being handled shall be grounded on either side of the crossover.

h) The conductors being strung or removed shall be kept under control by using adequate reels, guard structures, tie line, or any other appropriate means to prevent accidental contact with energized wires. The guard

structure shall have sufficient strength and have adequate dimensions and supported adequately. The wire ropes, come along clamps, anchors, guys and hoists shall have ample capacity to prevent failure / accident. The load rating specified by the manufacturers for stringing equipment, pulley blocks, all other load bearing hardware and tools shall not be exceeded during operations. These shall be inspected regularly and replaced/ repaired when found damaged or when its dependability is doubtful. Come along clamps designed for the specific conductor range only shall be used. During stringing or removal of conductors, no workmen shall be permitted to come directly under overhead operations, or on the crossarms. The dead end points of the conductors at section towers shall be adequately anchored before commencing operations. The grounds provided to the conductor shall be maintained intact until the conductors are hooked on to the insulators.

i) Vertical and horizontal clearance from the ground level from all adjacent structures.

j) Ensure safety devices are fixed on the line (cradle guard should be fixed below HT line in market places, busy roadside & crossing of roads).

k) Ensure generator line should not run in the poles of WBSEDCL lines.

l) All practical steps shall be taken to prevent operating the earth moving, lifting and housing machinery in dangerous proximity to a live overhead powerline. In particular, any part(s) of such machinery shall not be permitted to approach as per clause3.1.5.1 of IS7293:1974 within the following distance of overhead powerlines:

- 11 kV and below 1.40 m,
- 33 kV and below 3.60 m,
- 132 kV and below 4.70 m,
- 220 kV and below 5.70 m,
- 400 kV and below 6.50 m.

m) If it becomes necessary to operate the machinery with clearances less than those specified above, it shall be ensured that the overhead power lines are invariably shut off during the period of operating of the machinery. Location of any underground power cables in the area of operation shall also be ascertained and necessary safety precautions taken.

7.7 OTHER SAFETY REQUIREMENTS FOR OVERHEAD LINES

a) Material and strength

All conductors of overhead lines other than those provided in regulation 70 shall have a breaking strength of not less than 350 kgf. Where the voltage does not exceed 250 V and the span is less than fifteen metre and is drawn through the owners' or consumers' premises, a conductor having an actual breaking strength of not less than 150 kgf may be used. **[Ref: Regulation (57) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]**

b) Joints

No conductor or earthwire of an overhead line shall have more than one joint in a span provided that there shall be no joints in the conductor or earthwire in a span of crossing over the highways, express ways and railway lines.

[Ref: Regulation (58) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

c) Maximum stresses and factors of safety:

Overhead lines with the following supports shall have the minimum factors of safety as per the table given hereunder:

Description	Minimum factor of safety
metal supports	1.5
mechanically processed concrete supports	2.0
hand-moulded concrete supports	2.5
wood supports	3.0

[Ref: Regulation (59) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

d) Clearance in air of the lowest conductor of overhead lines

The minimum clearance above ground and across road surface of National Highway or Expressway or State Highway or other road or highest traction conductor of railway corridor or navigational or non-navigational river of the lowest conductor of an alternating current overhead line, including service lines, of nominal voltage shall have the values specified hereunder.

Nominal voltage of system	Clearance above ground			Clearance between conductor and road surface across Highway (m)	Clearance between conductor and rail level across Railway Corridor (m)		Clearance above HFL for River crossing	
	Across Street (m)	Along Street (m)	Elsewhere (m)		Normal OHE (where no double stack containers are to be run on railway tracks.)	High rise OHE for running of double stack containers on railway tracks	Navigational river (m)	Non-navigational river (m)
Up to 650 V	5.80	5.50	4.60	U/G Cable	U/G Cable	U/G Cable	16.50	5.80
11 kV	6.50	5.80	4.60	U/G Cable	U/G Cable	U/G Cable	19.00	6.50
22 kV	6.50	5.80	5.20	U/G Cable	U/G Cable	U/G Cable	19.00	6.50
33 kV	6.50	5.80	5.20	11.60 or U/G Cable	U/G Cable	U/G Cable	19.00	6.50

[Ref: Regulation (60) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

e) Clearance between conductors and trolley wires

No conductor of an overhead line crossing a tramway or trolley bus route using trolley wires shall have less than some specified clearances above any trolley wire.

[Ref: Regulation (61) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

f) Clearance from buildings of lines of voltage and service lines not exceeding 650 V

An overhead line shall not cross over an existing building as far as possible and no building shall be constructed under an existing overhead line.

[Ref: Regulation (62) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

g) Clearances from buildings of lines of voltage exceeding 650V

An overhead line shall not cross over an existing building as far as possible and no building shall be constructed under an existing overhead line.

[Ref: Regulation (63) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

h) Conductors at different voltages on the same supports

Where conductors of different voltages are laid on the same supports, the owner shall make adequate provision to guard against danger to linemen and others, from the lower voltage system being charged above its nominal voltage, by leakage from or contact with the higher voltage and the methods of construction and the applicable minimum clearances between the conductors of the two systems shall be as provided in regulation 71.

[Ref: Regulation (64) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

i) Erection or alteration of buildings, structures, flood banks and elevation of roads

If at any time subsequent to the erection of an overhead line, whether covered with insulating material or not or underground cable, any person proposes to erect a new building or structure or flood bank or to raise any road level or to carry out any other type of work whether permanent or temporary or to make in or upon any building, or structure or flood bank or road, any permanent or temporary addition or alteration, such person and the contractor whom he employs to carry out the erection, addition or alteration, shall give intimation in writing of his intention to do so, to the supplier or owner and to the Electrical Inspector and shall furnish therewith a scale drawing showing the proposed building, structure, flood bank, road or any addition or alteration and scaffolding thereof required during the construction.

[Ref: Regulation (65) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

j) Transporting and storing of material near electric lines

No rods, pipes or similar materials shall be taken below, or in the vicinity of any bare overhead conductors or lines or shall be brought within the flash over distance of bare live conductors or overhead lines and no flammable material shall be stored under the electric line.

[Ref: Regulation (66) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

k) General clearances

For the purpose of computing the vertical clearance of an overhead line, the maximum sag of any conductor shall be calculated on the basis of the maximum sag in still air and the maximum temperature as specified under regulation 59 and

computing any horizontal clearance of an overhead line the maximum deflection of any conductor shall be calculated on the basis of the wind pressure specified under regulation 59.

[Ref: Regulation (67) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

l) Routes in proximity to airport or aerodromes

Overhead lines shall not be laid in the vicinity of airports or aerodromes unless the Airport Authorities or concerned defence authorities have approved in writing the route of the proposed overhead lines as per relevant standards.

[Ref: Regulation (68) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

m) Maximum interval between supports

All conductors shall be attached to supports at intervals not exceeding the safe limits based on the ultimate tensile strength of the conductor and the specified factor of safety.

[Ref: Regulation (69) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

n) Conditions to apply where telecommunication lines and power lines are carried on same supports.

Where a telecommunication line is laid on supports of an electric line of voltage exceeding 650 V, a suitable arrangement shall be made to safeguard any person against injury resulting from contact, leakage or induction between such electric and telecommunication lines.

[Ref: Regulation (70) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

o) Lines crossing or approaching each other and lines crossing street and road

Where an overhead line crosses or is in proximity to any other overhead line or any telecommunication line, provisions, as laid down in CEA regulation (71) are to be followed.

[Ref: Regulation (71) of CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

7.8 SAFETY FOR HIGH VOLTAGE SWITCHING

General Safety of Switching Operation

(i) High Voltage Switching is done only with the permission of Authorized Engineer who is held responsible for that substation.

(ii) The concerned operator-in-charge will inform other Substations related to the switching operation.

(iii) Record in the book.

a) Place of Switching.

b) Part of Switching.

c) Who has given order for switching

d) What is the sequence of switching

e) Date & Time.

Safety during Switching Operation

(i) Do not make haste for switching operation.

(ii) Examine everything.

(iii) Take written permission before doing anything.

(iv) Use helmet and hand gloves.

(v) Circuit breaker should be switched off first, then open the isolator. During switching on, the isolator should be closed first then close the breaker.

(vi) After switching on operation, everything should be recorded.

Safety in Switchyard Operation and Maintenance:

(i) The maintenance gang in charge should take permit to work from the substation in charge.

(ii) Helmet, full body harness, hand gloves, toolkit to be used.

(iii) Line should be earthed at both sides of the work using discharge rod/ sorting chain & local earthing is to be done.

(iv) If feeder shutdown is to be taken, switchgear is to be tripped, truck should be downed & the shutdown notice is to be hanged to the concerned feeder & then PTW is to be taken from the substation operator.

(v) Do not operate the isolator when line is on load.

(vi) If switchgear is not there then gang isolator load should be opened by use of hand gloves. During opening of isolator, ensure all blades are in open condition and operating handle should be tied with chain & hang danger board.

(vii) The gang in-charge will take PTW from the substation.

(viii) After taking shutdown from the substation use discharge rod to discharge the line. Both the side all the conductors need to be shorted & make earthing by earth chain & use all PPE during work.

(ix) During working on DP & Four pole structure both the sides of the line should be disconnected from live line.

(x) To stop back-feed from any source, the operator should inform other end.

(xi) Take precaution regarding induced voltage from nearby charged line.

(xii) During lightning, stop work on line.

(xiii) Do not work in darkness.

7.9 SPECIAL SAFETY MEASURES FOR HILLY TERRAIN:

a) Pole Design and Placement

Use of M-Type Tower for HT Lines or galvanized steel Tubular poles with deep and secure foundations helps withstand soil movement. Pole placements should consider slope stability and avoid areas prone to landslides.

b) Adequate Ground Clearance

Maintaining standard vertical and horizontal clearances is crucial, particularly in sloped regions where sag can vary. Proper tensioning of conductors and use of anti-climbing devices are essential.

c) Lightning Protection and Earthing

Installing lightning arresters at key points and ensuring proper earthing using counterpoise or ring earthing systems is required to mitigate lightning-related surges and faults.

7.10 SAFETY REQUIREMENTS FOR TESTING ENGINEERS AT THE SITE OF MANUFACTURERS' WORKS

a) Pre-Test Preparation

(i) Permit to Work (PTW): Obtain a valid PTW and ensure it's signed by authorized personnel.

(ii) Isolation: Ensure the Job/Equipment to be tested is isolated from all power sources. Verify using a tester/meter.

(iii) Earthling: Properly earth all parts of the Job/Equipment to be tested to avoid build up of residual or induced voltages.

(iv) Tagging & Lockout:

- Apply lockout/tagout procedures on control panels and switches. Inspection:
- Visually inspect the Job/Equipment to be tested for oil leaks, damage, or foreign objects.

b) Personnel Safety

(i) PPE Requirements:

- PPE Equipments must be made available as per requirement of the manufacturing site.

(ii) Training:

- Ensure all personnel are trained and aware of emergency procedures and hazards.

(iii) Minimum personnel:

- Only authorized and trained personnel should be allowed near the testing area.

c) Equipment Safety of Test Instruments and Job to be tested:

(i) Cables & Leads:

- Insulated test leads. to be used of the required voltage/current level
- Sharp edges of the instrument and the job to be tested must be kept away from walkways

(ii) Grounding:

- All test equipment and job to be tested must be properly grounded.
- Use shorting/earthing rods if required.

(iii) During Testing Warning Signs:

- Placement of safety signage around the test area.
- Job to be tested should be isolated by providing proper protecting barricades during Testing

(iv) Job/ Testing Equipments subject to High Voltage

- Un-authorized personnel to be removed from the site of Test prior to start of Test
- Authorized Testing personnel are only allowed during the test
- Testing Personnel involved in the testing must maintain proper distance from the HV Equipments

(v) Emergency trip mechanism

- Instantaneous activation of Emergency Trip mechanism should be incorporated in case of any occurrence of fault or any mal-operation by testing personnel.

(vi) Communication:

- Ensure proper Communication between testing crew and control personnel at the manufacturing unit

(vii) Monitoring:

- Continuously monitoring of key parameters to be done during the testing process.

(viii) No Bypass:

- SOP as laid down for the testing should be strictly adhered to.
- Protection relays or interlocks must be in service during the testing process.

(ix) Discharge Mechanism:

- Proper discharge mechanism to be installed and used afters tests to eliminate residual charges.

d) Post-Test Procedures: De-Energize & Discharge:

Ensure the Equipments to be tested are fully discharged.

(i) Remove Grounds:

- Re-inspect the Job being tested for any abnormalities post-testing
- Carefully remove temporary grounding devices after testing. .

(ii) Restoration:

- Return the Equipment being tested to normal condition and remove lockout/ tagout with proper sign-off.

e) Emergency Protocol

First Aid: Keep first aid kit and trained personnel on site.

- **Fire Extinguisher:** Ensure appropriate fire extinguishers are available and accessible.
- **Evacuation Plan:** All team members should know the emergency exit and shutdown procedures.

7.11 SAFE OPERATING PROCEDURE IN CUSTOMER CARE CENTRE

a) To keep proper job register & ensure proper recording & maintenance of the same.

b) To ensure proper performance of duties by working & supervisory personnel as per duty roster & attendance register.

c) To avoid unauthorized engagement of outsider in repair, maintenance & other works in LT line.

d) To ensure use of Personal Protective Equipment (PPE).

e) To ensure use of safety devices and earthing equipment (discharge rod, earthing chain, earth spike etc.) during work.

f) To ensure periodic tree branches trimming from adjacent to LTOH line & line patrolling.

g) To issue Field Memo before each & every (O&M) work.

h) To organize awareness program on safety and its necessity among departmental staff, contract labours & general public.

i) To Label for duplicate/multiple source of supply in same structure/ pole.

j) To ensure standard span length & stringing in LTOH line & avoid loose connection.

k) To keep fire extinguishers in prominent places with valid refill date & carryout mock fire drill quarterly.

l) To ensure installation of spacers in LT line as per norms, after reducing sag of existing line.

m) To earmark and visibly display assembly point and escape route. Emergency numbers, exit signage etc. are also to be displayed at suitable locations.

7.12 SAFE PROCEDURE FOR HANDLING OF ELECTRIC SUPPLY LINES & APPARATUS

a) Before any conductor or apparatus is handled, adequate precautions are taken, by earthing or other suitable means, to discharge electrically such conductor or apparatus, and any adjacent conductor or apparatus if there is danger therefrom, and to prevent any conductor or apparatus from being accidentally or inadvertently electrically charged when persons are working thereon, and relevant standards are followed in this regard.

b) No person is allowed to operate and undertake maintenance work on any part or whole of an electrical plant or electric supply line or apparatus and no person is allowed to assist such person on such work, unless that person has been designated in that behalf and observes the safety precautions prescribed.

c) For the safety of operating personnel, all non-current carrying metal parts of switchgear and control panels are properly earthed and insulating floors or mats conforming to the relevant standards, of appropriate voltage level are provided in front and rear of the panels where such personnel are required to carry out operation, maintenance or testing work. Identification of all panels have been provided at front and at the rear.

d) Safe Working Distance:

The following safe working distances shall be observed for hot line maintenance operations:

Phase to Phase	Safe Clearance
kV	Metre
11	0.61
33	0.71
66	0.91
110	1.02
132	1.07
220	1.52
400	2.13

7.13 SAFE PROCEDURE FOR HANDLING ELECTRIC SUPPLY LINES & APPARATUS FOR CARRYING OUT SHUTDOWN WORK OR TESTING

Precautions to be observed:

(i) Before commencement of any shutdown work or testing in an electric supply line or apparatus, the Engineer or Supervisor in-charge of the work or testing shall identify the possible hazards, such as; electrocution, flash over, fall of person from height, fall of objects from height, failure of tools and plants, fire, and the like, that may been countered while carrying out the work or testing near charged area and take necessary precaution to protect the working personnel.

(ii) The Engineer or Supervisor in-charge of the work shall, before commencement of any work, brief the entire working group or gang regarding the hazards that may be encountered and the necessary precautions to be taken by them.

(iii) The Engineer or Supervisor in-charge of the work shall obtain proper permit-to-work from the concerned Operation In-charge(s) and ensure that the electric supply line or apparatus or section is isolated from all sources of energy, de-energised and earthed.

(iv) The Engineer or Supervisor in-charge of the work shall ensure that adequate and appropriate local earths are fixed at the zone of working, and the earthing rods remain connected to the isolated section of the electric supply line or apparatus or section till all men and materials have been moved away to safe zone and permit to work is returned on completion of the work.

(v) If the local earths are required to be removed for any testing purpose, the same shall be done only when all the working personnel are in the safe zone, on the ground or on the tower, and in the presence of the Engineer or Supervisor. If the working personnel are required to go up or approach the conductor(s) subsequently for any work, such as removal of test leads, tightening or adjustment, they shall be permitted to proceed only after re-fixing the local earths, as required.

(vi) The Engineer or Supervisor in-charge of the work shall positively confirm by suitable means that the electric supply line or apparatus or section is totally dead before giving clearance for the working personnel to approach the same.

(vii) The Engineer or Supervisor in-charge of the work shall, while carrying out the shutdown work or testing, ensure that working personnel are maintaining safe distance from the adjacent charged electric supply line or apparatus or section, and also, no objects, such as tools and plants, ladders, cranes, man-lifts, and the like, are moved, so as to infringe the safe distance, endangering the working personnel.

(viii) Mobile cranes, derricks, manlifts and wheel mounted ladders shall be effectively earthed when being moved or operated in close proximity with energized apparatus or section.

(ix) Portable ladders and poles shall be carried only in the horizontal position when being moved in close proximity with energised lines or equipment or area.

Further Precautions to be observed

(i) Adequate and effective supervision shall be ensured by the owner as well as the contractor for all activities while working or testing on electric supply lines and apparatus when any shutdown work or testing is done near charged electric supply line or apparatus or section.

(ii) Lone worker shall never be allowed to work on electric supply lines, equipment and apparatus or while testing.

(iii) Sufficient supervisory personnel shall be deployed for close monitoring while various type of works are under progress at the same or different locations. Supervising work shall never be delegated to the subcontractors' personnel.

(iv) The deployed Supervising Personnel shall not leave the working spot when shutdown work at height or testing is in the progress, as the working personnel may not be aware of the consequences of unsafe practices. No

other work, which requires them to move out of the location, shall be undertaken by supervising personnel, when shutdown work or testing is in the progress.

(v) Wherever shutdown activities are required to be carried out for more than one day on any electric supply lines, apparatus or section, earthing provided at the said work site shall be inspected by the Engineer or Supervisor everyday morning for their healthiness, fitness and proper tightening, before giving clearance for the working personnel to climb the tower or structure to resume the work.

[Ref: Part-II; Schedule I; CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

7.14 SAFE PROCEDURE FOR HANDLING HIGH VOLTAGE DIRECT CURRENT APPARATUS FOR CARRYING OUT SHUTDOWN WORK OR TESTING (>40V)

Precautions to be observed

(i) The Engineer or Supervisor in-charge of the work shall obtain proper Permit-To-Work (PTW) from the concerned Operation In-charge(s) and ensure that the electric supply line or apparatus or section is isolated from all sources of energy, de-energised and earthed.

(ii) Before commencement of any shutdown work or testing of high voltage direct current apparatus, the Engineer or Supervisor in-charge of the work or testing shall identify the possible hazards, such as; electrocution, flash over, fall of person from height, fall of objects from height, failure of tools and plants, fire, and the like, that may be encountered while carrying out the work or testing near charged area and take necessary precaution to protect the working personnel.

(iii) The Engineer or Supervisor in-charge of the work shall, before commencement of any work, brief the entire working group or gang of the hazards that may be encountered and the necessary precautions to be taken by them.

(iv) Attach warning labels to all neighboring installation parts (to be removed after the works have been carried out).

(v) The Engineer or Supervisor in-charge of the work shall allow access to the valve hall, DC Filter Area, AC Filter Area and DC hall (if any) only when the apparatus therein are completely de-energised and effectively earthed.

(vi) The work on AC/DC filter bank shall only begin after earthing the entire capacitor bank.

(vii) There shall be at least ten minutes time gap between earthing the entire capacitor bank and starting the work on bank. Thereafter unit must be short circuited.

(viii) The de-energised bushing shall be checked for stored charge by touching all the surfaces of both indoor and outdoor side composite insulators and all other parts of the bushing using a proper test instrument.

(ix) The gas pressure inside high voltage direct current through wall bushing shall be reduced to a level prescribed by the manufacturer before starting any work or handling of the bushing.

[Ref: Part-III; Schedule I; CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

7.15 SAFE PROCEDURE FOR HANDLING GAS INSULATED SWITCHGEAR (GIS) APPARATUS FOR CARRYING OUT SHUTDOWN WORK OR TESTING

Precautions to be observed:

(i) The Engineer or Supervisor in-charge of the work shall obtain proper Permit-To-Work from the concerned Operation In-charge(s) and ensure that the electric supply line or apparatus or section is isolated from all sources of energy, de-energised and earthed.

(ii) Operation, maintenance and repair must be carried out by trained and certified personnel only.

(iii) Before commencement of any shutdown work or testing of gas insulated switchgear apparatus, the Engineer or Supervisor in-charge of the work or testing shall identify the possible hazards, such as; electrocution, flash over, fall of person from height, fall of objects from height, failure of Tools and Plants, fire, and the like, that may be encountered while carrying out the work or testing near charged area and take necessary precaution to protect the working personnel.

(iv) The Engineer or Supervisor in-charge of the work shall, before commencement of any work, brief the entire working group or gang of the hazards that may be encountered and the necessary precautions to be taken by them.

(v) Wear hearing protection during operation.

(vi) Take care while touching the enclosure at any time as enclosures may heat up to the temperature of 70°C.

(vii) Observe the procedures for storage, transportation, and the use of filling equipment.

(viii) Wear the personal protective equipment: respirator mask (self-contained breathing apparatus if necessary), protective overall, protective gloves, safety shoes, safety glasses.

(ix) Attach warning labels to all neighboring installation parts (to be removed after the works have been carried out).

(x) Provide proper electrical clearance as required by interlocking rules. Mark e.g. main circuits and control circuits with appropriate tags.

(xi) Block off neighboring live parts with screens, insulating mats or spacer grids in order to prevent unintended contacts.

(xii) While working on any compartment in gas insulated switchgear, the immediate adjacent compartment(s) must be also depressurized for safety of the working person.

(xiii) SF_6 gas becomes contaminated and contains poisonous substances after events such as arc faults. Hence, handling of SF_6 in such cases must be done using proper PPEs and by a trained personnel, preferably from the original equipment manufacturer.

(xiv) The switchgear installation shall not be operated if the density of SF_6 gas indicated at the density monitors is not in the operating range.

(xv) Do not remove any protective covers if an assembly is energized.

(xvi) The Engineer or Supervisor in-charge of the work shall ensure that adequate and appropriate local earths are fixed at the zone of working, and the earthing rods remain connected to the isolated section of the electric supply line or apparatus or section till all men and materials have been moved away to safe zone and Permit-To-Work is returned on completion of the work.

[Ref: Part-IV; Schedule I; CEA Regulations on Measures Relating to Safety and Electric Supply, 2023]

7.16 SAFE HANDLING PROCEDURE FOR EQUIPMENT CONTAINING OR CONTAMINATED WITH PCB

Contaminated oil contains polychlorinated biphenyls (PCBs), man-made organic chemicals that are persistent, toxic, and can accumulate in the environment

and human tissues, causing adverse health effects and environmental damage. For avoiding any hazards to human and environment as well, precautionary measures need be taken to store, label, handling , transportation and disposal in a very careful manner following the National and International guidelines so far adopted.

Identification and Assessment

a) **Inventory**: Identify and label all equipment containing PCBs, noting the concentration of PCBs and the volume of oil involved.

b) **Testing**: Take oil samples using PPE (gloves, goggles, masks) to determine the PCB concentration.

c) **Classification**: Classify oil based on PCB concentration:

(i) Less than 50 ppm: considered fine, no immediate action needed.

(ii) 50-500 ppm: remove the oil, flush the unit, and refill to bring concentration below 50 ppm

(iii) More than 500 ppm: Immediately drain and remove the oil, flushing the unit multiple times to ensure the concentration is below 50 ppm.

Handling and Storage

a) **Restricted Access:** Restrict entry to areas with PCB-contaminated equipment and notify personnel of the risk

b) **PPE**: Workers must wear impervious gloves, chemical-resistant overalls, and eye protection. For higher exposure risks, a respirator may be necessary.

c) **Leakage** Containment: Place separate containers to collect any leaks from equipment or storage

d) **Labeling:** Containers must be clearly and permanently labeled with the contents (e.g., «DANGER: CHEMICAL [PCB] WASTE»), concentration, quantity, source, and date of storage.

e) **Segregation:** Never mix PCB-contaminated oil with other waste oils. Use separate, well-labeled containers for waste segregation.

f) **Safe Storage**:

(i) Store all PCB-containing equipment in dry, enclosed sheds to prevent leaks and soil contamination

(ii) Store PCB oil and equipment in a secure, covered area with a waterproof floor and continuous curbing.

(iii) **Weatherproofing:** The storage area should be located indoors or, if outside, covered to prevent rain from reaching the equipment and containers.

g) **Spill Management**:

(i) Wipe up spills with waste cloth, collect all cleaning materials (including gloves) in a designated plastic bag or drum, and store them safely for proper disposal.

(ii) Any spill should be contained immediately with absorbent materials. Contaminated wipes and PPE must be collected and treated as PCB waste.

h) **Access control:** The storage area should be accessible only to trained, authorized personnel.

i) **Authorized transporters:** Only licensed companies are authorized to transport PCB-contaminated oil. The transporter or sender must be authorized by the relevant environmental authority

Disposal:

a) **Licensed specialists**: Disposal of PCB-contaminated oil and waste should be done through specialist, licensed waste contracting firms only.

b) **Prohibited Practices**: PCB shall never be drained or discharged directly or indirectly on land, in surface water or in effluent treatment plant from defective, out of use/in use PCB containing or contaminated equipment.

c) Equipment containing PCB (Poly Chlorinated Biphenyl) are to be maintained properly in order to prevent leakage or release of PCB to environment.

Hygiene:

a) Wash hands thoroughly with soap and water after any contact with PCBs or contaminated surfaces.

b) **Skin Contact:** If skin contact occurs, wash the affected area with soap and water.

c) **Eye Contact:** In case of eye contact, flush with a saline solution or clean water and seek medical attention.

d) **Accidental Ingestion:** Do not induce vomiting. Seek immediate medical advice.

7.17 IMPORTANCE OF CONDITION MONITORING FOR SAFETY OF THE EQUIPMENT

WBSEDCL is practicing condition monitoring of various high voltage equipment installed at 33/11 kV Sub-Stations. The condition monitoring is health assessment of

the equipment with the help of modern instruments and techniques. The primary cause of equipment failure can be attributed to either insulation failure or abnormal temperature rise. If these two aspects, i.e, condition of insulation and temperature can be checked regularly, equipment failure can be restricted considerably. In other words if any equipment is found to be in vulnerable condition through health assessment or condition monitoring, it can be removed from circuit in time to avoid any catastrophic effect. As part of condition monitoring work Partial Discharge (PD) of any equipment is monitored as PD is widely regarded as best early warning indicator for insulation degradation. With the help of infrared Thermogram or Thermal Imaging temperature rise at junction points as well as at the body of the equipment can be observed and abnormality can be detected. Safety of costly & vital equipment in turn ensures reliability & safety of the power distribution system.

7.18 SAFETY PRECAUTIONS TO BE TAKEN DURING MAINTENANCE OF EQUIPMENTS IN PUMPED STORAGE PROJECT:

a) Readings of physical parameters like pressure, temperature, vibration, Run-out, oil levels, water levels, etc., are to be taken and monitored on a regular basis. Preventive actions are to be taken if any abnormality is found.

b) All annunciations, alarms, and tripping are to be closely monitored.

c) All equipment/installations like the Unit Generator-Turbines, Excitation Systems, Battery Banks, Auxiliary Motors, Circuit breakers, Transformers, Electrical Panels, Compressors, Oil Coolers, Mechanical Actuator Panels, Static Frequency Convertor system, piping networks, etc., are to be periodically checked.

d) Functionality of HVAC System Components, like Chiller Units, Ventilation Fans, etc., are to be periodically checked.

e) Functionality of Flood Alarm System, Fire Alarm System, and Fire Extinguishing System are to be periodically checked and replaced if found faulty.

f) The battery-powered Emergency lights present at evacuation routes, annex building, all floors are to be periodically checked and replaced if found faulty.

g) The readings of Strain gauges, Load Cells, V-Notches, Piezometers, Movement Measuring Devices, etc., related to structural stability of PPSP Power House, Lower Dam, Upper Dam, 400 kV Switchyard are to be regularly taken and monitored.

h) Mock drills are to be conducted at regular intervals as per Emergency Evacuation Plan to ensure readiness among persons working inside the Power House.

7.19 EMERGENCY EVACUATION PLAN FOR UNDERGROUND POWERHOUSE OF PUMPED STORAGE PROJECT:

a) Access Tunnel Collapse: PPSP is an underground powerhouse. Thereby susceptible to Tunnel Collapse. In case of main access Tunnel Collapse due to Earthquake or any other reason, all persons present inside the powerhouse at that point of time are to be evacuated immediately from the Emergency Access Tunnel and vice versa.

b) Flooding/water submergence: In the event of flooding/water submergence, all persons present inside the powerhouse at that point of time are to be evacuated immediately from the Emergency Access Tunnel.

c) Fire Hazard: In the event of any fire hazard, the smoke would tend to flow upwards and escape out of the powerhouse through the Emergency Access Tunnel. There would be a possibility of smoke entrapment if any person tries to evacuate through Emergency Access Tunnel. In that situation, all persons present inside the powerhouse at that point of time are to be evacuated immediately from the Main Access Tunnels.

d) Mono-rail escalator may be kept in-place in the Emergency Access Tunnel for smooth evacuation during emergent situations.

7.20 SECURITY MEASURES FOR IT-OT POSTURE

Physical, Environmental and Safety of People in DC, DRC, and SCADA Buildings

The safety measures implemented in **Data Centers (DC), Disaster Recovery Centers (DRC), and SCADA control rooms** are designed to **protect both critical infrastructure and personnel** working in these facilities. Below is a detailed overview of how these systems safeguard human life while ensuring operational continuity.

A. Protection of Personnel Through Power Redundancy

a) Automatic DG Set Activation (625 kVA with AMF Panel)

(i) Prevents sudden blackouts, ensuring continuous lighting and ventilation for staff safety.

(ii) Eliminates risks of accidents in darkness or loss of critical life-support systems.

b) Uninterrupted Power Supply (160 kVA & 10 kVA UPS)

(i) Ensures emergency lighting, fire alarms, and communication systems remain active during outages.

(ii) Allows safe evacuation if needed without panic due to sudden power loss.

B. Human-Centric Fire & Gas Safety Systems

a) Early Warning & Automated Suppression

(i) VESDA (Very Early Smoke Detection)

- Detects smoke at the **incipient stage**, allowing early evacuation before fire spreads.
- Alarms alert personnel **before toxic fumes or heat become dangerous**.

(ii) FM 200 Gas Suppression System

- **Non-toxic and safe for occupied spaces**—unlike CO_2, it does not displace oxygen.
- Rapid fire suppression **within 10 seconds**, minimizing risk to human life.
- **Automatic abort switch** allows manual override if personnel are inside.

b) Backup Firefighting Measures

(ı) **Portable Fire Extinguishers (CO_2 & ABC Type)**

- Placed in **exit routes, corridors, and non-FM 200 protected areas** for manual firefighting.
- CO_2 extinguishers for electrical fires (prevents electrocution risks).
- ABC-type extinguishers for general fires (paper, wood, flammable liquids).

(ii) Emergency Lighting & Exit Signage

- Illuminates escape routes during power failures.

C. Safe Climate Control for Personnel Health

(i) 12 Precision ACs (9.5 Ton PAC Units)

- Maintain **safe working temperatures** (22–24°C) to prevent heat stress.
- Regulate humidity to avoid respiratory discomfort or electrostatic risks.

D. Physical Security & Access Control for Staff Safety

(i) Restricted Access with Biometric/Card Systems

- Prevents unauthorized entry, reducing risks of sabotage or workplace violence.

(ii) 24/7 CCTV Surveillance

- Monitors for emergencies (e.g., medical distress, fires) to trigger rapid response.

(iii) Rodent Repellers

- Prevents disease-carrying pests from endangering staff health.

E. Emergency Preparedness & Staff Training

(i) Regular Fire Mock Drills

- Ensures all personnel know **evacuation routes, assembly points, and fire extinguisher use**.

(ii) First Aid & Emergency Response Training

- Staff trained in **basic first aid, emergency shutdown procedures, and gas release protocols**.

(iii) Water Leakage Detection

- Prevents **electrocution or slip hazards** from water ingress.

Cybersecurity Best Practices

(i) Network Segmentation: Strict separation between IT and OT networks with monitored communication channels

(ii) Access Control: Role-based access and multi-factor authentication (MFA) for all critical systems

(iii) Endpoint Detection and Response (EDR): Integration with the MITRE ATT&CK framework for real-time monitoring and response

(iv) Incident Response: Established protocols for handling cybersecurity incidents and recovery plans

Business Continuity and Disaster Recovery

(i) High Availability (HA) and redundancy in Data Centres, Disaster Recovery Centres, and cloud systems

(ii) Regular Disaster Recovery (DR) drills to test system resilience with Recovery Time Objective (RTO) and Recovery Point Objective (RPO) analysis

Incident Management

Reporting Cybersecurity Incidents

(i) All incidents will be recorded via the centralized helpdesk tool (helpdesk.wbsedcl.in).

(ii) Incidents will be routed to the relevant teams for investigation and resolution.

(iii) The Computer Security Response Team will coordinate the response, remediation, and follow-up actions.

Third-Party Security

Third-Party Security Audits

(i) WBSEDCL conducts periodic security audits for managing critical IT-OT infrastructure.

(ii) All IT-OT procurements undergo Factory Acceptance Testing (FAT) and Site Acceptance Testing (SAT) to ensure compliance.

IT-OT Redundancy

High Availability (HA) Setup

(i) Critical IT-OT systems (DC, DR, SCADA) operate in HA mode to ensure seamless failover in case of failure.

Infrastructure Redundancy

(ii) All non-IT infrastructure in DC, DR, and SCADA systems are configured redundantly to minimize single points of failure.

Awareness and Training

Employee Training and Awareness

(i) Regular cybersecurity training sessions for employees handling IT-OT assets.

(ii) Training programs for SCADA operators on secure operations to mitigate cyber risks.

(iii) Cybersecurity Awareness Programs to promote safe IT-OT system usage and reporting of suspicious activities.

Security Monitoring and Detection

Security Operations Centre (SOC)

(i) WBSEDCL is in the process of implementing a SOC for continuous real-time monitoring of security events and threats.

(ii) The SOC will provide 24/7 surveillance of critical IT-OT systems and respond promptly to detected threats.

Security Tools and Solutions

(i) EDR: Integration with the MITRE ATT&CK framework for threat detection and response.

(ii) Access Control: Enforcing role-based access controls and MFA for high-risk systems.

Cyber Crisis Management and Response

Cyber Crisis Management Plan (CCMP)

(i) WBSEDCL follows the Cyber Crisis Management Plan (CCMP) to ensure a coordinated response to cybersecurity incidents.

Incident Response Procedures

(i) All incidents are recorded via helpdesk.wbsedcl.in.

(ii) The Incident Response and Management Policy ensures swift identification, containment, eradication, and recovery from incidents.

Best Practices as per CERT-In Guidelines

Email Security

(i) Use email filtering software to block spam.

(ii) Avoid opening links/attachments from unknown sources.

(iii) Enable MFA for email accounts.

(iv) Never share credentials via unsolicited emails.

Mobile Security

(i) Download apps only from official stores (Play Store/App Store).

(ii) Avoid clicking on suspicious links received via SMS/WhatsApp.

(iii) Never share OTPs.

(iv) Enable MFA for mobile apps.

(v) Avoid USB charging in public places.

Social Media Security

(i) Avoid sharing personal information publicly.

(ii) Verify friend requests before accepting.

(iii) Use unique passwords for different accounts.

(iv) Enable MFA for social media profiles.

(v) Restrict profile visibility from public searches.

Aadhaar Security

(i) Lock biometrics via the m-Aadhaar app/UIDAI portal.

(ii) Use Virtual ID (VID) or Masked Aadhaar instead of full Aadhaar numbers.

(iii) Never share Aadhaar OTPs with strangers.

(iv) Delete Aadhaar copies from public computers after use.

Reporting Cybersecurity Incidents

To CERT-In

(i) **Website:** CERT-In Website

(ii) **Email:** incident@cert-in.org.in

(iii) **Toll-Free:** 1800-11-4949

(iv) **Fax:** 1800-11-6969

For Cyber Fraud & Crime (I4C)

(i) **Website:** Cyber Crime Portal

(ii) **Helpline:** 1930

For Vulnerabilities & Collaboration

(i) **Email:** vdisclose@cert-in.org.in (Vulnerability disclosure)

(ii) **Email:** collaboration@cert-in.org.in (Collaboration)

(iii) **Phone:** +91-11-22902600 (Ext: 1012)

CERT-In Free Security Tools

(i) **Bot Removal Tools:** eScan Antivirus, K7 Security, Quick Heal

(ii) **Mobile Security App:** M-Kavach 2

(iii) **Other Tools:** USB Pratirodh, AppSamvid, Browser JSGuard

Ensuring IT-OT security is a shared responsibility. Every WBSEDCL employee and stakeholder must adhere to these guidelines to mitigate risks and maintain uninterrupted operations.

7.21 SAFE WORK PRACTICES FOR HANDLING DISMANTLED ITEMS:

Handling dismantled materials from electrical systems requires strict adherence to safety protocols to prevent electric shock, injury from hazardous substances, and environmental contamination. This involves proper pre-planning, appropriate personal protective equipment (PPE), secure packaging, and responsible disposal through authorized channels.

a) Isolate and verify:

(i) Always shut off and lock out the power supply to the equipment before starting.

(ii) Use a non-contact voltage tester to confirm that the circuit is completely de-energized.

(iii) Post a "Men Working" or other warning sign to prevent accidental re-energization.

b) Create a safe work area:

(i) Use barricades and warning signs to restrict access to the work area.

(ii) If working at heights, use a stable ladder or mobile scaffolding with an attendant to oversee the activity.

c) Conduct a risk assessment:

(i) Survey the work area and equipment to identify potential hazards.

(ii) In case of dismantling of materials in Indoor work like old circuit boards, batteries, and fluorescent bulbs contain toxic substances such as lead and mercury that require special handling.

(iii) In case of dismantling of materials in Outdoor work like bare conductor, iron brackets, insulators, PVC Cables, AB Cable etc. this area should be barricaded and clearly marked to avoid human access and use of PPEs to be ensured to avoid any untoward incident while handling the dismantle materials.

d) Personal Protective Equipment (PPE)

Depending on the task, a variety of PPE is essential to protect the human being during handling the electrical and chemical dismantled items.

(i) Hand and arm protection: Use insulated rubber gloves rated for the appropriate voltage, and wear leather protectors over them to prevent cuts and abrasions.

(ii) Head protection: Wear an electrical-grade hard hat to protect against impact and shock.

(iii) Eye and face protection: Use safety glasses or goggles with side shields to protect against sparks, flying debris, and chemical splashes.

(iv) Foot protection: Wear safety footwear with non-conductive soles to provide insulation against electrical currents.

e) Handling and packaging

(i) Properly handle hazardous materials: Separate hazardous components, such as batteries, capacitors, and mercury-containing lamps, into designated, clearly labeled containers. Use anti-static bags and foam for sensitive electronic parts.

(ii) Use proper lifting techniques: For heavier materials, use mechanical aids like carts or dollies. If lifting manually, maintain good posture and avoid back injuries.

(iii) Securely package components: Place small parts in plastic bags to keep them organized and protected from dust.

(iv) Use appropriate packaging with shock-absorbent dividers for fragile items like circuit breakers.

f) Storage

(i) Store in a designated area: Transport all dismantled materials to a dedicated staging or storage area. This area should be barricaded and clearly marked.

(ii) Keep records: Maintain an updated inventory of all stored components. This

helps track parts for potential reuse and ensures compliance with disposal regulations.

g) Disposal

(i) Use authorized recyclers: Do not dispose of electrical waste in regular trash. Use certified and authorized e-waste recyclers who are equipped to handle hazardous materials safely and legally.

(ii) Comply with regulations: Adhere to federal, state, and local e-waste management and handling rules, such as the Indian E-Waste (Management) Rules, 2022, which mandate proper disposal and recycling.

(iii) Consider reuse and donation: If components are still functional, consider repairing, reusing, or donating them to extend their lifespan and reduce waste.

(iv) Avoid improper methods: Never engage in burning or illegally dumping electrical waste. These practices release toxic gases and chemicals into the environment, contaminating air, soil, and water.

7.22 WASTE MANAGEMENT AS A MEASURE TO MITIGATE THE RISKS FOR HEALTH AND SAFETY:

a). Safe practices for handling of Fuel, Oil, Chemicals:

(i) Fuel, oil and chemicals used to be kept under lock and key and stored in labeled, sealed containers on drip trays to provide secondary containment, ideally, they will be located on an impermeable surface and be under cover

(ii) Mounting of plant containing oil and diesel on drip trays to catch leaks.

(iii) Refuelling operations, equipment servicing and wash down to take place on an impermeable surface at least 25m from watercourses, and 50m from springs and wells, with drainage directed through oil and grease interceptors before being discharged into a settling pond prior to discharge offsite

(iv) Provide sufficient absorbent materials (e.g., sorbents, dry sand, sandbags) on-site for soaking up fuel, oil or chemical leaks/spills.

(v) Provision of an appropriate domestic solid waste and construction waste collection and disposal system.

(vi) Collect and segregate construction wastes, including scrap metal, oils, and solid waste.

(vii) Store all wastes in designated, labeled area in an environmentally sound manner e.g. oils to be stored in sealed drums on drip trays, solid wastes to be stored in an enclosed bin.

(viii) Recover recyclable wastes that could be reused or sold to recyclers.

(ix) Prohibit burning of construction wastes.

(x) Prohibit dumping of construction wastes into canals, rivers, agricultural fields etc.

(xi) on an impermeable surface at least 25m from watercourses, and 50m from springs and wells, with drainage directed through oil and grease interceptors before being discharged into a settling pond prior to discharge offsite

(x) Provide sufficient absorbent materials (e.g. sorbents, dry sand, sandbags) on- site for soaking up fuel, oil or chemical leaks/spills

b) Safe practices for Scrap Metal Waste :

(i) Scrap metal waste to be immediately removed off-site to the designated stores

(ii) Provision of an appropriate domestic solid waste and construction waste collection and disposal system.

(iii) Collect and segregate construction wastes including scrap metal, oils, and solid waste.

(iv) Store all wastes in designated, labeled area in an environmentally sound manner e.g. oils to be stored in sealed drums on drip trays.

(v) Solid wastes to be stored in an enclosed bin.

(vi) Recover recyclable wastes that could be reused or sold to recyclers.

(vii) Prohibit burning of construction wastes.

(viii) Prohibit dumping of construction wastes into canals, rivers, agricultural fields etc.

(ix) Scrap metal waste to be immediately removed off-site to the designated stores.

7.23 POSTERS FOR SUBSTATIONS

Do's & Don'ts
for Electrical Safety
Do's
Don'ts
Only authorized person should enter in electrical substation
Before starting work on electrical panel always wear Arc flash suit and electrical hand gloves
Check the gloves for leakages by inflating it and discard if damaged
Use potable tools with 30mA RCCB
Always use personal lock on VCS (Visual cutoff switch)
During welding ensure that earthing lead is connected to the job
In case Of electric shock raise an alarm for help and use rescue hook to withdraw the person from the live conductor
Never work on electrical equipment without LOTO and work permit
Do not operate the VCS, when equipment is running
Do not use Damage/ustested/ unchecked portable tools
DO not use gloves after expiry date
Do not cross the arc flash protection boundary without proper PPEs
Do not carry out welding job during wet condition (rain or perpitation)
Do not Directly come in contact with electrocuted person before isolating him from the supply

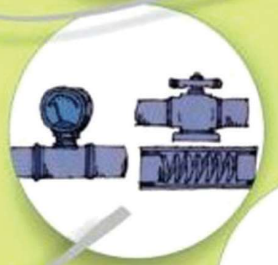
LOCKOUT - TAGOUT FOR SAFETY

LOCKOUT TAGOUT PRODUCTS INCREASE EFFICIENCY AND SAFETY LEVELS ARE USED IN CREATING A SAFER WORKER ENVIRONMENT

1.Identify
the types of energy sources used, potential hazards, and all control devices.

2.Notify
all affected employees.

3.Turn off
all operating controls.



4.Locate
all energy sources.

5.Isolate
all energy sources by blocking, bleeding and venting stored energy as found in springs, hydraulic systems and pneumatic systems.

6.Lock-out
all switches and energy controls in the 'off' or 'safe' position.

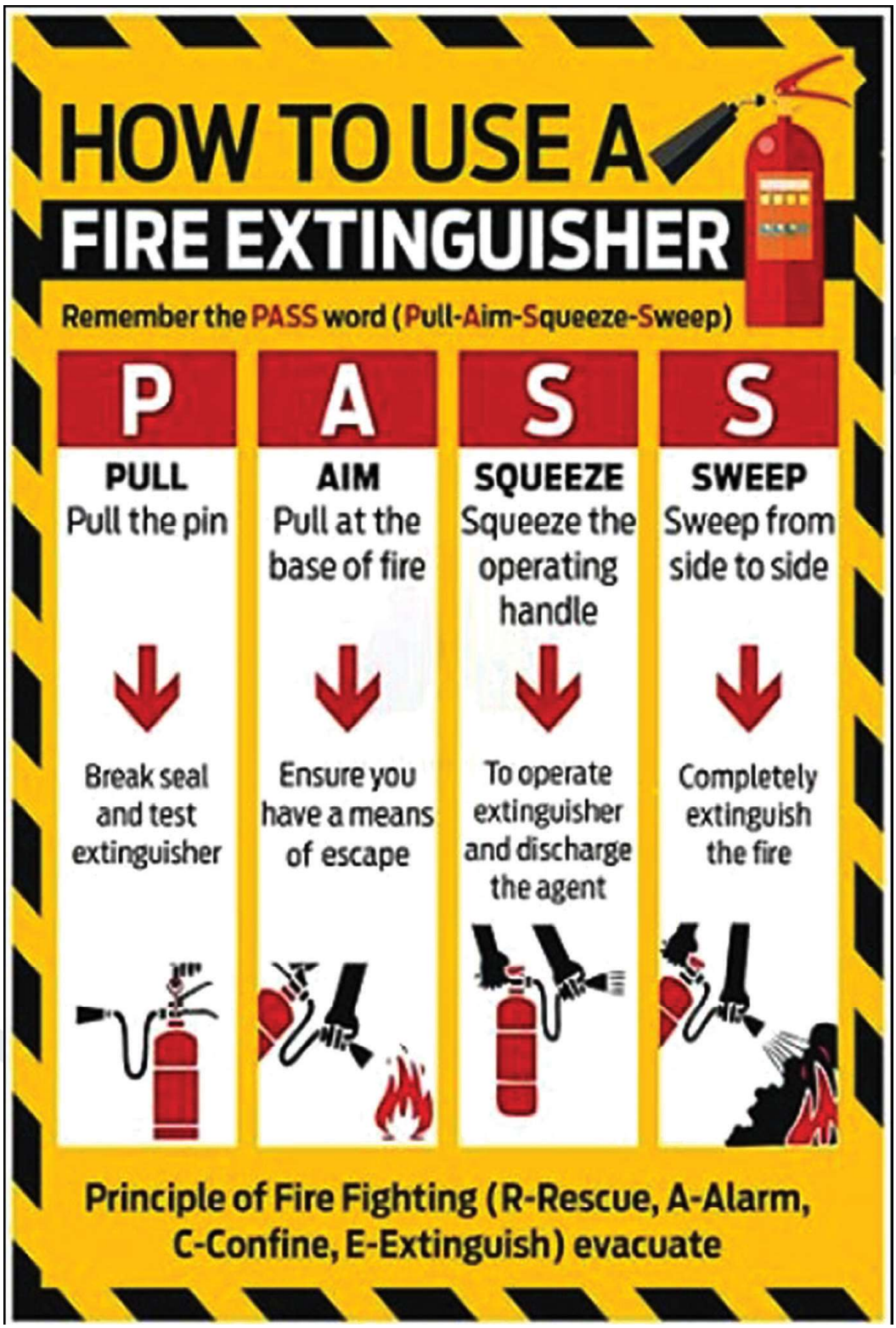
7.Test
the operating controls . Put all controls in the 'on' position. Be sure nobody can get hurt before testing.

8.Return
all operating controls to the 'off' position.

9.Perform
required task.

10.Remove
lock-out devices only after the equipment is fully assembled and all affected employees have been notified. Each lock-out device must be removed by the person who put it on .

Symbols found on fire extinguishers & what they mean	Water	Foam spray	ABC powder	Carbon dioxide	Wet chemical
Wood, paper & textiles	✓	✓	✓	✗	✓
Flammable liquids	✗	✓	✓	✓	✗
Flammable gases	✗	✗	✓	✗	✗
Electrical contact	✗	✗	✓	✓	✗
Cooking oils & fats	✗	✗	✗	✗	✓


HOW TO USE A
FIRE EXTINGUISHER
Remember the PASS word (Pull-Aim-Squeeze-Sweep)
P
PULL
Pull the pin
Break seal and test extinguisher
A
AIM
Pull at the base of fire
Ensure you have a means of escape
S
SQUEEZE
Squeeze the operating handle
To operate extinguisher and discharge the agent
S
SWEEP
Sweep from side to side
Completely extinguish the fire
Principle of Fire Fighting (R-Rescue, A-Alarm, C-Confine, E-Extinguish) evacuate

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WBSEDCL

CHAPTER-VIII

HOUSEKEEPING & UPKEEP OF ELECTRICAL INSTALLATION & ITS PREMISES

Housekeeping is the first defense against unsafe conditions

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CHAPTER-8

HOUSEKEEPING & UPKEEP OF ELECTRICAL INSTALLATION & ITS PREMISES

Workmen are frequently injured, by stumbling, stepping on, or bumping into tools, material and other objects left lying around, or by objects falling from above.

8.1 TO ENSURE GOOD HOUSEKEEPING & FOR UPKEEP OF ELECTRICAL INSTALLATIONS & ITS PREMISES FOLLOWING PRECAUTIONS SHOULD BE TAKEN:

a) Walkways, stairways, fire escapes and all other passageways shall be kept clear of all obstructions to maintain a clean and organized workspace to ensure safe and unhindered movement and access to all equipment.

b) Substation area as well as switch yard area are to be kept properly fenced to restrict unauthorised entry.

c) All electrical structures and yard of substations should be kept free from creepers and vegetations. These need be cleared regularly to prevent any fire hazard there.

d) Tools and materials should not be placed where they may cause tripping or stumbling hazards or where they may fall and strike any one below.

e) Oil leakage point to be rectified and arrested immediately.

f) Spillage of oil and water create slipping hazards and should be cleaned up promptly.

g) Oil drums should be placed away from control panels & other electrical installations and it is to be ensured that secondary containment measures like spill trays are installed beneath the drums to prevent leakage.

h) Nails in boards, such as those removed from scaffolds, and packing boxes, constitute hazards and should be removed. The boards should be carefully stacked or stored.

i) Dirty and oily waste rags should be deposited in approved containers and disposed of as soon as practicable to avoid fire hazard.

j) Broken light bulbs, glass metal and scrap and other sharp objects should be dumped in places or containers provided specially for them.

k) Discarded fluorescent and other gas filled tubes shall be disposed of safely.

l) All hazardous wastes should be identified and stored in a clearly earmarked area and disposal records should be maintained.

m) All potentially hazardous electrical installations should be fenced to restrict undesirable access.

n) Anti climbing devices, wherever needed, would be used to restrict unwanted access to electrical network.

o) Appropriate indication & caution board should be tagged in all hazardous electrical installations like double, triple & four pole supports, DTRs, RMUs etc.

p) Cable trenches of the substations should be neat & clean and may be filled with sand to minimize fire risks. The trenches must be completely covered with concrete slabs.

q) Glands for cable entry & exit in indoor & outdoor electrical installations are to be maintained and replaced whenever needed.

r) Effective pest control is to be done to safeguard electrical installations.

s) Corroded supporting structure should be reconditioned or replaced.

t) No unwanted food stuffs should be thrown inside the cable tranches.

u) No flammable materials should be kept near the electrical installations, panel boxes etc.

v) Combustible items to be stored in areas covered effectively with firefighting arrangement.

w) Earth isolator & mechanical interlocking should be maintained in 33 kV incoming bus in 33 / 11 KV substations.

x) Lightning mast is to be erected in substations to protect its installations against lightning.

y) The substation control room will have insulating mat of proper voltage grade placed in front of all switchgear panels.

z) Silica gel of transformer breather should be reconditioned / replaced.

aa) Specifications of conductors of all existing LT & HT lines are to be kept in registers, along with date of initial drawl.

ab) In consumer premises, earthing system should be installed as per CEA 2023 regulation

ac) Smoking inside office premises in other than designated area and in substations is strictly prohibited.

ad) Safety posters meant for erection, operation & maintenance workers, Cardio Pulmonary Resuscitation (CPR) chart, emergency phone numbers are to be displayed on walls of substation control room with clear visibility.

ae) SNAKE ALERT poster to be displayed in appropriate locations in control room and switch yard of the substation.

af) Breathing apparatus / Oxy pack along with sufficiently equipped First Aid Box to be maintained at each substation & mobile van.

ag) Regular inspection need be conducted by competent authority to ensure that the quantity and quality of PPE are maintained diligently. Findings of inspection are to be recorded in PPE register.

ah) In case any deformity or visible damage is observed in any PPE or if after inspection its condition is declared as poor, or its test result is found to be not satisfactory, the item is to be discarded and replaced by good one immediately.

ai) Rubber gloves, are to be tested once in every six months since its start of use to check its usability.

aj) Earth pits in every substation to be uniquely identified. Earth pit condition need be checked at least once in every six months. Earth resistance test value with date to be recorded for each of the pits.

ak) Firefighting arrangement should be in place. The fire extinguishers along with sand buckets should be routinely inspected and refilled or replaced whenever needed. Periodic inspection should be ensured & records should be maintained.

al) PPE register is to be maintained at every concerned field unit for inspection and tracking of PPE condition.

am) A register should be maintained in all concerned field units to check its tools & tackles regularly.

an) A register namely "Temporary Alternation" register must be maintained in substation to record with date & time all the temporary changes made in network, bus & switchgear arrangements, relay & instruments settings etc.

ao) Accidents, near miss incidents, lost time incidents and other dangerous occurrences are to be kept recorded religiously in "Incident Register" and must be duly communicated to competent authority for early rectification & need based sharing. Previous accident report has to be made available at concerned field units for sharing during tool box talk.

ap) Every available measuring instrument, like, multi meter, insulation resistance meter etc. used to obtain parametric values for ascertaining health and usability of any equipment, device or apparatus etc. need be calibrated annually and every time when it is brought back to service after repair & maintenance. Calibration Certificates are to be preserved scrupulously and kept accessible to the person(s) using the instruments.

aq) Proper arrangement should be there to regularly check availability of the first aid items including medicines. All expired items need be replenished with usable ones immediately.

8.2 TEMPORARY ALTERATION REGISTER:

Apart from a number of registers to be maintained, a register named "Temporary Alterations" is also to be maintained religiously in every substation to record with date, time and signature of authorized person, each & every temporary Alteration or modification in network, like feeder diversion, protection system bypassing, panel reorientation or allocation, change in relay and instrument settings etc. for the following reasons:

(i) load diversion or redistribution in need

(ii) optimizing uninterrupted power supply area minimizing the area of isolation during fault restoration

(iii) load sharing between two feeders due to various reasons outside substations.

(iv) combating emergency situations both in HT/LT and/or malfunctioning of protection system

(v) any other relevant reasons

a) The on duty substation operator would record the information in the register in details.

b) In case any such alteration involves feeders of different substations, records must be maintained in each of these substations. There has to

have exchange of useful information and coordination amongst these substations.

c) Supervising authority & the on duty substation operator with whose consent the changes would be made will have to sign jointly on the record entered in the register.

d) The register would be checked each time prior to any PTW is issued, for any information in connection with the work jurisdiction, both by the requisitioning person and the issuer of the PTW and the receiver of PTW will have to sign with date & time against the relevant information, if any, as a documentary evidence that information was passed on to him duly.

e) After withdrawal of temporary arrangement, suitable remark to be made against the record.

f) The register is to be maintained in an accessible location.

Keeping in mind the frequent unscheduled changes needed to maintain a vast network of electrical installations to combat critical and emergency situations, to avoid untoward incidents in consequence, due to inadvertent non exchange of critical information, this Temporary Alteration Register must be maintained with due diligence.

8.3 INCIDENT REGISTER:

A register named "Incident Register" is to be maintained religiously in every CCC and substation to record with date, time and signature of authorized person, each & every accident, near miss incident, lost time incident (LTI), minor injury and first aid case in its jurisdiction involving employee and agency workers.

A near miss is a close call at work where no injury or damage occurred, but one could have, while a Lost Time Injury (LTI) is a work-related injury or illness that keeps an employee from performing their regular duties for at least one full day or shift. Near misses serve as crucial safety indicators, signalling potential hazards that, if unaddressed, could lead to an LTI or other serious consequences. Reporting and investigating near misses helps identify and prevent future incidents and improve overall workplace safety.

The information as such must duly be communicated to competent authority for early corrective measures & need based sharing. Previous accident report is to be made available at concerned field unit for sharing during tool box talk.

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WBSEDCL

CHAPTER-IX

FIRE PREVENTION & PROTECTION

Fire destroys in seconds, safety lasts a lifetime

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CHAPTER-9

FIRE PREVENTION & PROTECTION

9.1 OBJECTIVES

The basic objectives of fire prevention and protection are:

a) To render firefighting & rescue services without any loss of time anywhere in and around the complex to save life & property.

b) To ensure fire prevention measures against all possible fire hazards to avoid chance of Fire & explosion.

c) To maintain & test periodically all firefighting and detection equipment in a state of reliability.

d) To undertake investigation of all fire incidents and to analyze the root cause and make recommendations to avoid future recurrence

e) To impart fire training to the employees to make them competent to extinguish the fire at the instance of their occurrence and to make them aware about the evils of fire.

9.2 DOS AND DON'TS FOR PROTECTION AGAINST FIRE:

a) At least two easily accessible exit paths in every building / premises are to be maintained for safe exit in case of fire.

b) An assembly point / area inside every building / premises is to be earmarked for enabling safe rescue from there in case of fire.

c) The escape routes from any building / premises and that of the assembly points are to be kept displayed at strategic locations.

d) All emergency contact numbers need be displayed visibly at strategic locations.

e) Running man signage as per IS 12349 along specified fire exit plan/route to be ensured.

f) All the fire / smoke alarms are to be checked regularly to keep those working.

g) There should have good number of trained persons available to use fire extinguishers.

h) Sufficient First Aid arrangements are to be kept at suitable locations.

i) In case of fire, electric supply lines as well as the elevators are to be switched / turned off.

j) During evacuation / escape only steps to be used and elevators must not be used.

k) Fire Brigade need be intimated from a safe place immediately after noticing fire, if needed.

l) Prior to opening any closed door during escaping in case of fire, the door should be touched first to check whether it is hot or not. In case it is detected to be hot, that door must not be opened.

m) Outward opening of exit route doors at places of mass gathering during emergency escape need be ensured.

n) If during escape, smoke is confronted, then the escapers need to sit down low on the floor.

9.3 CLASSIFICATION OF FIRE:

As per the Indian standard IS 15683:2018, fires are classified in four classes according to the fuel involved.

a) Class "A" Fires:

These are fires involving solid materials normally of an organic nature (compounds of carbon), in which combustion generally occurs with the formation of glowing amber, where the cooling effect of water is essential for extinguishing the fire.

b) Class "B" Fires:

These are fires involving flammable liquids or liquefiable solids. Foam or DCP is used to put out these fires.

c) Class "C" Fires:

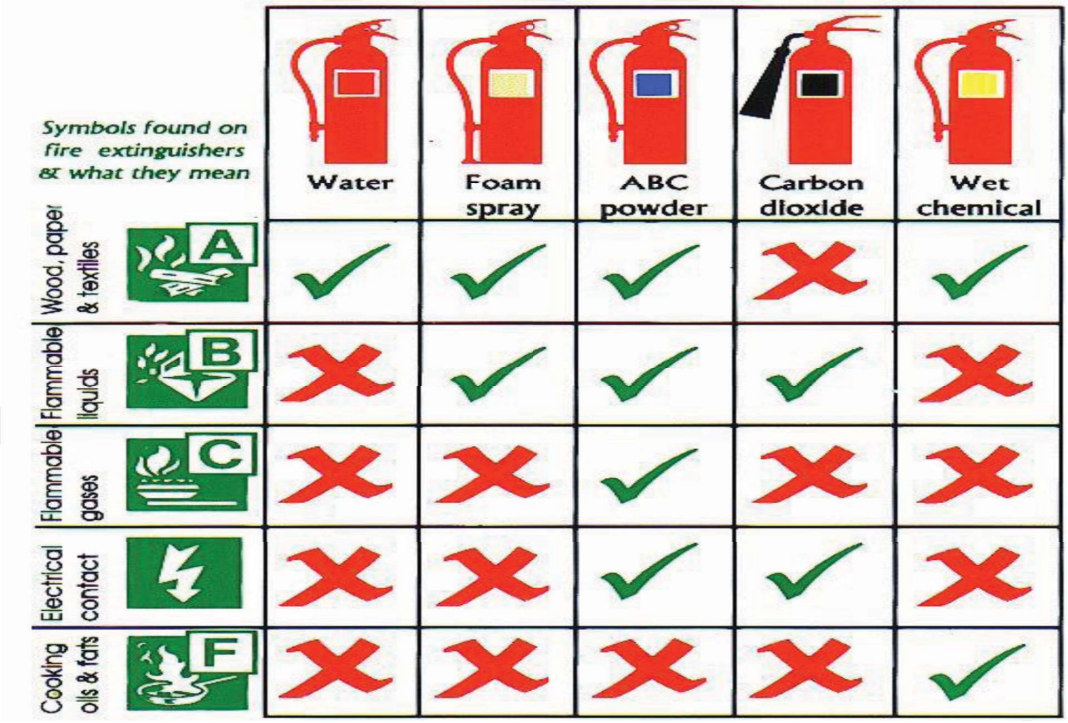
These are fires involving gases or liquefied gases under pressure, where it is necessary to isolate the burning gas. Dry chemical powder or vaporizing liquids are used for their extinguishment.

d) Class "D" Fires:

Fires that occur in combustible metals, such as magnesium, titanium, sodium, etc. are classified under this class. Specialized techniques, extinguishing agents and equipment are required to control such fires.

e) Electrical Fire:

Unlike the former ones, the electrical fire does not constitute a particular class. It is so because the electric current does not catch fire and does not burn, and what burns is some other material e.g. insulation and such. Any fire involving electrical equipment, therefore, must in fact be a fire of class A, B, C or D, depending on the substance that burns. The normal procedure in such fires is to cut off the electricity and use an extinguishing method appropriate to the burning material. Special extinguishing agents which are non-conductor of electricity and non-damaging to the electrical equipment should be used. Water in the form of a hose stream should not be used in electrical fire unless positive isolation of electric supply has been ensured. Finely divided water spray / fog being a non-conductive media is considered safe for the fighting activities in the vicinity of electrical current. Fog nozzles can protect switchgear, cable trays, transformers etc. Airspace present between water droplets prevents the back flow of current to the nozzle.



Classification of Fire extinguisher

9.4 SOURCES OF IGNITION

Ignition is defined as initial release of energy to trigger the combustion (fire). Factors for ignition include temperature, energy released, quantity of heat and direction of heat application.

It is necessary to understand sources of ignition to prevent fires / explosions. Such knowledge also enables elimination of substances that are potential sources of ignition. A general listing of primary ignition sources is given below and an effective fire protection program should include preventive measures with respect to all of them.

9.5 CAUSES OF FIRE:

The following are the main causes of fires.

Sources of Ignition	Examples	Preventive Measures
Electrical equipment	Sparks from motors, switches, lamps, hot elements and electrical defects	1) Use of approved equipment 2) To follow National electrical code like CEA regulation, Explosive regulation, Petroleum Rules etc. 3) Proper Maintenance
Friction	Hot bearings, misaligned, broken machine parts, choking, jamming of material, poor adjustment	Preventive maintenance and proper lubrication.
Open flames	Cutting and welding torches, gas and oil burners	Strict compliance of precautions stipulated in fire permit for hot jobs
Smoking, using matches, other flame producing devices in hazardous areas	Smoking prohibited areas.	1) Smoking is prohibited in refinery 2) Deposit matches & beedi /cigarette at entrance gate with security.
Hammering, Sanding, scraping, cutting and the like.	Sand blasting, Acetylene flame cutting, brazing.	Flushing of equipment, carrying out the operations away from combustible materials.

Sources of Ignition	Examples	Preventive Measures
Hot surfaces	Contact of combustible material with hot surfaces, heated lines etc.	Provide proper insulation and air circulation.
Static electricity	Occurs when there is no proper earthing during splash loading and loading at higher velocities.	1) Proper earthing of equipment 2) Do not resort to splash loading 3) Loading velocity should be controlled as permissible.
Lightning	Bad weather such as thunderstorm, cloud bursts.	Proper lightning arrestor and earthing continuity.
Auto ignition	Hydrocarbon mixtures in right proportion.	Oil soaked insulations etc. to be removed immediately and contact with air to be prevented.
External sources	Enemy attack, sabotage	Proper Security measures to be taken

9.6 Methods of extinguishing fires

Fire extinction usually consists of removal/limiting of one or more of three factors of fire triangle. However, a fourth factor i.e. chemical interference has also identified. This converts the 'Fire Triangle' into a 'Pyramid'. The four ways to fight fire are:

a) Starvation : Elimination of fuel.

b) Smothering : Limitation of oxygen.

c) Cooling : Removal of temperature. (Heat)

d) Chemical Interference : Interrupt in chain reaction

A. Starvation:

The extinction of fire by starvation is brought about in the following three ways:

(i) By removing combustible material from the neighborhood of the fire, such as the transfer of fuel from burning oil tanks, taking out of cargo at a ship fire; cutting of branches around forest fires, closing the valve on the gas/ fuel line etc.

(ii) By removing material on fire from the neighborhood of combustible material for instance, pulling apart a burning haystack or a thatched roof.

(iii) By sub-dividing the burning material to small sized isolated fires to break continuity.

B. Smothering:

If the oxygen content of the atmosphere in the immediate neighborhood of burning material can be sufficiently reduced, combustion will cease. The general procedure in methods of this type is to prevent or impede the access of fresh air to the seat of fire, and allow the combustion to reduce the oxygen content in the confined atmosphere until it extinguishes itself. This principle is ineffective in the cases where the burning material contain within itself the oxygen it requires for combustion in a chemically combined form, such as celluloid.

The principle of smothering is employed in a small scale in snuffing a candle, and, on a large scale in capping a burning oil well; two processes which are precisely analogous. Small fires, such as those involving a person's clothing, can be smothered with a rug/blanket. The use of sand or earth on small metal fire is also another example of smothering. In the petroleum refineries, foam is effectively used as a smothering agent for fighting oil fires.

C. Cooling:

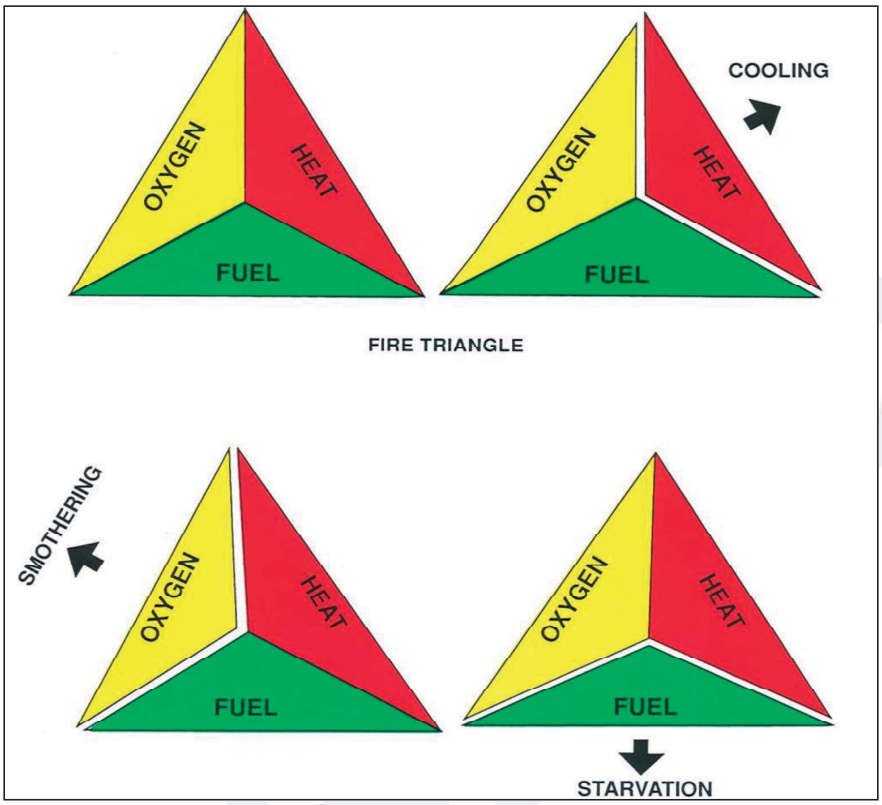
If the rate at which heat is generated by combustion is less than the rate at which it is dissipated through various agencies, the combustion cannot persist. In applying this principle of fire extinction the first step is to accelerate the speed with which heat is removed from the fire, thus reducing the temperature of the burning mass and as

a consequence the rate at which heat is produced. In due course the rate at which heat is lost from the fire exceeds the rate of heat production and the fire dies away. The application of a jet or spray of water to a fire is invariably based on this simple but fundamental principle.

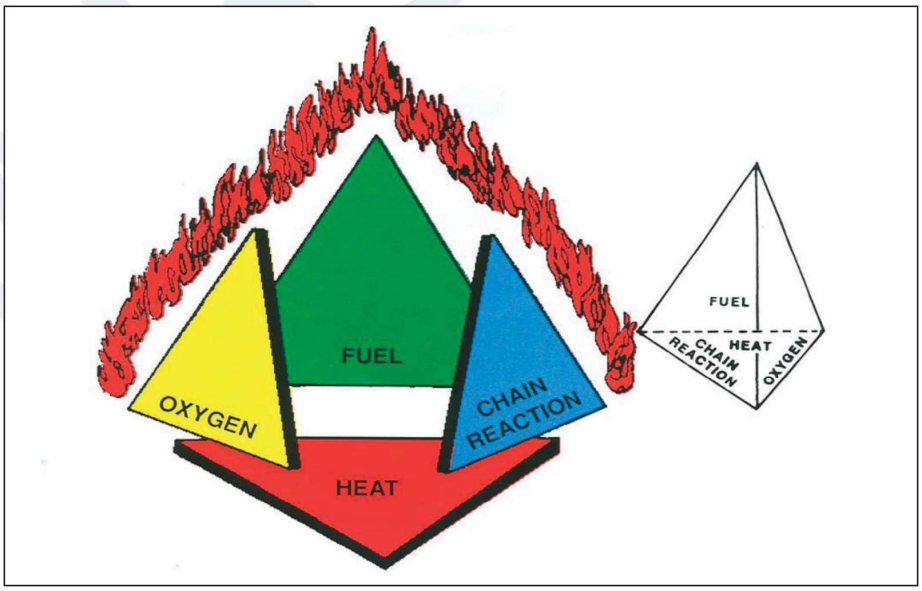
D. Chemical interference:

The combustion process actually takes place between free radicals and not between molecules as is usually depicted. Fire is an example of a free radical change reaction. Some chemicals, the bromo-fluroro-hydrocarbons, have a great affinity for free radicals and so injecting these into the fire has the effect of preventing the propagation of the chain reaction. In other words the fire goes out.

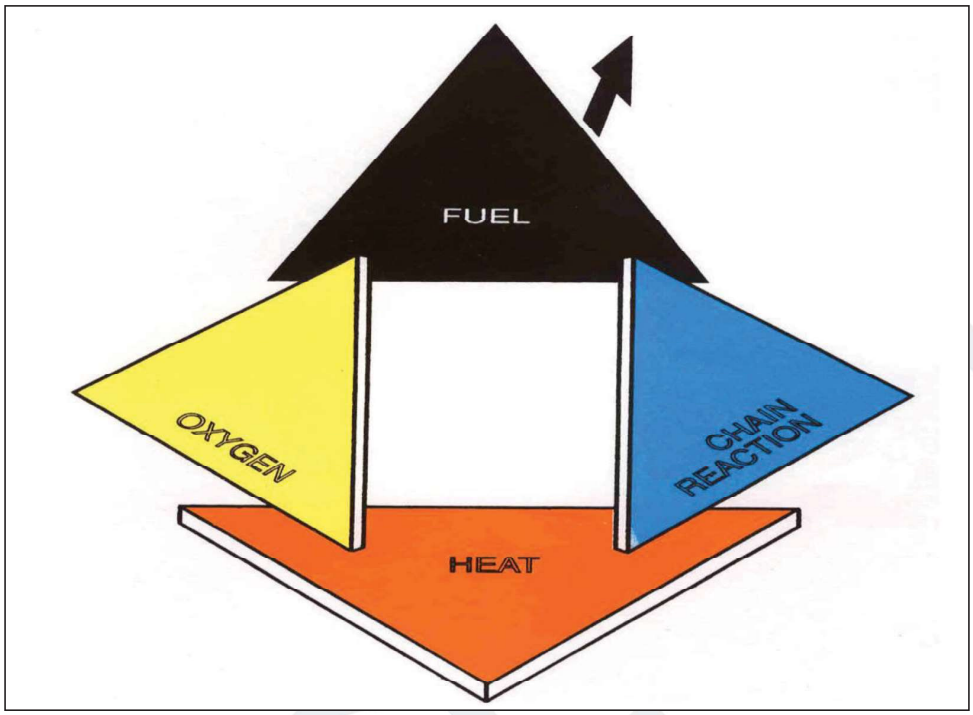
FIRE TRANGLE & METHODS OF EXTINGUISHING FIRE



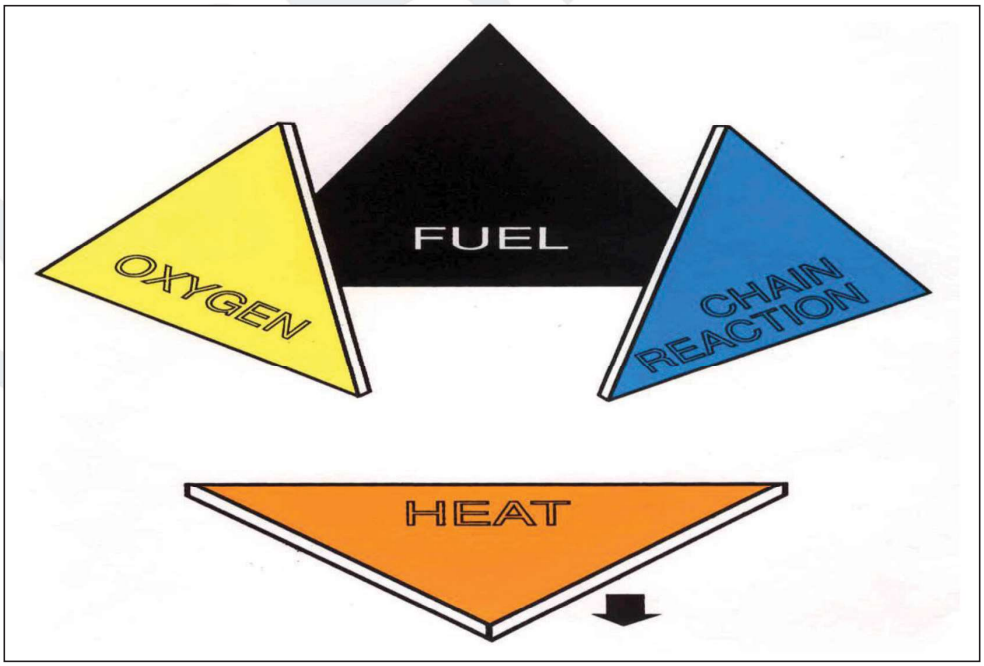
FIRE TETRAHEDRON (PYRAMID)



STARVATION



COOLING



9.7 FIRE EXTINGUISHING AGENTS

A. Water:

Water is the most efficient, cheapest and readily available medium for fighting the fires. It has the highest latent heat of vaporization. A variation in the application of water can be made by means of nozzles that produce jets or sprays ranging from large size droplets down to atomized fog effects.

B. Steam:

Steam in large quantities can be used to smother a fire. It is very convenient and effective mode of fire-fighting. Steam is particularly effective against high temperature hydrogen/hydrocarbon fires, the type of situation which may arise in our Catalytic Reforming Unit.

C. Mechanical foam:

One of the methods which can be employed for controlling oil fires is smothering with mechanical foam. Foam, relatively insoluble in most of the hydro-carbon products, floats on the surface of the oil and forms a blanket capable of covering the surface of the liquids thereby extinguishing the fire. It also has a cooling effect.

AFFF is synthetic Aqueous Film Forming Foam liquid. When proportioned with water and applied with conventional foam equipment, it generates white foam with the remarkable ability to make water float on flammable liquids, which are lighter than water. This double action foam and aqueous film enhances extinguishment of flammable liquid and prevents re-flash.

D. Dry chemical powder (DCP):

This again is smothering agent, which extinguishes fires mainly by interrupting the chain reaction. The basic ingredient of most chemical powders is either sodium bicarbonate or potassium bicarbonate. This, with the addition of metallic stearate as a water proofing agent, is widely used in extinguishers, not only in portable ones but also for general application in large quantities. Apart from stearate, other additives are sometimes used to decrease the bulk density and to reduce packing the cylinder.

Dry chemical powder is expelled from containers by gas pressure and by means of specially designed nozzles. It is directed in the form of concentrated cloud. This cloud also screens the operator from the flames and enables a relatively close attack on fire. Many other dry chemical powders such as Monex, ABC type etc. is also used for specific purposes.

E. Carbon dioxide and inert gases:

As a fire extinguisher CO2 acts in a manner similar to vaporizing liquids and is also non-conductor of electricity. The inert gases such as Nitrogen can be used for tackling fires by smothering.

F. Sand:

Where other smothering agents are not readily available, sand can also be use. It prevents spreading of burning liquids. On no account should sand be used for extinguishing fires in machineries, such as electric motors, since its use may well necessitate dismantling the entire machine for cleaning even though the fire damage is negligible.

G. Blanketing:

Another method, by which fire may be extinguished, especially for persons whose clothing is on fire, is by blanketing. The person should be laid down and covered or rolled in a rug, coat, jacket, woolen/other Fire resistant blanket, etc. For dealing with fires in small utensil, such as those containing cooking oils, the best method is to smother the fires with a lid, blanket, or similar material such as a doormat which has been wetted first.

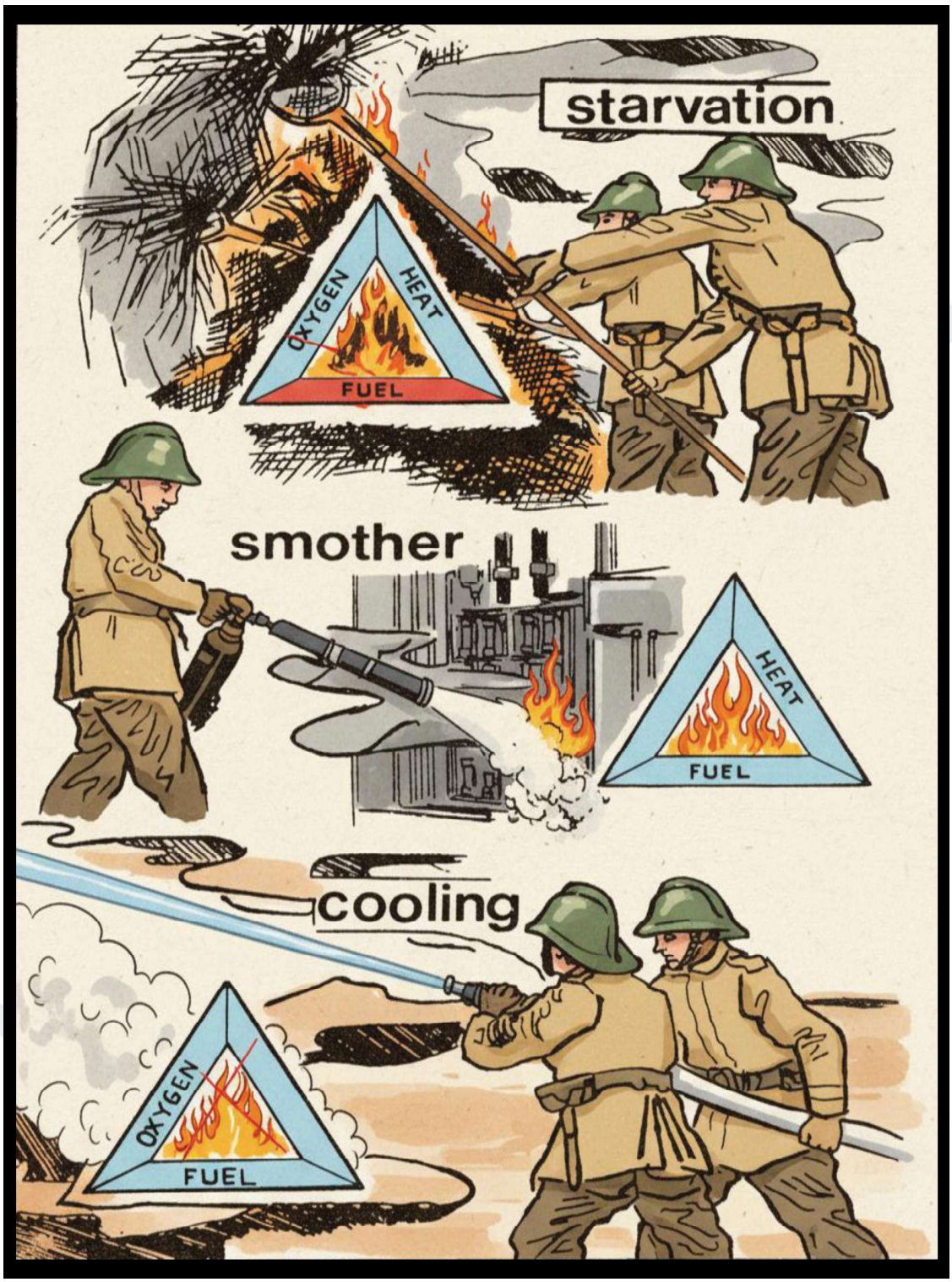
H. Beating out:

Small fires in materials such as grass, etc., may often be extinguished by beating them out, or by rolling up the burning materials tightly to exclude the air.

I. Fire water system

Water is an essential and the most important firefighting medium available for extinguishing fire. Water is used for fire extinguishment, fire control, cooling of equipment and protection of equipment and personnel from heat radiation. For these purposes water is used in various forms such as straight jet, water fog, water curtain, water spray, deluge/ sprinkler, for foam making etc.

The main components of the fire water system are fire water storage, fire water pumps and distribution piping network fitted with hydrants and monitors.



Methods of Extinguishing Fire

Correct way of using Sand

Correct way of using Blanket

9.8 HOW TO USE A PORTABLE FIRE EXTINGUISHER

Simply follow the steps **"P-A-S-S"**

P – Pull	Pull the Pin at the top of the extinguisher
A-Aim	Aim the nozzle or the outlet towards the base of the fire. Some hose assemblies are clipped to the body of the extinguisher then release the hose and point.
S -Squeeze	Squeeze the handle, lever to release the extinguishing agent. In some cases, valves are present. Before approaching the fire try a short burst
S -Sweep	Sweep from side to side at the base of the fire until it is out. After fire is doused, watch for the smoldering hot spots and possible re-flash. Put off the fire completely.

Fire extinguisher of appropriate type & size should be kept in strategic locations and should be maintained according to floor area. This portable fire extinguisher should be hydrostatically checked in every 5 year & serviceability should be checked every year & the due date for checking should be mentioned in the fire extinguisher. Proper inspection tags with history card should be maintained for fire extinguishers.

The access & egress of portable fire extinguisher should be free from obstacle & it should be easily available for operation in the event of emergency. Mock drill should be conducted periodically for knowhow of operation of the extinguisher. Sand bucket with standard IS-2546 should be kept in 33/11 kV Sub-Station near entrance of control room.

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CHAPTER-X

PERSONAL PROTECTIVE EQUIPMENT, SAFETY DEVICES & TOOLS

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CHAPTER-10

PERSONAL PROTECTIVE EQUIPMENT, SAFETY DEVICES & TOOLS

PERSONAL PROTECTIVE EQUIPMENT (PPE), SAFETY TOOLS & DEVICES

The company shall provide adequate & approved PPE & safety devices for various jobs depending upon the hazard. The procurement of all these items shall be made ensuring their quality as per respective standards.

10.1 TYPES AND USES OF PPE

a) Suitable protective equipment, like rubber hand gloves, electric shock resistant gloves, rubber mats, shock resistant safety shoes etc. are to be used wherever required as per instructions or wherever it provide greater safety.

b) All safety devices should be checked before starting of work.

c) Safety equipment should be tested at frequent intervals to ensure that equipment would provide the safety desired.

d) Protective gears such as safety helmets, safety shoes, full body harness, arc suit are to be issued to workmen, jointers, supervisors as applicable for their personal protection and usage is to be monitored.

e) It is the responsibility of the supervisor to ensure PPE usage.

f) The PPE shall not be carried or stored with tools etc. to avoid damage.

g) Only approved types of operating rods are to be used.

h) Discharge rods shall have to be kept as dry as possible. It should not be dropped or left lying on ground.

i) Rubber gloves are to be tested once in every six months since its start of

use to check its usability. The Test Certificates are to be preserved duly and kept accessible to all concerned. In case the test results are found to be not satisfactory, the item is to be discarded and replaced by good one immediately.

j) Any employee working on a level which is 6 feet above ground or floor level and unless working on a guarded platform shall use full body harness whose lanyard shall be anchored such that it offers fall restraint facility.

k) Every person working on an electric supply line or apparatus is to be provided with

(i) Personal protective equipment, tools and safety devices such as rubber gloves and safety footwear suitable for the working voltage, full body harness for working at height, nonconductive ladder, earthing devices of appropriate class, safety helmet, line tester, hand lines, voltage detector and hand tools as per the relevant standards; and

(ii) Other devices for protecting him from mechanical and electrical injury due to arc flash and such personal protective equipment, tools and devices which conform with the relevant standards and are maintained in sound working condition.

10.2 HAND TOOLS

a) Many accidents result from improper use of tools and use of defective tools and equipment. Employees should use only those tools and equipment, which are in good condition, and only of the purpose for which they are designed. Where proper and safe tools are not available for the work at hand, then employee shall report the fact to his supervisor.

b) Tools, which develop defects while in use, should be removed from the service, tagged and not used again until brought in good condition.

c) Impact tools such as chisels, drills, hammers and wedges with mushroom heads should not be used until they have been reconditioned.

d) Hammers, axes, shovels and similar tools should not be used if handles are loose, cracked or splintered.

e) Defective wrenches such as open end and adjustable wrenches with spread jaws or pipe wrenches with dull teeth should not be used, as they are likely to slip.

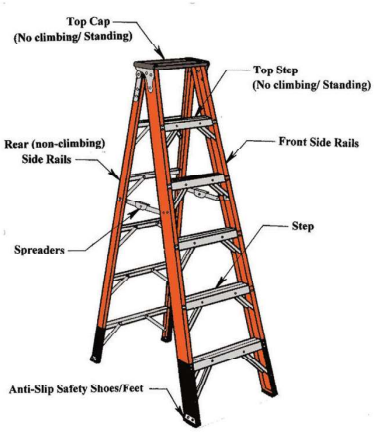
f) Pipe or other extensions should not be used on a wrench handle to increase the leverage unless the wrench is specifically designed for such an extension.

g) Portable electric tools should be equipped with 3 - wire cord having the ground wire permanently connected to the tool frame and is to be grounded at the other end.

h) Metal rulers, metal tape lines or lines containing wires shall not be used around electric conductors or equipment.

i) All tools carried on vehicles should be inspected each month. Defective tools should either be repaired or replaced by good ones.

Earth Rod/Discharge Rod



Ladder

Tool Box

10.3 LADDERS

a) Inspect the ladder before use.

b) Ensure firm footing.

c) Secure at top or have a man at the foot.

d) Ensure correct angle (75°) or position ladder 1 feet out at base for every 4 feet vertical height.

e) Ensure that the ladder rises 3 feet above landing point.

f) Face the ladder when ascending or descending.

g) Avoid make shift arrangement in lieu of ladder.

h) Ensure the rubber shoes at both arms and at each terminal i.e, at each end.

i) Before fixing confirm that no electrically charged conductor is passing nearby.

j) All hot sticks and ladders shall be cleaned and checked for integrity by the hot sticks tester.

10.4 PROTECTIVE BARRIER (TEMPORARY)

When the work is conducted along streets or highways, pedestrian and vehicular traffic shall be warned by signs and flags in daytime and by redlights or flares at night. Wherever necessary, signal men should be provided. The area should be barricaded using traffic cone & nylon rope.

10.5 EARTHING DEVICES

a) Only approved earthing devices shall be used in all work.

b) Care shall be taken to maintain earthing by ensuring condition of clamps.

c) Neon line tester may be used to check the bus bar area, cables, overhead conductors but prior to use it ensure that the tester is in good condition.

10.6 PROVISIONS OF PPE FOR EMERGENCY CONDITIONS

a) Every generating station, substation or switchyard is to be equipped with artificial respirator & gas mask. These are to be kept in good working condition for use in the event of fire or smoke. The locations where these have been kept are to be displayed in the control room and within operator's visibility.

b) Every generating station with capacity of five megawatt and above and every enclosed substation with transformation capacity of five megavolt-ampere and above has been are to be provided with at least two gas masks for use in the event of fire or smoke. These are to be kept conspicuously at readily accessible locations.

Rubber Hand Gloves (IS-4770)

Rubber Gum Boot (IS-5557)

Full Body Safety Harness (IS-3521)

Safety Helmet (IS-2925)

Foldable Discharge Rod with shorting chain

High Voltage Detector with foldable Rod

GI Earth Spike 1833x20 mm

Traffic Cone

Shorting Chain

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CHAPTER-XI

SAFETY COMMITTEE, ACCIDENTS & SAFETY AUDIT

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CHAPTER-11

SAFETY COMMITTEE, ACCIDENTS & SAFETY AUDIT

11.1 SAFETY COMMITTEE

With a view to ensure safety to all stake holders of WBSECL i.e, general public, consumers, employees, agency workers during generation and distribution of electricity safety committees have been formed at various levels. These committees are supposed to deliberate on various issues relating to deficiency in adopting safety procedures and use of safety equipment. Such committees are to inspect sites and suggest remedial measures for reducing accidents.

Safety Committees:

1) Divisional Safety Committee
2) Zonal Safety Committee
3) Rammam Hydel Project Safety Committee
4) Jaldhaka Hydel Project Safety Committee
5) Mini Micro Hydel Division Safety Committee
6) Teesta Canal Fall Hydel Project Safety Committee
7) PPSP Site and Massanjore HPS Safety Committee
8) Distribution Project Safety Committee
9) Apex Safety Committee

Constitution of the Committees:

Constitution and reconstitution of these committees and the needed periodicity of their meetings are declared through Office orders issued from time to time. As per the prevailing guideline, the constitution of the committees are as follows:

I) Divisional Safety Committee:

1) Divisional Manager – Chairman
2) HR&A representative of the Division – Member Convenor
3) All the Assistant Engineer(T) & Divisional Engineer(T) of the Division– Members
4) All the Station Managers of the Division – Members

II) Zonal Safety Committee:

1) Zonal Manager – Chairman
2) All the Regional Managers - Members
3) Sr. Manager(HR&A) of the Zone – Member Convenor
4) Sr. Safety Officer/Safety Officer of the Zone – Member
5) The Regional Safety Officers of the Regions - Members
6) All the Divisional Managers of the Zone – Members
7) HR&A representatives of all the Divisions of the Zone – Members

III) Rammam Hydel Project Safety Committee:

1) Project Manager – Chairman
2) HR&A Officer of the Project – Member Convener
3) Head of Civil Wing of the Project – Member
4) Head of Electrical Wing of the Project – Member
5) Head of Mechanical Wing of the Project – Member

IV) Jaldhaka Hydel Project Safety Committee:

1) Project Manager – Chairman
2) HR&A Officer of the Project – Member Convener
3) Head of Civil Wing of the Project – Member
4) Head of Electrical Wing of the Project – Member
5) Head of Mechanical Wing of the Project – Member

V) Mini Micro Hydel Division Safety Committee:

1) Head of Mini-micro Hydel Division – Chairman
2) HR&A Officer of the Division – Member Convener

3) Head of Civil Wing of the Division – Member
4) Head of Electrical Wing of the Division – Member
5) Head of Mechanical Wing of the Division – Member

VI) Teesta Canal Fall Hydel Project Safety Committee:

1) Project Manager– Chairman
2) HR&A Officer of the Project – Member Convener
3) Head of Civil Wing of the Project – Member
4) Head of Electrical Wing of the Project – Member
5) Head of Mechanical Wing of the Project – Member
6) Power Station In-charges of PS-I, PS-II & PS-III– Members

VII) PPSP Site and Massanjore HPS Safety Committee:

1) Project site in Charge of PPSP – Chairman
2) Inside Power House (IPH) in Charge of PPSP site – Member
3) Outside Power House (OPH) in Charge of PPSP site – Member
4) In Charge of Civil Wing of PPSP site – Member
5) Head of HR&A Wing of PPSP site – Member Convener
6) In Charge of Massanjore Hydel Power Station - Member

VIII) Project Safety Committee:

1) Executive Director (Projects) – Chairman
2) Chief Engineers of Project-I, Project-II, Project-III, Project IV, SPGD – Members
3) Additional Chief Engineer, Corporate Safety Cell - Member
4) Additional General Manager (HR&A), Project-II – Member Convenor
5) Concerned Project Managers/ Regional Managers – Invitee Members

IX) Apex Safety Committee:

1) Chief Engineer (Safety) – Chairman
2) Representative from Dist HQ– Member
3) All the Zonal Managers or their representatives– Members

4) Chief Engineer (Hydel) or his representative from Hydel HQ – Member
5) Chief Engineer (PPSP) or his representative from PPSP HQ – Member
6) Chief Engineer (Project I) or his representative from Project-I – Member
7) Chief Engineer (Project II) or his representative from Project-II – Member
8) Chief Engineer (Project III) or his representative from Project-III – Member
9) Chief Engineer (Project IV) or his representative from Project-IV – Member
10) Chief Engineer (SPGD) or his representative - Member
11) Additional Chief Engineer, HRDD – Member
12) Additional Chief Engineer, Corporate Safety Cell - Member
13) Chief Safety Officer - Member
14) All the Sr. Safety Officers/ Safety Officers of Zones - Members
15) Sr. Manager(HR&A), Corporate Safety Cell – Member Convenor

Purpose of the committees:

These Committees will deliberate on various issues relating to deficiency in safety system/ practices under their functional jurisdictions, inspect the sites, if required, and suggest remedial measures for reducing accidents. Apex Safety Committee will review the overall functioning of the safety committees, formulate safety Policies of the Company, infrastructural upgrade/ modification needs, safety training needs and decide upon the support services to be given to the site offices in this respect for ensuring safety in operation & maintenance.

Functions and duties of the Divisional Safety Committees:

The functions and duties of the committees include:

1. Random inspection & site visit to ensure safety at workplace.
2. Developing safety awareness among employees at Division level through periodic interaction & conference.
3. Conducting meetings (at least once in every month and as & when situation demands), seminars & workshops at Division level regarding safety at workplace.
4. Local members of the committee, under whose jurisdiction any accident takes place, to primarily follow up the enquiry for ensuring timely and effective investigation, retrieval & preservation of all relevant data & information for effective post-accident actions, root cause analysis of the

accident and formulating remedial measures.

5. To prepare & maintain statistics for safety related affair at Division level
6. Discussing at Division level reports on safety, environmental and occupational health surveys, safety audits and risk assessment for implementation of the recommendations made in the reports and conveying observations & suggestions to the concerned Region Office immediately.
7. Discussing at Division level onsite emergency plan and disaster control measures for effective implementation, carrying out health and safety surveys and identifying causes of accidents. Forwarding observations & suggestions to the concerned Region Office which subsequently would be sent to Zonal Safety Committee by the concerned Regional Manager for further needful actions.
8. Preserving all records & minutes of meetings for ease of accessibility and retrieval in need.

Functions and duties of the Zonal Safety Committee:

The functions and duties of the committee include:

1. Random inspection & site visit to ensure safety at workplace, apart from the regular site inspections to be carried out by Sr. Safety Officers / Safety Officers of the Zone.
2. Developing safety awareness among employees at Zonal level through periodic interaction/ conference.
3. Conducting meeting (at least once in every two months and as per necessity), seminars & workshops at Zonal level regarding safety at workplace.
4. Follow-up enquiry reports on accident, if situation demands.
5. Prepare & maintain statistics for safety related affair at Zonal level.
6. Assisting and cooperating with the management in achieving the aims and objectives outlined in the 'Safety Policy'.
7. Dealing with all matters concerning health, safety and environment and to arrive at practicable solutions to the problems encountered.
8. Undertaking educational, training and promotional activities towards health and safety.
9. Discussing at Zonal level the observations & suggestions received from the Divisional Committee(s) through the respective Region Office(s), the reports

on safety, environmental and occupational health surveys, safety audits and risk assessment for implementation of the recommendations made in the reports.

10. Discussing at Zonal level the observations & suggestions made by the Divisional Committees through their respective Regional Offices, on onsite emergency plan and disaster control measures for effective implementation, carrying out health and safety surveys and identifying causes of accidents.
11. Suggesting corrective measures and reviewing implementation of the remedial measures taken to eliminate the imminent danger.
12. Reviewing and analyzing the work and operational procedures to identify hazards and their elimination.
13. Forwarding all the observations, suggestions, proposals related to enhancing safety in system & workplace, implementing remedial measures, eliminating probabilities of accidents and adopting declared safety policy to the Apex Committee for their consideration & needful action.
14. Preserving all records & minutes of meetings for ease of accessibility or retrieval in need.

Functions and duties of the Hydel Project Safety Committee:

[For Rammam, Jaldhaka, Teesta Canal Fall Hydel Projects and Mini micro Hydel Division]

The functions and duties of the committee include:

1. Random inspection & site visit to ensure safety at workplace.
2. Developing safety awareness among employees at Project level through periodic interaction/ conference
3. Conducting (at least once in every month and as per need of the hour) meeting, seminars & workshops at Project level regarding safety at workplace.
4. Local members of the committee, under whose jurisdiction any accident takes place, will primarily follow up the enquiry for ensuring timely and effective investigation, retrieval & preservation of all relevant data & information for effective post-accident actions, root cause analysis of the accident and formulating remedial measures.
5. Prepare & maintain statistics for safety related affair at Project level.
6. Discussing at Project level reports on safety, environmental and occupational health surveys, safety audits and risk assessment for implementation of

the recommendations made in the reports and conveying observations & suggestions to Hydel HQ immediately.

7. Discussing at Project level onsite emergency plan and disaster control measures for effective implementation, carrying out health and safety surveys and identifying causes of accidents. Forwarding observations & suggestions to Hydel HQ which in turn would be forwarded by Chief Engineer(Hydel) to the Apex Committee.
8. Preserving all records & minutes of meetings for ease of accessibility or retrieval in need.

Functions and duties of the PPSP Site Safety Committee:

The functions and duties of the committees include:

1. Regular inspection to ensure safety at workplace,
2. Developing safety awareness among employees at PPSP Site through periodic interaction/ conference
3. Conducting meetings (at least once in every two months and as per necessity), seminars & workshops at PPSP Site regarding safety at workplace.
4. Follow-up enquiry reports on accident, if situation demands.
5. To prepare & maintain statistics for safety related affairs.
6. Assisting and cooperating with the management in achieving the aims and objectives outlined in the 'Safety Policy'.
7. Dealing with all matters concerning health, safety and environment and to arrive at practicable solutions to the problems encountered.
8. Undertaking educational, training and promotional activities towards health and safety.
9. Discussing the reports on safety, environmental and occupational health surveys, safety audits and risk assessment for implementation of the recommendations made in the reports.
10. Discussing on onsite emergency plan and disaster control measures for effective implementation, carrying out health and safety surveys and identifying causes of accidents.
11. Suggesting corrective measures and reviewing implementation of the remedial measures taken to eliminate the imminent danger.
12. Reviewing and analyzing the work and operational procedures to identify

hazards and their elimination.

13. Forwarding all the observations, suggestions &proposals related to enhancing safety in system & workplace, implementing remedial measures, eliminating probabilities of accidents and adopting declared safety policy to the Apex Committee for their consideration & needful action.
14. Preserving all records & minutes of meetings for ease of accessibility or retrieval in need.

Functions and duties of the Project Safety Committees:

[For Project-I, Project-II, Project-III, Project-IV & SPGD]

The functions and duties of the committees include:

1. Random inspection & site visit to ensure safety at workplace.
2. Developing safety awareness among employees at Project level through periodic interaction/ conference.
3. Conducting meetings (at least once in every two months and as & when necessary), seminars & workshops at Project level regarding safety at workplace.
4. Follow-up enquiry reports on accident, if situation demands.
5. To prepare & maintain statistics for safety related affair at Project level.
6. Assisting and cooperating with the management in achieving the aims and objectives outlined in the 'Safety Policy'.
7. Dealing with all matters concerning health, safety and environment and to arrive at practicable solutions to the problems encountered.
8. Undertaking educational, training and promotional activities towards health and safety.
9. Discussing at Project level the observations & suggestions received from the Divisional Committee(s) through the respective Region Office(s) and Project sites the reports on safety, environmental and occupational health surveys, safety audits and risk assessment for implementation of the recommendations made in the reports.
10. Discussing at Project level the observations & suggestions made by the Divisional Committees through their respective Regional Offices and Project sites on onsite emergency plan and disaster control measures for effective implementation, carrying out health and safety surveys and identifying causes of accidents.

11. Suggesting corrective measures and reviewing implementation of the remedial measures taken to eliminate the imminent danger.
12. Reviewing and analyzing the work and operational procedures to identify hazards and their elimination.
13. Forwarding all the observations, suggestions & proposals related to enhancing safety in system & workplace, implementing remedial measures, eliminating probabilities of accidents and adopting declared safety policy to PPSP HQ which in turn would be forwarded to the Apex Committee for their consideration & needful action.
14. Preserving all records & minutes of meetings for ease of accessibility or retrieval in need.

Terms of Reference of Apex Safety Committee:

1. To conduct meetings at least once in every 3 months and as & when situation demands.
2. To discuss on & analyze observation & suggestions received from Zonal Safety Committees, Hydel Project Safety Committees, PPSP Site Project Committee, Project level Safety Committees and any other resources
3. To analyze causes of different accidents
4. To analyze the reports on safety, environmental and occupational health surveys, safety audits and risk assessment for implementation of the recommendations made in the reports.
5. To propose remedial measures to be taken to arrest accidents
6. To formulate / update safety policy as well as safety manual of the Company
7. To assess safety training needs and make proposals to competent authority accordingly.
8. To recommend support services to be given to the site offices
9. To recommend infrastructural upgrading / modification to enhance safety
10. To follow up actions taken by the sites as proposed by the Apex Committee. Member Convenor of Safety Committee would have the responsibility to follow up in close coordination with Member Convenors of respective Safety Committee sand the ACE, Corporate Safety Cell.
11. To preserve all records & minutes of meetings for ease of availability & accessibility in need.

The Chairman of each Safety Committee is to issue an order constituting it as per above guidelines and endorse a copy of the order to all concerned including Chief Engineer(Safety) and Director(HR) for their perusal and subsequent needful actions towards sharing the Committee details to concerned department(s).

11.2 ACCIDENTS

Main causes of electrical accidents of workers

- Non conversance with system on which they are to work
- Use of improper and substandard PPE/other protective equipment/ tools & tackles
- Nonuse of required PPE/ protective equipment/ tools & tackles during work
- Carelessness/Overconfidence/ Indiscipline/Lack of interest in the job
- Unnecessary risk by working on live parts/wrong shutdown
- Bypassing protection system
- Not testing residual voltage before starting work
- Ignoring visual or available information about status
- Unauthorized entry in the substation
- Mutual/ prolonged duty in the substation
- Engagement of unauthorized workman at work
- In correct switching instruction
- Not following Permit to work system
- Improper display of hazard
- Un-insulated tools
- Lines are not discharged & locally earthed properly as per safety rule
- Non creation of safe working zone at work site
- Incorrect / missing feeder marking

Consequences of electrical accidents

- Loss of life
- Loss of property
- Loss of man-hours

- Bad reputation
- Industrial relation problem
- Penal action by authorities
- Tightening of laws making working more difficult/expensive
- Refusal by other employee store place victim's post

How to avoid Electrical accidents

- Proper usage of PPE during hazardous work.
- Permit to work system should be followed.
- Proper shutdown procedure should be maintained.
- On the job training should be imparted to the workman.
- Safety seminar cum workshop should be organized for the workman who are working in the electrical line.
- The certified licensed workman should be engaged for working in the electrical line. Safety posters should be fixed in public place, panchayat offices, schools for public awareness.

How to prevent electrical accidents,

A) ByDesign

- Select switchgear, cables etc. with extra capacity keeping in mind safety factor, over and above, plans for expansion.
- Select arc flash protected switchgear.
- Select switches for remote operation.
- Select only good quality products with relevant safety certification.
- Make good layout with adequate space for ingress and egress.
- Plan for adequate number of disconnecting switches for isolation.
- By providing suitable inked switch fuse unit or circuit breaker in the primary and secondary of transformer.
- By providing suitable interlocking to prevent back feeding in case of parallel operation of transformers.
- By providing suitable protective relays for forward and reverse power flow.

- Design sufficient and strategically located safety fences to keep out unauthorized entry by man and animal.
- Plan Proper routing for cables with proper ventilation and access
- Plan for proper measuring and testing instruments
- Use low voltage for controls
- Purchase adequate number and type of PPE.

B) By Installation

- Select eligible Contractors only
- Insist upon Contractor providing safety kit to their worker as part of contract. Take adequate managerial steps to ensure compliance.
- Make quality plan for each work, discuss with Contractor and inspect its implementation.
- Provide training in specific jobs to Contractor's workers.
- Maintain over all discipline at worksite.

C) By Operation

- Provide training to workers
- Provide safety kits and PPE
- Provide suitable tools and measuring instruments
- Provide competent supervision
- Make detailed plan for maintenance
- Use of predictive maintenance tools
- Provide good quality spares
- Make safety teams including workers and actively participate in their meeting plan for a rapid response team to provide rescue and medical aid in case accidents still do happen.

D) ByTraining

All electrical workers should be thoroughly trained in the following:

- 33/11 kV substation single line diagram
- Operation and maintenance of disconnect switches
- Operation and maintenance of interlocks

- Display of danger boards
- Emergency procedures
- Grounding procedures
- Tools and instrument training
- PPE training

11.3 Accident investigation

Constitution of Permanent Enquiry Committee (PEC):

Permanent Enquiry Committees (PEC), wing-wise, to be constituted comprising of the Officers mentioned hereunder to enquire into the causes of accidents under the jurisdiction of WBSEDCL directly or indirectly involving its infrastructure & workforce leading to loss or damage of life and property.

A. For Distribution Regions:

1) The Regional Manager of the Region : Chairman
2) The Regional Safety Officer of that Region : Member Technical
3) The Head of HR&A wing of that Region: Member Convenor

B. For Project works under supervision of Project Managers of Project Office:

1) The Project Manager of the concerned Project: Chairman
2) The Regional Safety Officer of the concerned Region where the accident occurred: Member Technical
3) The Head of HR&A wing of the concerned Region where the accident occurred: Member Convenor

C. For Project works under supervision of Regional Managers of Region Office:

1) The Regional Manager of the concerned Region: Chairman
2) The Regional Safety Officer of the concerned Region where the accident occurred: Member Technical
3) The Head of HR&A wing of the concerned Region where the accident occurred: Member Convenor

D. For Hydel Units:

1) The Project Manager / Project In-Charge of concerned Hydel Project: Chairman
2) The Head of HR&A wing of Hydel HQ: Member Convenor
3) One Engineer from Hydel HQ not below the rank of SE: Member

E. For Purulia Pumped Storage Project (PPSP) Site:

1) The Additional Chief Engineer & Project Site-In-Charge(PSIC): PPSP Site: Chairman
2) The Head of HR&A wing of PPSP HQ : Member Convenor
3) One Engineer from PPSP site not below the rank of SE: Member

In case the accident is severe in nature Corporate Safety Cell shall, if needed, conduct separate enquiry with the head of HR&A wing of the concerned region.

Based on the report of PEC, the Corporate Safety Cell may further review the Enquiry Report, if needed, on case to case basis.

Broad Terms of Reference of Permanent Enquiry Committee (PEC):

The PEC will enquire into every accident under its jurisdiction within stipulated time period ensuring collection of all the available & relevant evidences, samples, records etc. to enable preparation of fact based reporting and also to enable conduction of all the post-accident activities & root cause analysis to find out remedial measures.

PEC will:

1) Record chronological events leading to the accident.
2) For any accident of departmental or outsourced agency workers during execution of work, to enquire whether standard operating procedure was followed, proper PPEs, tools & tackles were utilized during execution of job.
3) To collect all the available evidences in the form of documents, records, photographs, written or if illiterate, recorded verbal depositions of eye witness/next of kin/ co-workers of the victim etc.
4) To additionally collect all relevant evidences like, information, documents, photographs etc. from the official(s)/ representative(s) of WBSEDCL who visited the accident spot immediately after getting the information of the incident.

5) To obtain deposition of eye witnesses, and, for worksite accident, deposition of co-workers also.

6) For any accident of departmental or outsourced agency workers, to ascertain from official records, evidences, reports of Police, Hospital including Post Mortem Report and any other relevant information in connection with the accident whether the accident arose out of & in course of employment.

7) For any accident of General Public, to ascertain from official records, evidences, statements of eyewitnesses, reports of Police, Hospital, Post Mortem Report and any other relevant information in connection with the accident whether the accident occurred due to defective infrastructure and/or malfunctioning of system and/or technical limitation of protection system of WBSEDCL.

8) If it is detected that the accident was due to unsafe infrastructure of WBSEDCL, identity of the person(s) responsible for leaving the system in unsafe condition, if any, to be recorded in the report. On the other hand, in case the accident takes place beyond WBSEDCL's meter and / or due to illegal attempt of handling WBSEDCL's system or due to coming in close proximity of hazardous electrical installations in protected or restricted area, that also to be recorded clearly in the report.

9) To analyze all the available information to infer the cause of accident and find out the remedial measure to avoid recurrence of same type of accident.

10) To find out the lapse(s), if any, on the part of any individual Departmental / Outsourced Technical workforce & fix up responsibility.

11) To find out whether the accident reveals any defect in the system and procedures of work and suggest remedial measures which may be adopted for prevention of recurrence of such accident in future.

12) Record & report all other mention-worthy relevant aspects, if detected in course of investigation.

13) Furnish additional information mandatorily in enquiry report related to accidents caused by conductor snapping.

14) The enquiry report of PEC is to be submitted as mentioned below:

    a) PEC of Distribution Region is to submit the report to the concerned Zonal Manager positively within thirty (30) days from the date of occurrence of the accident.

    b) PEC of Project works under supervision of Project Manager of the Project Office is to submit the report to the Chief Engineer of the concerned

Project Department having technical control of the Project positively within thirty (30) days from the date of occurrence of the accident.

c) PEC of Project works under supervision of Regional Manager of Region Office is to submit the report to the Chief Engineer of the concerned Project Department having technical control of the Project with a copy to the concerned Zonal Manager positively within thirty (30) days from the date of occurrence of the accident.

d) PEC of Hydel Unit is to submit the report to the Chief Engineer (Hydel) positively within thirty (30) days from the date of occurrence of the accident.

e) PEC of PPSP Site is to submit the report to the Chief Engineer (PPSP-HQ) positively within thirty (30) days from the date of occurrence of the accident.

Additional information to be furnished in accident enquiry reports related to conductor snapping:

The following information must be incorporated in the enquiry report:

i) Call docket number with date & time in case any call was docketed in WBSEDCL's system in this regard.

ii) Name of the employee with designation or name of contractual worker with name of agency who repaired or was assigned the job of repairing the conductor

iii) Type of snapped conductor with size &its position (upper, lower or middle)

iv) Existence of hooking / tapping on the conductor at the time of snapping

v) Condition of snapped conductor

vi) Presence of tree infringement

vii) Cause of conductor snapping

viii) Whether fuse blew / feeder tripped or not. If not, reason thereof.

Post Accident Activities:

Upon receipt of information about an accident, latest by the next working day, Accident Reporting Officer shall inspect the spot, collect requisite information, depositions and documentary evidences, take snapshot of the accident spot & lodge FIR with the local PS.

i) Intimation of any accident must be brought to the notice of Corporate Safety Cell with 24 hours. of occurrence of that incident.

ii) Intimation of accident to the Chief Electrical Inspector & submission of report through Rule-2 shall be guided by the Govt. Notification No. 342-Power/III dated 29.12.2003 with subsequent amendments.

iii) In case of every fatal accident, the respective Divisional Manager shall personally hold inspection on the spot, verify the report submitted under Rule-2 and submit the report to the Permanent Enquiry Committee within ten(10) days of occurrence of the accident.

iv) Chairman of the PEC is to submit enquiry report to the appropriate authority within 30 (thirty) days of occurrence of the accident, failing which he/she will be held responsible for delay in submitting enquiry report.

v) Upon receipt of the enquiry report, Accident Disposing Officer shall overview the report, make analysis for any shortcomings/ ambiguity in the report with the incident and subsequently accept or reject the report.

vi) If accepted, the Accident Disposing Officer shall dispose of the cases by fifteen (15) working days from the date of acceptance. However, the responsibility of finalizing the eligibility for solarium/ compensation & subsequent fixation of compensation amount after judging of legal aspects is on the HR&A Officer of the concerned Zone or Project Wing, as the case may be and subsequently, Disposal Order would be issued.

11.4 SAFETY AUDITS & RECORDS

I. The safety audit needs to be conducted as per the IS:14489:1998 guidelines, to effectively conduct a gap analysis and diagnosis of the current position against the scheme. The audit would provide clear guidelines for verifying the existence and implementation of elements of occupational safety and health system and for verifying the system's ability to achieve defined safety objectives. The aim of the audit would be to enable WBSEDCL to develop its own specific procedures for implementing this standard.

II. Evaluation of Safe operating and maintenance procedures in all of its Substations, Hydel Power Stations.

III. Field Inspection to identify unsafe condition, practices and acts and suggestion for remedial action thereof.

IV. The scope of work also contain all other activities not specifically mentioned above but may be required to carry out the safety audit as per requirements.

V. After conducting safety audit of allthe installations/facilities in all the regions covered under WBSEDCL, the observation and recommendations, shall be discussed with the Zonal Manager of that individual WBSEDCL

region on completion of audit in that region.

CHECK LIST FOR SAFETY AUDIT

A. L.T. Line installation:

1. To check the condition of conductor (years of use),
2. To check whether there is over loading or not,
3. To check whether standard span used or not,
4. To check whether due ground clearance is maintained or not,
5. To check the condition of jointing,
6. To check the condition of staying,
7. To check the condition of stringing & sagging of conductors in the line whether there is any loose/irregular sag,
8. To check the rating of LT. fuse used,
9. To check whether there is any provision of sectional fuse,
10. To check condition of safety device used,
11. To check whether the Safety devices are properly earthed or not and check condition of over all earthing system of the installations,
12. To check whether the earthing resistance is within permissible limit or not
13. To check condition of earthing & earthing resistance of neutral point of Distribution Transformer
14. To check whether the line is free from trees & tree branches and other obstacles
15. To check whether there is any hooking & tapping from the installation
16. To check whether there is any generator line running with the pole of WBSEDCL installation
17. To check the condition of supports/ poles
18. To check the spacing between conductors

B. Distribution Substation installation:

1. To check the overall earthing system, value of earthing resistance & its condition
2. To check the operation of Isolators;

3. To check the earthing condition of Isolator handle;
4. To check the rating of H.T. fuse used;
5. To check the loading of the transformer at peak load hours;
6. To check the condition of transformer oil;
7. To check the level of the transformer oil;
8. To check the condition of silica gel (wherever applicable);
9. To check the condition of insulators used;
10. To check the condition of fuse unit/main Switch used;
11. To check the condition of L.A.(wherever applicable);
12. To check the condition of load balancing including neutral current;
13. To check the condition of kiosk & cable connection including fuse arrangement & rating thereof.

C. H.T. line Installations(33 kV & 11kV)

1. To check the condition of conductor & size
2. To check the loading condition at peak load hours
3. To check whether standard span used or not, condition of sag
4. To check whether proper ground clearance is maintained or not
5. To check the existence & condition of Cradle Guard across the road, pond, river, canneletc.
6. To check the condition of jointing & no. of joints made in a span
7. To check whether sleeve jointing used or not
8. To check the condition of staying
9. To check the condition of brackets, insulators & other fittings
10. To check the condition of stringing & sagging of conductors in the line, whether there isloose/irregular sag
11. To check the operation of Isolators at Section point
12. To check whether cradle guards with proper earthing is used at highway crossing/ road crossing, power line crossing, telephone line crossing & wherever necessary ornot
13. To check whether G.I. wire/undersize conductor is used or not in place of rated conductor

14. To check the condition of the installation and value of earth resistance
15. To check whether the line is free from trees & tree branches and other obstacles to record no. oftripping occurred in bad weather & normal weather
16. To check whether the construction of the line installations done as per standards or not
17. To check whether appropriate indication board is used or not, as per the supports (single pole, D.P. four pole etc.) whether there are more than one source for supply
18. To check whether the line is controlled by individual breaker or not
19. To check the condition of supports used

D. H.T. Substation Installations (33 kV & 11 kV):

1. To check the condition of overall earthing system of the Substation, its value of earthing resistance etc.
2. To check the condition of protective arrangement and system,
3. To check condition of battery used,
4. To check the condition of gear oil/ transformer oil used,
5. To check the condition of breakers/ switchgear including operational condition,
6. To check the operation of Isolator,
7. To check the condition of clamps, connectors, insulator used,
8. To check the condition of battery charger,
9. To check the condition of cable joints & other terminal joints including connectors,
10. To check the condition of the entire feeder both at peak & off-peak hours,
11. To check the condition of silica gel of power transformers
12. To see that individual feeder is controlled by respective individual breakers/ switchgear & safe working clearances
13. To check whether safety clearances at different locations are maintained or not
14. To check the condition of firefighting equipment
15. To check the condition of personal protective equipment used,

16. To check the condition of tools & tackles used,
17. To check about the availability of First Aid Box and Artificial Respiration Chart,
18. To check the procedure of Work Permit system,
19. To check if the requisite numbers of Lightning Arrestors are installed or not and their conditions

E. Consumers' Premises

1. To check the condition of service line including bracket/support upto meter
2. To check earthing arrangement of consumer premises
3. To check condition of meter and metering arrangement
4. To check location of the meter(whether in safe & easy accessible position)
5. To check condition and location of consumer's starter, capacitor main switch, cut-out with their fuse rating
6. To record the consumer no., meter no. with meter reading (kwh) as found during checking of the premises

F. Hydro-Electric Power Generation Safety Check List

1. Cleaning Schedule of Gates at the upstream & downstream(HEAD RACE & TAIL RACE)
2. Maintenance of minimum water head for the Turbine
3. Periodic cleaning & inspection of Turbine including clearance of impeller
4. Periodic inspection of Generating Set including cleaning schedule
5. Safety Inspection record of Civil Construction at different level including Turbine & Generating Sets(From GRL to Lowest level)
6. Safety Inspection report of the overhead Crane with lifting Tools & Tackles
7. Report of Capital Maintenance
8. Control Room Temperature and Humidity
9. Housekeeping of Control Room & Battery Room
10. Relay Inspection Report
11. Synchronization
12. Calibration Report of the meters & Annunciation System

13. Earth pit inspection Report
14. Annunciation System
15. Earth pit Inspection Report
16. Lightning Arrestor Inspection Report
17. Illumination
18. Existence of necessary Safety Devices at tunnels, confined areas & other danger zone
19. Arrangement of sufficient PPEs including artificial oxygen masks to work under water
20. Work Permit
21. Shutdown & Startup Procedure
22. Break-down Analysis Report
23. Safety Training & Awareness
24. Maintenance of Switching yard including PT,CT, Isolator etc.
25. Fire Fighting System
26. Emergency Preparedness and response
27. Mock Drill
28. DG Set Maintenance Record
29. Daily routine checkup of consumption of Diesel oil
30. Temperature of Cooling Water
31. Lube-oil Pressure
32. De-Scaling Report
33. System diagram & Circuit Diagram
34. Isolation System
35. Circuit Breaker
36. Pollution Control Board Consent Letter of DG Sets
37. Stack Gas emission analysis Report
38. SPM Level
39. RPM Level
40. Noise Level monitoring
41. Battery Maintenance Report
42. System diagram & Circuit Diagram

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CHAPTER-XII

FIRST AID

A moment of care can last a lifetime

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CHAPTER-12

FIRST AID

12.1 HAZARDS OF ELECTRICITY

- Shock
- Bums
- Arc Flash
- Arc Blast
- Hazards from static charge
- Secondary Hazards
- All of above may cause injury and death

Shock:

Some of the major causes

- Failure to completely de-energize circuit prior to start of electrical work
- Unnecessary risk in working on energized systems
- Failure of earthing
- Not using PPE or using defective/unsuitable PPE

Factors which determine severity of shock

- Level and type of Voltage
- Current
- Path through body
- Duration of shock
- Skin condition
- Environment

Voltage as low as 50V (AC rms) or 120V DC or current as low as 30 mA can be fatal.

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Effect of current on human body

- Low current [< 3mA] – Tingling
- Slightly higher current [< 10 mA] - Mild to serve shock
- Still Higher [> 10mA] - Cause paralysis. Cannot release live part due to temporary muscular paralysis causing increased duration of shock
- Still Higher [> 30mA] - Respiratory paralysis
- Still higher [>75mA] - May affect electrical activity of heart like fibrillation
- Still higher [> 4A] -Tissue burning

Path through Body

- Through skin only - Less dangerous
- Through Heart -Can cause heart failure
- Through Respiratory system -Can cause breathing Paralysis

Duration

- Long duration can cause burning of tissues
- But even small duration of current passing through vital organs can cause its failure leading to death or permanent damage.

Shock Protection

Basic Shock Protection Policy is "Hazardous live parts shall not be accessible and accessible parts shall not be hazardous live".

Strategies that are used

- Reduce shock voltage to very low level by earthing non conductive metal body
- Reduce shock current by increasing resistance between man and touched part & between man and earth by use of PPE/ insulated item
- Sense leakage to earth and trip supply

Arc Flash

Causes

- Dielectric breakdown causes short circuit and limiting impedance is low
- When air becomes superheated and metal vaporizes

Dangers of Arc flash

- Can cause many of the following injuries of Shock, Burns, Blindness, Deafness
- Can burn almost any type of clothes increasing danger of secondary burns
- Can cause second degree burns at very large distance

Burn

Electric shock causes local burns but Arc Flash can cause extensive burns in face, chest, back and limbs

- First degree Burn -Superficial bums on the skin, painful but no permanent damage
- Second degree Burn - Skin and growth tissues are burnt. Growth of skin is slow. Usually leaves scar
- Third degree Burn -Skin and growth glands are also burnt. Skin cannot grow without grafting, leaves scar and some permanent damage.

12.2 FIRST AID INSTRUCTIONS

General

(i) Remove the patient from the source of accident or remove the cause of Injury.

(ii) Keep the injured person lying down in comfortable position, her/his head in level with her/his body. This is preventive against fainting. Never pick her/ him up by head and heel.

(iii) Severe hemorrhage must receive immediate attention, no matter what other injuries are present.

(iv) If the breathing has ceased, immediate measures must be taken to restore it. The patient should be in a position to breathe freely.

(v) Poisons swallowed should be got rid off or neutralized and should be hospitalized without delay.

(vi) If the patient has received bums, attend to them.

(vii) When the patient has fractured a bone, no attempt must be made to move the patient until the bone rendered as much immovable as practicable, unless life is in danger from some other cause.

(viii) Treat the patient for shock.

(ix) Send for medical help or ambulance where possible.

(x) Never give water or liquid to an unconscious patient.

(xi) Keep by-standers away from the patient.

(xii) Don't let the patient see his own injury. Keep the patient warm. Avoid application of external heat, but maintain normal body temperature.

(xiii) The first aider must on no account put upon himself the duties and responsibilities of a doctor.

(xiv) Air should not be blocked in a place where the patient has been kept.

Fainting

(i) Lower patient's head between knees, loosen tight clothing around neck. If impossible lower victim's head, elevate his lower limbs and keep him lying down until recovery seems assured. If unconsciousness persists cover patient and call for a doctor.

(ii) Sprinkle on the face hot and cold water alternately, and apply warmth to the pit of the stomach and over heart. Vigorous rubbing of the limbs upwards has a stimulating effect. Smelling salts may be held to the nose.

Sun stroke and Heat stroke

(i) Sun stroke and heat stroke have the same symptoms but the cause may be slightly different. Sun-stroke results from excessive direct exposure to the Sun's rays, while heat stroke results from excessive exposure to heat such as in boiler rooms.

(ii) Symptoms are red and flushed face, hot and dry skin, no sweating, rapid and strong poise, very high excessive, headache and usually unconsciousness.

Heat Exhaustion

(i) Heat exhaustion is caused by direct exposure to Sun's rays or by excessive indoor heat.

(ii) Symptoms are pale face, cool skin, profuse sweating, weak pulse, low temperature and fainting.

Fractures

(i) Do not move the patient unless absolutely necessary. Call a doctor to the scene of accident. If necessary to move the patient, always apply splint

before moving him. Handle him carefully to prevent sharp ends of bones cutting through flesh.

(ii) Do not hurry in moving an injured person. Always be careful in handling and transporting an injured person. Improper or careless methods frequently increase severity of injury and may even cause death. Acquaint yourself with the various methods of carrying and transportation.

Electrical Burns

Electrical burns occur:

(i) When current passes through the body burning or destroying tissues as it goes, it makes a deep third degree burn which may be smaller on surface than inside, and slow to heal.

(ii) Flash bums of the skin are not usually deep and first or second degree. Flash burns of the eye may not show up until sometime later. In first aid for flash burn of eye, light should be excluded by using a moist compress held lightly in place with a bandage.

(iii) Eye burns should have a doctor's attention as soon as possible.

(iv) First-aider's duties are to relieve pain, prevent infection and treat for shock. Death in a day or two after a bum is usually the result of shock. Death later is chiefly the result of infection.

(v) For burns of limited extent, apply Vaseline or bum ointment over the burnt area. Cover ointment with a layer or two, of fine mesh gauze and secure with a roller bandage. Take the patient to a doctor for further treatment.

(vi) Extensive burns may be much more serious. Shock is always present. Keep victim lying down with her/his head low and avoid exposure of cold. Leave her/his clothing on, cover her/him with blankets and get her/him to a hospital as quickly as you can.

(vii) If hospital is not nearby, remove all loose clothing from the burnt area unless it sticks to skin. Cut the adhering cloth around the bum and leave the remainder for the doctor to remove.

(viii) Do not break blisters. Dip strips of clean freshly laundered sheeting into a solution of baking soda of Epsom salt in warm water, and apply to burnt area.

12.3 TREATMENT OF ELECTRIC SHOCK

a) Act at once - delay is fatal.

b) Death from electric shock is rarely instantaneous.

c) Heart fibrillations (heart muscle tremors) persist as long as 30 minutes after an electric shock. Therefore life can be saved by immediate artificial respiration.

d) Send for, but never wait for a doctor.

e) Continue artificial respiration for four hours after apparent death.

Release from contact

(i) Switch off power supply immediately or send someone to do so. Do not attempt to remove a person from contact with high voltage unless suitable articles insulated for the system voltage are used for this purpose. When attempting to force a person from contact with low or medium voltage, use rubber gloves, boots mat or insulated stick, but if these are not available, use a loop of rope, cap or coat to drag the person free. Whatever is used should be dry and non conducting.

(ii) Keep the patient warm and provide artificial respiration.

12.4 ARTIFICIAL RESPIRATION

Mouth-to-Mouth method

(i) Place victim on his back. Place his head slightly downhill, if possible. A folded coat or similar object under victim's shoulders will help maintain proper position. Tilt head back so that the chin points straight upward.

(ii) Grasp victim's jaw and raise it upward until lower teeth are higher than upper teeth; or place fingers on both sides of jaw near ear lobes and pull upward. Maintain jaw position throughout period to prevent tongue from blocking air passage.

(iii) Take a deep victim's breath and place your mouth over victim's mouth making airtight contact. Pinch the victim's nose, shut with thumb and forefinger or close nostrils by pressing your cheek against them. If you hesitate at direct contact, place a porous cloth between you and victim. If an infant, place your mouth over it's mouth and nose.

(iv) Blow into victim's mouth (gently, if an infant) until his chest rises. Remove your mouth to let him exhale, turning your head to hear out rush of air. The first 8 to 10 breaths should be rapid as victim will respond, thereafter, rate should be slowed to about 12 times a minute (20 times if an infant).

(v) If air cannot be blown in, check position of victim's head and jaw and recheck mouth for obstructions, then try again more forcefully. If chest

still does not rise, turn victim's face down and strike his back sharply to dislodge obstructions.

(vi) Sometimes air enters victim's stomach evidenced by swelling of stomach. Expel air by gently pressing down on stomach during exhalation period.

12.5 CARDIO PULMONARY RESUSCITATION (CPR) & EMERGENCY INFORMATION

The electric shock treatment chart and instructions for CPR in Bengali and English are affixed in the Control room &/or in the Operator's cabin in every generating station, enclosed substation, and enclosed switchyard. Other emergency information displayed next to these are address and contact number of the nearest doctor, nearest hospital with facility for first-aid treatment of electric shock & burns, and of the ambulance service.

12.6 FIRST AID BOX

First-aid boxes with unexpired medicines, dressing materials and other essential contents are to be maintained in every generating station, enclosed substation, enclosed switching station, customer care center, in every other office unit and in vehicles used for maintenance of lines. These are to be kept in an accessible & visible place under the charge of a responsible person(s).

The contents of the boxes are:

- Small sized sterilized dressings,
- Medium sized sterilized dressings,
- Large sized sterilized dressings,
- Large sized sterilized burn dressings,
- Sterilized cotton wool,
- Bottle of antiseptic solution,
- Sealed packs of sterilized eye pads,
- Aspirin or any other analgesic tablet,
- Scissors,
- Medium sized adhesive plaster roll,
- Large sized adhesive plaster roll,
- 5 cm wide roller bandage,

- 10 cm wide roller bandage,
- Triangular bandage,
- Tourniquet,
- Splints,
- Safety pins,
- Kidney tray,
- Wash bottle (500 ml) for washing eyes,
- Snake bite lancet,
- Potassium permanganate,
- Sol-violate spirit (for smelling of unconscious person),
- Bottle of spirit,
- Boric powder,
- Silver Nitrate Gel etc.

12.7 FIRST AID ESSENTIALS

(i) Breathing Apparatus/Oxy pack

(ii) Artificial Respiration Chart

(iii) Rescue Stick (Preferably made of fibre and of at least 10′ in length) or any other similar type of apparatus

12.8 FIRST AID SUPPLY MAINTENANCE

Proper arrangement should be there to regularly check availability of the first aid items including the listed essential medicines. All expired items need be replenished with usable ones immediately.

12.9 FIRST AID TRAINING

First aid training including training on CPR from authorized institutes or individuals should be conducted regularly to have adequate no. of trained persons who can deliver the service in need.

DIFFERENT MEANS TO TAKE WATER OUT OF THE LUNGS OF A DROWNED PERSON
জলে ডোবা ব্যক্তির ফুসফুসের জল বের করার বিভিন্ন পদ্ধতি
FOR A BLEEDING NOSE PRESS THE BASE OF NOSE FIRMLY
নাক দিয়ে রক্তপাত হলে নাকের ডগা চেপে ধরুন
SUN STROKE
গর্মিতে আক্রান্ত ব্যক্তিকে ঠাণ্ডা ঘরে রাখা ও ঠিকভাবে পরিচর্যা করা
DISCONNECT SUPPLY & USE WOOD TO SAVE FROM ELECTRIC SHOCK
বিদ্যুৎ সংযোগ কেটে অথবা অপরিবাহী বস্তুর সাহায্যে শক লাগা ব্যক্তিকে বাঁচান
TO STOP BLEEDING, LIFT THE INJURED LIMB ABOVE THE HEART LEVEL AND PRESS FIRMLY
রক্তক্ষরণ বন্ধ করতে ক্ষতস্থান জোরে চেপে ধরুন অথবা হৃদযন্ত্র থেকে পা উঁচুতে তুলে ধরুন
FOR A POISONOUS INTAKE MAKE THE VICTIM VOMIT OR GIVE SALTY WATER
বিষপান ব্যক্তিকে বমি করানো অথবা নুন জল খাওয়ানো
REMOVAL OF FOREIGH BODY FROM THE EYE
চোখ থেকে ময়লা বার করা
ACCIDENT CALL AMBULANCE
দুর্ঘটনা ঘটলে অ্যাম্বুলেন্স ডাকুন
TELEPHON BOOTH
DIFFERENT BANDAGES ARE REQUIRED FOR DIFFERENT FRACTURES
শরীরের বিভিন্ন অংশের হাড় ভাঙার ক্ষেত্রে বিভিন্ন ধরনের ব্যাণ্ডেজ প্রয়োজন

FIRST AID CHARTS

ELECTRIC SHOCK AND TREATMENT

विद्युताघात तथा उपचार

घर के अन्दर

घर से बाहर

Stand on insulation material to touch victim.

विद्यु रोधक वस्तु पर खड़े होकर घायल का स्पर्श करें।

Effects of Shock

1. Paralysis of respiratory muscles.
2. Extensive and deep burns.
3. Muscle contraction.
4. Dehydration.
5. Clotting of blood.
6. Necrosis.
7. Heart failure.
8. Kidney failure.

Treatment

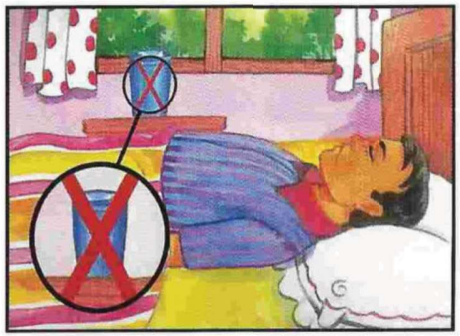
1. Do not move the victim.
2. Call for help.
3. Switch off or move the current source using insulating object.
4. Loosen tight clothing. Give artificial respiration and treat any burns.

विद्युताघात के परिणाम

1. श्वसन पेशीयों का पक्षाघात।
2. व्यापक व गंभीर दाह।
3. पेशीय संकुचन।
4. निर्जलीकरण।
5. रक्त के थक्के।
6. परिगलन।
7. हृद्पात।
8. गुर्दों की खराबी।

उपचार

1. मरीज को हिलायें नहीं।
2. मदद के लिए पुकारें।
3. विद्युतरोधक वस्तु की मदद से करंट बंद करें।
4. तंग कपड़ो को ढीला करें। कृत्रिम श्वास दीजिए और जला हुआ हो तो उसका उपचार कीजिए।



Never give any thing to drink to an unconscious person.

अचेत व्यक्ति को कोई भी वस्तु पीने के लिए न दें।

ARTIFICIAL RESPIRATION
कृत्रिम श्वास क्रिया

1. Apply moderate pressure to expel out air.
वहि : श्वसन के लिए धीर-धीरे दबाव डालिये।

2. Release pressure to take air in.
श्वास अन्दर भरने के लिए दबाव हटा लें।

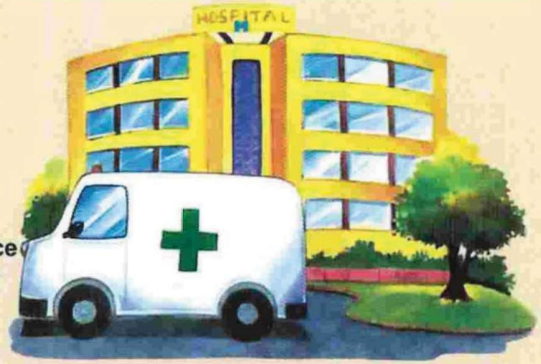
1. Raise the shoulders so that the head is tilted backwards. Bring the arms over chest and press.
घायल के कंधों को थोड़ा ऊपर करके सिर को पीछे की ओर लटका दें, कलाईयों को सीने पर रख कर दबाएं।

2. To inhale the air, hold the arms, bring them over the head.
दबाव को हटाइये और दोनों हाथों को तेजी से ऊपर तथा बाहर की ओर ले जाइए।

CONTACT IN EMERGENCY :
आपातकालीन स्थिति में सम्पर्क करें :

Doctor
डॉक्टर

Electricity Department
बिजली घर



Ambulance
रोगी वाहन

The speed with which casualty be brought under treatment is of utmost importance.
घायल का अतिशीघ्र उपचार परम आवश्यक है।

Fire Brigade अग्नि शमन सेवा

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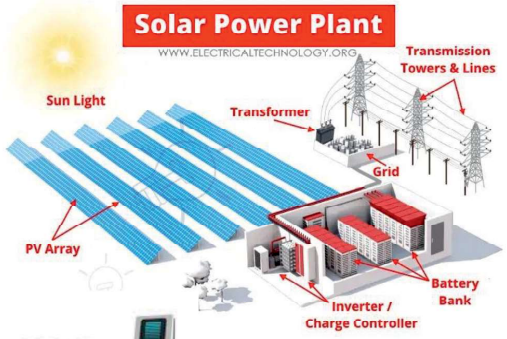
CHAPTER-XIII

SOLAR POWER PLANT & ELECTRIC VEHICLE



From solar panels to EV plugs—safety connects the future

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CHAPTER-13

SOLAR POWER PLANT & ELECTRIC VEHICLE

13.1 SOLAR POWER PLANT

Solar energy: Safety Risks and How to Prevent Them

Installing solar systems is a challenging business. Lifting and arranging unwieldy solar panels, the potential for falls of many-storied rooftops, panels that exposed in sunlight create & store energy which may cause serious hazards that solar workers face, if precautionary measures not taken. They're also subject to the risks of traditional roofing, plumbing and electrical trades – some of the most injury-prone occupations around.

Every Worksite Presents Different Risks

Before a solar installation begins, it's essential for the installer to visit the site, identify the safety risks and develop specific plans for addressing them. Plans should include:

1. Equipment to be used for safe lifting and handling of solar panels,
2. Type and size of ladders and scaffolding if needed,
3. Fall protection for rooftop work,
4. Personal protective equipment for each installer.

All equipment needed for the job should be inspected and verified to be in good working order before being brought to the worksite.

Lifting and Handling Solar Panels

Solar panels are heavy and awkward to lift and carry. Loading and unloading panels from trucks and onto roofs can cause strains, sprains, muscle pulls and back injuries as well as cumulative trauma that stresses the spine. The panels can also heat up quickly when exposed to sunlight, causing burns if not handled safely.

Safety measures for solar workers:

1. Lift each solar panel with at least two people while applying · safe lifting techniques.
2. Transport solar panels onto and around the work site using mobile carts or forklifts.
3. Never climb ladders while carrying solar panels. To get solar panels onto rooftops, use properly inspected cranes, hoists or ladder-based winch systems.
4. Once unpacked, cover panels with an opaque sheet to prevent heat buildup.
5. Always wear gloves when handling panels.

Effective lifting techniques

1. Left with your legs,
2. Do not exceed your physical ability,
3. Provide adequate recovery time for tasks that require frequent lifting.
4. Provide easy access so the load is in front of the person lifting.
5. Eliminate twisting by changing the start or end point of the lift.
6. Put items to be lifted between waist and shoulder height,
7. Lift items close to your body to reduce the stress on your lower back,
8. A safer method of getting panels onto difficult Canal tops is to hire an experienced crane operator. Crane lifts should always be performed by experienced personnel and hardhats must be worn.
9. Always use two people to lift and load large and heavy items.

Ladder Safety

Solar construction often involves working on roofs and on ladders. Choosing the right ladder and using it properly are essential.

Safety measures for solar workers:

1. Select the ladder that best suits the need for access-whether a stepladder, straight ladder or extension ladder. Straight or extension ladders should extend a minimum of three feet above the rung that the worker will stand upon.
2. Select the right ladder material. Aluminium and metal ladders are the most commonly used today and may have their place on the job, but they're a

serious hazard near power lines or electrical work. Use a fibre glass ladder with nonconductive side rails near power sources.

3. Place the ladder on dry, level ground removed from walkways and doorways, and at least 10 feet from power lines and secure it to the ground or rooftop.
4. Carry ladders horizontally when moving them into place.
5. Use two people to move and position ladders whenever possible.
6. The preferred method to rise an extension ladder into place involves two people. One person can hold the base of the ladder by placing their foot on the bottom rung. The second person raises the ladder, walking the ladder up one rung at a time until the ladder is vertical.
7. This process is similar to the one below, raising the ladder with one person. A wall is used to 'foot' the ladder instead of a person.
8. If you have difficulty raising a ladder into place it is sometimes possible to rest the ladder feet against the base of the wall you are working on.
9. Place the sliding 'fly' part of the extension ladder face down.
10. Starting at the top of the ladder, walk toward the wall while raising the ladder one rung at a time.
11. Once the ladder is in position the sliding 'fly' section will be up and facing in the correct position.
12. If it is not possible to secure a ladder using anchors then a person can stand at the base of the ladder and secure it manually against slipping.
13. Having a co-worker hold the ladder when you are climbing can be another good method of securing the ladder.
14. When working from the ladder, always work within an arm's reach from the ladder. While working, keep both feet on the rungs and use your belt buckle as a guide to keep your weight centered on the ladder during all times. With both feet on the rungs, your belt buckle should never stray outside of the side rails.

Trip and Fall

Trips and falls are a common hazard of all construction jobs, including solar. They can happen anywhere on the jobsite, especially, off roofs or ladders. Rooftops or Ground mounted solar installations are especially hazardous because the work space diminishes as more panels are installed, increasing the risk of falls.

Safety measures for solar workers:

1. Keep all work areas dry and clear of obstructions.
2. For fall distances of six feet or more, take one of three protective measures: install guardrails around ledges, sunroofs or skylights; use safety nets; or provide each employee with a body harness that is anchored to the rooftop to arrest a potential fall.
3. Cover holes on rooftops, including skylights, and on ground-level work surfaces.

Solar Electrical Safety

Solar electric (photovoltaic or PV) systems include several components that conduct electricity: the PV solar array, an inverter that converts the panel's direct current to alternating current, and other essential system parts. When any of these components are "live" with electricity generated by the sun's energy, they can cause injuries associated with electric shock and arc-flash. Even low-light conditions can create sufficient voltage to cause injury.

It's also important to recognize that with PV systems, electricity comes from two sources: the utility company and the solar array that is absorbing the sun's light. Even when a building's main breaker is shut off, the PV system will continue to produce power. This makes isolating the power source more difficult, and requires extra caution on the part of the solar worker.

Safety measures for solar workers:

1. Cover the solar array with an opaque sheet to "tum off' the sun's light.
2. Treat the wiring coming from a solar PV array with the same caution as a utility power line. Use a meter or circuit test device to ensure that all circuits are de energized before working on them.
3. Lock out the power on systems that can be locked out. Tag all circuits you're working on at points where that equipment or circuit can be energized.
4. Never disconnect PV module connectors or other associated PV wiring when it is under load.

GENERAL ELECTRICAL HAZARDS

Overhead power lines

Overhead power lines are not insulated - they can and do kill people that contact them. More than half of all electrocutions are caused by direct worker contact with energized power lines. When cranes, work platforms,

or other conductive materials such as ladders or aluminium mounting rails contact overhead wires, the equipment operator or other workers can be killed. If you do not maintain required clearance distances from power lines, you can be shocked and killed. You must always stay a minimum of 10 feet away from overhead power lines! (The minimum distance for voltages up to 50 kV is 10 feet. For voltages over 50 kV, the minimum distance is 10 feet plus 4 inches for every 10 kV over 50 kV.)

Electrical systems and tools those are not grounded or double-insulated

The most common OSHA (The Occupational Safety and Health Administration) electrical violation is improper grounding of equipment and circuitry. The metal parts of an electrical wiring system that are exposed (switch plates, ceiling light fixtures, conduit, etc.) should be grounded and at zero volts. If the system is not grounded properly, these parts may become energized. Metal parts of motors, appliances, or electronics that are plugged into improperly grounded circuits may be energized. When a circuit is not grounded properly, a hazard exists because unwanted voltage cannot be safely eliminated. If there is no safe path to ground for fault currents, exposed metal parts in damaged appliances can become energized. Extension cords with missing or broken ground terminals are unsafe and should never be used. If you contact a defective electrical device that is not grounded (or is grounded improperly), and you complete a path to ground, you will get shocked.

- Employers must not allow their workers to work near any part of an electrical circuit unless the circuit is de-energized (shut off) and grounded, or guarded in such a way that it cannot be contacted.
- Ground-fault protection must be provided at construction sites to guard against electrical shock. Voltage cannot be safely eliminated. If there is no safe path to ground for fault currents, exposed metal parts in damaged appliances can become energized. Extension cords with missing or broken ground terminals are unsafe and should never be used. If you contact a defective electrical device that is not grounded (or is grounded improperly), and you complete a path to ground, you will be shocked.

Overloaded circuits

Overloads in an electrical system are hazardous because they can produce heat or arcing.

A ground fault circuit interrupter, or GFCI, is an inexpensive lifesaver. GFCis detect any difference in electrical current between the two circuit wires. This difference in current could happen when electrical equipment is

not working correctly, causing leakage current. If leakage current (a ground fault) is detected in a GFCI-protected circuit, the GFCI switches off the current in the circuit, protecting you from a dangerous shock. GFCis are set at about 5 mA and are designed to protect workers from electrocution. GFCis also detect the loss of current resulting from leakage through a person who is beginning to be shocked. If this situation occurs, the GFCI switches off the current in the circuit. GFCis are different from circuit breakers because they detect leakage currents rather than overloads. Circuits with missing, damaged, or improperly wired GFCis may allow you to be shocked.

Wires and other components in an electrical system or circuit have a maximum amount of current they can carry safely. If too many devices are plugged into a circuit, or if a single device draws too much current, or if incorrect wire size is used, the electrical current can heat the wires to a very high temperature. Excessive heat can melt insulation and lead to arcing which can create ground faults or fires. In order to prevent too much current in a circuit, a circuit breaker or fuse is placed in the circuit. Breakers and fuses do the same thing: open the circuit to shut off the electrical current. If the breakers or fuses are too big for the wires they are supposed to protect, an overload in the circuit will not be detected and the current will not be shut off.

Electrical hazards can be made worse if the worker, location, or equipment is wet

You don't have to be standing in water to be electrocuted. Wet clothing, high humidity, and perspiration also increase your chances of being electrocuted. Working in wet conditions is hazardous because it increases the chance of completing a path for current flow. If you touch a live wire or other electrical component-and you are well-grounded because you are standing in a wet area-you will receive a shock.

Personal Protective Equipment

Personal Protective Equipment is an essential part of every solar installation. It's the employer's job to assess the workplace for hazards and provide the PPE deemed necessary for the employee's safety. Hard hats, gloves and steel-toed shoes with rubber soles are among the commonly required PPE for solar projects. Employees are in tum responsible for using PPE in accordance with their employer's instructions, maintaining it in a safe and reliable condition and requesting replacements when necessary. Risk is part of running any business, but when it comes to an employee's safety and health, risk should be avoided

at all cost. Proactive safety planning and its successful implementation on the job can help ensure that accidents don't happen.

Personal fall-protection systems.

The shock absorber of a fall-arrest system

13.2 SAFETY REQUIREMENT FOR SOLAR INSTALLATIONS

(1) General safety requirements:

(i) clear pathways of minimum seventy-five centimetre in width with hand rails for roof access and emergency exit shall be provided for roof top system;

(ii) there shall be clear pathways, walkways between the rows or columns of solar panels which is necessary for cleaning and maintenance;

(iii) cables shall be laid in trenches for ground based photovoltaic installation;

(iv) ground mounted solar installations shall be protected by fencing or other means not less than 1.8 metre in height so as to prevent unauthorised entry;

(v) disconnection switches or circuit breakers provided in combiner boxes to disconnect the photovoltaic system from all other conductors of the system shall be located at a readily accessible location;

(vi) three phases on the alternating current side, and positive and negative conductor on the direct current side shall be marked and identified with different colours;

(vii) inverter unit for solar photovoltaics shall be installed in the periphery of the building and as near as the solar panel:

Provided that the direct current cable shall be UV protected or routed through UV protected pipe;

(viii) there shall be a manual disconnection switch to isolate the system from grid and shall be situated outside the alternating current combiner box; and

(ix) protection shall be deployed (for both input and output) on site for overload, surge current, surge voltage, short circuit, high temperature, over voltage, under voltage and over frequency, under frequency, reverse polarity and lightning.

(2) Earthing requirements:

(i) solar earthing shall be as per the relevant standard;

(ii) the frame of inverter cabinet shall be connected with the earthing bus bar through the earthing terminals using flexible braided copper wire;

(iii) all metal casing and shielding of the plant, each array structure of the photovoltaic yard, equipment, inverters and control systems shall be earthed through proper earthing;

(iv) earthing system shall connect all non-current carrying metal receptacles, electrical boxes, appliance frames, chasis and photovoltaic module mounting structures in one long run and the earth strips shall be interconnected by proper welding and shall not be bolted;

(v) there shall be adequate number of interconnected earth pits provided in each location and minimum required gap shall be provided in between earth pits as per relevant standard.

(3) Protection, testing and interlocking requirement:

(i) the solar photovoltaic power plant shall be provided with lightning and over voltage protection by deploying required number of lightning arresters as per the relevant standards;

(ii) every combiner box shall be provided with suitable surge protective device with arc extinguishing capability as per the relevant standards to avoid any risk of fire;

(iii) the input circuits of combiner box shall be provided with over current protection as per the relevant standards;

(iv) the output circuits of combiner box shall be provided with isolation protection;

(v) earth fault protection and insulation monitoring for photovoltaic arrays and inverters shall be provided;

(vi) all photovoltaic modules safety qualification shall comply with relevant standards.

(4) Prevention of fire for solar installations: A fire detection system and automatic fire suppression system shall comply with the relevant standards.

(5) Additional safety requirements: for floating solar photovoltaic energy installations: Photovoltaic modules and associated structure of floating solar power plant shall comply with the relevant standards for tests such as salt mist, ammonia corrosion, environmental stress cracking of High Density

Poly Ethylene, stress cracking resistance of High Density Poly Ethylene and standard test method for tensile properties of plastics.

13.3 ELECTRIC VEHICLE

Safety requirements for electric vehicle charging

1. Safety requirements for electric vehicle charging station Safety provisions for charging stations **for electric vehicle** vide schedule XVI of the CEA (Technical Standards for Connectivity of Distributed Generation Sources) Regulations, **2013.**

(1) Electric vehicle charging stations shall be provided with separate protection against the overload of input supply and output supply as per relevant standards.

(2) Socket-outlet of supply of electric vehicle charging points shall be installed at least 800 milli metre above the finished ground level.

(3) A cord extension set or second supply lead shall not be used in addition to the supply lead for the connection of the electric vehicle to the charging point and shall not be used as a cord extension set.

(4) No adaptor shall be used to connect a vehicle connector to a vehicle inlet.

(5) The distance between the charging point and the connection on the electric vehicle shall not be more than five metre during charging.

(6) The portable socket outlets shall not be permitted for electric vehicle charging.

(7) The lightning protection system shall be provided for the electric vehicle charging stations as per relevant standards.

(8) The electric vehicle charging station shall be equipped with a protective device against the uncontrolled reverse power flow from electric vehicle to the charging point.

(9) One second after disconnecting the electric vehicle from the supply mains, the voltage between accessible conductive parts or any accessible conductive part and earth shall be less than or equal to 42.4 V_{peak} (30 V_{rms}), or 60 V DC, and the stored energy available shall be less than 20 J provided that, if the voltage is greater than 42.4 V_{peak} (30 V_{rms}) or 60 V DC, or the energy is 20 J ormore, a warning label shall be attached in a conspicuous position on the charging stations.

(10) A vehicle connector used for direct current charging shall be locked on the vehicle inlet if the voltage is higher than 60 V DC and in case of charging system malfunction, a means for safe disconnection shall be provided.

(11) The electric vehicle charging point shall disconnect supply of electricity to prevent overvoltage at the battery, if the output voltage exceeds maximum voltage limit permissible for the vehicle.

(12) The electric vehicle charging points shall not energise the charging cable when the vehicle connector is in unlock position.

(13) The electric vehicle connector shall not unlock if the voltage between the vehicle connector and the earth is more than 60 V.

(14) Safety clearance between the oil or gas dispenser and electric vehicle charging point shall be as per the order issued by the Authority.

(15) Only four core cable shall be used for charging points which require three phase power supply.

(16) Underground cables shall not cross the underground oil tank or oil pipeline.

(17) Underground cables through the charging area or vehicles passage shall be avoided and if provided shall be at a minimum depth of one metre from the finished ground surface.

(18) Charging points should be protected against mechanical damage by vehicle. They should be installed above ground level and be protected by Krebs, bollards or metal barriers. Charging points should also be protected against the ingress of water and foreign objects.

(19) Space for PCS must not be misused due to actions such as intentional / unintentional blocking of parking space by vehicles which are not being charged, utilization of space for other commercial purposes, displaying advertisements etc.

2. Earth protection system for the charging station:

(1) Each electric vehicle charging points shall be supplied individually by a dedicated sub-circuit protected by an overcurrent protective device complying with the relevant standards and the overcurrent protective device shall be part of a switchboard.

(2) Co-ordination of all protective devices in the charging stations shall be ensured.

(3) All electric vehicle charging stations shall be provided with an earth continuity monitoring system that disconnects the supply in the event of the earthing connection to the vehicle becomes ineffective.

(4) The charging lead shall be fitted with an earth-connected metal shielding and the cable insulation shall be wear resistant and maintain flexibility over the operating temperature range.

(5) A protective earth conductor shall be provided to establish an equipotential connection between the earth terminal of the supply and the conductive parts of the vehicle which shall be as per the relevant standards.

3. Prevention of fire for electric vehicle charging station

The enclosure of electric vehicle supply equipment shall be made of fire retardant material with self-extinguishing property and free from halogen. The fire detection, alarm and control system shall be provided as per relevant standards.

4. Testing of charging station

The owner of the charging station shall ensure that the tests as specified in the manufacturer's instructions for the RCDs and the charging station have been carried out.Electric Vehicle Supply Equipment (EVSE) shall be type tested by an agency / lab accredited by National Accreditation Board for testing and calibration laboratories (NABL) or any other regulatory authority for this purpose from time to time.

5. Maintenance of records

(1) The owner of the charging station shall keep records of design, construction and labelling to be compatible with a supply of standard voltage at a nominal frequency of 50 Hz of the charging station.

a) The owner of the charging station shall keep records of the relevant test certificates as indicated in these regulations and as per the relevant standards.

b) The owner of the charging station shall keep records of the results of every inspection, testing and periodic assessment and details of any issues observed during the assessment and any actions required to be taken in relation to those issues.

The owner of the charging station shall retain a copy of all records, as specified in sub-regulation (1), (2) and (3) of above, either in hard form or in electronic form, for at least seven years and shall provide a copy of the records to the officials during the inspection.

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WBSEDCL

CHAPTER-XIV

ON-SITE EMERGENCY

On-site emergencies need calm minds

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CHAPTER-14

ON-SITE EMERGENCY

14.1 INTRODUCTION

Emergencies are caused by incidents that take place in an industrial facility. The effects of the incidents should mostly remain confined within the facility premises. Such emergencies are termed as on-site emergencies. However, if emergencies are not controlled in time or are of very severe magnitude, their effects may extend well beyond the facilities premises thus creating an off-site emergency.

During on-site emergencies only the people working in the premises are affected. Also such on-site emergencies are handled by the resources available with the facility. Making the facility fully capable to handle its own on-site emergency is the responsibility of the facility management. While emergencies can never be totally prevented, their impacts may be mitigated by effective emergency planning and preparedness.

14.2 THE ROLE OF EMERGENCY ACTION PLAN

Emergency action plan aims to prepare for and mitigate the impacts of an emergency. Preparedness requires identifying what to prepare for and how to respond to. It therefore involves accumulating knowledge and skills, disseminating information about the management of potential emergencies, and providing and allocating facility resources and personnel to deal with the emergencies identified. Through emergency planning, facility personnel improve their understanding of the plant, equipment, processes and materials, and their possible impacts in emergency situations. They also develop an understanding of the roles of the emergency services and other external agencies that could be involved in responding to an emergency. This understanding provides a basis for determining the most effective ways of using facility resources, including the development of a management system identifying the functions required to respond automatically to an emergency. It also provides a basis for informed decision-making during the emergency and for effective working relations with external agencies.

Emergency management involves a cyclical process of four phases:

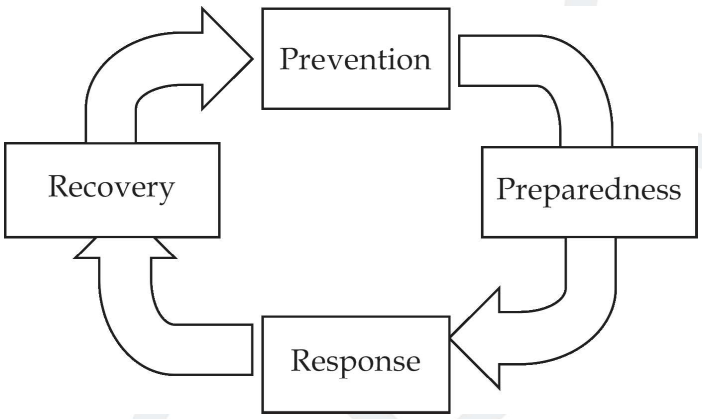
(i) Prevention- regulatory and physical measures to prevent emergencies or mitigate their impact;

(ii) Preparedness - arrangements to mobilize and deploy all necessary resources and services;

(iii) Response - actions taken during and immediately after an emergency to minimize the impact; and

(iv) Recovery - arrangements to restore the facility to normal as quickly and efficiently as possible and to assist the community to recover.

Emergency planning plays a key role in this cycle of emergency management, focusing primarily on the phases of preparedness and response.



14.3 EMERGENCY PLANNING CONSIDERATION

The main consideration of emergency planning is the protection of people, property and the environment from harm during an emergency situation. This is achieved by developing an emergency plan which implements a system able to respond automatically to any emergency and that leads to the most effective outcome possible under the circumstances. The plan should therefore be comprehensive, yet concise, simple and flexible. It should also be dynamic and interactive, ensuring ongoing relevance to the needs of the facility by continual monitoring, review and consultation. Emergency planning is therefore a cyclical process. All of the stages are inter-related and plan details should be continually evaluated, and revised as appropriate.

Consultation:

Consultation is the key to prepare and maintain an effective emergency action plan and should be conducted in all phases of the planning process. All stakeholders affected by the plan (including facility personnel, the community, and external agencies) should be consulted to ensure that each group knows what to expect of the other.

A coordinated and effective response to any emergency requires an understanding between the different parties involved. Consultation when developing the emergency action plan enables the development of this understanding before an incident occurs. It ensures that the roles, responsibilities, functions and needs of all agencies and groups are understood and accurately incorporated into the emergency plan. Once the plan is implemented, consultation during the management of the plan allows all stakeholders to contribute to the testing, monitoring, reviewing, and updating of the plan.

To ensure that consultation is comprehensive, the key stakeholders in the emergency planning process should be identified and on-going relationships with these groups developed. One method of achieving this is by forming an emergency planning working group which includes representatives from all interested parties. While much of the work in developing and managing the emergency plan can be performed by facility personnel, this working group can assist in developing concepts and ideas and also in verifying the emergency plan addresses their particular concern.

Consultation with Facility Personnel

All employees (including employee representatives) should be consulted extensively during the emergency planning process. Not only does this ensure that their intimate knowledge of the facility and its operations is incorporated into the development of the emergency plan; it also generates a sense of commitment and ownership. Each person within the organization has a responsibility to ensure that they are capable at all times of fulfilling their role in the event of an emergency. Ongoing consultation with facility personnel should be actively pursued. For example, staff should be involved in preparing and conducting exercises in order to test the capability of the plan. Debriefings following these exercises can allow participants to indicate the problems encountered and suggest possible solutions.

Consultations with Neighboring Facilities and the Community

Consultation with neighboring facilities and the community should result in a two-way flow of concerns and ideas. Community consultation not only results in a better prepared community; it can often lead to an improved understanding and acceptance of the industry by the wider community.

It is first necessary to identify all neighbors, including those that may have special requirements, such as:

a) neighboring hazardous facilities if any

b) local mutual aid groups if in existence

c) managers of sensitive environmental sites nearby

d) facilities accommodating large numbers of people (e.g. commercial establishments, recreational facilities); and

e) facilities provided for members of the community who may be more vulnerable to the consequences of an emergency (e.g. schools, hospitals, community religious places, etc.).With respect to neighboring hazardous facilities, it is essential that all parties gain an understanding of the potential impacts of an incident on other's operations or storage areas. This will enable procedures to be developed to prevent the escalation of an incident. Neighboring facilities may also be able to provide resources, including personnel, for responding to an emergency. Several industries of related types of operation from the locality may be involved in this type of cooperative arrangement, often referred to as a mutual-aid group.

In the event of an emergency, there needs to be an effective warning system for the neighboring community who could be affected by the emergency. Members of the community need to be aware of the actions to be taken when the warning is activated. Consultation should identify the needs of the community and address the difficulties likely to be encountered.

Local police, fire and other emergency services, Local Government, and safety, health and environmental agencies (government and non-government) should be consulted throughout the emergency planning process. Such consultation will help to define the circumstances when external agencies or other groups need to become involved. This consultation should also result in a clear understanding by all parties of the roles and responsibilities of each group at an emergency.

Emergency Functions and Organization

The system should include defined emergency functions which, like emergency planning in general, aim to protect people, property and the environment. The functions nominated should cover all areas of responsibility necessary to manage the types of emergencies identified. These functions should be defined, taking into account the facility's response requirements and capabilities (i.e. the nature of the operation, the types of emergencies identified, and the number of personnel available). Broad areas to be addressed by emergency functions should include:

a) responding to control the emergency;

b) limiting the spread and impacts of an emergency on adjoining processes, materials, property, and the environment;

c) protecting the safety and health of all personnel on site;

d) protecting the environment;

e) alerting people to the emergency and communicating adequately with all stakeholders during the emergency;

f) accessing the right information; and

g) controlling the entire emergency scene and the whole facility.

These areas may be addressed, by several functions. For example, the protection of the safety and health of all personnel on site may be addressed by functions relating to search and rescue, roll-call, and safeguarding measures such as evacuation

Various functions

The functions listed below are building blocks of Emergency response. Each function and its functionary would have defined roles, responsibilities, duties, and expectations that are associated with these functions. The basic functions are:

a) facility emergency response;

b) damage control;

c) facility emergency support;

d) operations control;

e) protecting people;

f) protecting the environment;

g) facility security and traffic control;

h) evacuation;

i) search and rescue;

j) communications;

k) Medical attention;

l) public relations and media relations.

m) communications;

14.4 FIRE DRILL

Emergency preparedness is an important aspect to prepare all employees mentally and physically in handling and management of an Emergency. Fire drills are initially conducted after informing all to familiarize with the plan and thereafter followed by surprise drills to check the effectiveness. Fire drills are practiced at frequent intervals:

a) To make everyone understand their functions during an emergency and act accordingly;

b) To create awareness among people about their desirable conduct during the emergency as to ensure that emergencies can be tackled effectively & efficiently;

c) To inculcate upon certain moral discipline among employees so that, they do not panic during the emergency;

d) To make all managers and supervisors learn that their workers are properly directed through carefully planned simulation procedures to go to the assembly points, in case not involved in emergency management.

Under emergency situations the following points need be taken care by all concerned duly:

a) Not to spread unauthenticated information;

b) Not to approach the scene as a spectator and make crowd;

c) Not to keep communication channels engaged unnecessarily;

d) To understand and be attentive to instructions.

Fire Mock Drill Format Particulars

a) Date & Time of Mock Drill

b) Location

c) Details of Emergency Scenario

d) Reported by (First person informed about scenario)

e) Response or Action initiated by reporting person

f) Response Time

g) Description of the Mock Drill (The narrative of the situation, all actions)

h) Total no. of persons present in the Drill

i) Time of Clearing the site after Mock Drill

j) Observations (including highlight of the positives of the Drill)

k) Recommendations

(Fire Mock Drill Standerd Format is avilable in Chapter-XVII)

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CHAPTER-XV

SAFETY PROMOTIONAL ACTIVITIES

Awareness today keeps accidents away

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CHAPTER-15

SAFETY PROMOTIONAL ACTIVITIES

WBSEDCL has introduced and is practicing multipronged initiatives in order to maintain a positive safety culture and to enhance its maturity with time. The initiatives have been designed to connect with almost every strata of the society.

Actions to be taken to keep common people out of hazards associated with electricity, build up safe work culture and maintain a healthy power distribution network with the goal to make WBSEDCL a Zero Accident Company:

15.1. AWARENESS CAMPAIGN FOR COMMON PEOPLE & AGENCY WORKERS:

- Safety messages are to be written on assets of WBSEDCL like, Poles, DTR, Kiosks.
- Safety posters are to displayed in front of cash counters in CCCs.
- In CCCs with considerable visitors, LED displays to be utilized to develop public awareness.
- In fairs, festivals and other public gatherings, leaflets & booklets are to be distributed.
- Banners & posters are to be displayed at strategic locations and administrative buildings of Govt. of West Bengal
- Safety awareness campaigns are to be conducted for farmers in villages.
- Safety awareness programs are to be conducted regularly at Educational Institutes. Films/ videos regarding safe usage of electricity may be shown to the students and quiz competitions may be conducted with awarding facility to attract the attention of the students.
- Special awareness campaigns jointly with forest department to be organized in villages within and in the vicinity of Elephant Corridors. Mode of communicating information on system vulnerabilities, like, lines with poor ground clearances, leaned poles etc. to be explained to the villagers.
- National Safety Day & Week are to be celebrated at all site offices

throughout the state of West Bengal. Various activities are to be carried out to celebrate the week viz. road march, safety seminar, safety workshop, film show, displaying of safety banners and posters etc.

- Safety seminars are to be conducted by site offices as well as by Corporate Safety Cell regularly for the agency workers.

15.2 SAFETY TRAINING FOR AGENCY WORKERS:

It is to be ensured that each & every agency worker attends safety training conducted by HRDD regularly. No worker should ever be engaged in any risky work in case he has not successfully completed this safety training.

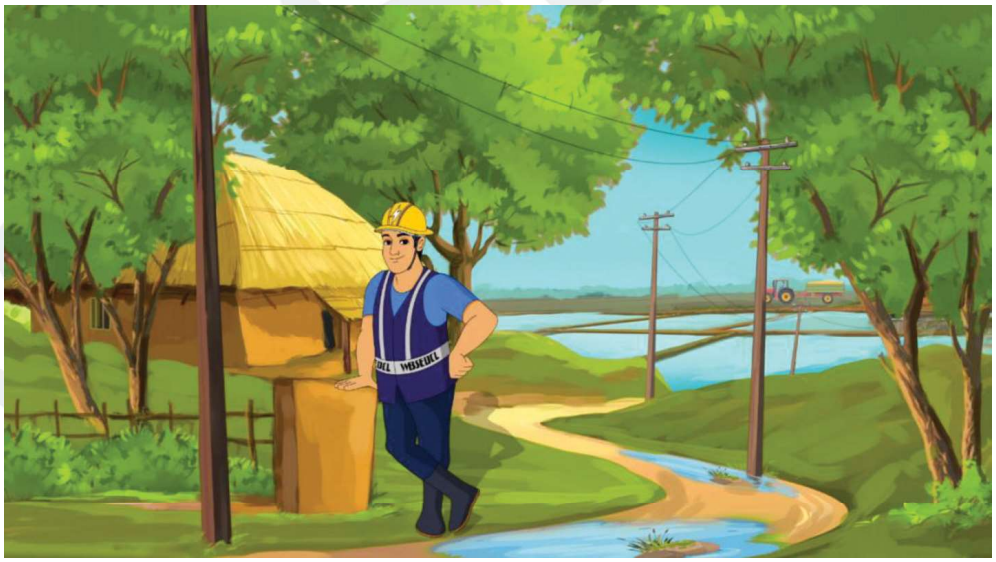
15.3 ACKNOWLEDGEMENT:

- Certificates of appreciation may be awarded to the officers of WBSEDCL for acknowledging appreciable performances towards keeping their jurisdictions accident free.
- Safety WhatsApp number has been kept active in WBSEDCL for receiving photographs and / or videos of vulnerabilities in electrical installation from common people. The relevant information are then forwarded to the concerned site offices for rectification. To encourage common people to send photographs / videos related to safety hazard using this mode of communication, the sender may be rewarded by site offices as a token of acknowledgement.

Digital LED Display in a Customer Care Centre meant for displaying messages for safe use of electricity

A Booklet in Bengali for distributing amongst common people in public gatherings

Banner to be displayed in strategic public places and Offices with foot prints are in large numbers.



Still frame of a video meant for building awareness of common people on safe handling of electricity

Safety seminar at Vidyut Bhawan

Community safety awareness campaign

afety awareness campaign at a school

National Safety Week celebration

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CHAPTER-XVI

SAFETY & DISASTER MANAGEMENT TRAINING

Knowledge in training, strength in safety

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CHAPTER-16

SAFETY & DISASTER MANAGEMENT TRAINING

16.1 TRAINING INSTITUTES:

a) **WBSEDCL's own training institutes:** EETI-Kolkata, EETI- Tribeni, EETI – Burdwan, EETI – Berhampore, EETI – Kharagpur, EETI – Coochbehar, EETI – Malda, EETI – Siliguri and Kalyani Junior Training Institute (KJTC)

b) **Training Partners of WBSEDCL:** IIM, Calcutta; CPRI, Bangalore; ACES INFOTECH, Kolkata; AIMA, New Delhi; ASCI, Hyderabad; ESCI, Hyderabad; PMI, NTPC, Noida; XLRI, Jamshedpur; IISWBM, Kolkata; NPTI, Durgapur; NPTI, FARIDABAD; NPTI, Shivpuri; RECIPMT, Hyderabad; CBIP, New Delhi; NIELIT, Jadavpur; National Safety Council, Kolkata etc.

c) Regional Labour Institute

16.2 SAFETY TRAINING PROVISIONS:

a) In Induction & Orientation Training curriculum, and, in a number of other programs training on personnel, equipment & infrastructure safety is imparted to various employee groups. A two days' safety training program is regularly conducted in WBSEDCL's own training institutes for both the skilled & unskilled agency workers & technical employees engaged in LT & HT line maintenance and Substation operation & maintenance. In order to ensure participation of the agency workers, stipend has been introduced for the participants. Valid certificate & identity card regarding successful completion of this safety training have been made mandatory for the agency workers in their list of possession in work place.

b) Syllabus of two days' Safety & Behavior Based Safety (BBS) Training:

A. Syllabus of Technical (O&M), Safety and BBS Training for Substation Operators : Syllabus for this training program is available in attached **Annexure– " Safety Training – I"**

B. Syllabus of Technical (O&M) , Safety and BBS Training for Workforce of HT & LT Mobile Van :- Syllabus for this training program is available in attached **Annexure – " Safety Training – II"**

c) **Faculty:** Empanelled experienced faculties are to impart the training for technical matters and internal faculties who have undergone the "Training Of Trainers" training curriculum conducted by National Safety Council are to impart Behavior Based Safety (BBS) training.

16.3 SCHEDULE OF SAFETY TRAINING:

Schedule of all the trainings conducted by HRDD is made available in Training Calendar published every year from HRDD.

16.4 BEHAVIOR-BASED SAFETY (BBS) TRAINING:

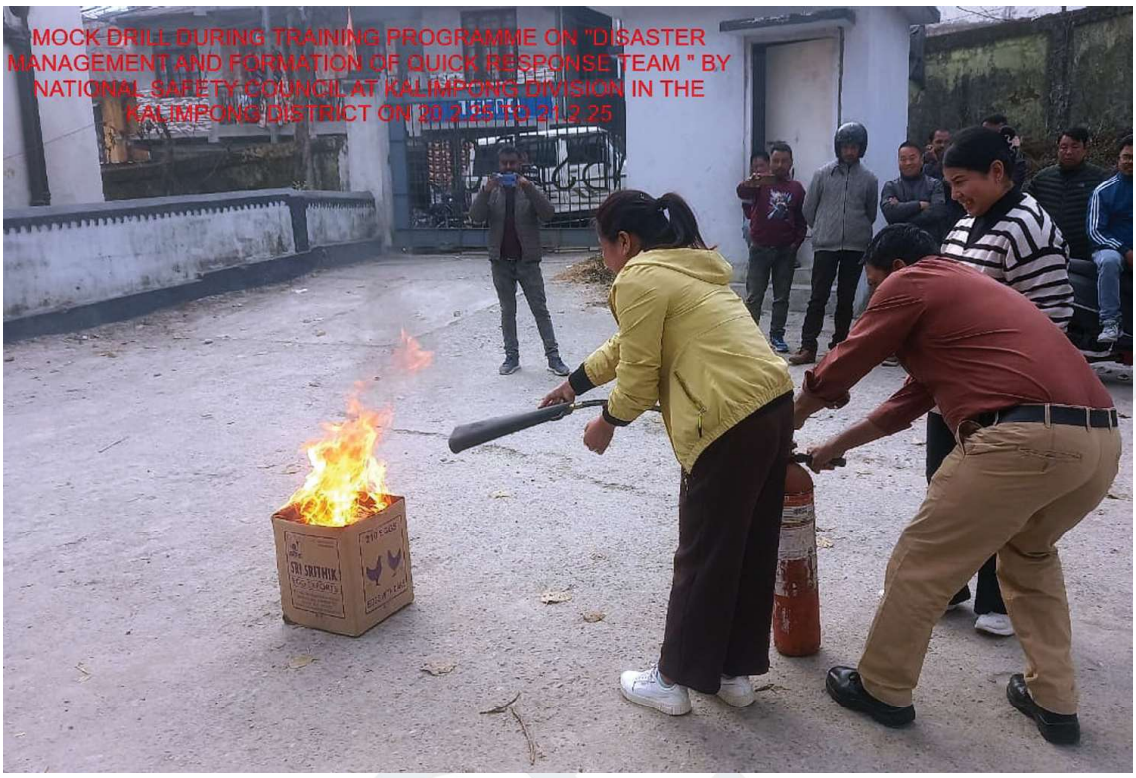
Behavior of workers influences accident probabilities. In workplaces, most injuries are caused by the improper behavior of the workers. Carelessness, overconfidence, lack of alertness, wrong predictions, erroneous interpretations, depression, frustration, uncontrolled emotions etc., along with lack of knowledge & awareness on hazards associated with their jobs, lead to untoward incidents. Behavior-based safety training (BBS) is a widely accepted method in bringing out behavioral changes in workers. Master Trainers amongst WBSEDCL employees, adequately trained in this regard, are to impart this BBS training to all concerned to bring forth maturity in workplace safety culture.

16.5 TRAINING ON DISASTER MANAGEMENT AND FORMATION OF QUICK RESPONSE TEAM:

This training is conducted for Outsourced staffs and Officials of WBSEDCL posted at Hydel Projects (North Bengal), PPSP at Purulia & other Offices situated at Hilly Region of WBSEDCL as well as Offices situated at plains in West Bengal. Course content is available in Training Calendar. WBSEDCL has formulated Disaster Mitigation Plan 2022 to manage (pre, post & during) disasters most effectively aiming at prevention of loss of life & property due to disasters.

Hands on Training of Agency Workers

Firefighting demonstration at EETI



Classroom Training Session of Agency Workers

Technical O&M & Safety with BBS Training at EETI under HRDD

"Safety Training – I"

Syllabus of Technical (O&M), Safety, and BBS Training for Substation Operators:

I. Familiarization:

i) Substation layout drawing

ii) Switchyard infrastructure:

- Power Transformer (PTR)
- Station Transformer
- Earth Isolator
- Isolator
- Lightning Arrestor (LA)
- Vacuum Circuit Breaker (VCB)
- Current Transformer (CT)
- Potential Transformer (PT)
- BUS Structure

iii) Control room infrastructure:

- Incomer, Outgoing & Bus Coupler Panel
- Current Transformer (CT)
- Potential Transformer (PT)
- Vacuum Circuit Breaker (VCB)
- Different types of Relay
- Battery Charger
- Battery
- Measuring Instruments
- Energy Meter etc

iv) Parameters/readings/status / indications related to the health of vital equipment & accessories

- Oil &Winding Temperature of PTR
- Colour of Silica Gel of Breather of PTR

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- Tap Changer of PTR
- BDV of Transformer Oil
- Specific Gravity of Battery Electrolyte
- Relay Settings
- Tripping & Closing of Breakers
- CT, PT Ratio
- Partial Discharge & Hot Spot in Equipment
- Buchholtz Relay Status
- Indicators & Alarms
- Readings of Measuring Instruments

II. Measuring:

Procedures for handling measuring instruments, measuring & recording of vital parameters of various equipment& accessories

III. Safe handling of equipment like PTR, VCB, Isolator, LA, CT, PT, Measuring Instruments, setting of Relays, acknowledgement of Alarm, Buzzer etc.

IV. Explaining Earthing System, Purpose of Earth Mat, Touch Potential, Step Potential etc.

V. Explaining Safe Working Procedures:

- Permit To Work and Field Memo – Its Application, Issue, &Withdrawal Procedure
- Safe Shutdown Procedure (Step-wise)

VI. Describing Personal Protective Equipment (PPE) & Safety Devices - Their purposes and handling procedures

VII. Hands-On Training on handling of PPE, Safety Devices, Fire Fighting at Offices, Network, Substation Yard & Control Room, First Aid, CPR, etc.

VIII. Explaining the need for Behavioral Change of erring workers.

IX. Playing Videos meant for awareness building on safe working practice &handling of electricity.

"Safety Training – II"

Syllabus of Technical (O&M), Safety and BBS Training for Workforce of HT & LT Mobile Van:

I. Familiarization:

- Single Line Diagram of Distribution Network (33 kV Incoming Feeder, Substation, 11 kV Outgoing Feeders, DTR, LT Distribution, Service Line, Consumer Connections, Switchgear Panel
- Distribution Network Infrastructure: Pole, DTR, Bare & Insulated Conductor, Insulator, Stay Set, Earth Wire, LT Kiosk, Feeder & DTR Meter, Lightning Arrestor, Isolator – Visual Identification, Purpose etc.
- Prevailing Rules& Regulations related to Pole Grouting, Span Length, Sectional Clearances, Ground Clearances, Placement of Guy Insulator in Stay Wire, Pole & DTR Earthing etc.

II. Vital Checks of Equipment during line patrolling – Pole & DTR Earthing, Leakage of Oil in DTR, Condition of Stay & Guy Insulator, Conductor Type, Span Length between Poles, Ground Clearance, Clearance from Constructions, Tree Infringement, Fuse of LT Kiosk, condition of Isolator Knives, Locking Arrangements, condition of Iron Bracket etc.

III. Measurement & Recording of Insulation Resistance, Earth Resistance, Feeder, DTR &Consumer Meter Reading etc.

IV. Safe Working Practices, like, straightening / erection of PCC and Rail Poles, jointing of Conductors, stretching of OH Bare & Insulated Conductors to improve Ground Clearance, working on Poles/Structures having multiple Feeders, etc.

V. Safe working practice to create a Safe Zone and prevent power back feed

VI. Safe working practice to be followed while working on slippery/ soft soils, on live snapped conductors, on poles having multiple feeders etc. in the monsoon season or after any cyclone/disaster, etc.

VII. Explaining Safe Working Procedures:

i) Permit To Work and Field Memo – Its Application, Issue, & Withdrawal Procedure

ii) Safe Shutdown Procedure (Step-wise)

VIII. Describing Personal Protective Equipment (PPE) & Safety Devices - Their purposes and handling procedures

IX. Hands-On Training on handling of PPE, Safety Devices, Fire Fighting at Network, First Aid, CPR etc.

X. Hand-On Training on Safe Working Procedure (Creation of Safe Zone, use of Discharge Rods, Earth Chains, Grouting Earth Spike, Removing LT Fuses in DTR etc.)

XI. Explaining the need for Behavioral Change of erring workers.

XII. Playing Videos meant for awareness building on safe working practice & handling of electricity.

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CHAPTER-XVII

STANDARD CHECK LIST

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CHAPTER-17

STANDARD CHECKLIST

17.1 SAFETY INSPECTION CHECKLIST FOR SUB-STATION / PROJECT/C.C.C./HYDRO ELECTRIC POWER GENERATION / SOLAR POWER PLANT / TESTING

Standard Checklist for Safety Inspection of 33/11 1 kV Sub-station:

Name of the sub/stn……

Zone/Division……Date of inspection……..

Name of the officer……..

Sl No	Items	Yes	No	Remarks
1.	Issue of "Permit to Work"			
2.	Duty Roster is signed by A.E			
3.	Approved shutdown format is introduced in Sub/Stn			
4.	Signature of on duty operator on attendance register with time in & out			
5.	Signature of contractor emp. on attendance register with time in &out.			
6.	I card of contractor emp. available			
7.	License of VIA &VIB cont. emp & competency certificate of contractor supervisor			
8.	Safety Poster			
9.	Signing of Log Sheet by A.E &operating personnel			
10.	Inspection Books for Sub/Stn. is updated			
11.	Availability of C P R chart at control room			

Sl No	Items	Yes	No	Remarks
12.	Availability of Emergency Telephone Number			
13.	Availability of Fire extinguisher			
14.	Discharge Rod with valid gauge of wire in HT Mobile Van			
15.	Availability of PPE's hand gloves, safety helmet, safety belt, gumboot with HT Mobile van			
16.	Availability of Ladder			
17.	Availability of Earth spike			
18.	Availability of First aid box			
19.	Housekeeping in yard, control room, battery charging room upto the mark			
20.	Feeder identification available on back side & front side of panel			
21.	Cable trench is covered in control room			
22.	Condition of Access road to switch yard			
23.	Rubber mat is placed in front of switchgear panel			
24.	Earth isolator in 33kV incoming side is fitted or not			
25.	Lighting arrester in 33 kV incoming side & HT/ LT side of power transformer is fitted or not			
26.	Good condition of Silica gel			
27.	Condition of Temperature recording meter in PTR			
28.	CT/PT are fitted in concerned places			
29.	Gravel is filled inside the switch yard			
30.	Shutdown/breakdown board available or not			
31.	Toilet/drinking water condition is as per standard or not			

Sl No	Items	Yes	No	Remarks
32.	Prescribed Shutdown format is used or not			
33.	Bushes inside yard			
34.	Access & Egress road inside the switch yard is upto the mark			
35.	Available Record of equipment & system earthing			
36.	Availability of Tripping register			
37.	Condition of Fencing of surrounding in the sub/stn.			
38.	Leakage of oil in PTR			
39.	Loose electrical connection inside control room			
40.	Proper Communication system			
41.	Authorisation letter of taking shut down to out sourced agency			
42.	illumination level at switch yard			
43.	Others if any			

Signature of the officer Designation

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Standard Checklist for Safety Inspection of Hydro-Electric Power Generation

Name of the sub/stn……

Zone/Division……Date of inspection……..

Name of the officer……..

Sl No	Items	Yes	No	Remarks
1.	Periodic cleaning & inspection of Turbine including clearance of impeller			
2.	Periodic inspection of Generating Set including cleaning schedule			
3.	Safety Inspection record of Civil Construction at different level including Turbine & Generating Sets (From GRL to Lowest level)			
4.	Safety Inspection report of the overhead Crane with lifting Tools & Tackles			
5.	Housekeeping of Control Room & Battery Room			
6.	Earth pit Inspection Report			
7.	Lightning Arrestor Inspection Report			
8.	llumination			
9.	Existence of necessary Safety Devices at tunnels, confined areas & other danger zone			
10.	Arrangement of sufficient PPE's including artificial oxygen masks to work under water			
11.	Work Permit			
12.	Shut down & Startup Procedure			
13.	Break-down Analysis Report			
14.	Safety Training & Awareness			
15.	Fire Fighting System			
16.	Mock Drill			
17.	D.G Set Maintenance Record			

Sl No	Items	Yes	No	Remarks
18.	Pollution Control Board Consent Letter of D.G Sets			
19.	SPM Level			
20.	Noise Level monitoring			
21.	Battery Maintenance Report			
22.	System diagram & Circuit Diagram			
23.	Safety seminar conducted or not			
24.	Safety committee meeting carried out			
25.	Others if any			

Signature of the officer Designation

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Standard Checklist for Safety Inspection of C.C.C. for Officers

Name of the sub/stn……

Zone/Division……Date of inspection……..

Name of the officer……..

Sl No	Items	Yes	No	Remarks
1.	Introduction of 'Field Memo'			
2.	Safety Induction form issued with LO 1			
3.	Part-VI / Workman Permit to Contractor Employee available or not			
4.	PPE's available in mobile van, No. of discharge rod, hand gloves, safety belt, gum boot, earthing chain sorting wire etc. are available or not			
5.	Availability of Ladder			
6.	Earthing Spike available for LT mobile van			
7.	Safety Poster fixed in offices			
8.	Fire Extinguisher fixed at offices			
9.	First Aid box in mobile van available or not			
10.	Housekeeping inside office			
11.	Attendance registers of Tech. / office staff			
12.	Fuse call register available or not			
13.	Safety Committee meeting carried or not			
14.	Surprise Safety Visit by station manager & officers			
15.	Safety Seminar Conducted or not			
16.	Spacer in fixed or not LT line			

Signature of the officer Designation

Standard Checklist for Safety Inspection of PROJECT for Officer

NNameofthesub/stn……

Zone/Division……Date of inspection……..

Name of the officer……..

Sl No	Items	Yes	No	Remarks
1.	Shutdown taken nearby energized line			
2.	Valid 3rd Party Certification of Lifting Tools I Tackles			
3.	Contractor project Safety Officer available or not?			
4.	Storing of Gas cylinder / Explosives & approval of Chief Inspector of Explosives			
5.	Available of PPE			
6.	Hand railings, ladder, scaffolding erected by competent person			
7.	ELCB / GFCI used in electrical circuit			
8.	'Permit to Work' available or not			
9.	Temporary electrical job is done by qualified full time licensed electricians			
10.	Accident reporting system is implemented or not			
11.	Safety meeting conducted in site			
12.	Safety Seminar conducted in site			
13.	Surprise visit done by Project Manager / Assistant Manager			
14.	Necked fuses nearby flammable material			
15.	First Aid arrangement available or not			
16.	Guarding of moving machinery			
17.	Certified Divers for under water working			
18.	Blaster license available or not			

Sl No	Items	Yes	No	Remarks
19.	Proper storing of blasting material is available or not			
20.	P.F./ Valid I.D. Card/ ESI coverage of the contractor employee is available or not			
21.	Availability of fire extinguisher at offices & flammable material storage area			

Signature of the officer Designation

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Safety Checklist for Solar Power Plant

Fire Extinguishers Safety System & Boundary fencing (Tick Mark ✓)						
Maintenance task	**Daily**	**Weekly**	**Monthly**	**Quarterly**	**Yearly**	**Remarks**
☐ Check for access to the extinguisher. If any obstruction is observed, clear it.						
☐ Inspect hose and horn. It must be in good condition and open Not choked)						
☐ Check for lock pin for its positioning. If not ok, set it.						
☐ Ensure looseness of sand in sand buckets. Stir properly if found solidified.						
☐ Check seal and pressure						
☐ Check all smoke detectors are fitted properly						
☐ Clean all Smoke detectors & Fire alarms						
☐ Check validity of Fire Extinguishers.						
☐ Visually inspect all concertina wire fences. If any man made big hole is observed fore theft.						
☐ Visually inspect bricks boundary. If any man made big hole is observed for theft.						

Safety Checklist for Electrical Equipment Testing during Commissioning, Troubleshooting & Condition Monitoring work

Sl. No.	Checklist	Status (Yes/no)	Remarks
1.	Helmet & Safety Shoes available		Required for testing personnel
2.	Discharge Rod with accessories available		Required for ensuring complete discharge of equipment after isolation from live system
3.	Equipment earthed properly		Equipment shall be earthed as per approved drawing or technical manual.
4.	Equipment made visible isolated from live system		Prior to troubleshooting of any equipment in live system the same need be isolated
5.	Isolated equipment has been discharged using discharge rod		The equipment, isolated from live system must be got completely discharged.
6.	Safe distance maintained during charging of any equipment or its condition monitoring		To avoid any injury of testing personnel
7.	Whether secondary circuit of CT in open condition		CT secondary must not be kept in open condition in any live system
8.	VCB made off and drawn out before opening of CT from cable chamber		VCB must be made off and taken out before opening of CT from Cable Chamber
9.	Insulated Mat available in Substation Control Room		This is a must to avoid electrical hazards to testing/operating personnel.

Safety Checklist for Electrical Equipment Testing during Factory Inspection or Store Testing

Sl. No	Checklist	Status (Yes/no)	Remarks
	Transformer		
1.	Helmet & Safety Shoes available		Required for testing personnel
2.	Discharge Rod with accessories available		Required for ensuring complete discharge of equipment after high voltage testing of the samples
3.	Requisite testing equipment are duly calibrated		All testing equipment shall be having valid calibration certificate
4.	All testing equipment are properly earthed		The testing equipment shall be earthed as per its technical mannual.
5.	Approved drawing & GTP of the item under testing available		
6.	Item under test is earthed properly		All test samples must be earthed as per approved drawing & technical specification.
7.	Tests samples have restricted access		Test samples must be kept isolated to avoid unauthorized access to the testing area
CT/PT:			
1.	Helmet & Safety Shoes available		Discharge rod with accessories
2.	Discharge rod with accessories available		Required for ensuring complete discharge of equipment after high voltage testing of the samples
3.	Requisite testing equipment are duly calibraed		All testing equipment shall be having valid calibration certificate
4.	All testing equipment are properly earthed		The testing equipment shall be earthe as per its technical manual
5.	Approved drawing & GTP of the item under testing available		
6.	Item under test is earthed properly		All test samples must be earthed as per approved drawing & technical specification
7.	Tests samples have restricted access		Test samples must be kept isolated to avoid unauthorised access to the testing area
8.	Whether secondary circuit of CT in open condition		CT secondary must not be kept inopen condition in any live system.

Sl. No	Checklist	Status (Yes/no)	Remarks
Switchgear			
1.	Helmet & Safety Shoes available		Required for testing personnel
2.	Discharge rod with accessories available		Required for ensuring complete discharge of equipment after high voltage testing of the samples
3.	Requisite testing equipment are duly calibrated		All testing equipment shall be having valid calibration certificate.
4.	All testing equipment are properly earthed		The testing equipment shall be earthed as per its technical manual
5.	Approved drawing & GTP of the item under testing available		
6.	Item under test is earthed properly		All test samples must be earthed as per approved drawing & technical specification
7.	Tests samples have restricted access		Test samples must be kept isolate to avoid unauthorised access to the testing area

17.2 AMENDMENT SHEET

Amendment no (if any):

Amendment		Insert		Discard	Remarks
No	Date	Chapter No	Revision No	Chapter No	

Fire Mock Drill Format	
1. Date & Time of Mock Drill	
2. Location	
3. Details of Emergency Scenario	
4. Reported by (First person informed about scenario)	
5. Response / action initiated by reporting person	
6. Description of the Mock drill (The narrative of the situation, all actions)	
7. Total persons present in the drill	
8. Time of Clearing the site after mock drill	
9. Observations (including highlight the positives of the drill)	
10. Recommendations	
Signature of Authorised person of WBSEDCL	

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ANNEXURE-I

CEA REGULATIONS, 2023 (Measures Relating to Safety and Electric Supply)

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The Central Electricity Authority
(Measures Relating To Safety And Electric Supply) Regulations, 2023

Frecuently referred regulations:

General Safety Requirement-

Regulation	Details
CEA Regulation 14 (replaces IE Rule 29)	**General safety requirements pertaining to construction, installation, protection, operation and maintenance of electric supply lines and apparatus. –** (1) All electric supply lines and apparatus shall be of sufficient rating for power, insulation and estimated fault current and of sufficient mechanical strength, for the duty cycle which they may be required to perform under the environmental conditions of installation, and shall be constructed, installed, protected, worked and maintained in such a manner as to ensure safety of human beings, animals and property. (2) Save as otherwise provided in these regulations, the relevant standards including National Electrical Code and National Building Code shall be followed to carry out the purpose of these regulations and where relevant Indian standards are not available, International standards shall be followed and in the event of any inconsistency, the provisions of these regulations shall prevail. (3) The material and apparatus used shall conform to the relevant standards. (4) All electrical equipment shall be installed above the Highest Flood Level and where such equipment is not possible to be installed above Highest Flood Level, it shall be ensured that there is no seepage or leakage or logging of water.
CEA Regulation 15 (replaces IE Rule 30)	**Service lines and apparatus on consumer's premises. –** (1) The supplier shall ensure that all electric supply lines, wires, fittings and apparatus belonging to him or under his control, up to the point of commencement of supply, which are on a consumer's premises, are in a safe-condition and in all respects fit for supplying electricity and the supplier shall take precautions to avoid danger arising on such premises from such supply lines, wires, fittings and apparatus. (2) The service lines placed by the supplier on the premises of a consumer which are underground or which are accessible shall be so insulated and protected by the supplier as to be secured under all ordinary conditions against electrical, mechanical, chemical or other injury to the insulation.

Regulation	Details
	(3) The consumer shall, as far as circumstances permit, take precautions for the safe custody of the equipment on his premises belonging to the supplier. (4) The consumer shall also ensure that the installation of the licensee under his control is kept in a safe condition.
CEA Regulation 17 (replaces IE Rule 32)	**Identification of earthed and earthed neutral conductors and position of switches and switchgear therein. –** Where the conductors include an earthed conductor of a two-wire system or an earthed neutral conductor of a multi-wire system or a conductor which is to be connected thereto, the following conditions shall be complied with, - (i) an indication of a permanent nature shall be provided by the owner of the earthed or earthed neutral conductor, or the conductor which is to be connected thereto, to enable such conductor to be distinguished from any live conductor and such indication shall be provided as per relevant standards, namely: – (a) where the earthed or earthed neutral conductor is the property of the supplier, at or near the point of commencement of supply; (b) where a conductor forming part of a consumer's system is to be connected to the supplier's earthed or earthed neutral conductor, at the point where such connection is to be made; and (c) in all other cases, at a point corresponding to the point of commencement of supply. (ii) no cut-out, link, switch or circuit breaker other than a linked switch arranged to operate simultaneously on the earthed or earthed neutral conductor and live conductors shall be inserted or remain inserted in any earthed or earthed neutral conductor of a two wire-system or in any earthed or earthed neutral conductor of a multi-wire system or in any conductor connected thereto: Provided that the above requirement shall not apply in case of a link for testing purposes, or a switch for controlling a generator or transformer.
CEA Regulation 18 (replaces IE Rule 33)	**Earthed terminal on consumer's premises**. – (1) The supplier shall provide and maintain on the consumer's premises for the consumer's use, a suitable earthed terminal in an accessible position at or near the point of commencement of supply as per relevant standards: Provided that in the case of installation of voltage exceeding 250 V the consumer shall, in addition to the aforementioned earthing arrangement, provide his own earthing system with an independent electrode and the same shall be interlinked with the earthed terminal mentioned in sub-regulation (1) through a suitable link. (2) The consumer shall take all reasonable precautions to prevent mechanical damage to the earthed terminal and its lead belonging to the supplier.

Regulation	Details
CEA Regulation 19 (replaces IE Rule 34)	**Accessibility to bare conductors**. – Where bare conductors are used in a building, the owner of such conductors shall, - (a) ensure that they are inaccessible to general public; (b) provide in readily accessible position switches for rendering them dead whenever necessary; and (c) take such other safety measures as are specified in the relevant standards.
CEA Regulation 20 (replaces IE Rule 35)	**Danger Notices**. – The owner of every installation of voltage exceeding 250 V shall affix permanently in a conspicuous position a danger notice in Hindi or English and the local language of the district, with a sign of skull and bones of a design as per relevant standards on, - (a) every motor, generator, transformer and other electrical plant and equipment together with apparatus used for controlling or regulating the same; (b) all supports of overhead lines of voltage exceeding 650 V which can be easily climbed upon without the aid of ladder or special appliances; and (c) luminous tube sign requiring supply, X-ray and similar high frequency installations of voltage exceeding 650 V but not exceeding 33 kV: *Explanation.* – For the purposes of clause (b) rails, tubular poles, wooden supports, reinforced cement concrete poles and pre stressed cement concrete poles without steps, I-sections and channels, shall be deemed as supports which cannot be easily climbed upon.
CEA Regulation 21 (replaces IE Rule 36)	**Handling of electric supply lines and apparatus. –** (1) Before any conductor or apparatus is handled, adequate precautions shall be taken, by earthing or other suitable means, to discharge electrically such conductor or apparatus, and any adjacent conductor or apparatus if there is danger therefrom, and to prevent any conductor or apparatus from being accidentally or inadvertently electrically charged when persons are working thereon shall be followed as per the relevant standards. (2) Every person who is working on an electric supply line or apparatus or both shall be provided with, – (a) personal protective equipment, tools and devices such as rubber gloves and safety footwear suitable for working voltage, safety belts for working at height, nonconductive ladder, earthing devices of appropriate class, helmet, line tester, hand lines, voltage detector and hand tools as per the relevant standards; and (b) any other device for protecting him from mechanical and electrical injury due to arc flash and such personal protective equipment, tools and devices shall conform to the relevant standards and shall always be maintained in sound working condition.

Regulation	Details
	(3) No person shall operate and undertake maintenance work on any part or whole of an electrical plant or electric supply line or apparatus. (4) Every telecommunication line on supports carrying an overhead line of voltage exceeding 650 V but not exceeding 33 kV shall, for the purpose of working thereon, be deemed to be a line of voltage exceeding 650 V: Provided that prior permission shall be taken from the concerned licensee before laying telecommunication lines on electric supports. (5) .For the safety of operating personnel, all non-current carrying metal parts of switchgear and control panels shall be properly earthed and insulating floors or mat conforming to the relevant standards, of appropriate voltage level shall be provided in front and rear of the panels where such personnel are required to carry out operation, maintenance or testing work. (6) All panels shall be painted with the description of their identification at front and at the rear.
CEA Regulation 26 (replaces IE Rule 41)	**Distinction of different circuits. –** The owner of every generating station, substation, junction box or pillar box in which there are any circuits or apparatus, whether intended for operation at different voltages or at the same voltage, shall ensure by means of indication of a permanent nature that the respective circuits are readily distinguishable from each other.
CEA Regulation 27 (replaces IE Rule 41-A)	**Distinction of the installations having more than one feed. –** The owner of every installation including substation, double pole structure, four pole structure or any other structure having more than one feed, shall ensure by means of indication of a permanent nature, that the installation is readily distinguishable from other installations.
CEA Regulation 28 (replaces IE Rule 42)	**Accidental charging. –** (1) The owners of all circuits and apparatus shall so arrange them that there shall be no danger of any part thereof becoming accidentally charged to any voltage beyond the limits of voltage for which they are intended. (2) Where alternating current and direct current circuits are installed on the same box or support, they shall be so arranged and protected that they shall not come into contact with each other.

Regulation	Details
CEA Regulation 29 (replaces IE Rule 43)	**Provisions applicable to protective equipment. –** (1) Fire buckets filled with clean dry sand and ready for immediate use for extinguishing fires, in addition to fire extinguishers suitable for dealing with fires, shall be conspicuously marked and kept in all generating stations, enclosed substations and enclosed switching-stations in convenient location. (2) Appropriate type of fire extinguisher conforming to the relevant standards, shall be installed, maintained, periodically inspected and tested as per the relevant standards for extinguishing and controlling fire and record of such tests shall be maintained. (3) Sufficient number of first-aid boxes or cupboards conspicuously marked and equipped with such contents as the State Government may specify or as per the relevant standards, shall be provided and maintained at appropriate locations in every generating station, enclosed substation, enclosed switching station and in vehicles used for maintenance of lines so as to be readily available and accessible at all the times and all such boxes and cupboards shall, except in the case of unattended substations and switching stations, be kept under the charge of responsible persons who are trained in first-aid treatment and one of such persons shall be available during working hours
CEA Regulation 30 (replaces IE Rule 44)	**Display of instructions for resuscitation of persons suffering from electric shock. –** (1) Instructions, in English or Hindi and the local language of the District and where Hindi is the local language, in English and Hindi for the resuscitation of persons suffering from electric shock, shall be affixed by the owner in a conspicuous place in every generating station, enclosed substation, enclosed switching station, mines and in every factory as defined in the Factory Act, 1948 (63 of 1952) in which electricity is used and in such other premises where electricity is used as the Electrical Inspector may, by notice in writing served on the owner, direct. (2) The owner of every generating station, enclosed substation, enclosed switching station and every factory or other premises to which these regulations apply, shall ensure that all designated persons or persons engaged or appointed to operate and maintain electrical plants or transmission or distribution systems are acquainted with and are competent to apply the instructions referred to in sub-regulation (1).

Regulation	Details
CEA Regulation 31 (replaces IE Rule 45)	**Precautions to be adopted by consumers, owners, occupiers, electrical contractors, electrical workmen and suppliers. –** (1) No electrical installation work, including additions, alterations, repairs and adjustments to existing installations, except such replacement of lamps, fans, fuses, switches, domestic appliances of voltage not exceeding 250 V and fittings as in no way alters its capacity or character, shall be carried out upon the premises of or on behalf of any consumer, supplier, owner or occupier for the purpose of supply to such consumer, supplier, owner or occupier except by an electrical contractor licenced in this behalf by the State Government and on its behalf under the direct supervision of a person holding a certificate of competency and by a person holding a permit issued or recognised by the State Government: (2) No electrical installation work which has been carried out in contravention of sub-regulation (1) shall either be energised or connected to the works of any supplier
CEA Regulation 32 (replaces IE Rule 46)	**Periodic inspection and testing of installations. –** (1) The periodic inspection and testing of installation of voltage above the notified voltage belonging to the owner or supplier or consumer, as the case may be, shall be carried out by the Electrical Inspector: Provided that the electrical installation below or equal to the notified voltage shall be self-certified by the owner or supplier or consumer, as the case may be. (2) The periodicity of electrical inspection by the Electrical Inspector or the self-certification by the supplier, owner or consumer shall be as directed by the Appropriate Government: Provided that the periodicity of electrical inspection and self-certification shall not exceed five years:
CEA Regulation 33 (replaces IE Rule 47)	**Testing of consumer's installation. –** (1) Upon receipt of an application for a new or additional supply of electricity and before commencement of supply or recommencement of supply after the supply has been disconnected for a period of six months, the supplier shall either test the installation himself or accept the test results submitted by the consumer when same has been duly signed by the licenced electrical contractor: Provided that in case of voltage level equal to or below the notified voltage, Chartered Electrical Safety Engineer can also test the installation on request of owner. (2) The testing and verifications shall be carried out as per relevant standards.

Regulation	Details
	(3) The testing equipment shall be calibrated by a Government authorised or National Accreditation Board for Testing and Calibration Laboratories accredited laboratory at periodical interval as per the periodicity specified by them. (4) The supplier shall maintain a record of test results obtained at each supply point to a consumer, as per the forms provided in Schedule III. (5) If as a result of such inspection and test, the supplier is satisfied that the installation is likely to be dangerous, he shall serve on the applicant a notice in writing requiring him to make such modifications as are necessary to render the installation safe and may refuse to connect or reconnect the supply until the required modifications have been completed.

General conditions relating to supply and use of electricity -

Regulation	Details
CEA Regulation 35 (replaces IE Rule 48)	**Precautions against leakage before connection. –** (1) The supplier shall not connect its works with the apparatus in the premises of any applicant seeking supply unless the supplier is satisfied that at the time of making the connection cause a leakage from that installation or apparatus of a magnitude detrimental to safety which shall be checked by measuring the installation's or apparatus' insulation resistance. (2) If the supplier declines to make a connection under the provisions of sub-regulation (1) the supplier shall convey to the applicant the reasons thereof, in writing for so declining.
CEA Regulation 36 (replaces IE Rule 49)	**Leakage on consumer's premises. –** (1) If the Electrical Inspector or the supplier has reasons to believe that there is leakage in the system of a consumer which is likely to affect injuriously the use of electricity by the consumer or by other persons, or which is likely to cause danger, he may give notice to the consumer in writing to inspect and test the consumer's installation. (2) If after such notice, the consumer fails to provide access to its installation for inspection and testing, or an insulation resistance of the consumer's installation is so low as to prevent safe use of electricity, the supplier may, and if directed so by the Electrical Inspector shall discontinue the supply of electricity to the installation but only after giving to the consumer forty eight hours notice in writing for disconnection of supply and shall not recommence the supply until he or the Electrical Inspector is satisfied that the cause of the leakage has been removed.

<table>
<tr><th>Regulation</th><th>Details</th></tr>
<tr><td>CEA Regulation 37

(replaces

IE Rule 50)</td><td>
Supply and use of electricity. –

(1) The electricity shall not be supplied, transformed, converted, inverted or used or continued to be supplied, transformed, converted, inverted or used unless the conditions provided in sub- regulations (2) to (8) are complied with.
(2) The following controls of requisite capacity to carry and break the current shall be installed as near as possible after the point of commencement of supply so as to be readily accessible and capable of completely isolating the supply to the installation, such equipment being in addition to any control switch installed for controlling individual circuits or apparatus, namely: –
<table>
<tr><th>Supplied at voltage</th><th>Control</th></tr>
<tr><td>Below 11 kV.</td><td>Switch fuse unit or a circuit breaker by consumers.</td></tr>
<tr><td>11 kV and above.</td><td>A circuit breaker by consumers.</td></tr>
</table>
(3) In case of every transformer the following shall be provided, namely:–
(i) on primary side of transformer, a linked switch with fuse or gang operated air break switch with fuse or circuit breaker of adequate capacity:
Provided that the linked switch with fuse on the primary side of the transformer may be of such capacity as to carry the full load current and to break only the magnetising current of the transformer:
Provided further that for transformer having capacity of 1000 kVA and above, a circuit breaker shall be provided:
Provided also that the linked switch with fuse or gang operated air break switch with fuse or circuit breaker on the primary side of the transformer shall not be required for the unit auxiliary transformer and generator transformer;
(ii) on the secondary side of all transformers a circuit breaker of adequate rating shall be installed:
Provided that for supplier's transformers of capacity below 1000 kVA, a linked switch with fuse or circuit breaker of adequate rating shall be installed on secondary side.
</td></tr>
</table>

Regulation	Details
	(4) Except in the case of composite control gear designed as a unit each distinct circuit is to be protected against excess energy by means of a suitable fuse link or a circuit breaker of adequate breaking capacity, suitably located and so constructed as to prevent danger from overheating, arcing or scattering of hot metal when it comes into operation and to permit for ready renewal of the fuse link without danger. (5) The supply of electricity to each motor or a group of motors or other apparatus meant for operating one particular machine shall be controlled by a suitable linked switch or a circuit breaker or an emergency tripping device with manual reset of requisite capacity placed in such a position as to be adjacent to the motor or a group of motors or other apparatus readily accessible to and easily operated by the person in charge and so connected in the circuit that by its means all supply of electricity can be cut off from the motor or group of motors or apparatus from any regulating switch, resistance of other device associated therewith. (6) All insulating materials shall be as per their application and their mechanical strength shall be sufficient for the purpose so as to maintain adequately their insulating property under all working conditions in respect of temperature, moisture, salinity and pollution. (7) Adequate precautions shall be taken to ensure that no live parts are exposed as to cause danger. (8) Every consumer shall use all reasonable means to ensure that where electricity is supplied by the supplier, no person other than the supplier shall interfere with service lines and apparatus placed by the supplier on the premises of the consumer.
CEA Regulation 38 (replaces IE Rule 50-A)	**Provisions for supply and use of electricity in multi-storied building more than fifteen metre in height. –** (1) The connected load and voltage of supply above which inspection is to be carried out by an Electrical Inspector for a multi-storeyed building of more than fifteen metre height shall be notified by the Appropriate Government. (2) Before making an application for commencement of supply or recommencement of supply after an installation has been disconnected for a period of six months or more, the owner or occupier of a multi-storeyed building shall give not less than thirty days notice in writing to the Electrical Inspector specifying therein the particulars of installation and the supply of electricity shall not be commenced or recommenced within this period, without the approval in writing of the Electrical Inspector.

Regulation	Details
	(3) The following safety measures shall be provided in the multi-storeyed buildings of more than fifteen metre height and other premises such as airports, hospitals, hotels, places of entertainment, places of worship, cultural centers, stadium, academic buildings, test labs, industrial installations, installation with explosive or flammable material, railway or metro stations and other public buildings, namely: – (i) the supplier or owner of the installation shall provide at the point of commencement of supply a suitable isolating device with cut-out or breaker to operate on all phases except neutral in the three-phase, four-wire circuit and fixed in a conspicuous position at not more than 1.70 metre above the ground so as to completely isolate the supply to the building in case of emergency; (ii) the owner or occupier of a multi-storeyed building shall ensure that electrical installations and works inside the building are carried out and maintained in such a manner as to prevent danger due to shock and fire hazards, and the installation is carried out as per the relevant standards; (iii) no other service pipes and cables shall be taken through the ducts provided for laying of power cables and all ducts provided for power cables and other services shall be provided with fire barrier at each floor crossing; (iv) the Fire Retardant Low Smoke and Low Halogen power cables shall be used in building of more than fifteen metre height as per relevant standards:Provided that Halogen Free Flame Retardant power cables as per the relevant standards shall be used in airports, hospitals and hotels irrespective of height; (v) distribution of electricity to the floors shall be done using bus bar trunking system; (vi) lightning protection of the building shall be as per the relevant standards; (vii) verification of electrical wiring of the building shall be carried out as per the relevant standards; and (viii) electricity meter shall not be installed in the passage of staircase.

Regulation	Details
CEA Regulation 39 (replaces IE Rule 51)	**Conditions applicable to installations of voltage exceeding 250 Volts. –** The following conditions shall be complied with where electricity of voltage above 250 V is supplied, converted, transformed or used, namely: **–** (i) all conductors, other than those of overhead lines, shall be completely enclosed in mechanically strong metal casing or metallic covering which is electrically and mechanically continuous and adequately protected against mechanical damage unless the said conductors are accessible only to a designated person or are installed and protected so as to prevent danger: Provided that non-metallic conduits conforming to the relevant standards may be used for installations of voltage not exceeding 650 V; (ii) all metal works, enclosing, supporting or associated with the installation, other than that designed to serve as a conductor shall be connected with an earthing system as per relevant standards and the provisions of regulation 43; (iii) every switch board shall comply with the following, namely: – (a) a clear space of not less than one metre in width shall be provided in front of the switchboard; (b) if there are any attachments or bare connections at the back of the switchboard, the space, if any, behind the switchboard shall be either less than twenty centimetre or more than seventy five centimetre in width, measured from the farthest protruding part of any attachment or conductor; and (c) if the space behind the switchboard exceeds seventy five centimetre in width, there shall be a passage way from either end of the switchboard, clear to a height of 1.8 metre; (iv) in case of installations provided in premises where inflammable materials including gases and chemicals are produced, handled or stored, the electrical installations, equipment and apparatus shall comply with the requirements of flame proof, dust tight, totally enclosed or any other suitable type of electrical fittings depending upon the hazardous zones as per the relevant standards; (v) where an application has been made to a supplier for supply of electricity to any installation, the supplier shall not commence the supply or where the supply has been discontinued for a period of six months or more, recommence the supply unless the consumer has complied with the relevant provisions in these regulations;

Regulation	Details
	(vi) where a supplier proposes to supply or use electricity at or to recommence supply of voltage exceeding 250 V but not exceeding 650 V after it has been discontinued for a period of six months, he shall, before connecting or reconnecting the supply, give notice in writing of such intention to the Electrical Inspector; and (vii) if at any time after connecting the supply, the supplier is satisfied that any provision of these regulations have not been complied with, the supplier shall give notice of the same in writing to the consumer and the Electrical Inspector, specifying the defects and to rectify such defects in a reasonable time: Provided that if the consumer fails to rectify such defects the supplier may discontinue the supply after giving the consumer a reasonable opportunity of being heard and recording reasons in writing and the supply shall be discontinued only on written orders of an officer duly notified by the supplier in this behalf and shall be restored with all possible speed after such defects are rectified by the consumer to the satisfaction of the supplier.
CEA Regulation 40 (replaces IE Rule 52)	**Appeal to Electrical Inspector in regard to defects. –** (1) If any applicant for a supply or a consumer is aggrieved by the action of the supplier in declining to commence, to continue or to recommence the supply of electricity to his premises on the grounds that the installation is defective or is likely to be dangerous, he may appeal to the Electrical Inspector to test the installation and the supplier shall not, if the Electrical Inspector intimates that the installation is free from the defect or danger complained of, refuse supply to the consumer on the grounds aforesaid, and shall, within twenty four hours after the receipt of such intimation from the Electrical Inspector, commence, continue or recommence the supply of electricity. (2) Any test for which application has been made under sub-regulation (1), shall be carried out within seven days after the receipt of such application.
CEA Regulation 41 (replaces IE Rule 59)	**Precautions against failure of supply and notice of failures. –** (1) The layout of the electric supply lines of the supplier for the supply of electricity throughout his area of supply shall under normal working conditions be sectionalised and so arranged, and provided with switchgear or circuit-breakers, so located, as to restrict within reasonable limits the extent of the portion of the system affected by any failure of supply.

Regulation	Details
	(1) The supplier shall take all reasonable precautions to avoid any accidental interruptions of supply, and also to avoid danger to the public or to any employee or designated person when engaged on any operation during and in connection with the installation, extension, replacement, repair and maintenance of any works. (2) The supplier shall send to the Electrical Inspector a notice of failure of supply of such kind as the Electrical Inspector may from time to time require to be notified to him, and such notice shall be sent by the earliest mode of communication after the failure occurs or after the failure becomes known to the supplier and shall be in the Form given in Schedule IV. (3) For the purpose of testing or for any other purpose connected with the efficient working of the supplier's installations, the supply of electricity may be discontinued by the supplier for such period as may be necessary, subject to not less than twenty four hours notice being given by the supplier to all consumers likely to be affected by such discontinuance: Provided that no such notice shall be given in cases of emergency

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Safety provisions for electrical installations and apparatus of voltage not exceeding 650 V

Regulation	Details
CEA Regulation 42 (replaces IE Rule 60)	**Test of insulation resistance. –** Where any electric supply line for use at voltages not exceeding 650 V has been disconnected from a system for the purpose of addition, alteration or repair, such electric supply line shall not be reconnected to the system until the supplier or the owner has carried out the test.
CEA Regulation 43 (replaces IE Rule 61)	**Connection with earth. –** The following conditions shall apply to the connection with earth of systems at voltage exceeding 50 V but not exceeding 650 V, namely: **–** (i) neutral conductor of a three phase, four-wire system and the middle conductor of a two-phase, three-wire system shall be earthed as per the relevant standards; (ii) neutral conductor shall also be earthed at one or more points along the distribution system or service line in addition to any connection with earth which shall be at the consumer's premises; (iii) in the case of a system comprising electric supply lines having concentric cables, the external conductor or armour of such cables shall be earthed by two separate and distinct connections with earthing system; (iv) in a direct current system, earthing and safety measures shall be as per the relevant standards; (v) every building shall have protective equipotential bonding by interconnecting the exposed and extraneous conductive parts as per the relevant standards; (vi) the alternating current systems which are connected with the earth as provided in this regulation shall be electrically interconnected: (vii) All electricity consuming apparatus of voltage exceeding 250 V but not exceeding 650 V shall be earthed by two separate and distinct connections with earth by the owner as specified in the relevant standards. (viii) The neutral point of every generator and transformer shall be earthed by connecting it to the earthing system not by less than two separate and distinct connections

Regulation	Details
CEA Regulation 44 (Replaces IE Rule 61-A)	**Residual Current Device. –** The use of electricity to electrical installation, shall be controlled by a residual current device to disconnect the supply having rated residual current and duration as per the relevant standards: Provided that in domestic installation, residual current device having residual operating current not exceeding 30 milliampere shall be used:

Safety provisions for electrical installations and apparatus of voltage exceeding 650 V

Regulation	Details
CEA Regulation 45 (replaces IE Rule 63)	**Approval by the Electrical Inspector and self-certification**. – Every electrical installation of notified voltage and below shall be inspected, tested and self-certified by the owner or supplier or consumer, as the case may be, of the installation before commencement of supply or recommencement after shutdown for six months or more for ensuring observance of safety measures specified under these regulations and such owner or supplier or consumer, as the case may be, shall submit the report of self-certification to the Electrical Inspector in the forms as provided under Schedule II of these regulations: Provided that the self-certified electrical installation shall be considered fit for the commencement of supply or recommencement after shutdown for six months only after the report of self-certification is duly received by the office of Electrical Inspector and if not acknowledged by the Electrical Inspector within three working days, it shall be deemed to be received: Provided further that the owner or supplier or consumer, as the case may be, has the option to get his installation inspected and tested by the Electrical Inspector of the Appropriate Government;
CEA Regulation 46 (replaces IE Rule 64)	**Use of electricity at voltage exceeding 650 V**. – (1) The Electrical Inspector where the supply voltage exceeds the notified voltage shall not authorise the supplier to commence supply or recommence the supply, where the supply has been discontinued for a period of six months or more, or the supplier, where the supply voltage is equal to or below the notified voltage but exceeds 650 V, shall not commence supply or recommence the supply where supply has been discontinued for a period of six months or more, to any consumer unless, – (a) all conductors and apparatus situated on the premises of the consumer are so placed as to be inaccessible except to the designated person;

Regulation	Details
	(b) the consumer has provided and agreed to maintain a separate building or a locked weather proof and fire proof enclosure of agreed design and location, to which the supplier at all times shall have access for the purpose of housing his apparatus and metering equipment, or where the provision for a separate building or enclosure is impracticable, the consumer has segregated the aforesaid apparatus of the supplier from any other part of his own apparatus: Provided that the segregation shall be made by the fire walls, if the Electrical Inspector considers it to be necessary: Provided further that in the case of an outdoor installation the consumer shall suitably segregate the aforesaid apparatus belonging to the supplier from his own; (c) all pole type substations are constructed and maintained in accordance with regulation 52. (2) Where electricity at voltage exceeding 650 V is supplied, converted, transformed or used, the owner shall, – (i) maintain safety clearances for electrical apparatus as per relevant standards specification so that sufficient space is available for easy operation and maintenance without any hazard to the operating and maintenance personnel working near the equipment and for ensuring adequate ventilation: Provided that in case of mines, the safety clearances for electrical apparatus to be as per relevant mining regulations; (ii) not allow any encroachment below such installation: Provided that where the Electrical Inspector comes across any such encroachment, he shall direct the owner to remove such encroachments; (iii) maintain minimum safety working clearances specified in Schedule V for the bare conductors or live parts of any apparatus in outdoor substations excluding overhead lines of installations of voltage exceeding 650 V; (iv) ensure that the live parts of all apparatus within the reach from any position in which a person may require to be, are suitably protected to prevent danger; (v) ensure that where the transformer is used, suitable provision shall be made, either by connecting with earth, a point of the circuit at the lower voltage or otherwise, to guard against danger by reason of the said circuit becoming accidentally charged above its normal voltage by leakage from or contact with the circuit at the higher voltage;

Regulation	Details
	(vi) not install a substation or a switching station with apparatus having more than 2000 litre of oil in the basement where proper oil draining arrangement cannot be provided; (vii) undertake the following measures, where a substation or a switching station with oil filled apparatus, such as transformer, static condenser, switchgear or oil circuit breaker having more than 2000 litre of oil is installed, whether indoor or outdoors,- (a) the separation wall or fire barrier walls of thickness and dimensions as specified in the relevant standards shall be provided between the apparatuses and between the apparatus and adjacent building if building wall adjacent to the apparatuses is not rated for four hours fire withstand rating; (b) provisions shall be made for suitable oil soakpit and where use of more than 9000 litre of oil in any one oil tank, receptacle or chamber is involved, provision shall be made for the draining away or removal of any oil which may leak or escape from the tank, receptacle or chamber containing the same, and special precautions shall be taken to prevent the spread of any fire resulting from the ignition of the oil from any cause and adequate provision shall be made for extinguishing any fire which may occur; (c) spare oil shall not be stored in the vicinity of any oil filled equipment in any such substation or switching station; and (d) all the transformers and switchgears shall be maintained in accordance with the maintenance schedules prepared in accordance with the relevant standards; (viii) without prejudice to the above measures, undertake adequate fire detection and protection arrangement for quenching the fire of the apparatus; (ix) ensure that every transformer of 10 MVA or reactor of 10 MVAR and above rating shall be provided with automatic fire fighting system as per relevant standards; (x) undertake the following measures, where it is necessary to locate the substation, or switching station in the basement, namely: – (a) the transformer room be in the first basement at the periphery; (b) the direct access to the transformer room be provided from outside and the surrounding walls of four hours fire withstand rating be provided as per relevant standards;

Regulation	Details
	(c) the entrances to the transformer room be provided with fire resistant doors of two hour fire rating and the door shall always be kept closed and a notice of this effect be affixed on outer side of the door; (d) a curb of a suitable height be provided at the entrance in order to prevent the flow of oil from a ruptured transformer into other parts of the basement; (e) the cables to primary side and secondary side have sealing at all floors and wall opening of atleast two hours fire withstand rating; and (f) Fire Retardant Low Smoke Low Halogen cable as per relevant standards be used; (xi) ensure that oil filled transformers installed indoors in other than residential or commercial buildings are placed on the ground floor or not below the first basement; (xii) ensure that only dry type transformer shall be used inside the residential and commercial buildings; (xiii) ensure that cable trenches inside the substations and switching stations containing cables are filled with sand, pebbles or similar non-inflammable materials or completely covered with non-inflammable slabs; and (xiv) ensure that unless the conditions are such that all the conductors and apparatus may be made dead at the same time for the purpose of cleaning or for other work, the said conductors and apparatus shall be so arranged that these may be made dead in sections, and that work on any such section may be carried on by the person designated or appointed or engaged or permitted under these regulations without danger. (3) The minimum clearances specified in Schedule VI shall be maintained for bare conductors or live parts of any high voltage direct current apparatus in outdoor substations, excluding high voltage direct current overhead lines. (4) There shall not be tapping of another transmission line from the main line for 66 kV and above class of lines: Provided that during natural calamities, tapping may be allowed to ensure emergency power supply to affected areas till normalcy is restored.

Regulation	Details
CEA Regulation 47 (replaces IE Rule 64-A)	**Inter-locks and protection for use of electricity at voltage exceeding 650 V.** (1) The owner shall ensure the following, namely: – (i) isolators and the controlling circuit breakers shall be inter-locked so that the isolators cannot be operated unless the corresponding breaker is in open position; (ii) isolators and the corresponding earthing switches shall be inter-locked so that no earthing switch can be closed unless and until the corresponding isolator is in open position; (iii) where two or more supplies are not intended to be operated in parallel, the respective circuit breakers or linked switches controlling the supplies shall be inter-locked to prevent possibility of any inadvertent paralleling or backfeed; (iv) when two or more transformers are operated in parallel, the system shall be so arranged as to trip the secondary breaker of the transformer in case the primary breaker of that transformer trips; (v) all gates or doors which provide access to live parts of an installation shall be inter-locked in such a way that these cannot be opened unless the live parts are made dead and proper discharging and earthing of these parts shall be ensured before any person comes in close proximity of such parts; and (vi) where two or more generators operate in parallel and neutral switching is adopted, inter-lock shall be provided to ensure that the generator breaker cannot be closed unless one of the neutrals is connected to the earthing system. (2) The following protection shall be provided in all systems and circuits to automatically disconnect the supply under abnormal conditions, namely: – (i) over current protection to disconnect the supply automatically if the rated current of the equipment, cable or supply line is exceeded for a time which the equipment, cable or supply line is not designed to withstand; (ii) earth fault or earth leakage protection to disconnect the supply automatically, if the earth fault current exceeds the limit of current for keeping the contact potential within the reasonable values; (iii) buchholz relay, pressure relief device and winding and oil temperature protection with alarm and trip contacts shall be provided on all transformers of ratings 1000 kVA and above; (iv) transformers of capacity 10 MVA and above shall be protected against incipient faults by differential protection;

Regulation	Details
	(v) all generators with rating of 100 kVA and above shall be protected against earth fault or leakage; (vi) all generators of rating 1000 kVA and above shall be protected against faults within the generator winding using restricted earth fault protection or differential protection or by both; (vii) high speed bus bar differential protection along with local breaker back up protection shall be commissioned and shall always be available at all 132 kV and above voltage substations and switching stations and generating stations connected with the grid: Provided that in respect of existing 132 kV substations and switching stations having more than one incoming feeders, the high speed bus bar differential protection along with local breaker back up protection, shall be commissioned and shall always be available; and (viii) in addition to above, all electrical protection system for generating stations, substations and transmission lines shall be as per the regulations notified by the Authority under clause (e) of sub-section (2) of section 177 of the Act.
CEA Regulation 48 (replaces IE Rule 65)	**Testing, Operation and Maintenance. –** (1) Before the approval is accorded by the Electrical Inspector under regulation 45, the manufacturer's test certificates shall, if required, be produced for all the type, acceptance and routine tests as required under the relevant standards. (2) No new apparatus, cable or supply line of voltage exceeding 650 V shall be commissioned unless such apparatus, cable or supply line are subjected to site tests as per relevant standards. (3) No apparatus, cable or supply line of voltage exceeding 650 V which has been kept disconnected for a period of six months or more from the system for alterations or repair, shall be connected to the system until such apparatus, cable or supply line are subjected to the site tests as per relevant standards. (4) Notwithstanding the provisions of this regulation, the Electrical Inspector may require certain tests to be carried out before or after charging the installations. (5) All apparatus, cables and supply lines shall be maintained in healthy conditions and tests shall be carried out periodically as per the relevant standards. (6) Records of all tests, trippings, maintenance works and repairs of all apparatus, cables and supply lines shall be duly kept in such a way that these records can be compared with the past records.

Regulation	Details
	(7) It shall be the responsibility of the owner of all installations of voltage exceeding 650 V to maintain and operate the installations in a condition free from danger and as recommended by the manufacturer or by the relevant standards. (8) Failures of any 220 kV and above voltage level transformer, reactor and transmission line towers shall be reported by the owner of electrical installation, within forty eight hours of the occurrence of the failure, to the Authority and the reasons for failure and measures to be taken to avoid recurrence of failure shall be sent to the Authority within one month of the occurrence in the forms provided in Schedule VII: Provided that in case of mines and oil-fields, the failure of 10 MVA or above transformers shall be reported to Electrical Inspector of mines.
CEA Regulation 49 (replaces IE Rule 66).	**Precautions to be taken against excess leakage in case of metal sheathed electric supply lines**. – The following precautions shall be taken in case of electric supply lines other than overhead lines, for use at voltage exceeding 650 V, namely: – (i) the conductors of the cable except the cable with thermoplastic or cross linked polyethylene, insulation without any metallic screen or armour shall be enclosed in metal sheathing which shall be electrically continuous and connected with earth, and the conductivity of the metal sheathing shall be maintained and reasonable precautions shall be taken where necessary to avoid corrosion of the sheathing; (ii) the resistance of the earth connection with metallic sheath shall be kept low enough to permit the controlling circuit breaker or cut-out to operate in the event of any failure of insulation between the metallic sheath and the conductor. *Explanation.* – For the purposes of this regulation; (a) in the case of thermoplastic insulated and sheathed cables with metallic armour, the metallic wire or tape armour shall be considered as metal sheathing; and (b) where an electric supply line as aforesaid has concentric cables and the external conductor is insulated from an outer metal sheathing and connected with earth, the external conductor may be regarded as the metal sheathing for the purposes of this regulation provided that the foregoing provisions as to conductivity are complied with. 50. Connection with earth for apparatus exceeding 650 V. – (1) The entire switchyard or substation equipment and buildings including all non-current carrying metal parts associated with an installation shall be effectively earthed to an earthing system or mat which shall, –

Regulation	Details
	(i) limit the touch and step potential to tolerable values as per relevant standards; (ii) limit the earth potential rise to tolerable values as per relevant standards, so as to prevent danger due to transfer of potential through ground, earth wires, cable sheath, fences, pipe lines or other such equipment; and (iii) maintain the resistance of the earth connection to such a value as to make operation of the protective device effective. (2) In the case of star connected system with earthed neutrals or delta connected system with earthed artificial neutral point,- (i) the neutral point of every generator and transformer shall be earthed by connecting it to the earthing system not by less than two separate and distinct connections: Provided that the neutral point of a generator may be connected to the earthing system through an impedance to limit the fault current: Provided further that in the case of multi-machine systems, neutral switching may be resorted to, for limiting the injurious effect of harmonic current circulation in the system; (ii) the generator or transformer neutral shall be earthed through a suitable impedance where an appreciable harmonic current flowing in the neutral connection causes interference with the communication circuits; and (iii) in case of the delta connected system, the neutral point shall be obtained by the insertion of a earthing transformer and current limiting resistance or impedance wherever considered necessary at the commencement of such a system. (3) In case of generating stations, substations and other installations of voltage exceeding 33 kV, the system neutral earthing and protective frame earthing may be, if system design so warrants, integrated into common earthing grid provided the resistance to earth of combined mat does not cause the step and touch potential to exceed the values as per relevant standards. (4) Single phase systems of voltage exceeding 650 V shall be effectively earthed. (5) In the case of a system comprising electric supply lines having concentric cables, the external conductor shall be connected with the earth.

Regulation	Details
	(6) Where a supplier proposes to connect with earth an existing system for use at voltage exceeding 650 V which has not hitherto been so connected with earth, he shall give not less than fourteen days notice in writing together with particulars of the proposed connection with earth to the telegraph authority established under the Indian Telegraph Act, 1885 (13 of 1885). (7) Where the earthing lead and earth connection are used only in connection with earthing guards laid under overhead lines of voltage exceeding 650 V but not exceeding 33 kV where they cross a telecommunication line or a railway line, and where such lines are equipped with earth leakage protective device, the earth resistance shall not exceed twenty five ohms and the project authorities shall obtain no objection certificate from Railway Authorities and Power and Telecommunication Coordination Committee before energisation of the facilities. (8) Every earthing system belonging to either the supplier or the consumer shall be tested for its resistance to earth on a dry day during dry season not less than once in a year and records of such tests shall be maintained and produced, if so required, before the Electrical Inspector.
CEA Regulation 50 (replaces IE Rule 67)	**Connection with earth for apparatus exceeding 650 V**. – (1) The entire switchyard or substation equipment and buildings including all non-current carrying metal parts associated with an installation shall be effectively earthed to an earthing system or mat which shall,– (i) limit the touch and step potential to tolerable values as per relevant standards; (ii) limit the earth potential rise to tolerable values as per relevant standards, so as to prevent danger due to transfer of potential through ground, earth wires, cable sheath, fences, pipe lines or other such equipment; and (iii) maintain the resistance of the earth connection to such a value as to make operation of the protective device effective. (2) In the case of star connected system with earthed neutrals or delta connected system with earthed artificial neutral point,- (i) the neutral point of every generator and transformer shall be earthed by connecting it to the earthing system not by less than two separate and distinct connections: Provided that the neutral point of a generator may be connected to the earthing system through an impedance to limit the fault current:

Regulation	Details
	Provided further that in the case of multi-machine systems, neutral switching may be resorted to, for limiting the injurious effect of harmonic current circulation in the system; (ii) the generator or transformer neutral shall be earthed through a suitable impedance where an appreciable harmonic current flowing in the neutral connection causes interference with the communication circuits; and (iii) in case of the delta connected system, the neutral point shall be obtained by the insertion of a earthing transformer and current limiting resistance or impedance wherever considered necessary at the commencement of such a system. (3) In case of generating stations, substations and other installations of voltage exceeding 33 kV, the system neutral earthing and protective frame earthing may be, if system design so warrants, integrated into common earthing grid provided the resistance to earth of combined mat does not cause the step and touch potential to exceed the values as per relevant standards. (4) Single phase systems of voltage exceeding 650 V shall be effectively earthed. (5) In the case of a system comprising electric supply lines having concentric cables, the external conductor shall be connected with the earth. (6) Where a supplier proposes to connect with earth an existing system for use at voltage exceeding 650 V which has not hitherto been so connected with earth, he shall give not less than fourteen days notice in writing together with particulars of the proposed connection with earth to the telegraph authority established under the Indian Telegraph Act, 1885 (13 of 1885). (7) Where the earthing lead and earth connection are used only in connection with earthing guards laid under overhead lines of voltage exceeding 650 V but not exceeding 33 kV where they cross a telecommunication line or a railway line, and where such lines are equipped with earth leakage protective device, the earth resistance shall not exceed twenty five ohms and the project authorities shall obtain no objection certificate from Railway Authorities and Power and Telecommunication Coordination Committee before energisation of the facilities. (8) Every earthing system belonging to either the supplier or the consumer shall be tested for its resistance to earth on a dry day during dry season not less than once in a year and records of such tests shall be maintained and produced, if so required, before the Electrical Inspector

Regulation	Details
CEA Regulation 52 (replaces IE Rule 69)	**Pole type substations. –** Where platform type construction is used for a pole type substation and sufficient space for a person to stand on the platform is provided, a proper hand rail shall be built around the platform and if the hand rail is of metal, it shall be connected with the earth: Provided that in the case of pole type substation on wooden supports and wooden platform, the metal hand rail shall not be connected with the earth.

Safety requirements for overhead lines and underground cables

Regulation	Details
CEA Regulation 57 (replaces IE Rule 74)	**Material and strength. –** (1) All conductors of overhead lines other than those provided in regulation 70 shall have a breaking strength of not less than 350 kgf. (2) Where the voltage does not exceed 250 V and the span is less than fifteen metre and is drawn through the owner's or consumer's premises, a conductor having an actual breaking strength of not less than 150 kgf may be used.
CEA Regulation 58 (replaces IE Rule 75)	**Joints. –** (1) No conductor or earth wire of an overhead line shall have more than one joint in a span: Provided that there shall be no joints in the conductor or earth wire in a span of crossing over the highways, expressways and railway lines. (2) The joint between conductors or earth wires of overhead lines shall be mechanically and electrically secured under the conditions of operation and the ultimate strength and the electrical conductivity of the joint shall be as per relevant standards.
CEA Regulation 59 (replaces IE Rule 76)	**Maximum stresses and factors of safety. –** (1) The load and permissible stresses on the structural members, conductors and earth wire of self supporting steel lattice towers or steel monopole towers for overhead transmission lines shall be as per relevant standards. (2) Overhead lines not covered in sub-regulation (1) shall have the minimum factors of safety as per the table given below:-

Description	Minimum factor of safety
metal supports	1.5
mechanically processed concrete supports	2.0
hand-moulded concrete supports	2.5
wood supports	3.0

Regulation	Details
	(3) The minimum factor of safety shall be based on such load as may cause failure of the support to perform its function, assuming that the foundation and other components of the structure are intact. (4) The load shall be equivalent to the yield point stress or the modulus of rupture, as the case may be, for supports subject to bending and vertical loads and the crippling load for supports used as strut. (5) The strength of the supports of the overhead lines in the direction of the line shall not be less than one-fourth of the strength required in the direction transverse to the line. (6) The minimum factor of safety for stay-wires, guard-wires or bearer-wires shall be 2.5 based on the ultimate tensile strength of the wire. (7) The tension limit for conductor and earth wire shall be as per relevant standards. (8) For the purpose of calculating the factors of safety in sub-regulation (2), the following conditions shall be observed, namely: – (i) the maximum wind pressure shall be as specified in the relevant standards; (ii) for cylindrical bodies the effective area shall be taken as full projected area exposed to wind pressure; and (iii) the maximum and minimum temperatures shall be as specified in the relevant standards. (9) (9) Notwithstanding anything contained in sub-regulation (2) to (8), in localities where overhead lines are likely to accumulate ice or snow, the load and permissible stresses on the structural members, conductors and earth wire of self supporting steel lattice towers and steel monopole towers for overhead transmission lines shall be as per relevant standards and in accordance with the specifications laid down, from time to time, by the Appropriate Government by order in writing.
CEA Regulation 60 (replaces IE Rule 77)	**Clearance in air of the lowest conductor of overhead lines. –** (1) The minimum clearance above ground and across road surface of National Highway or Expressway or State Highway or other road or highest traction conductor of railway corridor or navigational or non-navigational river of the lowest conductor of an alternating current overhead line, including service lines, of nominal voltage shall have the values specified in Schedule VIII A. (2) .The minimum clearances regarding high voltage direct current line shall be as per Schedule VIII B.

Regulation	Details
	(3) .In case of Electric lines of 33 kV and below passing through the protected areas (National Parks, Wildlife Sanctuaries, Conservation Reserves, Community Reserves), Eco-sensitive zones around the protected areas and Wildlife Corridors, only underground cable shall be used. (4) No tower footing or structure of an overhead line of voltage 33 kV or above or high voltage direct current, shall be closer than twenty five metre from the edge of the right of way of a Petroleum or Natural Gas pipeline. (5) Wherever overhead line of voltage 33 kV or above or high voltage direct current intending to cross the right of way of a Petroleum or Natural Gas pipeline, the angle of crossing of the overhead line with respect to the pipelines shall preferably be at right angles and, in any case, the crossing angle shall not be less than seventy five degrees.
CEA Regulation 61 (replaces IE Rule 78)	**Clearance between conductors and trolley wires**. – (1) No conductor of an overhead line crossing a tramway or trolley bus route using trolley wires shall have less than the following clearances above any trolley wire, namely: – (i) lines of voltage not exceeding 650 V - 1.2 metre: Provided that where an insulated conductor suspended from a bearer wire crosses over a trolley wire the minimum clearance for such insulated conductor shall be 0.6 metre. (ii) lines of voltage exceeding 650 V up to and including 11000 V - 1.8 metre; (iii) lines of voltage exceeding 11000 V but not exceeding 33000 V - 2.5 metre; (iv) lines of voltage exceeding 33 kV - 3.0 metre. (2) In any case of a crossing specified in sub-regulation (1), whoever lays his overhead line later, shall provide the clearance between his own overhead line and the overhead line which will be crossed in accordance with the provisions of the sub-regulation (1): Provided that if the later entrant is the owner of the lower overhead line and is not able to provide adequate clearance, he shall bear the cost for modification of the upper line so as to comply with sub-regulation (1).

Regulation	Details
CEA Regulation 62 (replaces IE Rule 79)	**Clearance from buildings of lines of voltage and service lines not exceeding 650 V. –** (1) An overhead line shall not cross over an existing building as far as possible and no building shall be constructed under an existing overhead line. (2) Where an overhead line of voltage not exceeding 650 V passes above or adjacent to or terminates on any building, the following minimum clearances from any accessible point, on the basis of maximum sag, shall be observed, namely: – (i) for any flat roof, open balcony, varandah roof and lean-to-roof, – (a) when the line passes above the building, a vertical clearance of 2.5 metre from the highest point; and (b) when the line passes adjacent to the building, a horizontal clearance of 1.2 metre from the nearest point; (ii) for pitched roof, – (a) when the line passes above the building, a vertical clearance of 2.5 metre immediately under the line; and (b) when the line passes adjacent to the building, a horizontal clearance of 1.2 metre. (3) Any conductor so situated as to have a clearance less than that specified in sub-regulation (2) shall be replaced with Aerial Bunched Cable and to be attached at suitable intervals to a bare earthed bearer wire having a breaking strength of not less than 350 kgf. (4) The horizontal clearance shall be measured when the line is at a maximum deflection from the vertical due to wind pressure. (5) The vertical and horizontal clearances shall be measured as per illustration provided in Schedule VIII C. *Explanation.* – For the purposes of this regulation, the expression "building" shall be deemed to include any structure, whether permanent or temporary.
CEA Regulation 63 (replaces IE Rule 80)	**Clearances from buildings of lines of voltage exceeding 650 V. –** (1) An overhead line shall not cross over an existing building as far as possible and no building shall be constructed under an existing overhead line. (2) Where an overhead line of voltage exceeding 650 V passes above or adjacent to any building or part of a building it shall have on the basis of maximum sag a vertical clearance above the highest part of the building immediately under such line, of not less than, – (i) for lines of voltages exceeding 650 V and up to and including 33 kV - 3.7 metre;

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(ii) for lines of voltages exceeding 33 kV - 3.7 metre plus 0.30 metre for every additional 33 kV or part thereof.

(3) The horizontal clearance between the nearest conductor and any part of such building shall, on the basis of maximum deflection due to wind pressure, be not less than, –
(i) for lines of voltages exceeding 650 V and up to and including 11 kV - 1.2 metre;
(ii) for lines of voltages exceeding 11kV and up to and including 33 kV - 2.0 metre;
(iii) for lines of voltages exceeding 33 kV - 2.0 metre plus 0.3 metre for every additional 33 kV or part thereof.

(4) For high voltage direct current systems, the vertical and horizontal clearances, on the basis of maximum deflection due to wind pressure, from buildings shall be maintained as below:
<table>
<tr><th>Sl. No.</th><th>High Voltage Direct Current</th><th>Vertical Clearance (metre)</th><th>Horizontal Clearance (metre)</th></tr>
<tr><td>1.</td><td>100 kV</td><td>4.6</td><td>2.9</td></tr>
<tr><td>2.</td><td>200 kV</td><td>5.8</td><td>4.1</td></tr>
<tr><td>3.</td><td>300 kV</td><td>7.0</td><td>5.3</td></tr>
<tr><td>4.</td><td>400 kV</td><td>7.9</td><td>6.2</td></tr>
<tr><td>5.</td><td>500 kV</td><td>9.1</td><td>7.4</td></tr>
<tr><td>6.</td><td>600 kV</td><td>10.3</td><td>8.6</td></tr>
<tr><td>7.</td><td>800 kV</td><td>12.4</td><td>10.7</td></tr>
</table>
(5) The vertical and horizontal clearances shall be as measured as illustrated in Schedule VIII C.

Explanation. – For the purposes of this regulation, the expression “building” shall be deemed to include any structure, whether permanent or temporary.
</td></tr>
<tr><td>CEA Regulation 64

(replaces IE Rule 81)</td><td>Conductors at different voltages on same supports. –
Where conductors of different voltages are laid on the same supports, the owner shall make adequate provision to guard against danger to linemen and others, from the lower voltage system being charged above its nominal voltage, by leakage from or contact with the higher voltage and the methods of construction and the applicable minimum clearances between the conductors of the two systems shall be as provided in regulation 71.</td></tr>
</table>

Regulation	Details
CEA Regulation 65 (replaces IE Rule 82)	**Erection or alteration of buildings, structures, flood banks and elevation of roads. –** (1) If at any time subsequent to the erection of an overhead line, whether covered with insulating material or not or underground cable, any person proposes to erect a new building or structure or flood bank or to raise any road level or to carry out any other type of work whether permanent or temporary or to make in or upon any building, or structure or flood bank or road, any permanent or temporary addition or alteration, such person and the contractor whom he employs to carry out the erection, addition or alteration, shall give intimation in writing of his intention to do so, to the supplier or owner and to the Electrical Inspector and shall furnish therewith a scale drawing showing the proposed building, structure, flood bank, road or any addition or alteration and scaffolding thereof required during the construction. (2) On receipt of such intimation, the supplier or owner shall examine,- (a) whether the overhead line or underground cable under reference was laid in accordance with the provisions of these regulations and any other law for the time being in force; (b) whether it is technically feasible; (c) whether it meets the requirement of right of way; and (d) whether such person was liable to pay the cost of alteration of the overhead line or underground cable and if so, issue a notice within a period of thirty days to such person together with an estimate of the cost of the expenditure likely to be incurred to alter the overhead line or underground cable and require him to deposit, within thirty days of the receipt of the notice, with the supplier or owner, the amount of the estimated cost. (3) If such person disputes the cost of alteration of the overhead line or underground cable estimated by the supplier or owner or even the responsibility to pay such cost, the dispute may be referred to the Electrical Inspector who shall after hearing both parties decide upon the issue in accordance with sub-regulation (4). (4) The Electrical Inspector shall estimate the cost of alteration of overhead line or underground cable on the following basis, namely:– (a) the cost of material used for the alteration after accounting for the depreciated cost of the material of the existing line or underground cable; (b) the wages of labour employed in the alteration; and (c) the supervision charge to the extent of fifteen per cent of the wages mentioned in clause (b) and charges incurred by the owner or supplier or consumer in complying with the provisions of section 67 of the Act, in respect of alterations.

Regulation	Details
	(5) Any addition or alteration to the building or structure shall be allowed only after the deposit of such estimated cost to the supplier or owner. (6) No work upon such building, structure, flood bank, road and addition or alteration thereto shall commence or continue until the Electrical Inspector certifies that regulations 60, 62 , 63, 66 and regulation 79 have not been contravened either during or after the construction: Provided that the Electrical Inspector may, if he is satisfied that the overhead line or underground cable has been so guarded as to secure the protection of persons or property from injury, certify within fifteen days that the work may be executed prior to the alteration of the overhead line or underground cable or in the case of temporary addition or alteration, without alteration of the overhead line or underground cable. (7) The supplier or owner shall, on receipt of such deposit, alter the overhead line or underground cable in such a way that it does not contravene the regulations 60, 62, 63 and 79 either during or after such construction within two months from the date of such deposit or within such longer period as the Electrical Inspector may permit for reasons to be recorded in writing.
CEA Regulation 66 (replaces IE Rule 82-A).	**Transporting and storing of material near electric lines. –** (1) No rods, pipes or similar materials shall be taken below, or in the vicinity of any bare overhead conductors or lines: Provided that if these materials contravene the regulations 62 and 63, such materials shall be transported under the direct supervision of a person designated or appointed or engaged or permitted under these regulations. (2) No rods, pipes or other similar materials shall be brought within the flash over distance of bare live conductors or overhead lines. (3) No material or earth work or agricultural produce shall be dumped or stored, no trees grown below or in the vicinity of bare overhead conductors or lines in contravention to the regulations 62 and 63. (4) No flammable material shall be stored under the electric line. (5) No fire shall be allowed below overhead lines and above the demarcated underground cables.

Regulation	Details
CEA Regulation 67 (replaces IE Rule 83)	**General clearances. –** (1) For the purpose of computing the vertical clearance of an overhead line, the maximum sag of any conductor shall be calculated on the basis of the maximum sag in still air and the maximum temperature as specified under regulation 59 and computing any horizontal clearance of an overhead line the maximum deflection of any conductor shall be calculated on the basis of the wind pressure specified under regulation 59. (2) No blasting for any purpose shall be done within three hundred metre from the boundary of a substation or from the electric supply lines of voltage exceeding 650 V or tower structure thereof without the written permission of the owner of such substation or electric supply lines or tower structures; and in case of mining lease hold area, without the written permission of the Electrical Inspector of mines. (3) No cutting of soil within ten metre from the tower structure of 110 kV and above voltage level shall be permitted without the written permission of the owner of tower structure. (4) No person shall construct brick kiln or other polluting units near the installations or transmission lines of 220 kV and above within a distance of 500 metre.
CEA Regulation 68 (replaces IE Rule 84)	**Routes in proximity to airport or aerodromes. –** Overhead lines shall not be laid in the vicinity of airport or aerodromes unless the Airport Authorities or concerned defence authorities have approved in writing the route of the proposed overhead lines as per relevant standards.
CEA Regulation 69 (replaces IE Rule 85)	**Maximum interval between supports. –** All conductors shall be attached to supports at intervals not exceeding the safe limits based on the ultimate tensile strength of the conductor and the factor of safety specified under regulation 59: Provided that in the case of overhead lines carrying conductors of voltage not exceeding 650 V when laid over, along or across any street, the interval shall not, without the consent in writing of the Electrical Inspector, exceed sixty five metre.
CEA Regulation 70 (replaces IE Rule 86)	**Conditions to apply where telecommunication lines and power lines are carried on same supports. –** (1) Every overhead telecommunication line laid on supports of an electric line shall consist of conductors each having a breaking strength of not less than 270 kgf. (2) Every telecommunication line laid on supports of an electric line shall be protected against lightning as per relevant standards. (3) Where a telecommunication line is laid on supports of an electric line of voltage exceeding 650 V, a suitable arrangement shall be made to safeguard any person against injury resulting from contact, leakage or induction between such electric and telecommunication lines.

<table>
<tr><th>Regulation</th><th>Details</th></tr>
<tr><td>CEA
Regulation 71

(replaces
IE Rule 87)</td><td>
Lines crossing or approaching each other and lines crossing street and road. – Where an overhead line crosses or is in proximity to any telecommunication line, the owner of the overhead line or the telecommunication line, whosoever lays his line later, shall arrange to provide for protective devices or guarding arrangement as per code of practice or guidelines issued by Power and Telecommunication Coordination Committee and shall observe the following provisions, namely: –
(i) when it is intended to lay a telecommunication line or an overhead line which will cross or be in proximity to an overhead line or a telecommunication line, as the case may be, the person proposing to lay such line shall give one month notice of his intention so to do along with the relevant details of protection and drawings to the owner of the existing line;
(ii) guarding arrangement shall be provided where a line of voltage not exceeding 33 kV crosses a road or street or telecommunication line;
(iii) where an overhead line crosses or is in proximity to another overhead line, guarding arrangements shall be provided so as to safeguard against the possibility of their coming into contact with each other; and
(iv) where an overhead line crosses another overhead line, minimum clearances as provided in the table below shall be ensured:-
Minimum clearances in metre between lines crossing each other
<table>
<tr><th>Sl. No.</th><th>Nominal System Voltage (kV)</th><th>11-66 kV</th><th>110-132 kV</th><th>220 kV</th><th>400 kV</th><th>765 kV</th><th>1200 kV</th></tr>
<tr><td>1.</td><td>Below 11</td><td>2.44</td><td>3.05</td><td>4.58</td><td>5.49</td><td>7.94</td><td>10.44</td></tr>
<tr><td>2.</td><td>11-66</td><td>2.44</td><td>3.05</td><td>4.58</td><td>5.49</td><td>7.94</td><td>10.44</td></tr>
<tr><td>3.</td><td>110-132</td><td>3.05</td><td>3.05</td><td>4.58</td><td>5.49</td><td>7.94</td><td>10.44</td></tr>
<tr><td>4.</td><td>220</td><td>4.58</td><td>4.58</td><td>4.58</td><td>5.49</td><td>7.94</td><td>10.44</td></tr>
<tr><td>5.</td><td>400</td><td>5.49</td><td>5.49</td><td>5.49</td><td>5.49</td><td>7.94</td><td>10.44</td></tr>
<tr><td>6.</td><td>765</td><td>7.94</td><td>7.94</td><td>7.94</td><td>7.94</td><td>7.94</td><td>10.44</td></tr>
<tr><td>7.</td><td>1200</td><td>10.44</td><td>10.44</td><td>10.44</td><td>10.44</td><td>10.44</td><td>10.44</td></tr>
</table>
Provided that no guarding arrangement is required when an electric line of voltage exceeding 33 kV crosses over another line of 250 V and above voltage or a road or a tram subject to the condition that adequate clearances are provided between the lowest conductor of the line of voltage exceeding 33 kV and the top most conductor of the overhead line crossing underneath the line of voltage exceeding 33 kV and the clearances as stipulated in regulation 60 from the topmost surface of the road maintained;
</td></tr>
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<table>
<tr><th>Regulation</th><th>Details</th></tr>
<tr><td></td><td>
(v) where an overhead direct current line crosses another overhead line, minimum clearances as provided in the table below shall be ensured:-

Minimum clearances in metre between AC and DC lines crossing each other
<table>
<tr><th>Sl. No.</th><th>System Voltage AC/ DC</th><th>100 kV DC</th><th>200 kV DC</th><th>300 kV DC</th><th>400 kV DC</th><th>500 kV DC</th><th>600 kV DC</th><th>800 kV DC</th></tr>
<tr><td>1.</td><td>Below 11 kV AC</td><td>3.05</td><td>4.71</td><td>5.32</td><td>6.04</td><td>6.79</td><td>7.54</td><td>9.04</td></tr>
<tr><td>2.</td><td>11-66 kV AC</td><td>3.05</td><td>4.71</td><td>5.32</td><td>6.04</td><td>6.79</td><td>7.54</td><td>9.04</td></tr>
<tr><td>3.</td><td>110-132 kV AC</td><td>3.05</td><td>4.71</td><td>5.32</td><td>6.04</td><td>6.79</td><td>7.54</td><td>9.04</td></tr>
<tr><td>4.</td><td>220 kV AC</td><td>4.58</td><td>4.71</td><td>5.32</td><td>6.04</td><td>6.79</td><td>7.54</td><td>9.04</td></tr>
<tr><td>5.</td><td>200 kV DC</td><td>4.71</td><td>4.71</td><td>5.32</td><td>6.04</td><td>6.79</td><td>7.54</td><td>9.04</td></tr>
<tr><td>6.</td><td>300 kV AC</td><td>5.32</td><td>5.32</td><td>5.32</td><td>6.04</td><td>6.79</td><td>7.54</td><td>9.04</td></tr>
<tr><td>7.</td><td>400 kV AC</td><td>5.49</td><td>5.49</td><td>5.49</td><td>6.04</td><td>6.79</td><td>7.54</td><td>9.04</td></tr>
<tr><td>8</td><td>400 kV DC</td><td>6.04</td><td>6.04</td><td>6.04</td><td>6.04</td><td>6.79</td><td>7.54</td><td>9.04</td></tr>
<tr><td>9.</td><td>500 kV DC</td><td>6.79</td><td>6.79</td><td>6.79</td><td>6.79</td><td>6.79</td><td>7.54</td><td>9.04</td></tr>
<tr><td>10,</td><td>600 kV DC</td><td>7.54</td><td>7.54</td><td>7.54</td><td>7.54</td><td>7.54</td><td>7.54</td><td>9.04</td></tr>
<tr><td>11.</td><td>765 kV AC</td><td>7.94</td><td>7.94</td><td>7.94</td><td>7.94</td><td>7.94</td><td>7.94</td><td>9.04</td></tr>
<tr><td>12.</td><td>800 kV DC</td><td>7.94</td><td>7.94</td><td>7.94</td><td>7.94</td><td>7.94</td><td>9.04</td><td>9.04</td></tr>
<tr><td>13.</td><td>1200 kV AC</td><td>10.44</td><td>10.44</td><td>10.44</td><td>10.44</td><td>10.44</td><td>10.44</td><td>10.44</td></tr>
</table>
(vi) a person laying or proposing to lay a line which may cross or be in proximity with an existing line, shall provide arrangements on his own line or require the owner of the other overhead line to provide guarding arrangements as referred to in clause (iii) and (iv);

(vii) in all cases referred to in this regulation the expenses of providing the guarding arrangements or protective devices shall be borne by the person whose line was laid later;

(viii) where two lines cross each other, the crossing shall be made as nearly at right angles as the nature of the case admits and as near the support of the line as practicable, and the support of the lower line shall not be placed in the right of way of the upper line; and

(ix) the guarding arrangement shall be carried out by the owner of the support on which it is made and he shall be responsible for its proper maintenance
</td></tr>
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Regulation	Details
CEA Regulation 72 (replaces IE Rule 88)	**Guarding. –** (1) Where guarding arrangement is required under these regulations, the following shall be observed, namely: **–** (i) every guard wire shall be connected with the earth at each point at which its electrical continuity is broken; (ii) every guard wire shall have an actual breaking strength of not less than 635 kgf and if made of iron or steel, shall be galvanised; and (iii) every guard wire or cross-connected system of guard wires shall have sufficient current carrying capacity to ensure them rendering dead, without risk of fusing of the guard wire, till the contact of any live wire has been removed.
CEA Regulation 73 (replaces IE Rule 89)	**Service lines from overhead lines. –** No service line or tapping shall be taken off an overhead line except at a point of support: Provided that the number of tappings per conductor shall not be more than six in case of connections at voltage not exceeding 650 V.
CEA Regulation 74 (replaces IE Rule 90]	**Earthing. –** (1) Earthing of support of overhead lines up to 33 kV shall be provided as under, namely: **–** (a) all metal supports and all reinforced and pre-stressed cement concrete supports of overhead lines and metallic fittings attached thereto, shall be permanently and effectively earthed by providing a continuous earth wire and securely fastened to each pole and connected with the earth at each support and the metallic fittings attached thereto shall also be permanently and effectively earthed; (b) metal cross arms and insulator pins for Plain Cement Concrete and Pre-Stressed Cement Concrete poles shall be bonded together and normally earthed at every pole; (c) for locations involving railways, electric line crossings and special structures, pipe or rod type earthing shall be provided; (d) all steel poles on which switches, transformers, fuses are mounted shall be earthed; (e) for poles of the electric lines below 650 V guarding arrangement with continuous earth wire or messenger wire in case of aerial bunched cable shall be provided and shall be connected to earth at three equidistant points in every km; and (f) each stay-wire shall be similarly earthed unless insulator of same voltage class as that of line has been placed in it at a height not less than three metre from the ground and shall be provided with insulated tube up to three metre height from the ground.

Regulation	Details
CEA Regulation 75 (replaces IE Rule 91(3))	**Anti-climbing devices. –** The owner of every overhead line of voltage exceeding 650 V shall make adequate arrangements as per relevant standards to prevent unauthorised persons from climbing any of the supports of such overhead lines which can be easily climbed upon without the help of a ladder or special appliances: Provided that the barbed wires conforming to relavant standards for a vertical distance of 30 to 40 cm, at a height of 3.5 metre to 4 metre from ground level or clamps with protruding spikes at a height of 3 to 4 metre shall be provided on each pole or tower of 11 kV line and above. *Explanation.* – For the purposes of this regulation, rails, reinforced cement concrete poles and pre-stressed cement concrete poles without steps, tubular poles, wooden supports without steps, I-sections and channels shall be deemed as supports which cannot be easily climbed upon.
CEA Regulation 76 (replaces IE Rule 91)	**Safety and protective devices. –** (1) Every overhead line which is not being suspended from a dead bearer wire, not being covered with insulating material and not being a trolley-wire, is laid over any part of a street or other public place or in any factory or mine or on any consumers premises shall be protected with earth guarding for rendering the line electrically harmless in case its conductor breaks. (2) An Electrical Inspector may, by notice in writing, require the owner of any such overhead line, wherever it may be laid, to protect it in the manner specified in sub-regulation (1) of this regulation. (3) To prevent bird dropping on the suspension insulator strings, suitable bird guards as per relevant standards, shall be provided on cross arms of suspension tower or suspension pole structures, over the suspension insulator strings.
CEA Regulation 77 (replaces IE Rule 92)	**Protection against lightning. –** (1) The owner of every overhead line, substation or generating station which is exposed to lightning shall adopt means as per relevant standards for diverting electrical surges to the earth due to lightning which may result into injuries. (2) The earthing lead for any lightning arrester shall be as short as possible and shall not pass through any iron or steel pipe, but shall be taken as directly as possible from the lightning arrester without touching any metal art to a separate vertical earth electrode or junction of the earth mat already provided for the substation of voltage exceeding 650 V subject to the avoidance of bends wherever practicable: Provided that a vertical earth electrode shall be connected to the junction of the earth mat.

Regulation	Details
CEA Regulation 78 replaces IE Rule 93)	**Unused overhead lines. –** Where an overhead line ceases to be used as an electric supply line, namely: **–** (i) the owner shall maintain it in a safe mechanical condition in accordance with regulation 59 or remove it; and (ii) the Electrical Inspector shall, by a notice in writing served on the owner, require him to maintain it in a safe mechanical condition or to remove it within thirty days of the receipt of the notice.
CEA Regulation 79	**Laying of cables**. – (1) No underground power cable of voltage level up to 33 kV shall be laid without a minimum underground depth from ground surface to top of the cable as per relevant standards: Provided that the underground power cable meant for use exceeding 33 kV shall be laid with a minimum underground depth of 1.2 metre. (2) No underground telecommunication cable shall be laid without a minimum separation distance of 0.3 metre and 0.6 metre to the underground power cable of voltage up to 33 kV and exceeding 33 kV, respectively.
CEA Regulation 80	**Protection against electromagnetic interference. –** The owner of every electric supply line of voltage level 11 kV or above shall obtain the clearance of Power Telecommunication Co-ordination Committee to ensure the safety of the personnel and telecommunication line as per the requirement of section 160 of the Act.

Additional safety requirements for electric vehicle charging station

Regulation	Details
CEA Regulation 124	**General safety requirements for electric vehicle charging station.–** (1) Electric vehicle charging stations shall be provided with separate protection against the overload of input supply and output supply as per relevant standards. (2) Socket-outlet of supply of electric vehicle charging points shall be installed at least 800 millimetre above the finished ground level. (3) A cord extension set or second supply lead shall not be used in addition to the supply lead for the connection of the electric vehicle to the charging point and shall not be used as a cord extension set. (4) No adaptor shall be used to connect a vehicle connector to a vehicle inlet. (5) The distance between the charging point and the connection on the electric vehicle shall not be more than five metre during charging. (6) The portable socket outlets shall not be permitted for electric vehicle charging.

Regulation	Details
	(7) The lightning protection system shall be provided for the electric vehicle charging stations as per relevant standards. (8) The electric vehicle charging station shall be equipped with a protective device against the uncontrolled reverse power flow from electric vehicle to the charging point. (9) One second after disconnecting the electric vehicle from the supply mains, the voltage between accessible conductive parts or any accessible conductive part and earth shall be less than or equal to 42.4 V_{peak} (30 V_{rms}), or 60 V DC, and the stored energy available shall be less than 20 J: Provided that, if the voltage is greater than 42.4 V_{peak} (30 V_{rms}) or 60 V DC, or the energy is 20 J ormore, a warning label shall be attached in a conspicuous position on the charging stations. (10) A vehicle connector used for direct current charging shall be locked on the vehicle inlet if the voltage is higher than 60 V DC and in case of charging system malfunction, a means for safe disconnection shall be provided. (11) The electric vehicle charging point shall disconnect supply of electricity to prevent overvoltage at the battery, if the output voltage exceeds maximum voltage limit permissible for the vehicle. (12) The electric vehicle charging points shall not energise the charging cable when the vehicle connector is in unlock position. (13) The electric vehicle connector shall not unlock if the voltage between the vehicle connector and the earth is more than 60 V. (14) Safety clearance between the oil or gas dispenser and electric vehicle charging point shall be as per the order issued by the Authority. (15) Only four core cable shall be used for charging points which require three phase power supply. (16) Underground cables shall not cross the underground oil tank or oil pipeline. (17) Underground cables through the charging area or vehicles passage shall be avoided and if provided shall be at a minimum depth of one metre from the finished ground surface.
CEA Regulation 125	**Earth protection system for the charging station. –** (1) Each electric vehicle charging points shall be supplied individually by a dedicated sub-circuit protected by an overcurrent protective device complying with the relevant standards and the overcurrent protective device shall be part of a switchboard. a) Co-ordination of all protective devices in the charging stations shall be ensured. b) All electric vehicle charging stations shall be provided with an earth continuity monitoring system that disconnects the supply in the event of the earthing connection to the vehicle becomes ineffective.

Regulation	Details
	c) The charging lead shall be fitted with an earth-connected metal shielding and the cable insulation shall be wear resistant and maintain flexibility over the operating temperature range. d) A protective earth conductor shall be provided to establish an equipotential connection between the earth terminal of the supply and the conductive parts of the vehicle which shall be as per the relevant standards.
CEA Regulation 126	**Requirement to prevent fire for electric vehicle charging station. –** (1) The enclosure of electric vehicle supply equipment shall be made of fire retardant material with self-extinguishing property and free from halogen. (2) The fire detection, alarm and control system shall be provided as per relevant standards.
CEA Regulation 127	**Testing of charging station. –** The owner of the charging station shall ensure that the tests as specified in the manufacturer's instructions for the residual current devices and the charging station have been carried out.
CEA Regulation 128	**Maintenance of records. –** (1) The owner of the charging station shall keep records of design, construction and labelling to be compatible with a supply of standard voltage at a nominal frequency of 50 Hz of the charging station. (2) The owner of the charging station shall keep records of the relevant test certificates as indicated in these regulations and as per the relevant standards. (3) The owner of the charging station shall keep records of the results of every inspection, testing and periodic assessment and details of any issues observed during the assessment and any actions required to be taken in relation to those issues. (4) The owner of the charging station shall retain a copy of all records, as specified in sub-regulation (1), (2) and (3) of above, either in hard form or in electronic form, for at least seven years and shall provide a copy of the records to the officials during the inspection.

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WBSEDCL

ANNEXURE-II

THE OCCUPATIONAL SAFETY, HEALTH AND WORKING CONDITIONS CODE, 2020

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THE OCCUPATIONAL SAFETY, HEALTH AND WORKING CONDITIONS CODE, 2020

An Act to consolidate and amend the laws regulating the occupational safety, health and working conditions of the persons employed in an establishment and for matters connected therewith or incidental thereto.

It shall come into force on such date as the Central Government may, by notification appoint; and different dates may be appointed for different provisions of this Code and any reference in any such provision to the commencement of this Code shall be construed as a reference to the coming into force of that provision..

Summary:

Key Provisions Relevant to WBSEDCL Safety Manual

1. Employer responsibility:

Employers are required to ensure the health, safety, and welfare of workers by:

- Providing and maintaining safe plants, systems, and places of work.
- Ensuring safety and removing risks to health during work.
- Offering information, instructions, training, and supervision as necessary.
- Implementing preventive and protective measures.
- Providing health surveillance where required.

2. Duties of Employee at workplace as per clause no. 6 in section 13 of Ch.III

- Take reasonable care for their own health and safety requirements specified in the standards.
- Cooperate with the employer in meeting the employer's statutory obligations under this Code.
- Report any unsafe or unhealthy situations to the employer, health and safety representative, or relevant safety officers as soon as practicable.
- Proper Use of Safety Appliances- Not wilfully hamper any appliance for securing health, safety, and welfare.

3. General right to Information and Representation of Employee as per clause no. 6 in section 14 (1-4) of Ch.III

- About health and safety at work & inadequate safety or health provisions
- Upon being notified of imminent danger, the employer or the concerned employee must take immediate remedial action if they are satisfied with the existence of the danger.
- Pay a lump sum amount for journey fare to the worker's native place annually.
- The appropriate government shall make schemes to provide— option to an inter-state migrant worker for availing benefits of public distribution system either in his native state or the destination state where he is employed.
- Government may conduct studies on inter-state migrant workers as prescribed.

4. Occupational safety & health standard

- All HIRA to be done- apprising the hazard to employees and users
- Related to specific symptoms – treatment
- Monitoring exposure to hazard by employees
- Necessary medical test by employer or at cost.
- Safety audit, HAZOP, FTA, etc.

5. Safety and occupational health survey

- Authority may do survey informing employer- afterward he will submit report to Government

6. Condition of safety committee with equal representation

- Appointment of safety officer(s)

7. Responsibility of employer for maintaining health, safety & working condition should be as per clause no. 23, of Ch.V.

- Cleanliness, hygiene, potable drinking water, lighting, toilets, waste disposal, etc.

8. Welfare facilities in the establishment, to be done by the employer as per clause no. 24 of Ch.VI.

- Washing, bathing place, locker room, place for clothing, siting, canteen > 150 employees
- Adequate first aid boxes, ambulance room > 500
- Medical facilities, halt station, uniform, raincoats, protection from rain, cold, etc. Night halt room > 50
- Appointment of welfare office > 250
- Creche > 50 working women, for children < 6 yrs.

9. Safety Committees:

The Code provides for the establishment of safety committees in establishments where directed by the appropriate government. These committees must include representatives from both management and workers, with the responsibility to review safety policies, investigate accidents, and recommend preventive measures. The committees are required to meet at regular intervals, maintain detailed minutes, and report findings to senior management. Their role includes conducting safety audits, addressing worker grievances related to safety, and ensuring the implementation of corrective actions to enhance workplace safety standards.

10. Emergency Preparedness

The Code mandates the development of comprehensive onsite emergency plans to address potential accidents, including fire outbreaks, electrical failures, or structural collapses. Employers must ensure the availability of first aid facilities, trained personnel, and emergency evacuation procedures. Accidents must be reported promptly to the appropriate authorities, and employers are required to provide immediate medical attention to injured workers. Regular drills and maintenance of emergency equipment, such as fire extinguishers and alarms, are essential to ensure readiness.

11. Training and Awareness

Safety training is a compulsory requirement under the Code for all employees and contractors, covering topics such as the safe use of equipment, hazard identification, and emergency procedures. Training programs must be conducted at regular intervals and tailored to the specific risks associated with the workplace. Employers are also encouraged to promote safety awareness through posters, manuals, and educational sessions, fostering a culture of safety consciousness among the workforce.

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ANNEXURE-III

POLYCHLORINATED BIPHENYLS (PCB) REGULATIONS, 2016

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MINISTRY OF ENVIRONMENT, FORESTS AND CLIMATE CHANGE NOTIFICATION

New Delhi, the 6th April, 2016

S.O. 1327(E).—Whereas, the Central Government had, in exercise of powers conferred under sub-section (1) of section 3 and clause (d) of sub-section (2) of section 6 of the Environment (Protection) Act, 1986 (29 of 1986), read with sub-rule (2) of rule 13 of the Environment (Protection) Rules, 1986, published the draft order on Regulation of Use, Handling and Disposal of Polychlorinated Biphenyls in the Gazette of India, Extraordinary, *vide* number S.O.219(E), dated the 12th August, 2015 for information of all persons and organisations likely to be affected thereby; and notice was given that the said draft notification would be taken into consideration by the Central Government on or after the expiry of a period of sixty days from the date on which copies of the Gazette containing this notification are made available to the public;

And whereas, the copies of the said notification were made available to the public on the 17th Sepemtember,2015;

And whereas, the suggestions or objections received in response to the above mentioned draft notification have been duly considered by the Central Government;

Now, therefore, in exercise of the powers conferred under section 3 and section 6 of the Environment (Protection) Act, 1986 (29 of 1986), read with rule 13 of the Environment (Protection) Rules, 1986, the Central Government hereby issue the following Order, namely:-

ORDER

1. (1) **Short title and commencement.–** This Order may be called, the Regulation of Polychlorinated Biphenyls Order, 2016.

   (2) It shall come into force on the date of its final publication in the Official Gazette.

2. **Manufacture, Import and ban of Polychlorinated Biphenyls.-** The manufacture and import of the Polychlorinated Biphenyls in India shall be banned from the date of final publication of this Order.

3. **Ban on Import of Polychlorinated Biphenyls containing equipment.-** The import of Polychlorinated Biphenyls containing equipment shall be banned from the date of final publication of this Order.

4. **Import, export or trade of Polychlorinated Biphenyls contaminated equipment.-** The import, export or trade of Polychlorinated Biphenyls contaminated equipment shall be regulated as per the provisions of the Hazardous Wastes (Management, Handling and Transboundary Movement) Rules, 2008.

5. **Prohibition of Polychlorinated Biphenyls**.- The use of Polychlorinated Biphenyls in any form shall be completely prohibited by 31st December, 2025.

6. **Safety measures.-** The use of Polychlorinated Biphenyls containing equipment shall be permitted for their certified life time or 31st December 2025, whichever is earlier, provided that they are maintained properly without possibility of leakage or release of Polychlorinated Biphenyls to environment.

7. **The number, quantity and equipment, etc.-** The occupier shall declare the total quantity of Polychlorinated Biphenyls, the number of Polychlorinated Biphenyls containing equipment and Polychlorinated Biphenyls contaminated equipment which are in use and their stockpiles, as per Polychlorinated Biphenyls inventory form as per Form 'A', to the Ministry of Environment, Forest and Climate Change, Government of India, within one year from the date of final publication of this Order.

8. **Precaution.-** The occupier shall not drain or discharge Polychlorinated Biphenyls directly or indirectly on land, in surface water or effluent treatment plant from defective, out of use Polychlorinated Biphenyls containing or contaminated equipment or in use Polychlorinated Biphenyls containing equipment.

9. **Disposal of Polychlorinated Biphenyls.-** The waste Polychlorinated Biphenyls or Polychlorinated Biphenyls contaminated equipment shall be disposed of as per the provisions of the Hazardous Wastes (Management, Handling and Transboundary Movement) Rules, 2008 by 31st December, 2028. Stockpiles of Polychlorinated Biphenyls if any shall be destroyed in environment beneficial manner by 31st December, 2028.

10. **Use of Polychlorinated Biphenyls for scientific purpose**.- Notwithstanding anything contained in the foregoing paragraphs, Polychlorinated Biphenyls may be used, sold and imported in quantities as required for research and development activities in Central Universities, Council of Scientific and Industrial Research Laboratories, Government Institutions or Central Power Research Institute after the concurrence of **the Ministry** of Environment, Forest and Climate Change, Government of India.

Explanation. **- For the purpose of this notification.-**

(a) Polychlorinated Biphenyls containing equipment shall mean any equipment that contains 0.05 per cent or more Polychlorinated Biphenyls by weight (i.e. Polychlorinated Biphenyls ≥ 500 mg/kg).

(b) Polychlorinated Biphenyls contaminated equipment shall mean any equipment that contains 0.005 per cent Polychlorinated Biphenyls by weight or more but less than 0.05 per cent. Polychlorinated Biphenyls by weight (i.e. 50 mg/kg ≤ Polychlorinated Biphenyls < 500 mg/kg).

(c) Polychlorinated Biphenyls contamination free equipment shall mean any equipment that contains Polychlorinated Biphenyls less than 0.005 per cent by weight (i.e. Polychlorinated Biphenyls < 50 mg/kg).

(d) A dielectric fluid or, mineral oil used in the power equipments (transformer, capacitor, etc.) shall continue to be used till 31st December, 2025 if, it contains Polychlorinated Biphenyls less than 0.005 per cent by weight (i.e. Polychlorinated Biphenyls < 50 mg/kg).

(e) If dielectric fluid or mineral oil used in power equipments contain Polychlorinated Biphenyls less than 2 mg/kg, it could be reprocessed, recycled or reused till 31st December, 2025.

(f) Polychlorinated Biphenyls waste means discarded materials or packaging that contains Polychlorinated Biphenyls or has been contaminated with Polychlorinated Biphenyls that are without any safe commercial, industrial, agricultural or economic usage.

[F. No. 22-3/2010 –HSMD]

BISHWANATH SINHA, Jt. Secy.

Form-A

DECLARATION FORM FOR POLYCHLORINATED BIPHENYLS

(See paragraph 7)

[Application by an occupier as per paragraph 7 of the Regulation of Polychlorinated Biphenyls Order, 2016.]

Date

From

_____

_____

_____

_____

To

HSM Division

The Ministry of Environment, Forest and Climate Change

Indira Paryavaran Bhawan

Jal Wing, Jor Bagh Road, New Delhi-110003 Sir,

I/We hereby disclose under paragraph 7 of the Regulation of Polychlorinated Biphenyls (PCBs) Order, 2016, the total quantity of PCBs, the number of PCBs containing equipment and PCBs contaminated equipment, which are in use, not in use or in my/our possession and their stockpiles as an occupier(s).

PART I

General

S. No.	Item details	Description
1 (a)	Name of Occupier	
(b)	Name and complete address, location, contact details of the unit (Telephone Nos., fax and e-mail).	
2.	Status of Occupier: Individual/Proprietary concern/ Partnership firm (whether registered or unregistered)/ Joint family concern/Hindu Undivided Family/Private Limited Company/Public Limited Company, Government Company: (State Government/Central Government/Union Territory/Public sector undertaking), Any other association or body.	

PART II

Information related to Polychlorinated Biphenyls, Polychlorinated Biphenyls containing equipment and Polychlorinated Biphenyls contaminated equipment in use and their stockpiles

S.No.	Item	Description
1.	Name of manufacturer(s) of PCBs and PCBs containing equipment : (in use)	
2.	Type (transformer, capacitor, etc.)	
3.	Year of manufacturing :	
4 (a)	Total quantity of Pure PCBs	
4 (b)	Number of PCBs containing equipment	
4 (c)	Number of PCBs contaminated equipment	
4 (d)	Total quantity of stockpiles of PCBs	
5.	Containers for PCBs	
(i)	Are containers leak–proof	❑ Yes ❑ No
(ii)	Is the place of storage of PCBs, PCBs contaminated equipments and PCBs contained equipments clearly marked	❑ Yes ❑ No
6	PCBS analysis performed.	❑ Yes ❑ No
6 (a)	If yes, method followed and when?	
7.	Address of factory (ies), premise(s) where PCBs, PCBs containing equipment and PCBs contaminated equipment is installed/stored (Survey No.,Khasra No., location as per revenue records of village/Town Tehsil, District Police Station, State, jurisdiction of the First-Class Magistrate).	

PART-III

Information on waste likely to contain Polychlorinated Biphenyls

8 (a)	Nature of wastes (e.g. transformer oil in drums, tankers or,reservoirs)	
(b)	Estimated quantity (in tonnes)	
(c)	Brief on how storage of PCBs waste is done	
(d)	Is the place of PCBs waste storage clearly marked	❑ Yes ❑ No

PART-**IV**

Polychlorinated Biphenyls elimination Action Plan

9 (a)	Action Plan for PCBs elimination prepared	❑ Yes ❑ No
	If yes, provide a copy along with time frame for elimination	
(b)	Brief history of any previous remediation efforts, e.g removal of PCBs containing equipment, PCBs contaminated equipment and PCBs waste for disposal (when, by whom,where to, etc.)	

PART-**V**

Other relevant information

Place, Date Name and signature of applicant

Note: 1. Strike out which is not applicable / relevant.

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ANNEXURE-IV

STANDARD CODES OF PRACTICE

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Standard Codes of Practice

Sl. No.	IS No.	Part	Reviewed In	IS Title
1	IS 18347:2023			Hazard Identification and Risk Assessment (Other Than Chemical Industries) - Code of Practice
2	IS 15656:2006		2023	Hazard Identification and Risk Analysis - Code of Practice
3	IS 14489:2018		2023	Occupational Health and Safety Audit - Code of Practice (First Revision)
4	IS 9457:2005		2021	Safety Colours and Safety Signs - Code of Practice (First Revision)
5	IS 9457:2025			Safety Colours Safety Signs and Accident Prevention Tags - Code of Practice (Second Revision)
6	IS 3043:2018			Code of Practice for Earthing (Second Revision)
7	IS 10028	(Part 1):1985	2021	Code of Practice for Selection, Installation and Maintenance of Transformers: Part 1 Selection
		(Part 2):1981	2021	Code of Practice for Selection, Installation and Maintenance of Transformers: Part 2 Installation
		(Part 3):1981	2018	Code of Practice for Selection, Installation and Maintenance of Transformers: Part 3 Maintenance
8	IS 10118	(Part 2):1982	2016	Code of Practice for The Selection, Installation and Maintenance of Switchgear and Controlgear: Part 2 Selection
		(Part 3):1982	2021	Code of Practice for Selection, Installation and Maintenance of Switchgear and Controlgear: Part 3 Installation
		(Part 4):1982	2016	Code of Practice for Selection, Installation and Maintenance of Switchgear and Controlgear: Part 4 Maintenance

Sl. No.	IS No.	Part	Reviewed In	IS Title
9	IS 2026	(Part 7):2009	2025	Power Transformers: Part 7 Loading Guide for Oil-Immersed Power Transformers
		(Part 8):2009	2025	Power Transformers: Part 8 Application Guide
		(Part 10/Sec 1): 2018	2023	Power Transformers Part 10 Determination of Sound Levels Section 1 Application Guide
		(Part 12):2018	2023	Power Transformers: Part 12 Loading Guide for Dry-Type Power Transformers
		(Part 19):2018		Power Transformers Part 19 Rules for The Determination of Uncertainties In The Measurement of The Losses On Power Transformers and Reactors
10	IS 18284:2023			Repair of Distribution Transformers - Code of Practice
11	IS 4146:1983		2024	Application Guide for Voltage Transformers (First Revision)
12	IS 4201:1983		2024	Application Guide for Current Transformer (First Revision)
13	IS 8061:1976		2024	Code of Practice for Design, Installation and Maintenance of Service Lines Up To and Including 650 V
14	IS 5613	(Part 1/ Sec 1):1985	2018	Code of Practice for Design, Installation and Maintenance of Overhead Power Lines: Part 1 Lines Up To and Including 11 kV Sec 1 Design (First Revision)
		(Part 1 /Sec 2):1985	2018	Code of Practice for Design, Installation and Maintenance of Overhead Power Lines: Part 1 Lines Up To and Including 11 kV Sec 2 Installation and Maintenance
		(Part 2 /Sec 1):1985	2018	Code of Practice Or Design, Installation and Maintenance of Overhead Power Lines: Part 2 Lines Above 11 kV and Up To and Including 220 kV: Sec 1 Design (First Revision)
		(Part 2 /Sec 2):1985	2018	Code of Practice for Design, Installation and Maintenance of Overhead Power Lines: Part 2 Lines Above 11 kV and Up To and Including 220 kV: Sec 2 Installation and Maintenance (First Revision)
15	IS 1255:1983		2016	Code of practice for installation and maintenance of power cables up to and including 33 kV rating (Second Revision)

Sl. No.	IS No.	Part	Reviewed In	IS Title
16	IS 14782:2000		2020	Code of Practice for Maintenance and Testing of Large Lead Acid Batteries for Generating Stations and Substations
17	IS 3961	(Part 2):2017	2024	Recommended Current Ratings for Cables: Part 2 PVC Insulated and PVC Sheathed Heavy Duty Cables (First Revision)
		(Part 6):2016	2024	Recommended Current Ratings for Cables Part 6 Crosslinked Polyethylene Insulated PVC Sheathed Cables
		(Part 7):2017	2024	Recommended Current Ratings for Cables: Part 7 Crosslinked Polyethylene Insulated Thermoplastic Sheathed Cables
18	IS 15429:2004		2019	Storage, Installation, and Maintenance of DC Motors - Code of Practice
19	IS 900:2019		2024	Code of Practice for Storage, Installation and Maintenance of Induction Motors (Third Revision)
20	IS 15707:2006		2016	Testing, Evaluation, Installation, and Maintenance of A.C. Electricity Meters - Code of Practice
21	IS 6711:1972		2020	Code of Practice for Maintenance of Wood Poles for Overhead Power and Telecommunication Lines
22	IS 7321:1974		2020	Code of Practice for Selection, Handling and Erection of Concrete Poles for Overhead Power and Telecommunication Lines
23	IS 4247	(Part 1):1993	2023	Structural Design of Surface Hydroelectric Power Stations: Part 1 Data for Design - Code of Practice (Third Revision)
		(Part 2):1992	2023	Code of Practice for Structural Design of Surface Hydroelectric Power Station: Part 2 Superstructure (Second Revision)
		(Part 3):1998	2023	Code of Practice for Structural Design of Surface Hydel Power Stations: Part 3 Sub-Structure (Second Revision)
24	IS 12967	(Part 1):1990	2025	Analysis of Hydraulic Transients In Hydro - Electric and Pumping Plants - Code of Practice: Part 1 Criteria for Analysis
25	IS 4461:1998		2023	Code of Practice for Joints In Surface Hydroelectric Power Stations (Second Revision)

Sl. No.	IS No.	Part	Reviewed In	IS Title
26	IS 4720:2020		2025	Ventilation of Surface Hydel Power Stations – Code of Practice (Second Revision)
27	IS 4721:2014		2024	Code of Practice for Drainage and Dewatering of Surface/ Underground Hydroelectric Power Stations (Second Revision)
28	IS 9859	(Part 1):2023		Code of Practice for Installation and Maintenance of Outdoor Power Line Carrier Equipment Part 1 Line Traps
		(Part 2):2023		Installation and Maintenance of Outdoor Power Line Carrier Equipment - Code of Practice Part 2 Coupling Devices (First Revision)
		(Part 3):2023		Installation and Maintenance of Outdoor Power Line Carrier Equipment -- Code of Practice Part 3 Coupling Capacitors/Capacitor Voltage Transformers (First Revision)
29	IS 2974	(Part 1):1982	2023	Code of Practice for Design and Construction of Machine Foundations: Part 1 Foundation for Reciprocating Type Machines (Second Revision)
		(Part 2):1980	2023	Code of Practice for Design and Construction of Machine Foundations: Part 2 Foundations for Impact Type Machines (Hammer Foundations) (First Revision)
		(Part 3):1992	2025	Code of Practice for Design and Construction of Machine Foundations: Part 3 Foundations for Rotary Type Machines (Medium and High Frequency)
		(Part 4):1979	2025	Code of Practice for Design and Construction of Machine Foundations: Part 4 Foundations for Rotary Type Machines of Low Frequency
		(Part 5):1987	2023	Code of Practice for Design and Construction of Machine Foundations: Part 5 Foundations for Impact Machines Other Than Hammer (Forging and Stamping Press, Pig Breaker, Drop Crusher and Jolter) (First Revision)
30	IS 1641:2013		2023	Fire Safety of Buildings (General): General Principles of Fire Grading and Classification - Code of Practice (Second Revision)
31	IS 1642:2013		2023	Fire Safety of Buildings (General): Details of Construction - Code of Practice (Second Revision)

Sl. No.	IS No.	Part	Reviewed In	IS Title
32	IS 1643:2013		2023	Fire Safety of Buildings (General): Exposure Hazard - Code of Practice (Second Revision)
33	IS 1644:2013		2023	Fire Safety of Buildings (General): Exit Requirements and Personal Hazard - Code of Practice (Second Revision)
34	IS 1646:2015		2020	Fire Safety of Buildings (General): Electrical Installations - Code of Practice (Third Revision)
35	IS 3034:1993		2022	Fire Safety of Industrial Buildings: Electrical Generating and Distributing Stations - Code of Practice (Second Revision)
36	IS 9109:2000		2021	Fire Safety of Industrial Buildings - Paint and Varnish Factories - Code of Practice (First Revision)
37	IS 8758:2013		2023	Fire Precautionary Measures In Construction of Temporary Structures and Pandals - Code of Practice (Second Revision)
38	IS 3594:2024			Fire Safety of General Storage and Warehousing Including Cold Storages - Code of Practice (Second Revision)
39	IS 12456:2004		2020	Fire Protection of Electronic Data Processing Installation - Code of Practice (First Revision)
40	IS 2189:2008		2023	Selection, Installation and Maintenance of Automatic Fire Detection and Alarm System Code of Practice (Fourth Revision)
41	IS 5896	(Part 1):1970	2020	Code of Practice for Selection, Operation and Maintenance of Special Fire Fighting Appliances: Part 1 Combined Foam and Cop Crash Tender
42	IS 14851:2000		2020	Maintenance of Fire Hose - Code of Practice
43	IS 15519:2020			Water Mist Fire Protection Systems - System Design, Installation and Commissioning - Code of Practice (First Revision)
44	IS 3844:1989		2020	Code of Practice for Installation and Maintenance of Internal Fire Hydrants and Hose Reels On Premises (First Revision)
45	IS 15301:2003		2023	Installation and Maintenance of Fire Fighting Pumps - Code of Practice

Sl. No.	IS No.	Part	Reviewed In	IS Title
46	IS 15105:2021			Design and Installation of Fixed Automatic Sprinkler Fire Extinguishing Systems- Code of Practice (First Revision)
47	IS 15908:2021			Selection, Installation, and Maintenance of Control and Indicating Equipment for Fire Detection and Alarm Systems - Code of Practice First Revision
48	IS 2190:2024			Selection, Installation and Maintenance of First-Aid Fire Extinguishers - Portable and Mobile - Code of Practice (Fifth Revision)
49	(Part 1):2021			Design and Installation of Fixed Foam Fire Extinguishing System - Code of Practice Part 1 Low Expansion Foam (First Revision)
50	IS 15496:2004		2020	Inspection and Maintenance of Gaseous Fire Extinguishing Systems - Code of Practice
51	IS 3016:1982		2021	Code of Practice for Fire Precautions In Welding and Cutting Operations (First Revision)
52	IS 13039:2014		2024	External Hydrant Systems - Provision and Maintenance - Code of Practice (First Revision)
53	IS 15325:2020			Design and Installation of Fixed Automatic High and Medium Velocity Water Spray System — Code of Practice (First Revision)
54	IS 17893:2023			Work Permit System - Code of Practice
55	IS 4648:1968		2025	Guide for Electrical Layout In Residential Buildings
56	IS 3362:1977		2023	Code of Practice for Natural Ventilation of Residential Buildings (First Revision)
57	IS 732:2019		2024	Code of Practice for Electrical Wiring Installations (Fourth Revision)
58	IS 11431:1985		2025	Code of Practice for The Packaging of Electrical Indicating and Recording Instruments
59	IS 1197:1970		2022	Code of Practice for Laying of Rubber Floors (First Revision)
60	IS 13367	(Part 1):1992	2025	Safe Use of Cranes - Code of Practice: Part 1 General
61	IS 3177:2020		2025	Electric Overhead Travelling Crane and Gantry Crane for All Applications — Code of Practice (Third Revision)

Sl. No.	IS No.	Part	Reviewed In	IS Title
62	IS 4014	(Part 1):1967		Code of Practice for Steel Tubular Scaffolding: Part 1 Definitions and Materials
		(Part 2):2013	2023	Steel Tubular Scaffolding - Code of Practice: Part 2 Safety Provisions for Scaffolding (First Revision)
63	IS 3973:2022			Code of Practice for The Selection, Installation, and Maintenance of Wire Ropes, Second Revision
64	IS 8940:1978		2018	Code of Practice for Maintenance and Care of Industrial Safety Equipment for Eye and Face Protection
65	IS 8990:1978		2023	Code of Practice for Maintenance and Care of Industrial Safety Clothing
66	IS 6519:2021			Code of Practice for Selection, Care and Repair of Safety Protective and Occupational Footwear First Revision
67	IS 15886:2010		2017	Road Vehicles - Battery Operated - Vehicles - Code of Practice
68	IS 1881:1998		2022	Code of Practice for Indoor Installation of Public Address Systems (First Revision)
69	IS 1882:1993		2022	Outdoor Installation of Public Address System - Code of Practice (First Revision)
70	IS 4453:2009		2025	Subsurface Exploration By Pits, Trenches, Drifts and Shafts - Code of Practice (Second Revision)
71	IS 4631:1986		2022	Code of Practice for Laying of Epoxy Resin Floor Toppings
72	IS 5572:2009		2024	Classification of Hazardous Areas (Other Than Mines) Having Flammable Gases and Vapours for Electrical Installation (Third Revision)
73	IS 4645:2024			Storage of Paper and Board -- Code of Practice (Second Revision)

Sl. No.	IS No.	Part	Reviewed In	IS Title
74	IS 7356	(Part 1):2002	2021	Code of Practice for Installation, Maintenance and Observation of Instruments for Pore Pressure Measurements In Earth Dams and Rockfill Dams - Part 1: Porous
		(Part 2):2003	2023	Installation, Observation and Maintenance of Instruments for Pore Pressure Measurements In Earth and Rockfill Dams - Code of Practice: Part 2 Twin Tube Hydraulic Piezometers (Second Revision)
75	IS 8629	(Part 13):1977	2021	Code of Practice for Protection of Iron and Steel Structures from Atmospheric Corrosion
76	IS/ISO/ IEC 27050-3:2020		2024	Information Technology - Electronic Discovery Part 3: Code of Practice for Electronic Discovery (First Revision)
77	IS/ISO/IEC 27018:2019		2023	Information Technology - Security Techniques - Code of Practice for Protection of Personally Identifiable Information (PII) In Public Clouds Acting as PII Processors (First Revision)
78	IS/ISO/IEC 27017:2015		2024	Information Technology - Security Techniques - Code of Practice for Information Security Controls Based On ISO/IEC 27002 for Cloud Services
79	ISO 45001		2024	Occupational Health and Safety Management Systems- Requirements with guidance for use

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ANNEXURE-V

WBSEDCL'S CIRCULARS AND ORDERS RELATED TO SAFETY

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WBSEDCL's circulars and orders related to safety are available on the website

Link: https://portal.wbsedcl.in/irj/go/km/docs/internet/new_website/Safety_Circular.html

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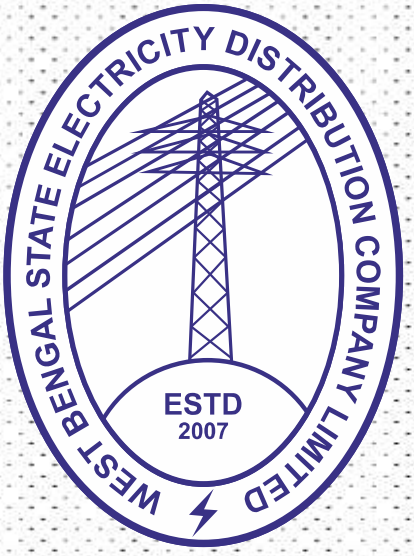
WEST BENGAL STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED
ESTD
2007

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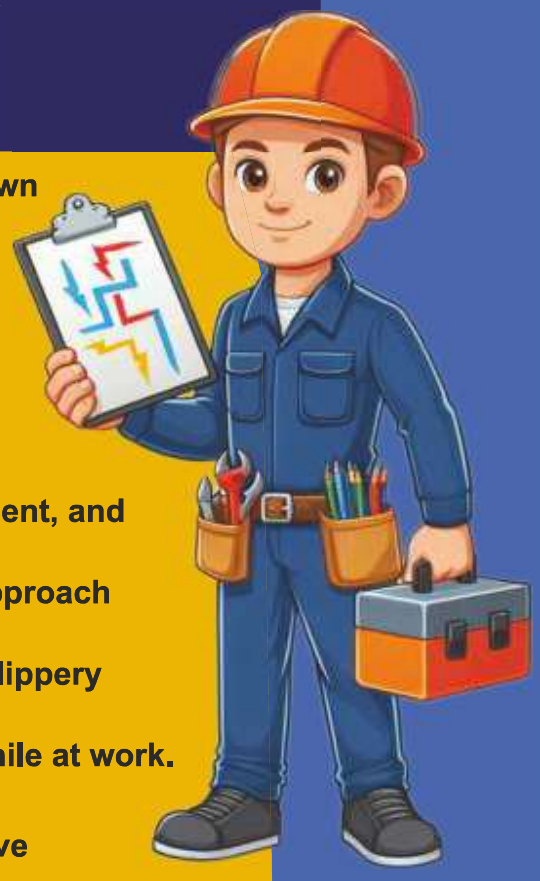
10 RULES FOR WORKPLACE SAFETY

1. You are responsible for your own safety and the safety of others.
2. All accidents are preventable.
3. Do not take shortcuts; always follow the rules.
4. If you are not trained, don't do it.
5. Use the right tools and equipment, and use them correctly.
6. Assess the risks before you approach your work.
7. Never wear loose clothing or slippery footwear.
8. Do not indulge in horseplay while at work.
9. Practice good housekeeping.
10. Always wear personal protective equipment (ppe).



West Bengal State Electricity Distribution Company Limited
Vidyut Bhavan, Bidhannagar, Kolkata-700091